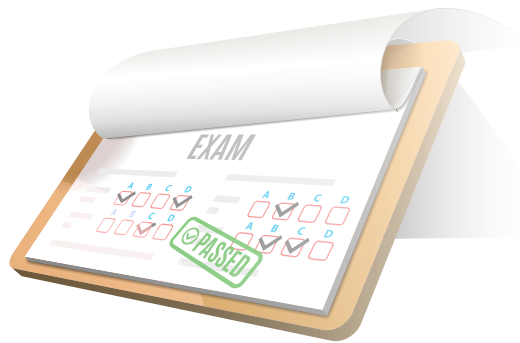




# NATIONAL SENIOR CERTIFICATE 2022

## DIAGNOSTIC REPORT PART 1: CONTENT SUBJECTS



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Department:  
Basic Education  
REPUBLIC OF SOUTH AFRICA



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# Foreword from the Minister

The Grade 12 Class of 2022 is extraordinary – a description reserved for only the remarkable!

Since the pandemic began in 2020, the well-being of young people enduring unprecedented challenges was one of the major concerns of the Department of Basic Education (DBE).

The year 2020, when the Class of 2022 faced great uncertainty in their Grade 10 year, is most certainly not a year on which we will look back with undiluted pleasure.

Yet the Class of 2022 forged ahead, and despite the myriad of challenges, these champions sat for the November 2022 National Senior Certificate (NSC) Examinations and have produced results that attest to resilience and fortitude.

As we reflect, we recall the words of President Nelson Mandela who reminded us that *‘everyone can rise above their circumstances and achieve success if they are dedicated to and passionate about*

*what they do’*. The Class of 2022 mustered the courage and they have proven to the South African nation that success can be a reality amidst great adversity.

The NSC remains one of the most important barometers of the health of the South African education system.

The DBE is in the process of integrating more holistic indicators of success across the various components of the system and one of them is the model for Systemic Evaluation, which will provide a more in-depth and comprehensive report on the performance of the system.



All indications prior to COVID, confirmed that we are a system on the rise. COVID has been a setback but the Class of 2022 affirmed that we are clawing back the gains made during the pre-COVID years.

The Class of 2022 were the beneficiaries of significant learner support programmes implemented to ensure that quality education is offered to all our learners. The Grade 12 class of 2022 benefited from differentiated targeted learner support programmes; such as extra tuition over weekends, vacation classes, the provision of additional learning and teaching support material (LTSM), revision guides, School Based Assessment (SBA) exemplars, broadcast lessons, online and radio lessons, content, and pedagogy workshops for teachers.

The Learning Recovery Plan of the DBE, which resulted in a trimming of the curriculum and a revision of the assessment programme for all grades, has allowed the system to focus on core content and skills and explore other forms of assessment, which are more relevant.

At the centre of the Learning Recovery Plan, was an assessment of the learning deficits and the development of learning programmes to address these deficits. The system has also over the years, established a strong data-driven focus of monitoring and feedback; and hence, there is a continuous identification of gaps and weaknesses in the system, and the implementation of initiatives to address these gaps.

The DBE is committed to a comprehensive curriculum strengthening process that will take into cognisance learners' competencies that they need to thrive socially and economically in a fast-changing world.

Efforts are afoot to infuse a skills-based focus into the existing curriculum. In 2022, Marine Sciences was offered for the second time as an examinable subject in the Grade 12 NSC Examinations. Various technical vocational specialisations, including Civil Technology, Electrical Technology, Mechanical Technology, Technical Mathematics and Technical Sciences were offered for the fifth year, since their introduction in 2018.

The ultimate intention is to make schools not only sites of academic achievement, but also sites of vocational and occupational skills development.

As a Department, we will work without ceasing to support candidates who do not obtain their NSC certificate after their first attempt. All provinces and districts will encourage these candidates to enrol for the Second Chance Matric Programme.

We are making the attainment of the NSC, with its vocational specialisations, a national endeavour.

No matter the failure or the poor performance, there is a second chance, as expressed in Sesotho, '*Leboela le a ja*' – '*perseverance pays*'.

The Executive Committee (EXCO) of the Umalusi Council concluded that the 2022 NSC Examinations were administered largely in accordance with the Regulations Pertaining to the Conduct, Administration and Management of the National Senior Certificate Examinations.

In view of this, the Quality Assurance Council, Umalusi, approved the release of the results for the October/November 2022 Examinations.

This approval was preceded by a holistic and comprehensive verification of all examination-related processes. Umalusi further confirmed that the irregularities identified during the writing and marking of the examinations were not systemic and therefore did not compromise the overall credibility and integrity of the October/November 2022 Examinations administered by the DBE.

The nation owes its gratitude to parents, teachers, principals, teacher unions, communities, district and provincial officials, and social partners for their commitment to supporting the Class of 2022.

The work we do in basic education has a multiplier effect. A country's economy becomes more productive, as the proportion of educated individuals increases.

I am confident that by working together yet again, we can support the Class of 2023 in the manner that we supported their predecessors.



**MRS AM MOTSHEKGA, MP**  
**MINISTER OF BASIC EDUCATION**  
**19 JANUARY 2023**



# CHAPTER 1

## INTRODUCTION

### 1.1 INTRODUCTION, SCOPE AND PURPOSE

The Class of 2022 entered the formal schooling system in Grade 1 in January 2011 and they became the largest cohort to sit for the NSC Examinations. The class has had the advantage of being the ninth cohort to be exposed to examinations based on the Curriculum Assessment Policy Statement (CAPS).

Hence, we can infer that this class benefitted from the developments of the last nine years relating to the CAPS, which would have created a high degree of stability in the system with regard to the teaching, learning and assessment. The last three years have been unprecedented, and this called for extraordinary measures to be implemented to ensure teaching, learning and assessment were maintained in schools.

Against this background, the Class of 2022 is possibly the worst affected class in relation to the effects of the Covid-19 pandemic. These candidates bore the brunt of COVID-19 in that they were confronted with disrupted academic years in Grades 10 and 11. There was a change in the assessment in Grades 10 and 11 with a greater focus on SBA, which constituted 60% of the promotion mark. Controlled tests replaced formal examinations in these grades. Subsequently, candidates sat for formal examinations only in Grade 12. Moreover, the impact of severe load-shedding cannot be discounted.

These learners are the beneficiaries of the educational transformation that heralds the move towards an innovative and technologically enhanced curriculum. The Class of 2022 is the third cohort to be subjected to a revised Section 4 of the CAPS, including revisions and amendments to examination guidelines.

In addition, this is the second cohort to sit for the newly introduced Marine Sciences and the third cohort to sit for two question papers in Accounting and Business Studies. The Class of 2022 is also the fifth cohort to offer South African Sign Language at Home Language level (SASL HL), Technical Mathematics, Technical Sciences, Civil Technology (Construction /Civil Services/Woodworking), Electrical Technology (Digital Systems/Electronic/Power Systems) and Mechanical Technology (Automotive/Fitting and Machining/Welding and Metal Work).

The policy on progression and promotion, which states that a candidate should not spend more than four years in a phase, has had a direct impact on the Class of 2022. Hence, candidates who did not meet the promotion requirements in either Grade 10 or Grade 11, would be progressed to Grade 12 on condition that they pass four of the seven subjects in Grade 11, inclusive of the Language of Learning and Teaching (LoLT), attend school on a regular basis and comply with SBA requirements in all subjects. This is the third cohort that is not subject to the policy on Multiple Examination Opportunity. This implies that all progressed candidates in this class had to sit for examinations in all six subjects in the November 2022 Examinations.

The 2022 Diagnostic Report delves into key observations in candidates' performance. Therefore, it serves as a primary resource for teaching and learning in the ten high enrolment subjects, Afrikaans First Additional Language, English First Additional Language, the twelve official home languages, the Technologies, Technical subjects and Engineering Graphics and Design. This report needs to be used in conjunction with the 2016 to 2021 diagnostic reports.

The seven reports extrapolate key subject didactic principles and content matter that can be used effectively in the classroom in 2023.

The 2022 diagnostic report is formulated on qualitative reports compiled by chief markers, internal moderators, and subject specialists post the NSC marking process. In the ten key subjects and English First Additional Language, quantitative data was gathered from the analysis of 100 scripts per question paper, per subject, randomly selected from each province. This qualitative and quantitative data highlighted the areas of weakness in each of the identified subjects and articulated the remedial measures which should be adopted at school level to improve performance in these subjects. As a result, this National Diagnostic Report on candidates' performance provides teachers, subject advisors, curriculum planners and curriculum implementers with a picture of weaknesses and strengths in each of the key subjects.

Each subject's diagnostic report commences by presenting comparative data on the performance trends observed over a five-year period in the subject, from 2018 to 2022. It also provides an overall performance of candidates per question, in the respective question papers, in each subject. In all subjects, common errors, misinterpretations and misconceptions identified during marking and suggestions for improvement are also outlined.

It cannot be ignored that content coverage was compromised due to the pandemic and that teaching methodology needs targeted intervention in 2023. It is imperative that Subject Advisory Services devise intervention measures to address recurrent areas of weakness. This diagnostic report serves as a source of valuable baseline information for intervention in 2023. This also suggests that continued reference to previous diagnostic reports is essential since the areas of weakness identified in previous years may still be applicable in certain cases.

## 1.2 METHODOLOGY

In the ten high-enrolment subjects and English First Additional Language, 100 scripts per question paper were randomly selected from each province during the marking process. These scripts included samples of low, medium and high achievement scores.

The internal moderators and chief markers analysed and noted candidates' responses to each question. This entailed recording the marks obtained by candidates from the 100 scripts on a per question basis. The individual scripts were scrutinised to provide an in-depth understanding of the range of different responses and to note the strengths and weaknesses. Particular attention was given to common errors and misconceptions identified in the candidates' responses.

Based on the analyses, a detailed explanation is provided per question/subquestion under the following three main titles:

### **Section 1: Performance trends (2018–2022)**

A comparative analysis of the performance of candidates over the last five years in terms of the number of candidates who wrote, the number and percentage of candidates who achieved at 30% and above, and the number and percentage of candidates who achieved at 40% and above, is presented in this section. The information is represented by tables and graphs to enable easier interpretation of any trends, especially on changes over the medium term, as well as changes from year to year.

Performance distribution curves are also provided to graphically present the distribution of candidates' scores in the last three examinations. Any improvement or decline in the

performance can be observed from the position of the 2021 graph, relative to previous years. If the 2022 graph lies to the right of the two previous graphs, this suggests an improvement in performance, while a slant to the left indicates a decline in performance.

## **Section 2: Overview of candidates' performance**

This section summarises the performance of candidates in the question paper as a whole. It makes reference to generic areas, the extent of good performance or weakness and the possible reasons for these observations.

## **Section 3: Diagnostic question analysis**

This includes the following:

- A graphical representation of the average percentage marks obtained per question;
- An analysis of the performance of learners in each specific question, stating whether the question was answered well or poorly (and the reason);
- Common errors and misconceptions that were identified in candidates' responses;
- Suggestions for improvement in relation to teaching and learning, content and methodology, subject advisory support and provision, and utilisation of LTSM.

The internal moderators' reports from the nine provinces for each question paper in each subject, were consolidated. The findings are summarised in this report. It is recommended that this report be read in conjunction with the November 2022 NSC question papers and marking guidelines since particular references are made to specific questions, in the respective question paper. This will enable teachers to establish a baseline for the new cohort of Grade 12 learners in 2023; develop strategies for differentiated learning and provide a frame of reference for the development and design of school-based assessment during the course of the year.

### **1.3 LIMITATIONS**

The diagnostic analysis of candidates' performance in this publication is limited only to the ten subjects with high Grade 12 enrolments, Afrikaans First Additional Languages, English First Additional Language and the twelve official home languages, the Technologies and Technical subjects. The remaining subjects will be covered in reports compiled by the provincial chief markers and internal moderators during the marking process. The DBE will endeavour to broaden the scope of the subject coverage in future.

It needs to be noted that areas of weakness could be unique to each district and each school. This report therefore provides a national summary of the general areas of weakness. However, district specialists are encouraged to develop a district diagnostic report. Ultimately, there should also be a school diagnostic report, which focuses specifically on the areas of weakness at school level.

While further quantitative data would have been useful in providing feedback for the purpose of test development, this is not the intention of this report. The focus of this report is more qualitative than quantitative. The quantitative aspects are limited to the performance trends in each subject and the average performance per question in the 2022 question papers.

## 1.4 GENERAL FINDINGS AND AREAS OF CONCERN

The 2022 diagnostic reports for the ten key subjects covered in this publication (Part 1), indicate that the pass rate has improved in six of these subjects (Accounting, Agricultural Sciences, Economics, Geography, Life Sciences, Mathematical Literacy and Physical Sciences) at the 30% levels. The pass rate has however declined to varying degrees at the 30% level in the remaining key subjects. The pass rate for English First Additional Language increased at both the 30% level and at the 40% level. In the home languages (Part 2) the pass rate remained the same in isiNdebele, Tshivenda and Xitsonga, improved to varying degrees in three home languages (English, isiZulu and Sepedi) and declined in the rest. The most notable decline in the pass rate at 40% was observed in South African Sign Language Home Language.

The following areas of concern were identified in past diagnostic reports and they are, once again, highlighted as concerns in the 2022 NSC Examinations.

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The following areas of concern were identified in past diagnostic reports and they are, once again, highlighted as concerns in the 2022 NSC Examinations.

- Teachers and subject advisors need to be fully familiar with the changes in the CAPS and the content of the 2021 Examination Guidelines. In some instances, it was observed that the new topics were not addressed in teaching and learning.
- It was evident in candidates' responses that there is a general lack of reading and comprehension skills. It was noted that many candidates did not understand the vocabulary used in questions, extracts and comprehension texts. Districts and schools are encouraged to initiate reading projects/reading periods/vocabulary/dictionary exercises to expand their learners' vocabulary. In addition, teachers are encouraged to expose learners to a wide array of texts to build their vocabulary, improve their comprehension skills and sharpen their critical thinking skills.
- A significant percentage of the 2022 candidates responded adequately to questions that required lower-order thinking skills. However, many candidates performed poorly in questions that demanded analytical, evaluative, and problem-solving skills. In view of this, teachers are encouraged to expose learners to a wide array of exercises that also include questions that assess higher order thinking skills.
- In most home languages, it was noted that candidates were not able to recognise the demands of abstract topics and higher order questions. There is therefore a need to encourage abstract thinking in languages. Challenging topics need to be included in the classroom and homework exercises to allow learners to get accustomed to employing the critical language skills required to think analytically and critically.
- In all languages, it was evident that many candidates were not aware of the various formats for transactional texts. It is advised that teachers revise the required formats

on a regular basis throughout the academic year. Informal writing must be taught as per the CAPS prescripts. Learners' writing skills will improve if informal writing activities become part of classroom and homework activities. This pertains to all grades and all subjects. A few suggestions to encourage writing in schools is to launch a letter writing competition, establish school newspapers and allow learners to write articles.

- There was a remarkable improvement in Accounting, Economics, Geography, Mathematical Literacy, Physical Sciences, Technical Sciences and Technical Mathematics but it can be deduced from weaker candidates' responses that focused intervention strategies need to be initiated and implemented to ensure that candidates have a solid understanding of the more complex topics.
- In 2022, performance declined significantly in SASL HL. It was evident that candidates were not familiar with the subject matter. Although dialectical skills are acknowledged, a worrying trend is that many candidates could not sustain a coherent conversation in SASL. Moreover, it was observed that candidates did not give sufficient responses in the writing papers and literature. It is imperative that literature needs to be studied in detail and candidates need to be accustomed to the register, style and conventions for transactional writing texts.

## 1.5 KEY RECOMMENDATIONS

### 1.5.1 Cognitive levels

Each assessment task and question paper must reflect the prescribed cognitive levels. School-based assessment tasks need to be in line with the CAPS requirements. Teachers are urged to:

- encourage discussion about what is being taught;
- ask learners to reflect on their experiences;
- help learners to find new solutions to problems;
- assist learners to explore and understand how ideas are connected;
- guide learners to substantiate their statements with solid arguments;
- use technology, visual demonstrations, practical activities and visualisation to improve learners' understanding of subject matter.

### 1.5.2 Examination Guidelines and the CAPS

It is imperative that teachers and subject specialists fully familiarise themselves with the changes introduced in the CAPS, and subsequently the 2021 Examination Guidelines. This requires teachers to be well informed of the new topics, format of question papers and restructured mark allocation affecting various topics.

### 1.5.3 Diagnostic Reports from 2015 to 2021

The diagnostic reports published from 2015 to 2021 are pertinent to gain a holistic grasp of candidates' performance and to identify weaknesses in the teaching and learning of the ten key subjects (Part 1). Part 2 of this diagnostic report, published for the first time in 2017, will serve as a teaching and learning tool in the language classroom, while Part 3 covers Technical subjects. Part 1, Part 2 and Part 3 must be used in preparing the Class of 2023 for the NSC Examinations.

#### 1.5.4 Past question papers

Teachers are discouraged from teaching to the paper. However, past question papers should be used as a teaching and learning resource and they serve as one of the resources for revision purposes. It must be stressed that the CAPS and the Examination Guidelines for each subject must be followed to ensure that all topics are covered. It must also be noted that in instances where the format of a question paper has changed, candidates need to be made aware of such changes.

#### 1.5.5 Language in teaching

Language Across the Curriculum (LAC) acknowledges the fact that language education does not only take place in specific subjects explicitly defined and reserved for it, such as the home languages, first additional languages and second additional languages. Language learning also takes place in each subject in school, whether teachers or learners are conscious of it or not.

In an effort to build learners' language proficiency and their confidence in decoding both the Language of Learning and Teaching (LoLT) and the language of assessment, teachers are encouraged to add their own language aspects, as these apply within the context of their schools or classrooms. The following points serve as a guide to teachers:

- It was observed in the marking of the October/November 2022 Examinations that candidates have a good grasp of subject matter and display an ability to respond appropriately to questions but are often restricted by inadequate language skills to formulate responses appropriately.
- Coupled with language skills, a lack of a solid understanding of examination techniques impacts negatively on candidates' performance. There needs to be greater emphasis on aspects of language competence and examination technique.
- In view of the point above, it is imperative that learners must have a firm understanding of action verbs that are used in the phrasing of questions. It is also essential that learners understand the meaning of each action verb in its context and in terms of the cognitive demand that is expected.
- Subject terminology and definitions must be clearly understood by learners. A firm understanding of subject matter can only be guaranteed if learners understand terminology and concepts used in the subject. It is suggested that a glossary of subject-specific jargon and their definitions is provided to learners.

#### 1.5.6 Integrated intervention strategies

Integrated intervention strategies are used to address gaps in teaching and learning. Such strategies could include:

- At the start of the academic year, teachers should provide their learners with the topics to be covered during the year and the relevant websites that they could explore, as part of independent reading.
- Online and virtual study groups are valuable in facilitating revision activities and examination preparations as they promote communication and conversations around subject content. Platforms such as *Microsoft Teams*, *Zoom*, *Google Classroom*, *YouTube* and *WhatsApp* could be used to good effect.

- Learners and teachers can gain access to online learning platforms such as *YouTube* that offer visual presentations and explanations of challenging topics. Teachers and subject specialists can source video clips and incorporate these in their lessons to give learners a clear understanding of subject matter.
- Learners who cannot log on to digital portals could be issued with DVDs/USB drives, with the information from various websites. Hard copies of the information should be provided to learners who do not have access to electricity and/or technology.
- Teachers from different schools in each circuit or district could collaborate to support one another in presenting challenging topics to learners. In view of this, different schools can build an item bank of higher order questions which can be used as a resource for revision purposes.
- Challenging topics must be revisited regularly during the course of the academic year through extension activities. These topics should form the basis of extra classes where stronger candidates can be paired with weaker candidates in completing more complex assignments.

## **1.6 RESPONSIBILITIES**

### **1.6.1 Provincial Education Departments**

Given that the target audience of this report includes the teacher and learner, this report must be cascaded from the provincial to the district level and finally to the school.

### **1.6.2 Subject Advisors and district officials:**

- Subject specialists should initiate a baseline assessment of the 2023 Grade 12 cohort, to establish learners' understanding of Grade 10 and Grade 11 topics that serve as foundation for Grade 12 topics.
- Subject advisers are encouraged to convene meetings/workshops that aim to mediate this diagnostic report. It is further suggested that the use of this diagnostic report must be encouraged during on-site support visits.
- The improvement plans of teachers need to be monitored, looking specifically for the inclusion of recommendations emanating from the individual subject reports.
- District officials should closely monitor curriculum coverage to ensure that all the topics in a subject have been covered according to the Revised Annual Teaching Plan (ATP). This would ensure that all topics receive due attention, allowing candidates to be better prepared for the examination.
- The monitoring process also needs to focus on the standard and quality of the assessment tasks used for SBA, as these tasks prepare learners for the NSC Examinations. They also provide an opportunity for the teaching and learning interventions to gain traction well before the NSC Examinations.
- Subject Advisers should direct teachers to websites that will enhance teaching and learning.

### **1.6.3 Teachers**

- Teachers should ensure that learners are provided with adequate resources to facilitate self-regulated learning. Learners must be in a position to study independently.
- In order to develop learners' holistic understanding and applied competence, teachers must prepare learners adequately by creating learning opportunities to reflect, analyse and evaluate the content.
- Teachers should ensure coverage of the curriculum and the full range of cognitive levels in their teaching and assessment strategies. The mere recall of procedures or specific content on the part of learners will not enable them to respond fully to the demands of question papers.

# CHAPTER 2

## ACCOUNTING

The following report should be read in conjunction with the Accounting Paper 1 and Paper 2 final examination papers for the NSC November 2022 examinations.

The Accounting examination focuses on two Accounting disciplines:

- Paper 1: Financial Reporting and Evaluation
- Paper 2: Managerial Accounting, Internal Auditing and Control

### 2.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who wrote the Accounting examination in 2022 decreased by 1 096 compared to that of 2021.

The table below indicates stability in the pass rate at 30% (Level 2) and at 40% (Level 3), particularly over the past three years.

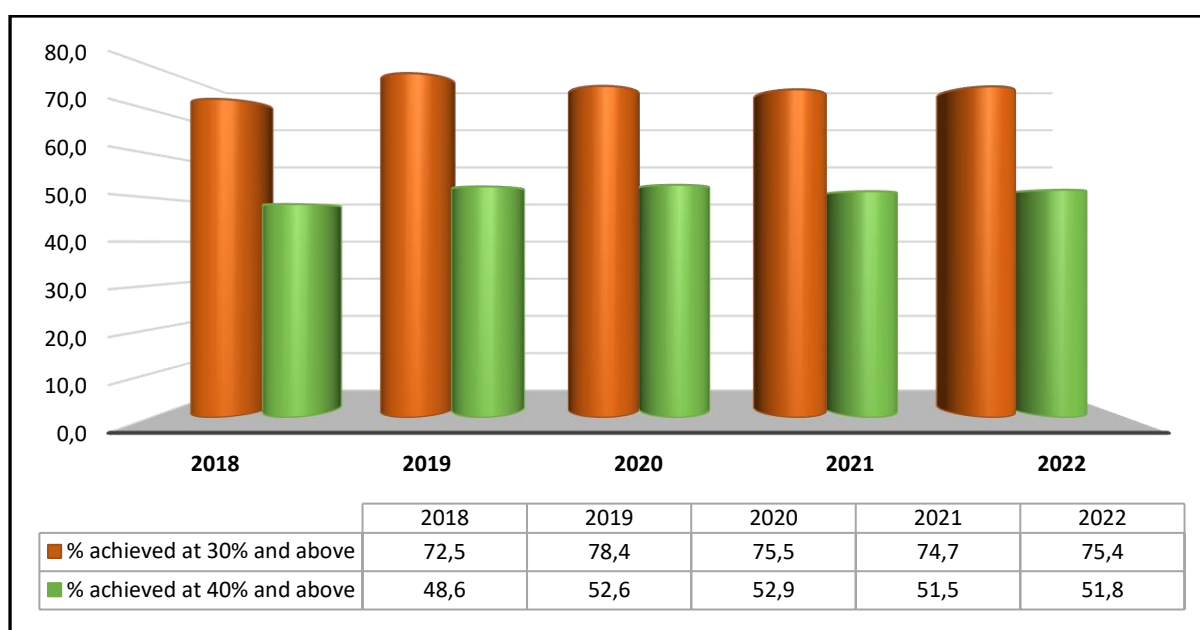
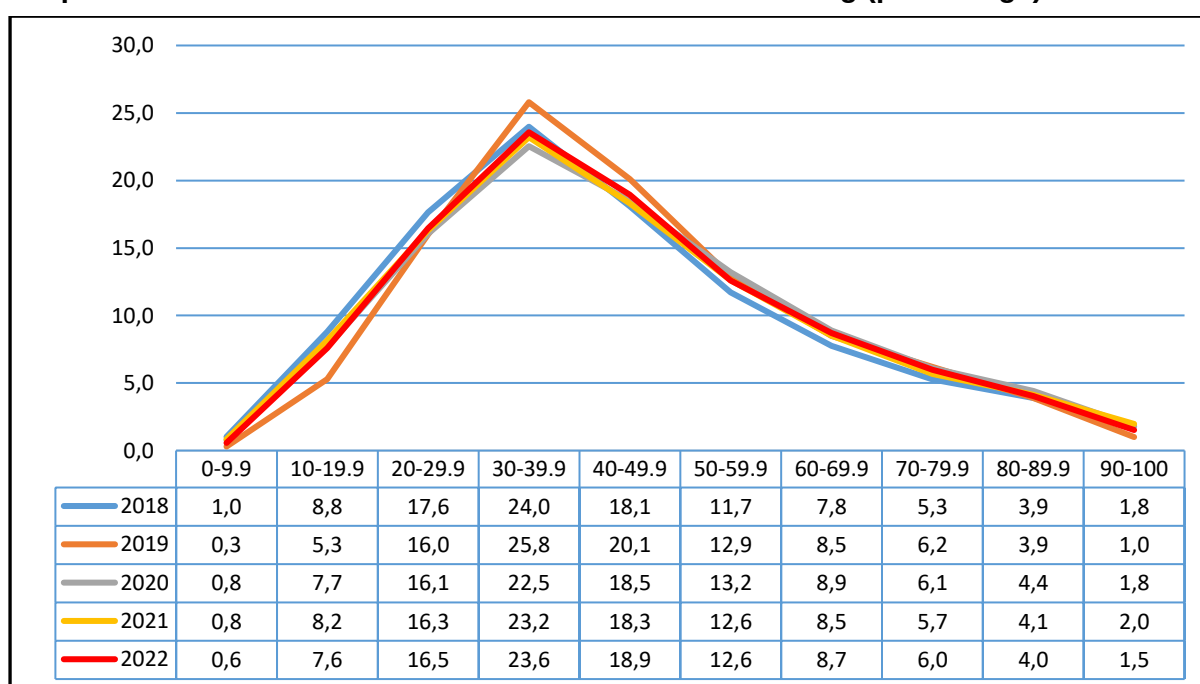
There was a marginal improvement in the pass rate this year. Candidates who passed at the 30% (Level 2) improved from 74,7% in 2021 to 75,4% in 2022. There was a corresponding improvement in the pass rate at the 40% (Level 3) over the past two years from 51,5% to 51,8%.

There was a marginal change in the percentage of distinctions over 80% (Level 7), which decreased from 6,1% in 2021 to 5,5% in 2022. Given the lower number of candidates in 2022, this converts into a total of 5 764 distinctions compared to 6 460 in the previous year.

In view of disruption to academic programmes in 2020 and 2021, and despite the slight decline in the number of distinctions, the results achieved by this cohort are commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the favourable overall performance.

**Table 2.1.1 Overall achievement rates in Accounting**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 90 278    | 65 481                        | 72,5                        | 43 831                        | 48,6                        |
| 2019 | 80 110    | 62 796                        | 78,4                        | 42 113                        | 52,6                        |
| 2020 | 92 767    | 70 014                        | 75,5                        | 49 103                        | 52,9                        |
| 2021 | 105 894   | 79 093                        | 74,7                        | 54 518                        | 51,5                        |
| 2022 | 104 798   | 78 993                        | 75,4                        | 54 291                        | 51,8                        |

**Graph 2.1.1 Overall achievement rates in Accounting (percentage)****Graph 2.1.2 Performance distribution curves in Accounting (percentage)**

## 2.2 OVERVIEW OF CANDIDATES' PERFORMANCE: PAPERS 1 AND 2

It was evident that the 2022 cohort had generally prepared themselves well for these papers, resulting in improved quality of responses across most questions. Extensive intervention initiatives and the reintroduction of June examinations in certain provinces, appeared to have benefited the candidates in their preparation and examination-writing techniques.

Despite the positive trend, some candidates appeared not to have managed their time effectively and struggled to complete certain subsections. The wide achievement gap between capable and weaker groups of candidates is still noticeable. This is possibly influenced by the

varying pace of adjusting to normal schooling and the extent of access to appropriate resources.

The general comments in this report include issues that surfaced from a detailed analysis of the 2022 Accounting NSC papers. Teachers are advised to refer to comments in the Diagnostic Reports of recent years, as the underlying issues continue to require attention.

Accounting is a dynamic subject that requires regular practice, financial insight and general knowledge of current trends in the business world. Teachers are therefore expected to keep up with new trends in the subject. Recent examples included the repurchase of shares, the introduction of online banking and the discontinuation of cheques which significantly affected bank reconciliations. A factor that is currently topical is that of *Corporate Governance*. In this regard, teachers are referred to the *2021 Diagnostic Report* and the *Examination Guidelines*, as well as the comments in this report relating to Q4 in Paper 1.

Teachers are also expected to productively use a variety of resources in supporting learners. These include the *CAPS*, revised teaching plans (*ATP*), the current *Examination Guidelines*, previous *Diagnostic Reports* and past NSC papers (particularly from 2015), in addition to study guides and online resources such as education websites, video presentations and updates from publishers.

#### **Factors covered in previous Diagnostic Reports that still remain relevant:**

- **Inability of weaker candidates to deal effectively with lower- and middle-order questions:** This limitation exposes weaknesses in the teaching and learning process as questions are structured to be accessible to all candidates, at least in part.
- **Ineffective formative assessment:** Regular formative tasks or tests must be skilfully used to build confidence in all topics. To be effective, these must be short, practical and aimed at achieving targeted outcomes. Feedback to learners is essential.
- **Lack of meaningful revision of relevant Grade 10 and 11 content:** Approximately 20% of an Accounting examination paper is devoted to integration of expected prior knowledge. In order to manage teaching time effectively, it is imperative that teachers factor revision of prior content into their teaching of Grade 12 topics.
- **Poor mathematical and arithmetical ability:** A thorough understanding of the logic of arithmetical calculations is a prerequisite for an Accounting learner. This skill includes the correct use of signs, rands and cents. Continued practice is the only solution to this issue.
- **Language barriers linked with poor comprehension skills:** These continue to be cited as a reason for poor performance of weaker candidates. This, and previous diagnostic reports, contain specific advice on preventing misinterpretation of question requirements and motivating learners to provide explanations or comments.
- **Inability to identify relevant information to answer specific subquestions:** Candidates are expected to engage with the information and extract what is needed to answer each subquestion. Weaker candidates find this challenging, and often use inappropriate information to support explanations.

#### **Further recommendations to improve exam-writing skills in Accounting:**

The following strategies can be time-consuming, but the benefits are very rewarding:

- **Interpreting questions and identifying evidence:**  
Questions often require figures or other evidence to be quoted from the information provided. This can be achieved by teachers analysing examination questions with learners at strategic times during the academic year, preferably after a topic or section. Learners should be advised to adopt this approach:

1. Read the instructions (under 'Required') and identify or underline key words.
2. Identify the relevant evidence in the source information provided.
3. Verbally explain what must be done to answer the question, in line with the mark allocation.

Such open forums would expose learners to views of others and assist in constructing valid responses to interpretive or open-ended questions.

- **Providing explanations:**

Answering skills must focus on explanations being concise and to the point. Unnecessary repetition of specific requirements of questions often leads to poor or incorrect responses. Explanations may be presented in bullet-point form. Lengthy explanations are time-consuming and might not earn additional marks. However, depending on the mark allocation, partial, simple or single-word responses may not always be sufficient. Learners should be cautioned that if several points are required, each point must be different and not a mere re-wording of previous items.

- **Use of the structured answer book:**

Learners must be reminded that the answer book should not be relied upon to replicate the question paper and that the spaces provided suggest the expected length of responses. This will enable candidates to manage their time for each question more effectively.

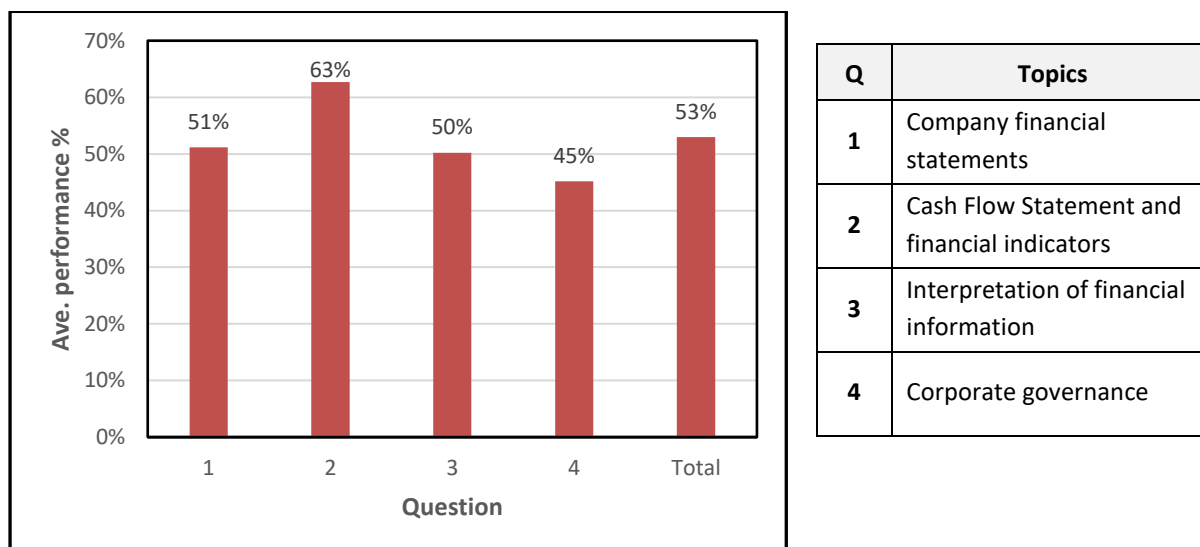
- **Ensuring effective time management:**

In the context of examinations, this remains a challenge for many learners. Learners must regularly practise the skill of adhering to the time allocations provided in the paper. A practical strategy is to divide past NSC papers into separate parts to be completed during a teaching period or under specific time constraints. Such activities, at regular intervals, should train learners to focus on the specific demands of each subquestion and to formulate efficient ways of expressing themselves without wasting time on unnecessary or irrelevant comments. Although teachers might advise learners to start with questions in which they feel most comfortable in accumulating marks, the process by the learner in deciding what sequence to follow must not create a time delay.

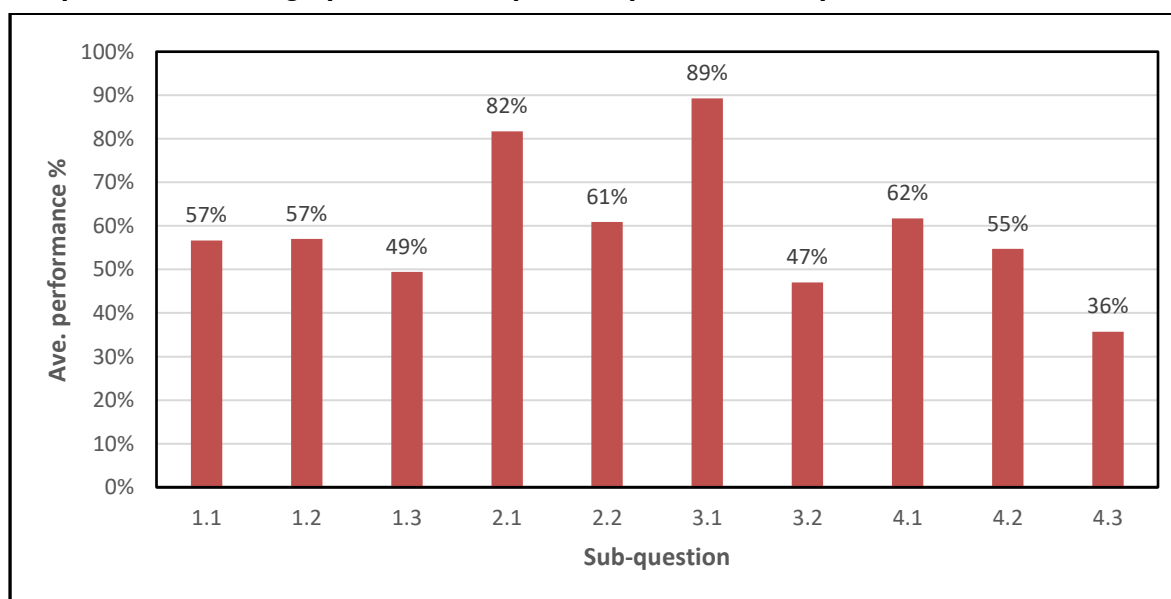
## 2.3 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 1

The graphs presented below are based on data from a random sample of candidates in the different provinces. They provide a clear picture of the general performance in each topic (Graph 2.3.1). This is useful in assessing the relative degrees of challenge of each question as experienced by candidates (Graph 2.3.2).

**Graph 2.3.1 Average performance per question in Paper 1**



**Graph 2.3.2 Average performance per subquestion in Paper 1**



| Sub-Q | Topic                                    | Sub-Q | Topic                             |
|-------|------------------------------------------|-------|-----------------------------------|
| 1.1   | Calculate: Value of stock written off    | 3.1   | Concepts: Financial statements    |
| 1.2   | Calculate: Asset disposal & depreciation | 3.2   | Interpret: Company financial info |
| 1.3.1 | Statement of Comprehensive Income        | 4.1   | JSE ethics for listed companies   |
| 1.3.2 | Statement of Financial Position          | 4.2   | Independent Audit reports         |
| 2.1   | Financial concepts                       | 4.3   | Concerns of shareholders at AGM   |
| 2.2   | Retained Income, Cash Flow & Indicators  |       |                                   |

Paper 1 reflected a general improvement in the quality of candidates' responses. Although there were subquestions where poor performance was noted, it was evident that most candidates had been well prepared. Capable candidates displayed proficiency across all topics.

Innovations in the paper comprised two open-ended subquestions; Q3.2.5 and Q4.3. It was encouraging that most learners were willing to contribute creative responses to the scenarios presented in these sub-questions.

However, below average candidates were not able to take full advantage of the lower- and middle-order questions. The main deficiencies noticed were the poor calculation of financial indicators, the inability to support financial indicators with valid explanations, and superfluous or misplaced items in various statements.

There are still a number of centres where candidates answer factual parts of questions yet avoid those which require explanations. Much of this might be due to the style of teaching or teachers' preferences, which focus on certain types of questions to the exclusion of others. To this end, a shift in thinking and planning must focus on the needs of the 20<sup>th</sup> century learner.

## **2.4 ANALYSIS OF LEARNER PERFORMANCE IN EACH QUESTION IN PAPER 1**

### **QUESTION 1: STATEMENT OF COMPREHENSIVE INCOME (SOI) AND STATEMENT OF FINANCIAL POSITION (SOFI)**

This question covered the Statement of Comprehensive Income (SOI) and a portion of the Statement of Financial Position (SOFI), i.e. the equity and liabilities section. It integrated calculations relating to valuation of trading stock and fixed assets in addition to a variety of adjustments applicable to a trading business.

As most of the standard calculations and adjustments have been examined extensively in the past it was anticipated that most candidates would cope well. This did not prove to be the case in a large number of centres.

It remains a concern that many weaker candidates seem to have a poor understanding of the format and structure of these major financial statements, and the logic of how to process increases and decreases to income, expenses, assets, liabilities and equity. However, they were, at times, able to earn part marks for pre-adjustment figures, and method marks for post-adjustment and balancing figures in the SOI and SOFI.

#### **Common errors and misconceptions**

- (a) In Q1.1 (5 marks), a large number of candidates provided inappropriate and illogical methods to calculate the closing-stock figure using the weighted average method despite this topic being well-covered in textbooks and past papers. Some candidates also neglected to subtract the stock value from the pre-adjustment figure in the books in order to reflect the stock loss as an expense in the SOI.
- (b) In Q1.2 some candidates proved to be incapable of calculating the profit/loss on the sale of a vehicle (Q1.2.1; 5 marks) and depreciation (Q1.2.2; 4 marks). This content represents relatively predictable Grade 10 and Grade 11 topics that are repeated annually in the context of sole traders, partnerships or companies.
- (c) A lack of understanding of concepts in the basic accounting equation and hence in these financial statements was reflected through the items misplaced by weaker

candidates. For example, in the SOCI (Q1.3.1; 28 marks) some income items were shown as expenses, while in the SOFP (Q1.3.2; 18 marks) some expense or asset items were shown as liabilities.

- (d) Well-prepared candidates admirably addressed all the adjustments in this question. However, in preparing the SOCI, the following items challenged less capable candidates despite having been covered in previous grades and past papers:
- **Provision for bad debts adjustment:** Many candidates did not adjust for bad debts correctly before calculating the amount of the provision. Others neglected to subtract this amount from the opening provision to arrive at the decrease in provision for bad debts figure.
  - **Security contract prepayment and rent received in advance:** Many candidates were not able to apportion these expense or income items across the relevant months.
  - **Salaries and wages:** Although candidates were generally able to earn most of the 6 marks available for this adjustment, there appeared to be several centres where this content had obviously not been effectively revised. The pre-adjustment list of balances specified that contributions are included in the salaries and wages line item, however, a fair percentage of candidates inappropriately used the sundry expenses line item to reflect some of these adjustments.
  - **Income tax for the year:** Weaker candidates did not realise that they should multiply by  $\frac{30}{70}$ . This is a basic arithmetic calculation which is frequently examined. Some candidates also did not subtract the provisional tax paid from their tax calculation to reflect the balance owed to SARS in the SOFP.
- (e) The preparation of a section of the SOFP reflected variable performance. The following items were not well handled by weaker candidates:
- **Current portion of the loan:** This caused difficulty for candidates who did not appreciate that the effect of interest capitalised (i.e. R142 250) would already have been reflected in the difference in loan balances at the beginning and end of the year (i.e. R121 750). Some candidates also inappropriately reflected a figure for the current portion of the loan under *Current Liabilities* that differed from the amount deducted from the loan under *Non-Current Liabilities*.
  - **Shareholders' equity:** Many candidates did not realise that the *net asset value* (NAV) per share could be used to determine the total value of the 1 150 000 shares in issue.
  - **Dividends payable to shareholders:** Some candidates neglected to deduct *interim dividends* from *total dividends* to arrive at the amount due to shareholders.
  - **Trade and other payables:** Many candidates omitted to transfer the rent received in advance from the SOCI to the SOFP. They also inappropriately including other items from the SOCI in the SOFP e.g. salary deductions that had already been settled.

### Suggestions for improvement

- (a) It was expected that the topics covered in Q1.1 (*stock valuation*) and Q1.2 (*fixed assets*) would provide realistic opportunities for relatively high marks to be achieved by the entire 2022 Accounting cohort. Stock valuation is relatively straight-forward, and procedures relating to fixed assets are covered incrementally from Grade 10. These topics are applicable to both papers. Teachers are reminded that these topics should be incorporated and revised while teaching *Companies* in Grade 12. Short formative tasks on items such as these are advised as a possible solution to this problem.

- (b) The marking guide for the SOCI (Q1.3.1; 28 marks) and the SOFP (Q1.3.2; 18 marks) reflects 6 marks for entering pre-adjustment figures and 16 method marks for reflecting post-adjustment figures and sub-totals. This indicates that 48% of the total of 46 marks on these subquestions are available to weak candidates who experienced difficulty in calculating any of the 10 adjustments correctly. The reasons for many weaker candidates not coming close to achieving these marks needs to be seriously investigated by teachers to understand the type of remediation that is required.
- (c) The most obvious assumption based on evidence of actual scripts is that weaker candidates simply do not fully understand the logic and structure of the two major financial statements (i.e. SOCI and SOFP). This is probably due to their poor grasp of the *Accounting Equation* and the six concepts contained therein i.e.

**A**ssets + **E**xpenses + **D**rawings (dividends) = **O**wner's Equity + **I**ncome + **L**iabilities

When revising past NSC papers with such learners, teachers may find it productive to give them the correct figures for each adjustment and request them to simply enter the figures in the SOCI and SOFP in the appropriate places. This strategy could well impress upon weaker learners that easy marks in processing figures on the financial statements are within their reach. To enhance understanding, learners should also be required to verbalise the concepts in the *Accounting Equation* and to identify where they would be reflected in the financial statements.

- (d) Obviously, the additional problem that requires attention is the calculation of the adjustments which in this question accounted for 14 marks in Q1.2 and 24 marks in Q1.3. All the possible adjustments in the curriculum have been covered in previous NSC papers, and some are very predictable, e.g. prepayments, accruals, taxation, dividends and loan repayments.

In enhancing the confidence of weaker learners in revision classes, teachers are advised to focus on the calculation of each adjustment in isolation without compiling the financial statements.

- (e) The calculation of adjustments would offer easy, moderate or difficult degrees of challenge to learners. None of these calculations would be more complex than working with fractions, ratios, percentages and solving for x in an equation. However, poor interpretation of the information given for each adjustment is a perennial problem. Teachers would have to devote time at regular intervals to the skill of breaking down or analysing the specifics of the information given so that the appropriate operation is used by the learner.
- (f) Additional points of advice on specific items in the question:
- Learners must be made aware that certain financial indicators may be integrated in the question to calculate missing figures. In this case, the indicator was the *Net Asset Value*.
  - In processing salary adjustments in a SOCI, learners must be aware that the *Salaries* line item should include gross salary before deductions, and employer's contributions. Although the bookkeeping entries cover separate accounts for employers' contributions, the principle of *materiality* should be applied in preparing financial statements. Use of extra line items for contributions is not appropriate, particularly as it was stated in the list of balances that this was the case.
  - In identifying the current portion on the loan, learners could draft a summary of the loan statement to calculate missing figures as follows. Once the capital portion for

the past year is identified, it is then a simple task to calculate the portion of the loan that will be repaid in the following financial year, i.e.  $121\,750 \times 1,1 = R133\,925$ .

|                                  |                  |                                              |
|----------------------------------|------------------|----------------------------------------------|
| Loan at beginning of year        | 1 280 750        | = Interest 142 250 + Capital portion 121 750 |
| <b>Add:</b> Interest capitalised | 142 250          |                                              |
| <b>Less:</b> Repayments          | (264 000)        |                                              |
| Loan at end of year              | <u>1 159 000</u> |                                              |

## QUESTION 2: RETAINED INCOME, CASH FLOW STATEMENT (CFS) AND FINANCIAL INDICATORS

The majority of candidates performed relatively well in this question, as is evident by Graph 2.3.1 above. This was mainly due to the marks earned on the basic concepts (Q2.1; 3 marks), the Retained Income Note (Q2.2.1; 7 marks) and calculations for the Cash Flow Statement (Q2.2.2; 13 marks).

However, the calculation of financial indicators (Q2.2.3; 12 marks) was generally poor, despite the fact that candidates had access to a formula sheet.

### Common errors and misconceptions

- (a) Although the 'confidence-boosting' subquestion (Q2.1; 3 marks) which required identification of basic concepts was well answered, it remains a serious concern that not all candidates were able to earn full marks. These concepts form the foundation of Accounting terminology from Grade 10. The lack of understanding of these terms indicates poor basic understanding on the part of the candidates which has major ramifications for their understanding of more detailed content in the curriculum.
- (b) The Retained Income Note (Q2.2.1; 7 marks) was generally well done by most candidates although weaker candidates did not correctly calculate the excess over the average issue price on the shares repurchased. They were able to earn method marks and part-marks.
- (c) In Q2.2.2 (13 marks) candidates were required to calculate three figures for the CFS. Although this was consistent with questions asked in many previous papers, basic errors were still noted across a number of centres. These were:
  - **Income tax paid:** Some candidates did not recognise that the debit figure on SARS at the beginning of the year represents an overpayment of provisional tax which should be deducted in calculating the net payment to SARS. Others also incorrectly added the amount owed to SARS at the year-end.
  - **Funds used to repurchase shares:** Many candidates were unable to identify the average share price from the figures given in the question.
  - **Net change in cash and cash equivalents:** This should be a relatively simple calculation but, except for the high achievers, candidates were not able to achieve well on this question. Many used illogical methods in trying to calculate the net change.

The use of brackets to indicate outflows of cash was not required in this question as only the calculations were asked. Many candidates, however, presented inappropriate use of brackets, which would have been a concern had the preparation of the CFS been the focus of the question.

- (d) In Q2.2.3 (12 marks) most candidates calculated the debt-equity ratio correctly. However, the calculation of the other two indicators was very poorly done, with weaker candidates earning few marks on these items:

- **Return on capital employed (ROCE):** The provision of the formula sheet appeared not to help the majority of weaker candidates. Many copied the formula incorrectly. It also appears that there has been inadequate focus on this calculation in classrooms as several candidates did not answer this question. Others attempted to use the formula provided but did not include all the factors in the numerator or denominator.
- **Dividends per share (DPS):** In this question the change in the number of issued shares during the year created difficulty for most candidates. This is a serious concern as this type of question has been asked in past papers and has been addressed specifically in the 2020 and 2021 Diagnostic Reports. Most candidates attempted to divide total dividends by the total number of shares in issue at either the beginning or the end of the year. This is inappropriate as dividends are declared at discrete intervals i.e. at mid-year and at the year-end.

In calculating DPS, the applicable number of shares on those specific dates would apply. The appropriate method in calculating the total DPS during a year would therefore simply be: Interim DPS + Final DPS.

It appears that this calculation has not been regularly practised in class. Some high achieving candidates in certain centres used their initiative in creative methods such as a weighted-average method which would also be valid in arriving at the correct answer and earning the full 4 marks. However, these complex methods are unnecessary as they would have demanded more time to complete what should be a very simple calculation.

### Suggestions for improvement

- (a) Teachers must ensure that learners are able to verbalise what is meant by basic concepts such as *solvency*, *liquidity*, *profitability*, *returns*, *gearing* and *risk*. This skill would enable learners to earn full marks on such ‘appetiser questions’ and to select appropriate indicators in answering evaluation sub-questions.
- (b) Weaker learners will require further practice through short formative tasks in the repurchase of shares which affects retained income (Q2.1.1) and the CFS (Q2.2.2).
- (c) Teachers should also provide regular practice in compiling a more detailed CFS than the parts required in Q2.2.2. Refer to Q2.2.2 in the Sept 2021 Preparatory (trials) Paper 1 for an appropriate example. Learners should be reminded of the variations in other calculations that affect operating, investing and financing operations in the CFS, e.g. income tax paid (in Q2.2.2) and other items that were not included in this question such as dividends paid, changes in loans, fixed assets, investments and shares issued.
- (d) Q2.2.2 also required a calculation for the net change in cash and cash equivalents. Weaker candidates often find this concept confusing. It would therefore be in their interests to be advised to draft the three lines found at the bottom of the CFS, as an easy alternative, as follows:

|                                              |                  |               |
|----------------------------------------------|------------------|---------------|
| Net change in cash & cash equivalents        | Missing figure → | 169 700       |
| Cash & cash equivalents at beginning of year | –133 000 + 8 000 | (125 000)     |
| Cash & cash equivalents at end of year       |                  | <u>44 700</u> |

To enhance understanding, it is however, important to use the difference between the opening balance and the closing balance to illustrate a net inflow or outflow of cash.

- (e) Below-average learners will need support in engaging with the financial indicators. The extremely poor responses in calculating two of the financial indicators in Q2.2.3 revealed a serious lack of understanding of the logic and purpose of these indicators, i.e. the return on capital employed (ROCE) and dividends per share (DPS).
- (f) Teachers must impress upon learners that the ROCE % serves to indicate the overall returns earned by the two significant groups of investors in a company, i.e. shareholders (who contribute share capital) and lenders (who contribute borrowed capital). Learners should be led to appreciate that:
  - *Capital employed*, the term for the denominator, refers to *shareholders' equity* (SHE) plus *non-current liabilities* (NCL). Average capital employed for the consecutive years should be used.
  - The corresponding returns for the investors that should be reflected in the numerator would consequently be *net profit before tax* (in the case of SHE) and *interest on loans* (for NCL).
  - The % return ultimately calculated would then reflect whether the combined financial resources of the two major groups of investors are earning a combined return that exceeds the cost of borrowing (i.e. the interest rate). This is relevant to the evaluation of *operating profit* and *gearing*.
- (g) In the case of DPS, teachers must instruct learners to use the simplest method of calculating DPS when there has been a change in the number of shares during a financial year, i.e.

$$\begin{aligned}\text{Interim DPS} &= \text{Interim dividends} \div \text{number of shares in issue on that date} \\ \text{Final DPS} &= \text{Final dividends} \div \text{number of shares entitled to final dividends} \\ \text{Total DPS} &= \text{Interim DPS} + \text{Final DPS}\end{aligned}$$

This item was also covered in the Nov 2021 NSC Paper 1 (Q2.2). Teachers must illustrate to learners that when there is a change in share capital during a financial year, dividing total dividends by any number of shares is not likely to lead to the correct answer.

### QUESTION 3: INTERPRETATION OF FINANCIAL INFORMATION

This question covered evaluation of financial information of a company.

Some teachers might be of the opinion that high school students do not respond positively to interpretation and evaluation questions. However, increased willingness amongst weaker candidates to attempt such questions was noticed this year. Improvement in the general quality of responses on this topic was also evident and most candidates proved to be capable of earning at least part-marks in the various sub-questions.

Q3.2.5 (9 marks) was an innovative open-ended subquestion which focused on the matter of retaining the services of a key member of management, i.e. the CFO. This scenario introduced elements of critical thinking and problem-solving and it was therefore pleasing that most candidates from specific centres were able to offer valid points and achieved exceptionally well in this question.

### Common errors and misconceptions

- (a) Q3.1 (3 marks) was intended as a confidence-boosting easy question. However, weaker candidates in some centres were not able to link the titles of the financial statements to their purpose. Such candidates will inevitably be limited in the extent to which they can engage positively in more detailed questions on the financial statements.
- (b) While candidates earned marks for quoting figures for relevant financial indicators in supporting their opinions, many neglected to mention the trends from one year to the next. This is not a difficult task. Many candidates also tended to quote a financial indicator and trends without qualifying this with an explanation, which also prevented them from earning full marks in the sub-question.
- (c) Candidates generally performed well in Q3.2.1 (4 marks) with relevant comparisons on profitability. Many correctly identified the % operating expenses on sales as one indicator which reflected good control over wasteful expenditure. The other relevant indicator in this case is % net profit on sales, but some candidates tended to choose the mark-up % which was not accepted in this context. Mark-up is a decision taken by senior management which affects pricing and is not a relevant indicator to assess control of expenses.
- (d) Some candidates were not able to provide appropriate comments on the dividend pay-out rate (Q3.2.2; 6 marks). The question specifically required an assessment on whether the change in the pay-out rate would benefit the company and whether a specific shareholder would be satisfied with the dividends earned by her despite the change in policy. The correct response (quoting the correct rate and trends) should have been that the pay-out rate declined by a large margin (which increases retained income and therefore benefits the future growth of the company), and that the shareholder's DPS increased significantly from one year to the next.
- (e) The predictable topics of *gearing* and *risk* were covered in Q3.2.3 (6 marks). Most candidates recognised that the debt/equity ratio and the % ROCE were the relevant indicators. However, some candidates did not offer an explanation on why they disagreed with the shareholders' opinion that the CFO was reckless. Others failed to mention that the debt/equity ratio reflected low risk despite the fact that it increased, or that positive gearing was indicated by the % ROCE being higher than the interest rate on loans.
- (f) In Q3.2.4 (12 marks) although most candidates were able to calculate the % shareholding of the specific shareholder some opted to use inappropriate calculation methods or failed to take into the account the additional shares bought by the shareholder (i.e. the rights issue).

While most candidates were able to recognise that the combined shares of two shareholders would enable them to exercise the majority of the votes at the AGM, some incorrectly assumed that the idea of their combining votes at the AGM was not acceptable.

The question also required candidates to provide reasons why they would be concerned about the strategy of the two shareholders to combine their votes. Weaker candidates were able to provide valid points which earned part-marks but failed to give complete explanations. This was an open-ended question in which capable candidates achieved full marks for their insight or explanations.

- (g) In Q3.2.5 (9 marks) weaker candidates, limited themselves to quoting financial indicators in retaining the services of the CFO without further explanation, thereby earning part-marks. This was an open-ended question in which candidates generally responded well as they had free reign in deciding what information to use.

### Suggestions for improvement

- (a) Developing learners' skills on how to respond to questions requiring interpretation and evaluation must become a priority for teachers.

As a first step, learners should be made aware of the following principles or conventions stated in the question paper, answer book and marking guide:

- The over-riding instruction for this question states: 'Provide figures, trends, financial indicators or calculations to support your comments and explanations.'
- Most of the subquestions on this topic stipulate the word 'explain' in both the question paper and the answer book, e.g. 'Explain why ...' or 'Explain whether ...'
- The answer book reflects reminders to candidates to 'Explain ...' and 'Quote figures and trends.'
- The marking guide for each subquestion states: 'Part-marks for incomplete/partial/unclear response'.

It must be made clear to learners that, unless they comply with each of the four requirements listed above, they will be limiting themselves in achieving full marks in a subquestion.

- (b) Learners must realise that a response such as 'The operating expenses on sales decreased from 27,8% to 14,8%' is a statement of the figures and the trends and does not constitute an *explanation* of the substance of the figures, the reason or the circumstance applicable to the scenario in a subquestion. Learners must attempt to explain the point required by the question in addition to providing figures and trends.
- (c) In communicating this message through to learners, teachers should endeavour to devote time to classroom discussions at regular intervals on what perfect responses would comprise of in the context of an interpretation or evaluation question such as Q3. The time spent on this type of activity will greatly benefit learners.
- (d) With regard to the dividend pay-out rate (Q3.2.2), learners should be aware that:
- A low pay-out rate would free up funds for capital growth;
  - A high pay-out rate might satisfy shareholders who prefer dividends although this would negatively affect capital growth and cash resources;
  - Market prices of shares might improve or decline depending on the preferences of potential shareholders for long-term capital growth or immediate dividends.
- (e) In assessing whether the shareholder in question would be satisfied with her dividends (Q3.2.4) the simplest consideration would be the change in the value of the DPS from one year to the next. However, learners should also be aware of alternative valid responses such as calculating the total dividends earned by the shareholder and comparing to that of the previous year.

It was pleasing to note that some learners used the dividend yield as a valid response in this question, i.e. calculating DPS as a % of the market price of the share or the price paid by the shareholder to purchase the share. This reflects a creative way of

thinking, particularly as this indicator is not currently in the curriculum. Teachers are encouraged to introduce such enrichment concepts via the project on a published financial report.

- (f) Q3.2.3 was phrased in a way that questioned whether the CFO was reckless or not. Learners must be made aware that simply rephrasing questions as responses will not earn marks unless they provide the standard evidence of financial indicators and trends. Their opinions on whether they agreed or not were also essential for a complete answer to the question.
- (g) When calculating % shareholding of a particular shareholder (Q3.2.4), learners should be advised to apply the most common method, i.e. number of shares owned by the shareholder as a % of the total number of shares in issue, in the company. The additional shares bought by the shareholder in the current year must form part of the numerator. It is not necessary for learners to waste valuable exam time in experimenting with more complex methods such as applying a ratio between the two shareholders mentioned in the scenario.
- (h) Learners should be reminded that if they make calculation errors in one part of a question, they must persist in answering the follow-up questions as they will be credited for valid responses based on their figures. For example, in the case of Q3.2.4, learners might not identify a majority shareholding, but they would earn marks on a valid comment such as providing the shareholders with a greater voting presence in the AGM.

The aspect of concern over the coalition between the two shareholders highlighted learners' use of general knowledge on political coalitions. The wide diversity of responses showed that learners relate well to real-world issues and teachers are encouraged to use such topical scenarios as a motivational measure in the classroom.

- (i) Teachers should also impress upon learners that open-ended questions, in this case Q3.2.5 (9 marks), may be answered by applying their own opinions and creativity, provided they are relevant to the question. The problem of trying to gain evidence to justify the retention of the services of a valuable and skilled CFO should be seen as a practical issue that learners could relate to.

Learners must be urged to not simply quote the relevant indicators without making the point about why the indicator reflects well on the performance of the CFO or makes a positive difference to the company.

Although examiners will anticipate a number of possible valid answers which will be reflected in the marking guide for open-ended questions, the list is not definitive. Markers are advised to be alert to alternative valid responses. Learners must not assume that their creativity will be ignored. Again, classroom discussions on scenarios involving problem-solving would serve to open the minds and increase the confidence of learners in new ways of formulating responses.

## QUESTION 4: CORPORATE GOVERNANCE

The topic of Corporate Governance is highly relevant in the context of the financial and ethical challenges facing countries worldwide and in its impact on economic resources and service delivery which affect all citizens. Consequently, the CAPS requires that corporate governance issues be integrated across all topics in the curriculum.

The introduction of stand-alone questions on this topic in NSC papers reinforces the importance of sensitizing high school students to the concepts of business ethics that they will inevitably encounter in their daily lives and future careers.

Q4.1 (4 marks) covered the role and standards of the JSE, while Q4.2 (2 marks) focused on types of independent audit reports. These short questions provided context for the scenario which followed.

(Q4.3; 9 marks) was intended as a higher-order question within the ability of most candidates with an element of practical problem-solving. It was encouraging that, due to the open-ended nature of the question, many candidates, including those achieving below-average marks, endeavoured to provide three valid questions that should be raised at an AGM based on the scenario provided. This observation should serve as a catalyst in improving the general standard of learning both inside and outside the classroom.

While it is evident from the responses of candidates to all three subquestions that teachers at many centres have devoted quality time to this topic, it is also evident that a sizable number of centres have yet to reach a satisfactory level.

### Common errors and misconceptions

- (a) Although a sizable majority of candidates was able to explain why companies want to be listed on the JSE and why the JSE would not tolerate false reporting (Q1), it was alarming that a large number of candidates in certain centres could not earn marks by explaining the difference between a qualified and disclaimer audit report (Q2).
- (b) Q4.3 required candidates to provide three questions that they would raise at the AGM together with a reason for each question. However, in contrast to the positive points mentioned above, it appeared that a large number of centres had not effectively exposed their candidates to the nature or context of a company or an AGM. This is evidenced by the inappropriate responses from candidates in these centres.

### Suggestions for improvement

- (a) It is imperative for teachers to motivate learners to engage with open-ended questions such as Q4.3. However, to ensure that learners provide appropriate responses they will need some background information on how shareholders, directors and auditors interact with each other and why AGMs are necessary. This is well covered in the CAPS and textbooks. This theoretical knowledge should be covered at the outset when the topic of Companies is introduced in Grade 12. It should also be reinforced at regular intervals in class when learners are compiling or evaluating financial statements and engaging with other items or scenarios that affect corporate stakeholders.
- (b) Sections of the formal assessment task of evaluating published financial reports must be used productively to make learners aware of the responsibilities of shareholders, directors, auditors, the JSE and their impact on the wider community and the environment. This project should not merely require assessment of financial information as this skill is extensively covered in regular tests or examinations.

- (c) Currently there are six examples of questions specifically devoted to Corporate Governance in past NSC Paper 1 examinations. Refer to papers of Nov. 2020, 2021 and 2022, June 2021 and 2022, and the common trials paper of Sept. 2021.

Instead of requiring learners to engage with these questions independently, it is advised that teachers use the questions at regular intervals to conduct 20-minute group discussions in class, followed by general feedback from members in each group. This should lead them to appreciate that fellow learners are able to creatively offer a variety of valid suggestions which would earn marks in open-ended questions such as Q4.3. In addition, verbalising information on companies in this way would benefit learners' understanding of the subtopics and increase their confidence in engaging with such questions.

- (d) In conducting formative SBA activities, teachers must make a point of including open-ended questions or tasks to expose learners to the possibilities of being credited for their opinions and creativity provided these have been formulated in a valid and logical manner.
- (e) Despite the natural tendency of some Accounting teachers to lack confidence in presenting topics that do not have definitive right or wrong answers, the willingness of candidates to provide their own opinions in Q4 in this paper indicates that learners in general would like to be heard and recognised.
- (f) Topical scenarios of positive governance are often evident in annual reports of leading companies, e.g. their contribution to the immediate community or the environment, job creation, skills development, housing or disease control.

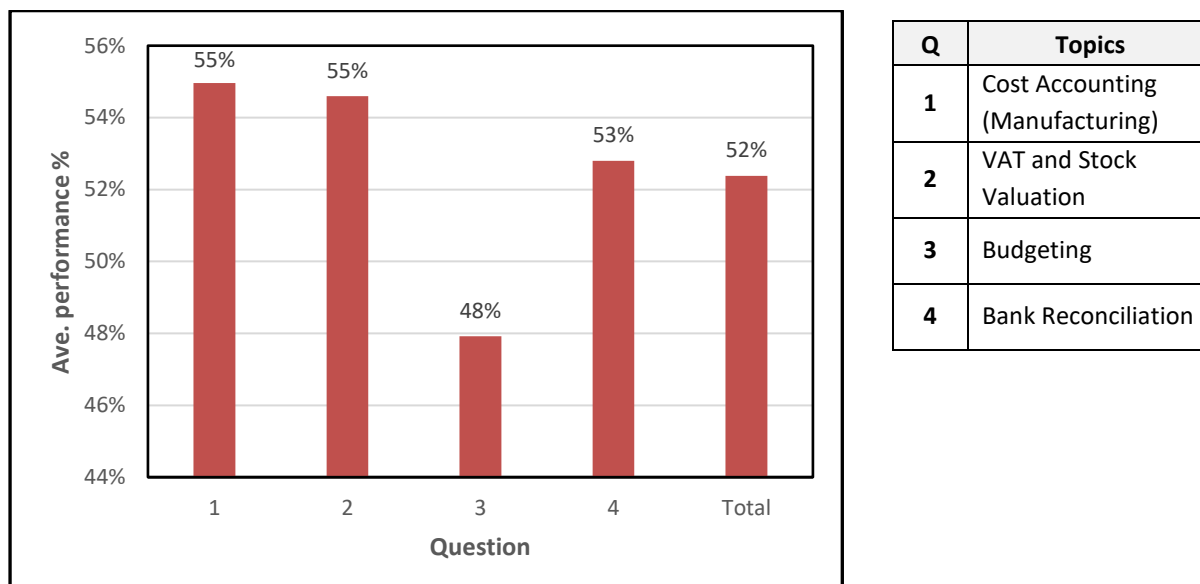
Negative factors in the business environment also need to be confronted, e.g. corruption, auditing irregularities, reckless decisions, lack of accountability, tender fraud, weak internal controls, cybercrime and bribery.

Opening Accounting classes up to the possibility of controversial areas such as *Business Ethics* and *Corporate Governance* is likely to revolutionise discourse and communication in the classroom in a positive way. This is also likely to promote critical thinking and problem-solving attitudes in learners.

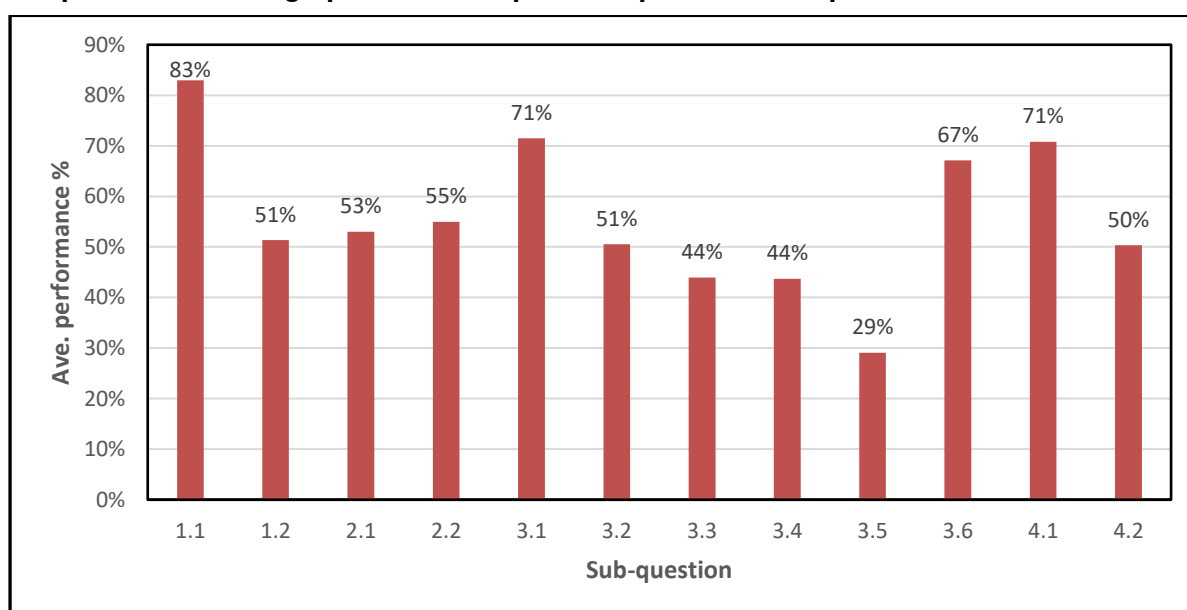
## 2.5 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 2

The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degree of challenge of each question as experienced by candidates.

**Graph 2.5.1 Average performance per question in Paper 2**



**Graph 2.5.2 Average performance per sub-question in Paper 2**



| Sub-Q | Topic                                     | Sub-Q | Topic                             |
|-------|-------------------------------------------|-------|-----------------------------------|
| 1.1   | Cost Accounting – matching                | 3.3   | Budgeting – purchase of vehicle   |
| 1.2   | Cost Accounting – analysis and evaluation | 3.4   | Budgeting – decision on mark-up % |
| 2.1   | VAT calculation                           | 3.5   | Budgeting – actual vs budget      |
| 2.2   | Inventory Valuation                       | 3.6   | Budgeting – online shopping       |
| 3.1   | Budgeting – Creditors Payment Schedule    | 4.1   | Bank Reconciliation – concepts    |
| 3.2   | Budgeting – Budget calculations           | 4.2   | Bank Reconciliation preparation   |

Prepared candidates managed well across all questions while weaker candidates were also able to take advantage of the lower- and middle-order subquestions. This is testament to the initiatives of teachers to adequately address the specific topics and the extensive revision programmes at all levels.

Poor performances were noticed especially in the interpretative or evaluative aspects in various topics. Candidates continued to show weaknesses in reading and understanding, and often provided inappropriate and irrelevant clichés from previous marking guidelines to these subquestions. It is evident that candidates prefer using predictable questions, as presented in past papers, in preparation for the examinations.

## 2.6 ANALYSIS OF LEARNER PERFORMANCE IN EACH QUESTION IN PAPER 2

### QUESTION 1: COST ACCOUNTING (MANUFACTURING)

The majority of candidates were able to take advantage of the easier marks on offer in various subquestions, as this topic generally covers content that is predictable and manageable. Subtle changes in context and different questioning styles, however, lifted the challenge in some subquestions. The well-prepared candidates seemed to handle this with ease while weaker candidates struggled to interpret the requirements possibly due to their stereotypical approach to preparation. They seemed to prefer basic application and neglect the analysis sections in their revision.

#### Common errors and misconceptions

- (a) In calculating the direct labour cost (Q1.2.1; 7 marks) a fair percentage of candidates did not realise that the overtime hours comprised a total for all workers in production and not per worker. The calculation of factory overhead cost proved to be challenging for many because they had to correct an incorrect amount given. This involved adjusting the error by either adding to or subtracting from the given total. Many omitted to include the relevant signs (+ and –) and lost the method mark for the operation.
- (b) Q1.2.2 (4 marks) required an explanation of *economies of scale*. The majority of the candidates simply stated that fixed costs increased because of an increase in production, thereby likening this cost to a variable cost. It was clear that these candidates did not understand that fixed costs are independent of the level of production and that the *cost per unit* would drop when production increases.
- (c) The scenario of whether the production staff deserved the production bonus (Q1.2.3; 6 marks) was generally well answered as it offered an open-ended element. Some candidates neglected to make reference to improvements in *unit costs* or *percentages* in their comparisons. Part-marks were earned by many candidates for incomplete responses provided.
- (d) Using the *contribution per unit* to calculate additional units that must be produced to achieve a targeted profit (Q1.2.4; 4 marks) was once again poorly answered by the majority of candidates. This was disappointing since this was highlighted as a concern previously and a detailed explanation was provided in the current *Examination Guidelines*. Candidates provided inappropriate, lengthy calculations making it difficult for markers to determine where part-marks could be allocated.

### Suggestions for improvement

- (a) Innovations to extend the challenge and to maintain relevance in this topic is a more recent trend due to the historically predictable nature of the content. It is therefore necessary to use a variety of examples to expose learners to different questioning styles. Questions from past examination papers are useful as a basis. Teachers must adapt these and create other scenarios to enhance the level of understanding of operations within a manufacturing environment. A variety of short informal case studies can be useful. These can be peer-marked and class discussions must follow.
- (b) Although the Production Cost Statement might not be tested every year, emphasis must be placed on calculations relevant to the components of this statement. Activities must be designed to focus on improving arithmetical and interpretative skills, particularly for weaker learners. Fractions, ratios and/or percentages, errors/omissions and adjustments are commonly used to apportion specific expenses to the different cost components. These must be mastered in a piecemeal fashion before addressing the whole statement. Formative tests on specific aspects must be used to reinforce the content covered in previous grades, before moving to more complex examples.
- (c) Interpretation of unit costs and understanding of *economies of scale* will improve if more attention is placed on the different cost categories. The Grade 12 curriculum is based on the assumption that direct material, direct labour and selling and distribution costs are variable in nature and that factory overheads and administration costs are fixed in nature. Teachers must relate these costs to the level of production when illustrating the difference between fixed and variable costs. Practical examples provided in previous diagnostic reports must be used to demonstrate the differences.
- (d) Break-even analysis and interpretation of unit costs are generally assessed using the same set of information. Teachers are requested to use the tables of information from past examination papers in lessons and for classwork so that learners become familiar with this presentation style. Emphasis must be placed on making meaningful comparisons using the unit costs or converting differences to percentages. Actual amounts would not be useful, especially if production levels change from one year to the next. Teachers are encouraged to use actual data to illustrate the concept.
- (e) The deep problem-solving potential of this topic has been more prevalent in recent NSC papers. This paper focused on production bonuses. Learners could skilfully use their knowledge of break-even analysis to offer valid alternative responses. Such scenarios are normally considered as higher-order but accessible to all learners.
- (f) Although calculations and interpretation of direct material costs were not tested in the paper, teachers are advised to focus attention on this direct cost, including the use of inventory valuation methods, calculating wastage and the impact on profits.
- (g) Average and below-average learners may appreciate more practical or visual stimuli since they are familiar with manufacturing concepts and calculations introduced in Grade 10. Teachers could make use of a variety of resources available on educational websites. Presentations and video clips, if used effectively, could assist in managing time allocated for the topic according to the ATP. Short class tests can be used to develop their skills to perform the more complex calculations as well as to identify problems related to decision making and internal controls. Robust class discussions must be encouraged, and learners should feel comfortable to express and share their ideas and opinions.

## QUESTION 2: VAT AND INVENTORY VALUATION

The subquestion on VAT (Q2.1; 9 marks) focused on a calculation. A majority of candidates was proficient in calculating the amount payable to SARS. The marking guideline made provision for a number of calculation options. Markers had to inspect workings carefully to ensure maximum benefit to candidates.

Inventory valuation questions focused on specific identification and first-in-first-out methods. Calculations were confined to closing stock figures and the stockholding period. The majority of candidates generally achieved high marks on these predictable calculations but struggled on the interpretative aspects of this question. These were pitched at the higher cognitive level and required a clear understanding of concepts and insight.

### Common errors and misconceptions

- (a) It was disappointing to note that a fair percentage of candidates still struggled to distinguish between *output* VAT and *input* VAT and used incorrect signs (+ or –) next to each amount. Some also could not perform relatively simple calculations involving *inclusive* and *exclusive* VAT amounts. It is possible that not much revision on this topic was done, and there appears to be little consistency in the manner in which different centres approach this topic.
- (b) Q2.2.1 (9 marks) required the value of the closing stock according to the Specific Identification method. A small percentage of candidates struggled to identify the closing number of units while others did not realise that the cost per unit had to be used. In the case of the Gama model, this had to be calculated.
- (c) In Q2.2.2 (4 marks), it was clear that many candidates could not provide appropriate examples of *division of duties*. This exposed the fact that they simply rote-learned this concept without having a clear understanding of what it entailed.
- (d) In calculating the closing stock using FIFO (Q2.2.4; 6 marks) many of the candidates neglected to take into account the carriage per unit. A few also did not consider the goods returned from the last purchase. The marking guideline provided for part-marks in these cases. The fact that certain goods were returned where the supplier did not cover the carriage costs, also confused some candidates.
- (e) Although many candidates correctly opted to use their closing stock calculated in Q2.2.4, in calculating the holding period in Q2.2.5, some candidates decided to use the average stock which was not required in this case. It was pleasing to note that a number of candidates used the number of units given in the question paper instead of the rand amounts.
- (f) Candidates who did not get an answer greater than 58 days in Q2.2.5 were challenged to comment appropriately in Q2.2.6 because this question was based on a perishable product with a limited shelf-life. They were, however, credited for comments that matched their calculation, such as ‘clear the stock that are close to expiry before ordering additional stock’ or ‘extend the target market by approaching local gyms to move stock before they reach their sell-by date’.
- (g) Q2.2.7 was a higher-order question of moderate challenge that required candidates to identify the portion of the closing stock that had passed the expiry date. Very few candidates were able to identify this amount.

- (h) Some candidates did not see Milo's proposal of selling the expired stock at half price as an ethical issue (Q2.2.8; 4 marks) and did not see this as a social problem that would affect the health of consumers and place the business at risk.

### Suggestions for improvement

- (a) VAT calculations may appear to be elementary but can be time-consuming. Practising many examples is necessary to master the basic arithmetic skills required. Short class tests to address prior knowledge are suggested, especially percentages for VAT inclusive and exclusive amounts. Although not tested in this examination, more complex examples such as correcting errors, dealing with zero-rated goods and commenting on ethical issues must also be addressed during normal classwork and revision sessions.
- (b) Calculations on stock valuation are generally predictable. Emphasis must be placed on the closing stock value of each method as this is the distinctive factor. It is advised that the commonly used calculation methods recommended in most textbooks be followed as they are generally the most efficient. Methods using cost of sales as the basis for the calculation are lengthy and not necessary.
- (c) Before introducing the different stock valuation methods, teachers must ensure that learners have a clear understanding of the *perpetual* and *periodic inventory systems*. A significant feature of the periodic system is that the value of the closing stock can only be determined by a physical count, and that stock losses will only be noticed after valuing the closing stock. This is relevant in the context of this question because returns or damaged stock should not be considered when assigning value to the actual stock on hand. It is also understandable that the specific identification method would naturally follow the perpetual inventory system, as the actual value or cost is assigned to individual expensive items that are either sold or are in stock.
- (d) Teachers must draw attention to the types of products used in stock questions during lessons and revision. Essentially, the product would either be a durable item that can easily be sold in the future or a perishable item that has a limited shelf-life. Additional clues are often provided, such as 'expiry date' or 'sell-by-date'. A clear understanding of this will assist with appropriate comments on stock holding periods.
- (e) Questions from past NSC papers represent a reliable resource that will expose learners and teachers to different ways in which this topic can be tested. Innovative questioning will always be introduced to assess the deeper understanding of learners, such as unpacking the concept of division of duties. Short class tests and discussions must be an ongoing exercise to sharpen insight on internal control and ethical issues. This would encourage learners to express themselves more clearly rather than simply quoting vague statements extracted from marking guidelines.
- (f) Although stock valuation is set as a separate question, learners must be made aware of the integration of stock valuation with other topics such as Financial Statements and Cost Accounting. Appropriate activities of this nature must be used at appropriate times during the course of the academic year.

### QUESTION 3: BUDGETING

It was once again observed that candidates are more proficient in performing basic calculations, but struggle with analysis and interpretation of budget information. The easy application processes are covered in Grade 11 and the majority of candidates took full advantage of the marks on offer. It was also pleasing to note that the concept of online sales

was well understood by the majority. However, the simplistic responses of candidates to the interpretative subquestions are an indication that this aspect of the topic is not given enough attention during class time or revision.

### Common errors and misconceptions

- (a) Q3.1 (7 marks) was well answered by the majority of candidates. The fact that this paper consisted of a Creditors' Payment Schedule did not deter candidates because the principles for calculations are similar to that of a Debtors' Collection Schedule.
- (b) Of the calculations tested in Q3.2, parts (ii) and (iii) appeared to be quite challenging. Many candidates scored part-marks for workings. These were complex calculations that required candidates to read with understanding and perform their calculations in sections, such as the additional rent income for a specified surface area or working backwards to calculate the budgeted salary for November.
- (c) In calculating the deposit of the company vehicle (Q3.3.1; 5 marks), many of the candidates did not adjust for the projected interest. A large percentage of candidates also did not realise that the total repayments constituted 75% of the total cost and that they had to calculate the 25% portion which excluded the interest.
- (d) Q3.3.2 required candidates to add back (i.e. reverse) the budgeted payments to the December deficit figure as the purchase of the vehicle would only take place in January. Some candidates used inappropriate signs (i.e. + or –) for some of the amounts. They were penalised only on the final answer because they were able to identify the specific items to be included in this calculation.
- (e) Many candidates found it difficult to provide two valid points or to quote relevant figures to support the decision to adjust the mark-up % in Q3.4. Many simply provided the improved variance for sales and for advertising as given in the question paper and earned part-marks.
- (f) In Q3.5.1 (2 marks) a few candidates did not comment on the benefits of regular maintenance and merely viewed the amount underspent as a savings for the business without considering the practical long-term effects.
- (g) A large percentage of candidates neglected to make meaningful comparisons using percentages (Q3.5.2; 8 marks). They compared only the actual expenditure amount against the budgeted amount and concluded that the expense was either well controlled or not. Only the more capable candidates were able to identify the connections to sales, convert to percentages and provide valid and insightful explanations.

### Suggestions for improvement

- (a) The Grade 12 CAPS emphasises analysis and interpretation of the budget. Teachers are encouraged to thoroughly assess prior knowledge which includes basic concepts and calculations, before addressing the more complex calculations and interpretation of the Grade 12 curriculum. Short formative and summative activities and informal class tests can be used in this regard. Such activities must be designed to identifying weaker learners who require further assistance so that appropriate support material can be provided.
- (b) To improve interpretative skills, it is recommended that various examples and different scenarios be extracted from past NSC examination papers for use in lessons.

Teachers can use these tasks as part of the informal assessment programme or adapt them for assignments or case studies.

- (c) Analysing budgeted vs actual figures is a key aspect of this topic and more time needs to be spent on clearly explaining terminology such as *overbudgeted*, *underbudgeted*, *overspent*, *underspent*, *over the budget* and *under the budget* in the context of whether budget items are well controlled or not. It must be stressed that it would make no sense to quote actual figures or the variance as the impact of such a difference must be assessed against what it is intended to achieve. Marking guidelines would insist that changes in percentage be used to demonstrate whether an expense item is well controlled or not. Learners must specifically be taught how to use these percentages to motivate their opinions.

#### QUESTION 4: BANK RECONCILIATION

This question was of easy to moderate challenge and was placed as Q4 in Paper 2 because of the changes to the content emanating from the introduction of EFTs replacing cheques, and the increase in online/internet banking. This topic was not examinable in 2021, hence the question was pitched at the application level more in line with the Grade 11 curriculum.

#### Common errors and misconceptions

- (a) Very few candidates experienced difficulty in placing the relevant figures in the correct Cash Journals (Q4.2.1; 10 marks). Those who struggled placed the given figures in the incorrect columns or included additional irrelevant amounts and were penalised for foreign items.
- (b) Q4.2.3 (8 marks) was also effectively managed by a large percentage of candidates. The marking guideline made provision for the two-column and the one-column methods. Weaker candidates could not appreciate the difference, often repeating figures in the incorrect columns of the reconciliation statement.
- (c) While many candidates were able to take advantage of the easier entries which included bank charges, debit order payments and direct deposits, many had difficulty with the treatment of errors either on the statement or in the journals.
- (d) The calculation of the closing bank account balance was generally well done although weaker candidates did not use the journal totals, instead entering individual figures of all the transactions. The reason for this is puzzling as it should have been obvious to candidates that the purpose of journals is to consolidate the receipts in the CRJ and the payments in the CPJ. Alternative methods for calculations were acceptable, including a ledger account format. There were some cases of candidates not being able to determine whether their final answer was a favourable or unfavourable balance, when transferring this amount to the reconciliation statement.
- (e) It was disappointing that some candidates did not realise that the balance as per bank statement was a balancing figure in the Reconciliation Statement. To get this amount, it was necessary to ensure that the totals in the two-column format are equal.
- (f) Many candidates were also penalised for misplaced and superfluous entries in the journals and the Bank Reconciliation Statement. This indicates that they do not have a clear understanding of the basic logic of the reconciliation process.

### Suggestions for Improvement

- (a) Reconciliations comprise important monthly internal control processes with structured procedures or steps. The basic applications are covered in Grade 11 but must be further reinforced in Grade 12 before addressing the interpretative aspects. Bank reconciliations will require more revision material and regular practice because of the changes in the banking sector.
- (b) Teachers should continually refer to the internal control benefits of reconciliations. The steps involve identifying differences and taking corrective action to address errors and omissions. Practical examples using external documents (i.e. bank statements) would be the first logical step. More complex examples must follow once learners acquire the skills of determining where the errors were made, and how they should be treated.
- (c) Source documents supporting the entries in cash journals and their effect on the *Accounting Equation* form a necessary starting point. A comparison between documents and the information on bank statements is likely to highlight possible errors and expose any mismanagement, incompetence or fraud.
- (d) Teachers must ensure that in-depth revision of preparation in all forms of reconciliations (including debtors and creditors) are included in their teaching plans.
- (e) Although this question did not focus on analytical problem-solving scenarios, teachers must adapt and use questions from past papers and various other sources to ensure that learners become familiar with the different ways in which this topic can be tested.
- (f) The informal assessment programme must include short, formative class tests that can be self- or peer-marked. Learners will then take responsibility for their shortcomings and appreciate how marks can be earned (or lost) in examination situations.
- (g) Teachers are encouraged to keep up with current trends in all matters affecting the Accounting content. All sectors in the economy are moving towards the electronic/online systems. Learners are exposed to the media and many are familiar with the internet. This trend is also transforming the banking sector. These changes must be incorporated into the teaching and learning process.
- (h) Class discussions on cyber-crime, ethics, fraud and human error must be an ongoing process. Learners must be allowed to express their views freely. Teachers must be open to differing views and appreciate the open-ended nature of such discussions. They need to explore various avenues to source relevant, up-to-date information. Newspaper articles and financial publications should be used more extensively in class discussions to enhance the teaching of this topic.
- (i) Teachers and subject advisors should explore the true purpose of the 2-column method and whether it serves a positive purpose or not. This method is not commonly used in practice and it tends to encourage learners to apply a rote-learning approach by placing certain figures in certain columns with no appreciation for the logic of the process. In a 2-column approach, learners consequently perform badly if they confuse the two columns.

Learners should be more likely to appreciate the logic of the 1-column method which involves increases (+) and decreases (–) to adjust the bank balance either from the bank statement or in the books of the business. The simple logic is: outstanding deposits will improve the bank balance on the bank statement when they are ultimately recorded, while unrecorded transfers out of the business will worsen the balance.

# CHAPTER 3

## AGRICULTURAL SCIENCES

The following report should be read in conjunction with the Agricultural Sciences Paper 1 and Paper 2 question papers for the NSC November 2021 examination.

### 3.1 PERFORMANCE TRENDS (2017–2021)

The number of candidates who sat for the Agricultural Sciences examination in 2022 increased by 1 363 in comparison to 2021.

The table below indicates a positive trend in the pass rate over the past three years.

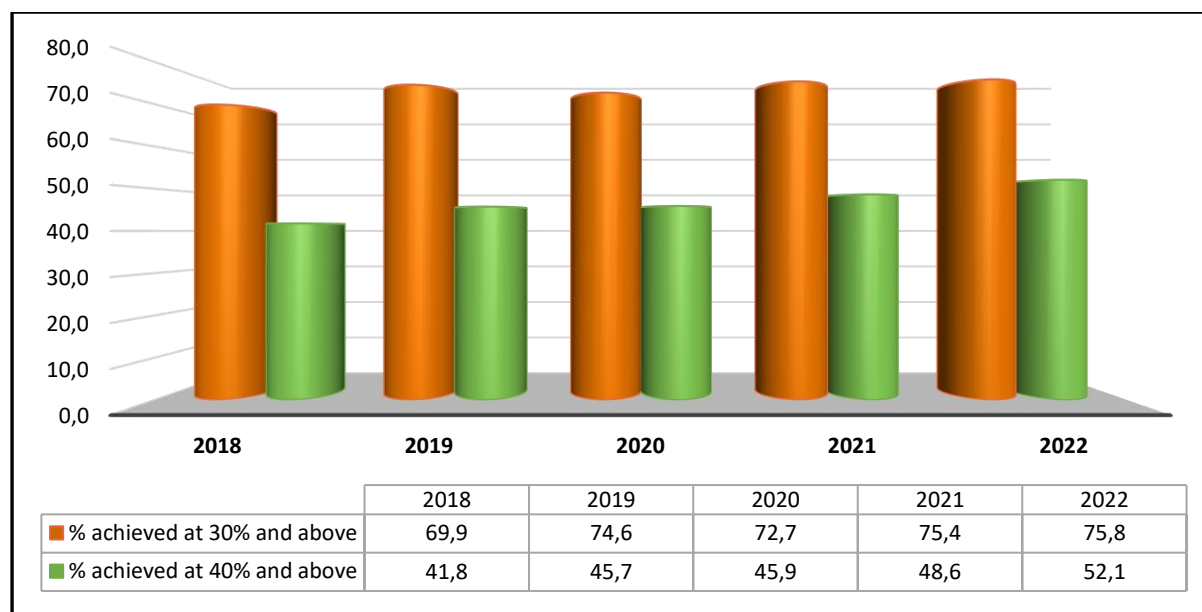
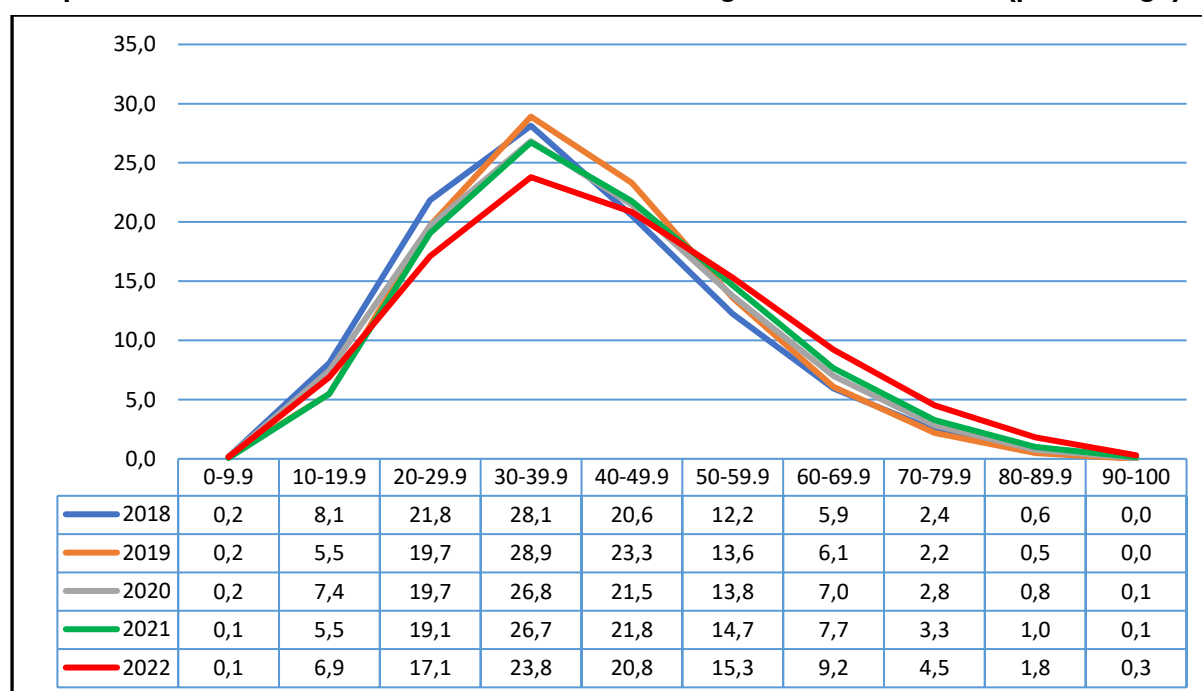
Although there was only a marginal increase in the pass rate at 30% (Level 2) in 2022, there was a very pleasing increase in the pass rate at 40% (Level 3). Candidates who passed at 30% (Level 2) improved marginally from 75,4% in 2021 to 75,8% in 2022. The pass rate at 40% (Level 3) improved over the past two years from 48,6% to 52,1%.

There was also an encouraging increase in the percentage of distinctions over 80% (Level 7) from 1,1% to 2,1%. This converts into an increase of 1 268 in the number of distinctions.

In view of disruption to academic programmes in 2020 and 2021, the results achieved by this cohort are particularly commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) have ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the favourable overall performance.

**Table 3.1.1 Overall achievement rates in Agricultural Sciences**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 95 291    | 66 608                        | 69,9                        | 39 800                        | 41,8                        |
| 2019 | 92 680    | 69 132                        | 74,6                        | 42 385                        | 45,7                        |
| 2020 | 96 155    | 69 916                        | 72,7                        | 44 114                        | 45,9                        |
| 2021 | 123 990   | 93 447                        | 75,4                        | 60 308                        | 48,6                        |
| 2022 | 125 353   | 95 070                        | 75,8                        | 65 251                        | 52,1                        |

**Graph 3.1.1 Overall achievement rates in Agricultural Sciences (percentage)**

**Graph 3.1.2 Performance distribution curves in Agricultural Sciences (percentage)**


## 3.2 OVERVIEW OF CANDIDATE PERFORMANCE: PAPERS 1 AND 2

### General comments

All stakeholders embarked on extensive intervention initiatives throughout the year to make up for the lost time due to Covid in order to improve the performance of the 2022 class. Continuous content delivery was compromised by disruptions, unavoidable curriculum adjustments and a reduced assessment programme. The plight of the current cohort must be recognised, in that they never sat for a formal examination; instead, they were subjected to tests only to determine their progression. Despite these challenges, a fair percentage of candidates were proficient in addressing the requirements of all subquestions.

However, the widening of the achievement gap between capable and weaker groups of candidates was noticeable this year. This should not be surprising in the context of the two abnormal years that this cohort experienced in Grades 10 and 11. Capable or motivated candidates with appropriate resources might have appreciated the opportunity to study independently, while less proficient candidates, the majority of this group, or those lacking resources might have benefitted from having greater access to teachers.

The inability of weaker candidates to deal effectively with even the less challenging parts of questions is a clear indication of the impact of the disruptions on teaching and learning processes within the educational environment. The unsatisfactory performance in Q1.3 and Q1.4 of both papers, which deal with terminology and concepts, is indicative of the above-mentioned issue.

Language barriers linked to poor comprehension skills and spelling are continuing to challenge most candidates because they tend to give responses that have no meaning and are far removed from the correct concept. Sometimes they write words that have different meanings, for example *seamen* instead of *semen*, *electrocuted* instead of *electroporation*.

Despite the subject-specific language used in Agricultural Sciences, many candidates continue to experience difficulty in understanding the requirements of questions. Weaker candidates tend to provide incomplete or unclear responses to questions requiring explanations or comments. They often rely on the commonly used responses offered in the previous marking guidelines, and use them inappropriately in different contexts.

Candidates were unable to analyse the relevant information provided in answering specific subquestions. This was especially relevant in the Calculations in Animal Nutrition, Marketing and Genetics sections of the papers, where only the more capable candidates were able to extract and respond to the relevant information. Weaker candidates found this process challenging, as they were often unable to comprehend and to respond as expected.

Although there was a slight improvement in learners' ability to draw a graph, which was evident in both Papers 1 and 2, candidates could still not successfully identify the independent and dependent variables, including the units in both Paper 1 and Paper 2, where the X- and Y-axes were swapped around. This was more apparent in Paper 2 which was quite challenging for candidates.

### General suggestions for improvement

There are several factors that contributed to the poor performance of a large percentage of candidates in the 2022 papers. The following general recommendations for improvement are applicable to both papers in 2023:

- (a) **Basic concepts and terminology:** Learners need to be exposed to the basics of each topic for them to engage effectively with the content in that topic. The process of conceptualising and understanding these concepts is more than mere rote-learning of the definitions. Terminology should form an integral part of teaching and learning and needs to be emphasised on a regular basis. Teachers are advised to make the teaching of terminology interesting by engaging learners in the identification of key concepts for each topic. The learners should then be taught to formulate shorter definitions, based on the context. Teachers can also make use of short question tests on these terms using previous examination papers.

Teachers are advised to use the following strategies to improve the teaching of basic concepts and terminology:

- Engage learners in the identification of new terms and then find an explanation from the textbooks.
  - Use new concepts and terms in sentences and in short scenarios to illustrate their understanding.
  - Learners should be directed to first identify the new concepts for each topic and compile a glossary of terms in their notebooks on completion of the topic with a brief but clear definition for each term or concept. A separate notebook for this purpose may also be kept. By the end of the year, all learners should have a comprehensive glossary of all the relevant terms and concepts pertaining to each topic.
  - Agricultural Sciences terminology should be assessed frequently using different forms of informal activities.
  - Challenging or confusing terminology could be explained by using illustrations and/or posters. These posters can be pinned on notice boards in the classroom so that learners are exposed to them on a regular basis.
  - Spelling tests and word cards can be utilised to train learners on how to spell complicated terms.
- (b) **The importance of formative and topic testing:** Tests should assess learning after every topic has been covered to provide remedial measures where necessary. Self-assessment and peer assessment, with immediate feedback on errors, provides learners with an opportunity to increase their understanding of the problem. They also become exposed to valid alternative responses and different, easier approaches to solving problems.
- (c) **Enhancing learners' skills in accurately interpreting specific subquestions and using information that is relevant:** It is essential that learners have a good understanding of the instructional verbs as emphasised in the *Examination Guidelines*. Teachers must explain the context in which key verbs such as 'deduce', 'justify', 'explain' and 'suggest' are used and the expected depth required by each question. The marking guidelines of past examination papers can be used to show how the responses to similar questions can differ because of the key verb used in the question. A variety of instructional verbs must be used in both informal and formal assessment tasks. It is recommended that these informal tasks lean mostly towards developing conceptual skills, as this will enable learners to develop a better understanding of the requirements of each question.
- (d) **Skills to be assessed:** Assessment should be of such a nature that it challenges the learners' ability to think beyond what is presented in the textbooks or by the teacher. Learners need to be guided on how to process data presented in different forms, such as tables, graphs, calculations or scenarios. These areas have proven to be

challenging for the learners. Teachers need to sharpen their learners' analytical skills by exposing them to challenging informal and formal tasks.

- (e) **Real-life scenarios:** Learners show serious deficiencies in the processing of application questions and this is an indication of a lack in the depth of their subject knowledge. Learners need to be exposed to more real-life agricultural situations through visits to sites of practice. Where a practical demonstration is not possible, the use of videos that simulate the actual practice is recommended to enhance intensive learning.

Teachers are advised to include sources such as pictures, scenarios, case studies and short statements in their informal and formal assessment tasks, and to demonstrate to learners how to approach such questions. These tasks should test the application of theoretical knowledge into real farming practices. This could be done by first reading and/or analysing the source, guiding learners on how to find clues and thereafter associating the key information discovered, before finally attempting to answer the actual questions. In some instances, learners can be requested to formulate their own questions based on the source. This practice will allow learners to analyse the source critically. Teachers can then develop follow-up questions to extend learners' understanding of the content.

- (f) **Enhancing the interpretation of calculated values:** Examination papers in Agricultural Sciences contain some simple mathematical processes, e.g. drawing of graphs, calculating percentages, conversion of values, expression into relevant units, use of formulae and substitution of values. Learners seem to lack appreciation of the magnitude of the various units, such as the difference between tons and kilograms. They also seem uncertain when to divide or multiply, how to convert monthly to daily needs or vice versa or change group needs to individual needs and vice versa.

Teachers are advised to give regular informal tasks on calculations incorporating the different versions. Teachers should not assume that learners have successfully engaged with these skills in other subjects or can successfully transfer these skills from other subjects to the study of Agricultural Sciences. Teachers are advised to first indicate to learners the importance of the various calculations to farming before showing them the actual skill of performing the calculations by applying the information that is given. Moreover, teachers are also advised to mark such calculations accurately emphasising the conversions, units, substitutions of values and formulae.

- (g) **Use of past NSC papers:** Learners must have access to past examination papers, but they should also be alerted to the limitations of past papers. It should be noted that although questions in past papers may cover the same content, they may have different foci, e.g. a question which asks for a *comment* requires a different response to a question which asks for a *justification* or *suggestion*. Teachers are advised not to engage in whole question paper revision; it is better to consolidate questions from various papers into a bank of questions for each topic and then engage with question revision.
- (h) **Reference to the CAPS, Examination Guidelines and previous Diagnostic Reports:** Teaching and assessment must be informed by the content prescribed in the CAPS and the approach outlined in the *Examination Guidelines*. Aspects of the content that have never been assessed in recent question papers proved to be a challenge in P1 Q3.6 and P2 Q4.7. A holistic understanding of all topics is essential. It is also important that teachers use a variety of the prescribed textbooks to source information and then consolidate it for learners. It is also imperative that teachers take cognisance of comments and recommendations made in previous diagnostic reports.

### 3.3 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 1

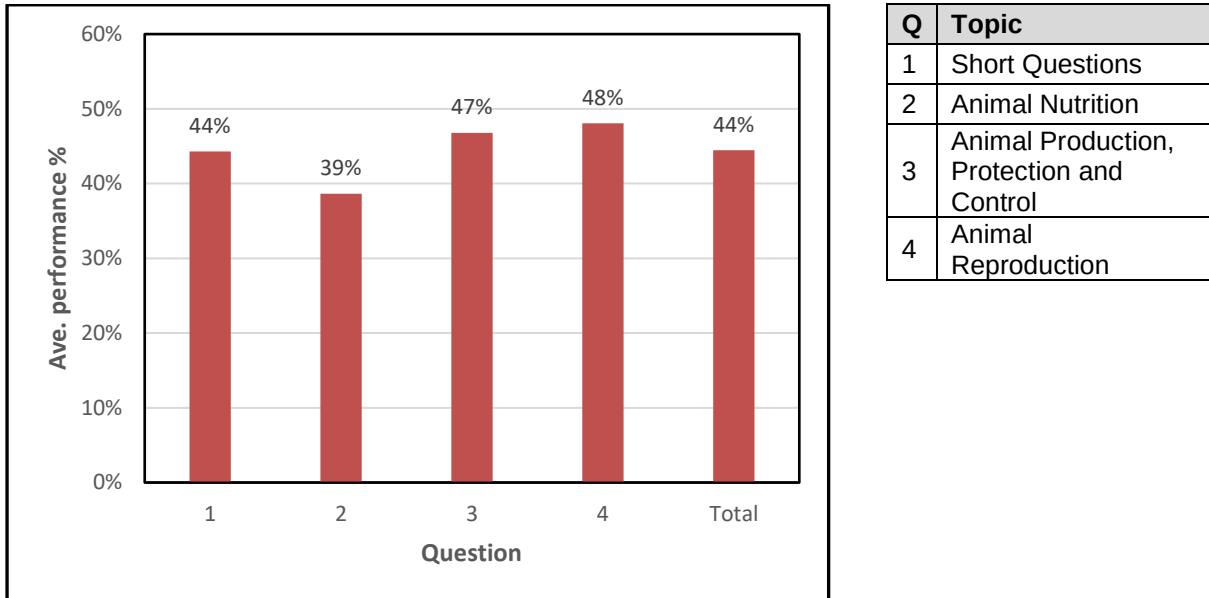
#### General comments

- (a) There is a decline in the candidates' performance in Paper 1 compared to 2021. The decline could be attributed mostly to a poor performance in Q1 and Q3 where performance declined by 3% and 7% respectively.
- (b) Answering as per instruction is still a challenge for some candidates. In Q1.2 the word 'ONLY' was omitted. In Q1.4 some candidates wrote TRUE/FALSE when they were expected to replace the incorrect underlined word with the correct word, and in Q2.1.3 they wrote the names of stomach parts instead of the letters representing those parts.
- (c) Q4 was the best-answered question, followed by Q3. Even though performance in Q2 was still poor, there was an improvement of 6% compared to 2021.
- (d) The following challenges are still persistent: writing of the correct formula for the *Nutritive Ratio*, where candidates did not indicate the formula as a ratio but instead presented it as a percentage. In addition to this problem, a large number of candidates failed to show all the steps, especially the calculation of the TDN. They failed to abide by the instruction which indicated that they must show all their calculations.
- (e) Analysis and application questions are still a problem, for example in the case of interpretation of the calculated values of the digestibility co-efficiency. Most candidates failed to identify the animal that was used in the trial based on its poor digestion of wheat straw, which is roughage. Some identified the animal based on its age whereas the question did not ask for stage of development. Only the type of animal was required.
- (f) The majority of candidates are now mastering the drawing of a graph but are still challenged by its interpretation. For example, they struggled to make deductions in Q4.4.2 (a) and 4.4.3 based on the information presented on the graph. Some candidates could not distinguish between the dependent and the independent variable.
- (g) Questions requiring reasoning, motivation or justification were still poorly answered by most candidates. This is an indication that candidates are not sufficiently exposed to these types of questions in the classroom.
- (h) The language of learning and teaching proved to be a challenge for most candidates. They were not able to respond appropriately to the instructional verbs used in questions. They also experienced difficulty with the spelling of terms or concepts.

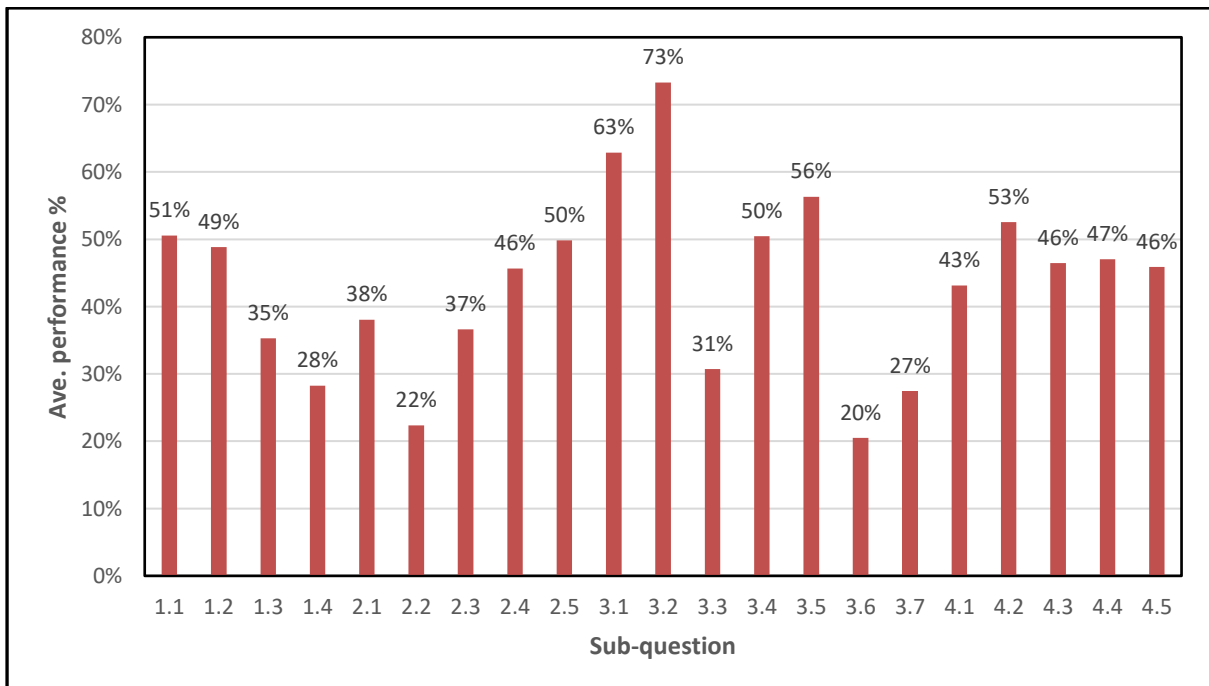
### 3.4 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 1

The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degree of challenge of each question as experienced by candidates.

**Graph 3.4.1 Average performance per question in Paper 1**



**Graph 3.4.2 Average performance per subquestion in Paper 1**



| Sub-Q | Topic                                   | Sub-Q | Topic                                                                                     |
|-------|-----------------------------------------|-------|-------------------------------------------------------------------------------------------|
| 1.1   | Multiple choice                         | 3.3   | Intensive chicken production system                                                       |
| 1.2   | Match columns                           | 3.4   | Guidelines in handling animals                                                            |
| 1.3   | Terminology                             | 3.5   | Parasite infestation                                                                      |
| 1.4   | Replace incorrect words                 | 3.6   | Life cycle of tapeworm and liver fluke                                                    |
| 2.1   | Ruminant stomach and regurgitation      | 3.7   | Symptoms of diseases in animals                                                           |
| 2.2   | Nutrient deficiencies                   | 4.1   | Accessory sex glands in bulls                                                             |
| 2.3   | Digestibility co-efficiency             | 4.2   | Female reproductive system & sperm cell                                                   |
| 2.4   | Calculation of nutritive ratio          | 4.3   | Artificial Insemination                                                                   |
| 2.5   | Energy value of feed                    | 4.4   | Reproductive processes in dairy cows                                                      |
| 3.1   | Growth rate against temperature changes | 4.5   | Techniques in animal reproduction<br>Oestrus synchronisation, Embryo transfer and cloning |
| 3.2   | Equipment in broiler production         |       |                                                                                           |

### 3.5 ANALYSIS OF LEARNER PERFORMANCE IN INDIVIDUAL QUESTIONS IN PAPER 1

#### QUESTION 1: SHORT QUESTIONS (ANIMAL SCIENCES)

##### Common errors and misconceptions

- (a) In Q1.1.4 candidates seemed not to have understood the term *re-insalivation* in the sequence of the *process of rumination* and as a result, they opted for C.
- (b) In Q1.1.9 choosing a primary sex organ from among the organs was a challenge. This reflected a lack of understanding between primary and secondary sex organs.
- (c) In Q1.1.10 most candidates struggled with the roles of different hormones in the reproductive processes, hence they opted for D (oestrogen and progesterone) instead of C (oxytocin and oestrogen) as the hormones that are responsible for the contraction of uterine muscles.
- (d) Many candidates did not recognise their spelling mistake in Q1.3.4, i.e. *seamen* instead of *semen*.
- (e) In Q1.2.1 some candidates failed to read the question statement to the end to put emphasis on the enlarged first part of the large intestines.
- (f) In Q1.2.4 many candidates displayed a lack of knowledge and understanding of the chemicals that kill ticks and mites and gave A (ointments) as an answer.
- (g) In Q1.3.1 the majority of candidates failed to differentiate between 'along' and 'against' *concentration gradient*.
- (h) In Q1.3.3 some candidates confused the hormone which inhibited milk ejection, *adrenalin*, with the hormone responsible for the milk ejection, *oxytocin*. Candidates did not understand the meaning of 'inhibiting'.
- (i) In Q1.3.4 candidates gave *sperm cell* as an answer instead of *semen*, showing that they were unable to distinguish between these terms.
- (j) In Q1.3.5 some candidates confused *vagina* with *artificial vagina*, thus showing a lack of understanding that *vagina* is an organ and *artificial vagina* is an apparatus used to collect semen.

- (k) In Q1.4.1 many candidates provided *improper* which is the direct opposite of the distractor *proper* which was to be replaced.
- (l) In Q1.4.2 some candidates overlooked the key word in the question, *bleeding*, and gave *Burddizzo* as the answer, while this tool does not cause bleeding at all.
- (m) In Q1.4.3 the incorrect answer, *chorion*, was more popular than *ectoderm* which is the correct answer, indicating that candidates were confusing germ layers with membranes surrounding the foetus.

### **Suggestions for improvement**

- (a) Teachers should compile a glossary of terms/concepts for each topic to be taught in order for the learners to master section A on terminology.
- (b) Teachers also need to compile questions for each topic thus encouraging learners to study and understand the glossary of terms. More testing on terms will improve the learners' performance.
- (c) Consider using interesting games, word puzzles and PowerPoint presentations for the teaching of key concepts and improving the spelling of these concepts. Spelling tests on concepts that are not easy to pronounce or write out should be incorporated in weekly activities.
- (d) The use of electronic technology, such as smart boards, internet and visual aids should be encouraged to improve learners' enthusiasm for the subject. It is hoped that this will stimulate learners to explore other material pertaining to the subject.
- (e) Teachers should train learners on how to answer questions by guiding them on what responses are required in order to meet the demands of the question.
- (f) Teachers are advised to realign the content and integrate related content. This will allow the learners to draw parallels between similar/related topics.
- (g) Teaching should be guided by the *CAPS* more than textbooks. However, textbooks and other resources may be used for enrichment.
- (h) Teachers are encouraged to do more in-depth research on various aspects in the *CAPS* so that they can further elaborate on the content.
- (i) Teachers need to be exposed to current practices in the subject.
- (j) Teachers should form a cohesive unit in their clusters. They can support one another by addressing challenging topics and by suggesting different strategies/methodologies to make such topics more interesting. These cluster units could also set common assessment tasks collectively.
- (k) Provincial coordinators, together with subject advisors and teachers, need to prepare revision packs that cover all the topics. Teachers should use activities from these packs as informal tasks, in class revision sessions and as a mock examination.

## QUESTION 2: ANIMAL NUTRITION

### Common errors and misconceptions

- (a) In Q2.1.1 some candidates still provided the class of the animal, viz. a *Ruminant* even though the question asked for the name like 'sheep'/'cattle'/'goat', thus showing a lack of understanding of the difference between the name and the class or type.
- (b) In Q2.1.2 (a) candidates did not know where foreign objects accumulate. The most common incorrect response was A *Omasum* instead of C *Reticulum*.
- (c) In Q2.1.3 many candidates wrote only the adaptation feature (the presence of micro-organisms) and left out their role (synthesis of vitamins) whereas the question asked for an adaptation feature and its role. This shows that they did not read the question in full. Selective reading was a problem in this question.
- (d) In Q 2.1.4 some candidates ignored the instruction to use the letters indicating the stomach parts; instead they wrote the names of the stomach parts.
- (e) In Q2.2.3 candidates struggled to classify the vitamin; instead they wrote the name of a vitamin. This is an indication that they lack understanding of classes of vitamins.
- (f) In Q2.2.4 most candidates gave general methods of supplementing deficiencies whereas the question was looking for methods specifically used to supplement vitamins. Some confused the equipment with the role, e.g. *drenching gun* which is a tool that is used for drenching/dosing.
- (g) In Q2.3.1 candidates were unable to differentiate between the name and the type of the animal as some wrote the name whereas the question required a type.
- (h) In Q2.3.2 most candidates could not utilise the data provided to give a reason for their choice of animal. Some wrote 13 kg was digested instead of 13%.
- (i) In Q2.3.4 the majority of candidates wrote the methods of improving digestibility of feeds in general, instead of those applicable to roughages, which are mainly grasses or straws, with *soaking* and *roasting* not being applicable.
- (j) In Q2.4.1 the common errors identified when calculating the *nutritive ratio* were as follows:
  - Failure to show how TDN was calculated. They just came up with an answer without showing how they got to the answer.
  - The formula was not presented in the form of a ratio, for example  $\frac{\text{TDN} - \text{DP}}{\text{DP}}$
  - If the ratio was only shown in the answer, 1 mark was awarded instead of 4.
  - Expressing the answer as a percentage instead of as a ratio.
- (k) In Q2.4.2 some candidates wrote 'suitable' without indicating what the feed was suitable for.
- (l) In Q2.4.3 candidates wrote 'less than 1 : 5' instead of 'less than 1 : 6' which is a benchmark for comparison of NR.
- (m) In Q2.5.4 many candidates confused the aims of calculating *energy value* with the importance of a *fodder flow*.

### Suggestions for improvement

- (a) Posters indicating the classification of each animal as well as their names could be a useful teaching tool to indicate what a *class* is. In addition, diagrams of the *alimentary canals* alongside the classes and names would give learners the opportunity to establish the similarities and differences between the *alimentary canals* of animals belonging to the various classes.
- (b) Various textbooks and the dissection of real animals for observation of the parts could be used to draw the similarities and differences and the roles of each part in digestion.
- (c) Carefully planned practical investigations and questionnaires will assist learners in developing an in-depth understanding of the content.
- (d) Giving more exercises with different scenarios is imperative to make learners aware of how to respond to questions.
- (e) Teachers are encouraged to conduct regular informal assessments on calculations, providing guidance on the use of correct formulae and following the correct steps when substituting the formula with values. This will develop the learners' ability to make the correct calculation.
- (f) The implications of the calculated values should be explained to learners. This will enable learners to have a better understanding of the application of the results.

### QUESTION 3: ANIMAL PRODUCTION, PROTECTION AND CONTROL

#### Common errors and misconceptions

- (a) In Q3.1.2 the majority of candidates failed to refer to both the growth rate and temperatures in justifying the choice of animal in Q3.1.1. The common answer was 'high or low growth rate', without mentioning the influence of temperature on the growth rate.
- (b) In Q3.1.3 few candidates achieved full marks in the graph question. The following are common errors:
  - Writing only 1 unit
  - Incorrect placement of variables
  - Incorrect calibration: for example figures taken as they are in the question paper
- (c) In Q3.3.1 some candidates incorrectly identified picture A as a backyard system and picture B as a free range. This shows that they could not analyse the picture to find clues that would lead them to an answer, for example the presence of a woman with a basket, feeding the chickens and a fence to contain them from moving, compared to those on an open field picking up insects.
- (d) In Q3.3.2 few candidates wrote 'nutrition' as one of the answers. They did not realise that the question asked for factors that increased production *except* for nutrition. This was an indication that they did not read the question in full.
- (e) In Q3.4 candidates showed a lack of understanding of the basic guidelines applied when handling different animals. They referred to sheep in Q3.4.1 and cattle in Q3.4.2.

- (f) Instead of classifying the parasite as internal in Q3.6.1, some candidates wrote the name. This again shows that they could not differentiate between the class and the name.
- (g) Some candidates in Q3.6 incorrectly referred to a *liver fluke* as a *round worm*.
- (h) In Q3.6.3 the majority of candidates, including the top achievers, could not state the biological measures to control the parasite. The common responses offered were: veld burning and rotational grazing, which are not biological as such but general veld management measures that are used to control parasites without the use of chemicals. The problem here was a lack of understanding of a 'biological control measure'.
- (i) Most candidates wrote *FMD/RVF* in Q3.7.1 which are viral diseases whereas the symptoms in the data referred to bacterial and protozoan diseases. Linking the symptoms to a specific disease has been a problem for some time, regardless of the guidance provided in the examination guidelines on how diseases and parasites will be assessed.
- (j) In Q3.7.2 and Q3.7.3 some candidates failed to see that the question directed them to the table with ANIMAL 1 and ANIMAL 2. They wrote 'cows' and 'sheep' instead of ANIMAL 1 and ANIMAL 2.
- (k) In Q3.7.4 candidates misunderstood the vector that transmits red water. It seems as if they are not aware that tick-borne diseases are tick-specific. Answers such as 'bont tick/mosquito' were provided instead of 'blue tick' which is the only tick that can cause Redwater.

### **Suggestions for improvement**

- (a) Teachers need to train learners not only to draw graphs but to consider accuracy as indicated in the current marking criteria/rubric based on the examination guidelines. The learners should also be able to interpret graphs.
- (b) Learners should be encouraged to use the internet to access information about the tools, equipment and facilities used in different farming enterprises.
- (c) PowerPoint slides with pictures and videos of the facilities and tools could be prepared and used in the classroom, to arouse the interest of the learners.
- (d) Excursions or visits to farms could be organised in collaboration with local extension officers from the Department of Agriculture.
- (e) Collaboration among subject teachers in preparation and teaching will be very beneficial. These sessions could build capacity in the concepts and in the teaching of diseases and parasites on production enterprises (by making use of mind maps that group diseases/parasites based on the pathogens/vectors), modes of transmission, key symptoms as well as preventative and control measures.
- (f) Teachers should assist learners to distinguish between the classification and the name of a parasite.
- (g) Intensive revision of work using charts, scenarios and tables on diseases and parasites is necessary as the information is vast and likely to cause confusion.

## QUESTION 4: ANIMAL REPRODUCTION

### Common errors and misconceptions

- (a) In Q4.1.1 to Q4.1.3 many candidates failed to link accessory glands to the statements. It became evident that they did not know the location of accessory glands and their functions.
- (b) In Q4.2.1(c) and (d) a number of candidates described the process (*ovulation* and *fertilisation*) even though they were only required to identify the processes. Some candidates incorrectly labelled *mid-piece* (l) as *mitochondrion*. This is an indication that they could not differentiate between the mitochondrion and the mid-piece.
- (c) In Q4.2.2 some candidates stated *oestrogen* instead of *luteinizing hormone*. It became evident that they were not aware that oestrogen is not directly responsible for ovulation; rather, it stimulates the secretion of *LH* which causes rupturing of the *Graafian follicles*.
- (d) In Q4.2.3 a few candidates confused *production* of an ovum with the *release* of an ovum.
- (e) In Q4.2.5 the majority of candidates did not respond to the question fully. They only mentioned the presence of the enzyme without describing how the enzyme assisted the sperm cell to penetrate; as a result, they lost a mark.
- (f) In Q4.3.1 candidates explained what happens during the *oestrus phase* instead of mentioning the *oestrus phase*.
- (g) In Q4.3.2 a few candidates incorrectly mentioned *thermometer* as the answer instead of *pedometer*.
- (h) In Q4.4, answers were available from the graph but candidates failed to interpret the source.
- (i) In Q4.4.1 some candidates incorrectly referred to the curve as *milk production* instead of *lactation curve*.
- (j) In Q4.4.2 (b) candidates could not deduce the foetus stage from the graph; instead they extracted the description of what occurs during foetus stage directly from the graph. A common response was: 'Foetus grows very fast.'
- (k) In Q4.4.3 candidates were unable to refer to the graph to identify the month when pregnancy normally ends. Some referred to *12 months* instead of *month 12*. Most of those who indicated *9 months* did not base their response on the graph; they simply used their knowledge that pregnancy in cows lasts for 9 months 9 days.
- (l) In Q4.4.4 candidates confused causes of abortion with pregnancy problems, hence they gave responses like *maceration* and *mummification*.
- (m) Q4.5.1 required different techniques used in animal reproduction and the majority of candidates opted for the processes within the techniques. They confused *ET* with *embryo flushing*, which is a process within the technique and also confused *nuclear transfer* with *enucleation*. Some candidates wrote *synchronisation without oestrus*, which resulted in the loss of a mark because synchronisation alone is too general. This is a great concern because the issue of synchronisation alone was addressed in the previous diagnostic report.

- (n) In Q4.5.2 a few candidates stated a device (*CIDR*) which is used to administer a hormone instead of providing a hormone.
- (o) In Q4.5.3 some candidates did not understand that there are no differences between a *donor* and *superior* animal, and an *inferior*, *recipient* or *surrogate* animal. They wrote *donor* and *superior* as separate animals and lost a mark because that refers to the same animal; similarly, *recipient* and *inferior* animals are the same animals.

### **Suggestions for improvement**

- (a) Teachers are encouraged to contact institutions where the various processes are practised so that learners can observe them to enhance their understanding of the reproductive processes.
- (b) In presenting the various processes such as *oestrus cycle*, *synchronisation of oestrus*, *artificial insemination*, *stages of pregnancy*, *embryo transfer*, *parturition* and other reproductive processes, teachers should use flow diagrams, schematic representations, projections and videos to indicate the characteristics and the effects of various hormones in the different processes.
- (c) It is also suggested that teachers cover all the reproductive processes simultaneously in order to show the synergy between the processes. This will assist learners to understand these processes better.
- (d) Enrichment and enhancement material should be sourced from various resources such as the slides from the ASAAE software.
- (e) Standardised formal tasks should be prepared with the aim of raising the level of questioning in order to prepare the learners for questions that require analysis and reasoning.
- (f) The importance of learning the subject terminology must be emphasised, for example *synchronisation of oestrus* and not just *synchronisation*.
- (g) Teachers should expose learners to questions that require interpretation of data in the form of graphs and diagrams.
- (h) Learners need to be assisted with different reproductive techniques and the processes involved in each technique.

## **3.6 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 2**

### **General comments**

- (a) Generally, candidates' performance in this paper has improved in comparison with 2021.
- (b) In 2022 candidates' responses indicated that most of them still struggled with questions that require application of knowledge, as well as questions based on sources.

### General suggestions for improvement

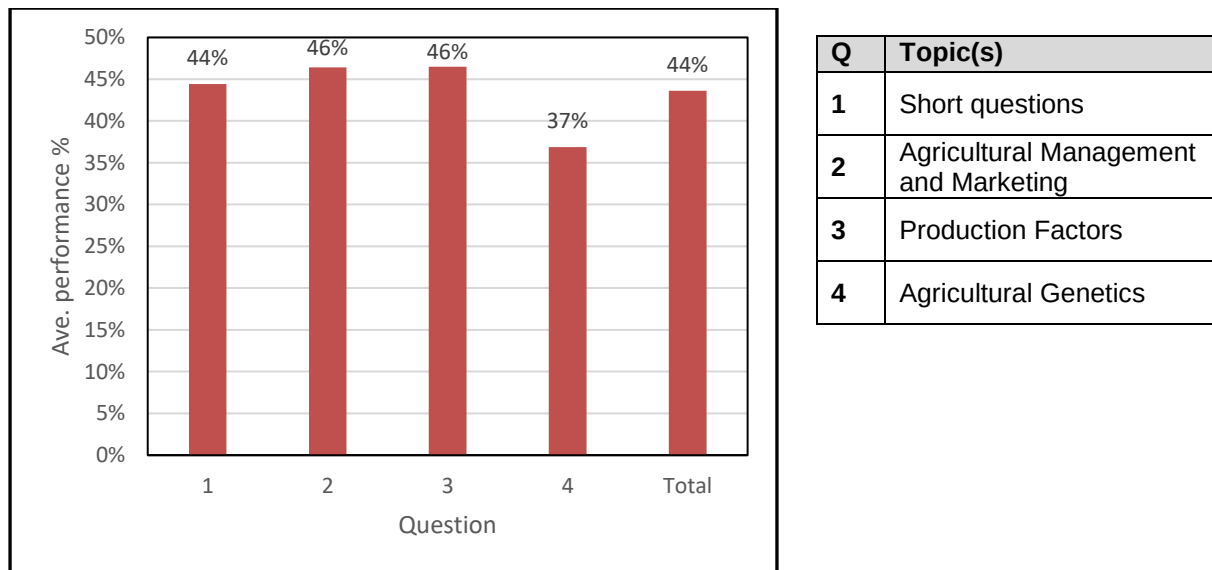
The following suggestions for improvement are proposed to address the quality of performance of candidates:

- (a) In addition to textbooks, resources such as agricultural magazines or the internet should be used to develop a comprehensive understanding of subject terminology.
- (b) Teachers should expose learners to regular and consistent informal assessment tasks or activities that will improve their confidence in dealing with the subject content.
- (c) Setting of quality-assured common tasks on more data response questions, such as graphs and case studies, to enhance interpretation and application is recommended.
- (d) Expose learners to simple mathematical calculations involving percentages, ratios, working out the effect of polygenes, mass, height and length, as well as their respective units. Calculations, pictures, graphs and tables is an integral part of the subject. It is recommended that all calculations commence with the formula/formulae, then the correct substitution should be done, followed by the actual calculation and ultimately the correct answer. The final answer should also be re-checked, if time allows.
- (e) Although learners might have access to only one textbook, teachers should acquaint themselves with and use several available textbooks in their lesson preparation. Teachers and learners will thus be exposed to a wide range of possible activities.
- (f) Teachers need to broaden their knowledge and practical experience in certain areas of the curriculum so that they are able to expose learners to practical situations; for example *Production Factors* in Paper 2 required candidates to understand different farm records, management skills, types of labour legislation, and to interpret tables.
- (g) Responses to short questions show that many learners still lack basic conceptual knowledge and teachers should use various approaches to expose and explain terminology and concepts to learners. Teachers must ensure that learners are exposed to the language in which they will be writing the examination, as many learners struggle with reading, understanding and interpreting questions. Such learners also find it challenging to express their responses correctly.

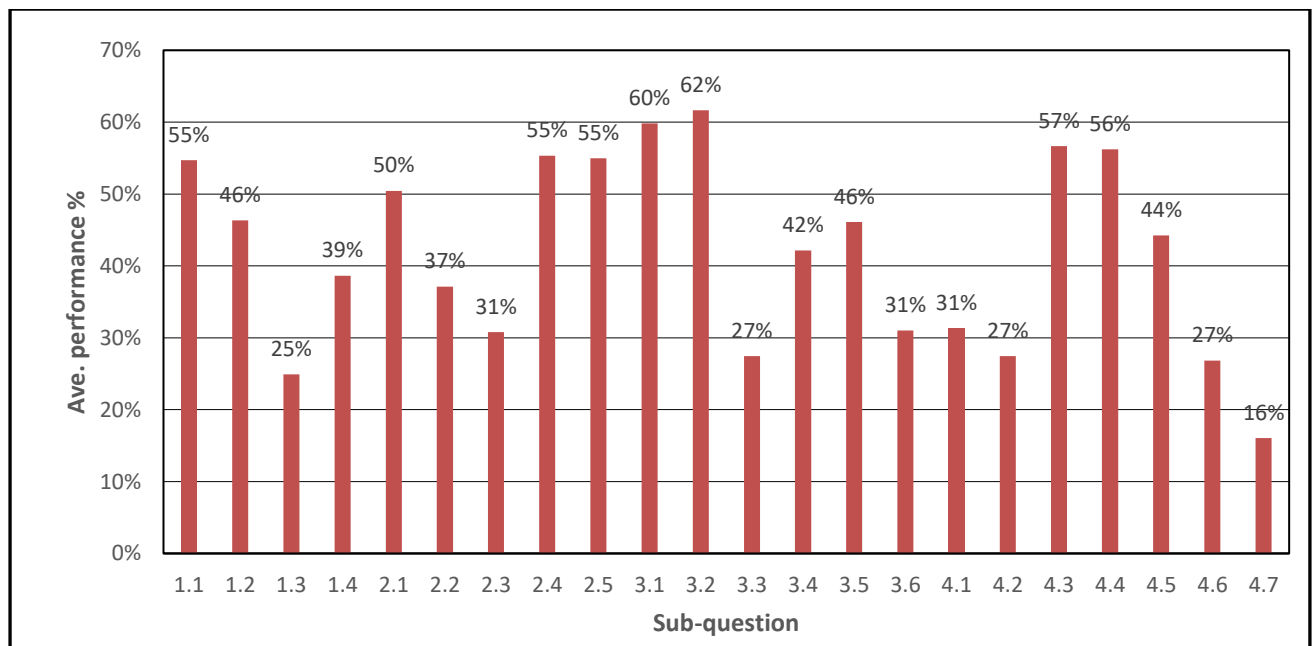
### 3.7 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 2

The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 3.7.1 Average performance per question in Paper 2**



**Graph 3.7.2 Average performance per subquestion in Paper 2**



| Sub-Q | Topic                                | Sub-Q | Topic                                    |
|-------|--------------------------------------|-------|------------------------------------------|
| 1.1   | Multiple choice                      | 3.3   | Labour legislation                       |
| 1.2   | Match columns                        | 3.4   | Assets and liabilities                   |
| 1.3   | Terminology                          | 3.5   | Types of capital in a farming enterprise |
| 1.4   | Replacement of incorrect words       | 3.6   | Management principles and risk           |
| 2.1   | Quantities of product in the market  | 4.1   | Breeding systems                         |
| 2.2   | Free market system                   | 4.2   | Variation                                |
| 2.3   | Entrepreneurship                     | 4.3   | Methods of selection                     |
| 2.4   | Quantities supplied over 5 months    | 4.4   | Di hybrid crossing                       |
| 2.5   | Marketing chain                      | 4.5   | Sex-linked chromosomes and autosomes     |
| 3.1   | Land as a production factor          | 4.6   | EBV and heritability                     |
| 3.2   | Labour output as a production factor | 4.7   | Genetic modification                     |

### 3.8 ANALYSIS OF CANDIDATES' PERFORMANCE IN INDIVIDUAL QUESTIONS IN PAPER 2

#### QUESTION 1: SHORT QUESTIONS (AGRICULTURAL MANAGEMENT AND GENETICS)

##### Common errors and misconceptions

- (a) Both Q1.1.4 and Q1.1.9 had a combination of distractors. One question was on SWOT analysis and the other was on Mendel's Laws of Inheritance. Most candidates struggled to answer questions with combinations of multiple distractors, which has been a problem for some time.
- (b) In Q1.3.1 most candidates confused *controlled marketing system* with a *free marketing system*.
- (c) Some candidates could not indicate that the practice of combining pieces of land with maximise production is *consolidation* in Q1.3.2.
- (d) In Q1.3.3, candidates were required to indicate the type of dominance where none of the parents' characteristics are visible in their offspring. They wrote *complete dominance* instead of *incomplete dominance*.
- (e) A few candidates stated *electric injection* or *electrocution* as the correct answer for Q1.3.4 instead of *electroporation*.
- (f) In Q1.3.5, most candidates could not identify that an organism that has more than two sets of chromosomes is a *polyploidy*, instead referring to it as a *diploid* or *polygene*.
- (g) Many candidates opted for *fixed price* in Q1.4.1 instead of identifying that the setting of prices for goods and services with no bargaining is *price fixing*. There seemed to be great confusion and misunderstanding.
- (h) In Q1.4.2 some candidates could not explain that the money which is paid back to a financial institution over and above the money borrowed is *interest* and not a *loan*.
- (i) A few candidates interpreted the use of statistics as *biolistic* instead of *biometrics* in Q1.4.3.
- (j) In Q1.4.5 most candidates referred to a small piece of the DNA that carries hereditary information as an *allele* instead of a *gene*.

##### Suggestions for improvement

- (a) There needs to be a greater emphasis on the teaching and learning of appropriate subject terminology related to various topics, together with the correct spelling of these terms.
- (b) Encourage learners to write terms properly, as incorrectly spelt terms may have a different meaning in the context of the question. This normally leads to a loss of marks.
- (c) Teachers should compile a list of concepts based on the CAPS document and present these concepts first whenever introducing a new topic.

- (d) The use of word banks, word puzzles and other terminology exercises is still strongly recommended.
- (e) Teachers need to reinforce the understanding of the terms and concepts by compiling a bank of questions from 1.1 to 1.4, and extract questions for each topic, as activities for learners.

## QUESTION 2: AGRICULTURAL MANAGEMENT AND MARKETING

- (a) In Q2.1.4 many candidates struggled to identify the factor that could hamper the marketing of product 1, as *seasonal fluctuation*.
- (b) Some candidates confused factors influencing *supply* with those influencing *demand* in Q2.1.5. This is an indication that they are unable to differentiate between *supply* and *demand*.
- (c) Many candidates confused the *marketing approaches* with *marketing systems*. It was for this reason that they could not provide the correct market system in Q2.2.1, namely the *free market system*.
- (d) In Q2.2.2 most candidates displayed a lack of knowledge of the channels of a free market system. They provided answers randomly.
- (e) In Q2.2.3 a few candidates could not differentiate between the advantage of *farm gate marketing* for the farmer on the one hand and the consumer on the other.
- (f) Most candidates could not identify phases of entrepreneurship based on the scenario that was given. They extracted responses directly from the scenario. However, they were supposed to use the information in the scenario to give the phases in Q2.3.1.
- (g) In Q2.3.3 many candidates struggled to identify a *success factor* that enabled a young entrepreneur to perform, as reflected by given statements.
- (h) Many candidates struggled with identifying the axis on which to place quantities offered and the axis on which to place the price in Q2.4.1. They still confused independent and dependent variables.
- (i) Some candidates could not deduce the law that was applicable to the information provided on the table (law of supply), responses like 'the higher the price, the lower the demand'. This was not the response required as the information was based on the *Law of Supply*.
- (j) Many candidates knew the factors that affect supply but responded poorly. Although Q2.4.3 was based on the table, factors affecting the supply were not in the table.
- (k) In Q2.5.2(b) some candidates could not interpret this question on a *marketing chain*. They were expected to concentrate on the stage that needed improved road infrastructure before processing stage which was the *farm*. Most gave *stage 5* as an answer. This showed that the candidates could not respond to questions based on the interpretation of data.

### Suggestions for improvement

- (a) Learners should be taught channels of marketing by using flow charts, pictures or drawing for them to have a broad visualisation of the marketing channels. Teachers need to give learners practical exposure to marketing channels on local farms and in towns.
- (b) Learners must be taught to understand the context and be able to interpret the data before they respond to questions. They should be shown how the heading is derived from the given data and how to compose the heading.
- (c) Learners must be made to understand that if data or a scenario is given, unless indicated otherwise, the questions will be based on the data or scenario. Furthermore, it is important to interpret the leading statement correctly, which will guide learners on the questions to follow.
- (d) Learners must be taught how to label and calibrate the axes correctly. This will help avoid the wrong curve being drawn. Teachers could use a number of examples to demonstrate the skills of labelling and calibrating axes practically.
- (e) When dealing with graphs on supply and demand, learners should be taught that the *Price* is always on the Y-axis and *Quantity offered/demanded* is always on the X-axis. This will prevent confusion when drawing such graphs.

### QUESTION 3: PRODUCTION FACTORS

#### Common errors and misconceptions

- (a) Most candidates performed poorly in Q3.1.1 on the economic characteristics of land and in Q3.1.3 on functions of land. This was caused by the fact that they could not differentiate between these two aspects of land. Some provided limitedness even though the question had specified that other factors that would exclude limited availability. Some provided *availability* as a response, although the question specified that factors other than *limited availability* should be given.
- (b) Some candidates in Q3.2.1 could not respond to this question correctly because of poor interpretation of graphs. A guide on the number of people employed was provided but candidates still stated the *law of diminishing returns* instead of *labour*.
- (c) In Q3.2.2 many candidates interpreted the graph as a depiction of the *law of diminishing returns*; the deduction was based on the *law of diminishing returns* and not on the relationship between the number of people employed and the output. This showed a lack of conceptual abilities.
- (d) In Q3.2.4 many candidates did not realise that the question required ways to improve the *economic conditions* of farm workers, instead of providing ways to improve *labour productivity*.
- (e) In Q3.3.1 to Q3.3.3 some candidates were not able to indicate and link a description to a piece of labour legislation. Candidates may be conversant with legislation governing farm workers but are not able to link laws to the scenarios and situations. Some gave incorrect descriptions of the Basic Conditions of Employment Act and provided responses such as the as Basic Conditions of Unemployment Act or Basic Employment Act and sometimes the Occupational Health and Safety Act. The laws should be referred to correctly.

- (f) Some candidates could not distinguish between *assets* and *liabilities* in the table in Q3.4.1. Instead they defined *assets* and *liabilities*. Certain candidates confused *assets* with *liabilities*. In some instances, the formula for net worth was expressed in reverse ( $\text{Net worth} = \text{Liabilities} - \text{Assets}$ ), since candidates became confused when they realised that the value of liabilities was higher than the value of assets. They could not comprehend that the business was not viable.
- (g) In some instances, the formula was not correctly presented in Q3.4.2; common mistakes like  $\text{Assets} - \text{Liabilities}$  leaving out  $\text{Net Worth} =$  to make it a correct formula.
- (h) In Q3.4.3 many candidates responded on *profitability* instead of *viability* of the enterprise. This might indicate a lack of knowledge of subject terminology.
- (i) Candidates in Q3.5.1 were expected to identify types of capital in the picture; instead they gave examples of capital. They seemed to ignore the key word in the question.
- (j) In Q3.5.2 some candidates could not associate depreciation with movable capital item F in the question paper. Candidates are conversant with problems associated with capital but could not apply them to the picture.
- (k) Many candidates tried to define management in Q3.6.1, but could not include important aspects of the concept. This may be attributed to the fact they concentrated on the three other production factors.
- (l) In Q3.6.2 some candidates performed poorly due to a lack of knowledge of the risk management strategies; instead they provided management success factors.

### **Suggestions for improvement**

- (a) Emphasis should be placed on terminology each time a topic is introduced to learners. Informal tests can be used assess the terminology.
- (b) Learners should be taught about labour legislation with acronyms in order to assist recall.
- (c) Learners should be taught that movable capital, like tractors and machinery in this category, has a limited life span and has to be replaced over time due to depreciation as a result of wear and tear.
- (d) Case studies, diagrams and scenarios need to be included in assessment tasks at regular intervals during the school year, with the aim of exposing learners to activities that will improve their skills in answering these types of questions. Such interventions could assist learners by improving their reading and understanding skills, the application of knowledge and an awareness of how to follow instructions.
- (e) Content knowledge is vital. Learners should be familiar with risk management strategies. Once this knowledge is in place, case studies can be used as a tool to help them to practise applying the knowledge learnt in a specific situation and thereby identify the risk management strategy in any given scenario.

## QUESTION 4: BASIC AGRICULTURAL GENETICS

### Common errors and misconceptions

- (a) Q4.1.1(a) was very poorly attempted. Many candidates confused *crossbreeding* with *outbreeding*. The same applied to Q4.1.1(b) where *inbreeding* was also confused with *linebreeding* or *upgrading*. Candidates seemed to struggle with following the generation crossing using the same bull. They mistakenly confused *inbreeding* with *linebreeding* even though there was no animal from another family of the same ancestral line that was used.
- (b) Some candidates were not able to respond appropriately to Q4.1.2. This question was based on Q4.1.1(b). The responses depended on whether *linebreeding* or *upgrading* was the answer or *upgrading* applied in Q4.1.1(b).
- (c) In Q4.1.3 many candidates' responses reflected that they could not separate the advantages from the disadvantages of *inbreeding*. They explained that the offspring produced would be uniform. That would be an advantage rather than a disadvantage.
- (d) Most candidates in Q4.1.4 indicated that one advantage of crossbreeds was that they had the desirable characteristics. As with any breeding system, the intention is to achieve the desirable characteristics, however, with crossbreeding the intention is more on the *improvement* of the desirable characteristics.
- (e) Some candidates were not able to respond appropriately to Q4.2.1. They could not provide the internal causes of *variation*; instead they confused internal causes with external factors such as climate, feeding and diseases. The correct responses should have been *mutation*, *meiosis* and *fertilisation*.
- (f) In Q4.2.2 most candidates explained *variation* by emphasising that it 'results in different organisms' instead of explaining how those differences are utilised in selection and breeding, for example in 'the development on new breeds/cultivars'.
- (g) In Q4.3.1 a few candidates displayed poor knowledge of *selection* methods. This was evident when they swapped family selection with pedigree selection. It showed that knowledge of content is inadequate, particularly in genetics.
- (h) Many candidates lost marks in Q4.4.1 on a *dihybrid cross*. They were unable to interpret the *alleles* as presented. This could have been a reason they could not come up with the *gametes*, for the offspring numbered 17; they were also unable to give the genotype of a female parent in Q4.4.1(a).
- (i) In Q4.4.2(a) some candidates could not determine the *phenotype* of the individual which was *polled* and *roan*. They did not understand the concept *polled*.
- (j) Many candidates in Q4.5.1 confused the *sex chromosomes* with *autosomes*. They stated 60 instead of 2. They also failed to remove the *sex chromosomes* from the total number of chromosomes to get the correct number of *autosomes* in horses. They responded with 32 pairs instead of 31 pairs.
- (k) In Q4.6.1 most candidates failed to differentiate between *heritability* and *heredity*. Instead of defining *heritability*, the ability to transfer those characteristics, they defined *heredity*, the transfer of characteristics from parents to offspring. Most candidates could not answer this question correctly.

- (l) In Q4.6.3 the majority of candidates were unable to giving the correct response of 60% environmental effect if the genetic effect on milk production was 40%. They did not realise that the environment played a greater role than genes. This was another challenging question for even the high flyers.
- (m) Some candidates struggled to show the relationship between *heritability* and the *EBV* of the trait in Q4.6.4. It was evident that they did not understand which one of the two influenced the other. As a result, responses such as 'the higher the *EBV*, the higher the *heritability*' were common. Candidates could not understand that it is the % heritability of a trait that will determine the predictable estimated performance of the progeny for the trait before it is even born hence the higher the heritability for the trait the more the chances the progeny will acquire that characteristic (*EBV*).
- (n) In Q4.7.2 more general information on the negative effects of *GM* crops other than those directly affecting the environment were given.

### **Suggestions for improvement**

- (a) The key to mastering basic genetics is the understanding of terminology. Learners should be able to describe the concepts and provide practical examples to illustrate their understanding of the terms and concepts.
- (b) Teachers should give special attention to basic crossing, genetic concepts and terminology in their teaching of this topic. Teachers must teach learners *monohybrids*, *di-hybrid*, and *F1* and *F2* crosses as included in the *CAPS*.
- (c) Teachers need to specify that any letter of the alphabet can be used to represent the *alleles*. However, if letters are given in a question, learners are advised to use them. It is important for learners to ensure that they read and understand the preceding statements before answering questions on genetics.
- (d) When teaching breeding systems, learners should be taught examples of crossings that could have negative implications, as well as examples with desirable outcomes. Diagrams representing animal breeding systems should be used to illustrate basic types of breeding methods.
- (e) Responses to questions on genetics suggest that there could be an underlying content gap amongst teachers. Subject advisers should conduct workshops to address the shortcomings in content knowledge in this regard.
- (f) The teaching of genetics should be enhanced by providing practical examples at the learning site where it applies to plants, flowers and livestock.
- (g) There should also be integration with Life Sciences, as genetics is taught comprehensively in Life Sciences.

# CHAPTER 4

## BUSINESS STUDIES

The following report should be read in conjunction with the Business Studies Paper 1 and Paper 2 question papers for the NSC November 2022 examinations.

### 4.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who sat for the Business Studies examination in 2022 decreased slightly by 1 854 candidates.

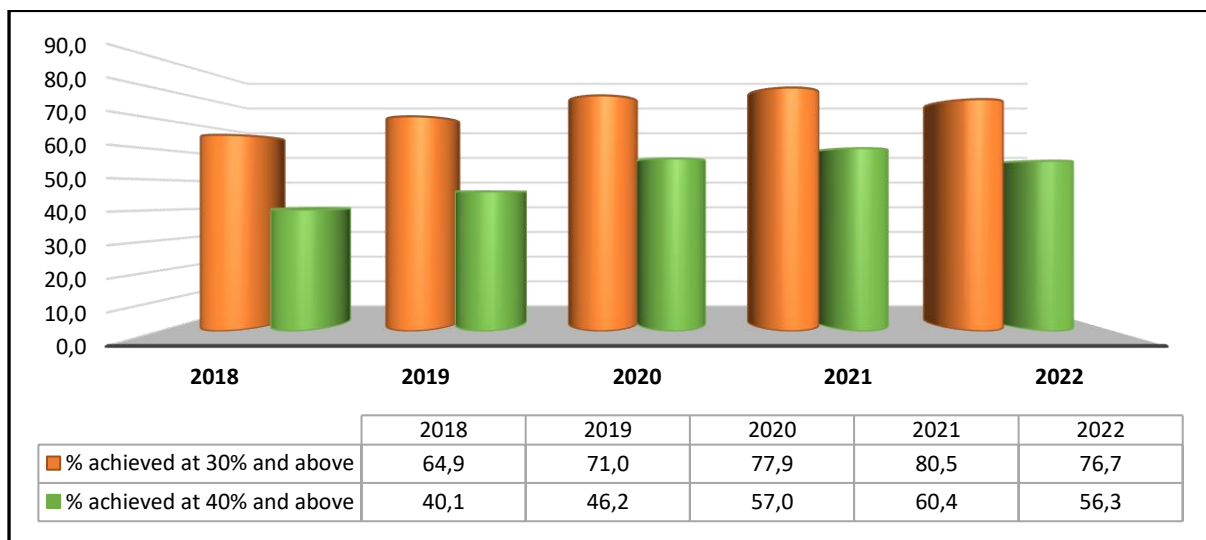
While the table below indicates a positive trend in the pass rate over the period 2018-2021, a decline in the pass rate was evident this year. Candidates who passed at 30% (Level 2) declined from 80,5% in 2021 to 76,7% in 2022. There was a corresponding decline in the pass rate at 40% (Level 3) over the past two years from 60,4% to 56,3%. An encouraging feature, however, was the improvement in the percentage of distinctions over 80% (level 7), which increased from 5,2% in 2021 to 6,0% in 2022. This converts into an increase of 1 839 in the number of distinctions.

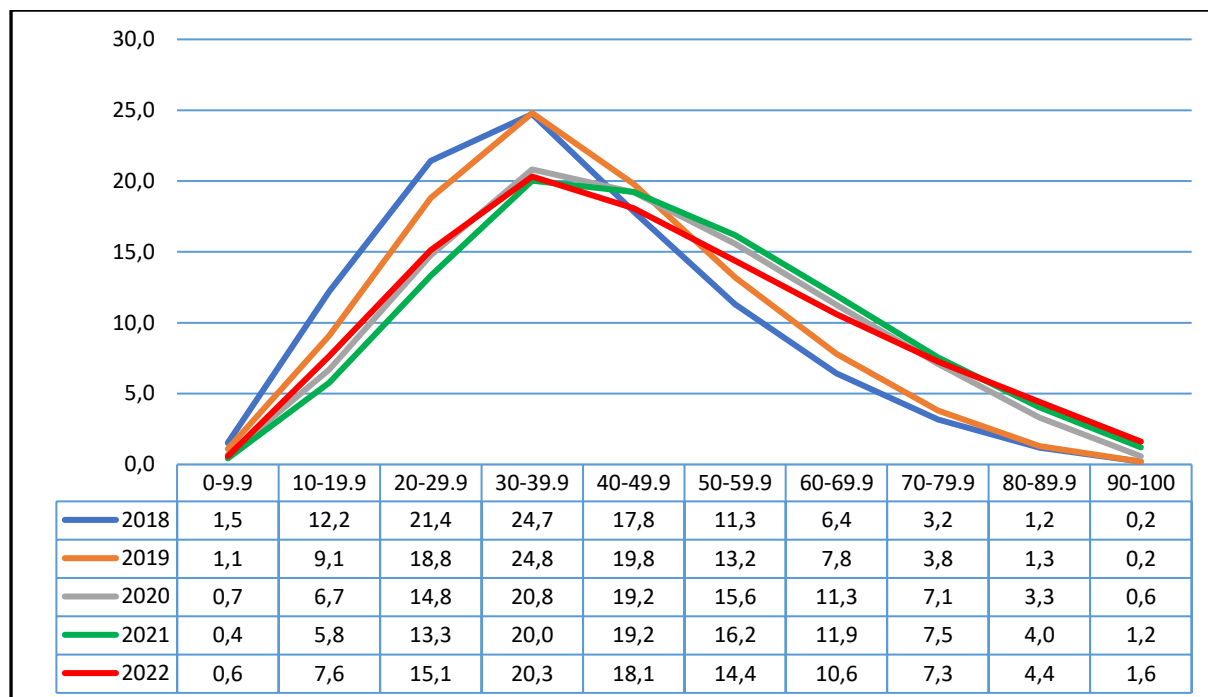
In view of disruption to academic programmes in 2020 and 2021, and despite some of the factors mentioned above, the results achieved by this cohort are still commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the overall performance.

**Table 4.1.1 Overall achievement rates in Business Studies**

| Year | No. wrote | No. achieved at 30% and above | % Achieved at 30% and above | No. achieved at 40% and above | % Achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 192 139   | 124 618                       | 64,9                        | 77 105                        | 40,1                        |
| 2019 | 186 840   | 132 571                       | 71,0                        | 86 242                        | 46,2                        |
| 2020 | 207 045   | 161 224                       | 77,9                        | 118 100                       | 57,0                        |
| 2021 | 243 843   | 196 233                       | 80,5                        | 147 398                       | 60,4                        |
| 2022 | 241 989   | 185 503                       | 76,7                        | 136 330                       | 56,3                        |

**Graph 4.1.1 Overall achievement rates in Business Studies (percentage)**



**Graph 4.1.2 Performance distribution curves in Business Studies (percentage)**

## 4.2 OVERVIEW OF CANDIDATES' PERFORMANCE: PAPERS 1 AND 2

### General comments

- (a) The general performance of candidates declined in Paper 1 compared to that of the performance in Paper 2. Candidates appeared to experience difficulties with the topics *Legislation* and *Quality of Performance* which were assessed in Sections B and C of Paper 1. These topics continue to pose challenges to many learners despite the recommendations to address these topics suggested in past national diagnostic reports on learner performance.
- (b) It was also disappointing to note that candidates' performance declined in Section A in both Paper 1 and Paper 2 even though some of the subquestions in this section appeared many times in past NSC November examinations.
- (c) Candidates performed well in the subquestions on *Business Ventures* which were covered in Q2 and Q5 in Paper 2. It is pleasing that recommendations made on this topic in the 2020 and 2021 national diagnostic report were implemented through various provincial and districts intervention programmes. It must be pointed out that many learners found the content on this topic challenging, as evident by the poor performance in 2021.
- (d) Many candidates were able to provide relevant quotes and/or write the essence in response to questions based on scenarios. They were also able to respond appropriately to questions requiring concept identification from given scenarios and statements and to write full sentences on middle- and higher-order questions.
- (e) However, the inappropriate responses provided by certain candidates to middle- and high-order questions continue to expose deficiencies in language proficiency. They did, however, earn part-marks for such a shortcoming.

- (f) Candidates were not able to provide relevant and valid examples of the latest developments in Business Studies to support their responses in Q5 and Q6 and subsequently lost the two marks allocated for originality in essay questions. This challenge persists despite recommendations made in the 2021 national diagnostic report on learner performance.

### General suggestions for improvement

- (a) When approaching *Legislation*, an in-depth analysis of each Act must be done with specific reference to the purpose, impact, compliance, and penalties for non-compliance. Learners must be given the opportunity to conduct research on businesses that must or should comply with these Acts and present their findings in the classroom. Furthermore, teachers must ensure that each Act is adequately taught, assessed, and remediated during the academic year.
- (b) Subject advisors must conduct capacity building workshops on *Quality of Performance* using practical examples and demonstrations. They must also invite the Grade 10 and 11 teachers to attend these workshops as this topic progresses into Grade 12. Learners need to know all aspects covered in this topic, and more especially the differences between *quality* concepts and *total quality management* as outlined in the *2021 Examination Guidelines*.
- (c) Learners must be exposed to a variety of Section A type direct and indirect questions that focus on concepts, terminology and application of knowledge. These questions must form part of informal and formal assessment tasks and they must cover aspects of all topics that were taught during the academic year.
- (d) Learners must be encouraged to provide complete responses to questions that require middle- and higher-order thinking skills. They must also be made aware of the marking principles that will be applied to these types of questions as outlined in the 2022 Business Studies November NSC notes to markers.
- (e) All subtopics for both Paper 1 and 2 must be adequately taught and assessed during the academic year. It is important to alert learners about potential contextual and essay questions and the impact of this on future performance.
- (f) It is suggested that teachers should construct a 5-year report using the national diagnostic data on learner performance from 2018-2022. It is necessary to bring together all misconceptions, errors, and recommendations for each topic in the two papers. Learners must be made aware of these errors so that they avoid repeating them.
- (g) Learners must be given opportunities to conduct research on topics that form part of essay questions so that they may be able to offer relevant examples on these topics. It is suggested that examples given by learners are not older than two (2) years and they are based on recent information, current trends, and developments.
- (h) Subject advisors and teachers are advised to carefully inspect the addendums of Paper 1 and Paper 2 and take note of facts that have been revised in the 2022 NSC marking guidelines. These facts must also be updated in the DBE notes and other credible resource materials to prevent learners from losing marks for quoting incorrect or invalid information.
- (i) In placing greater emphasis on the learning of appropriate terminology related to the various topics, teachers are advised to include the following strategies:

- Introduce new terms in every lesson, elaborate on the meaning and context of each, and create a glossary.
- Illustrate the meaning of new terms by using them in context, in sentences and in short scenarios.
- Create an environment to keep learners attentive during lessons, to spot new terms and to find their meanings using a dictionary, a textbook or Google. This may form the basis of an informal class 'competition'.
- Always strive to include subject-specific terminology in all informal assessment tasks, as well as during teaching.
- The meanings and expectations of verbs that are commonly used in Business Studies should be pasted in learners' books.
- Copies of the *2021 Examination Guidelines* with specific reference to 'elaborated content' must be shared and mediated with learners. They must also be instructed on the requirements or expectations of key verbs in each subtopic.

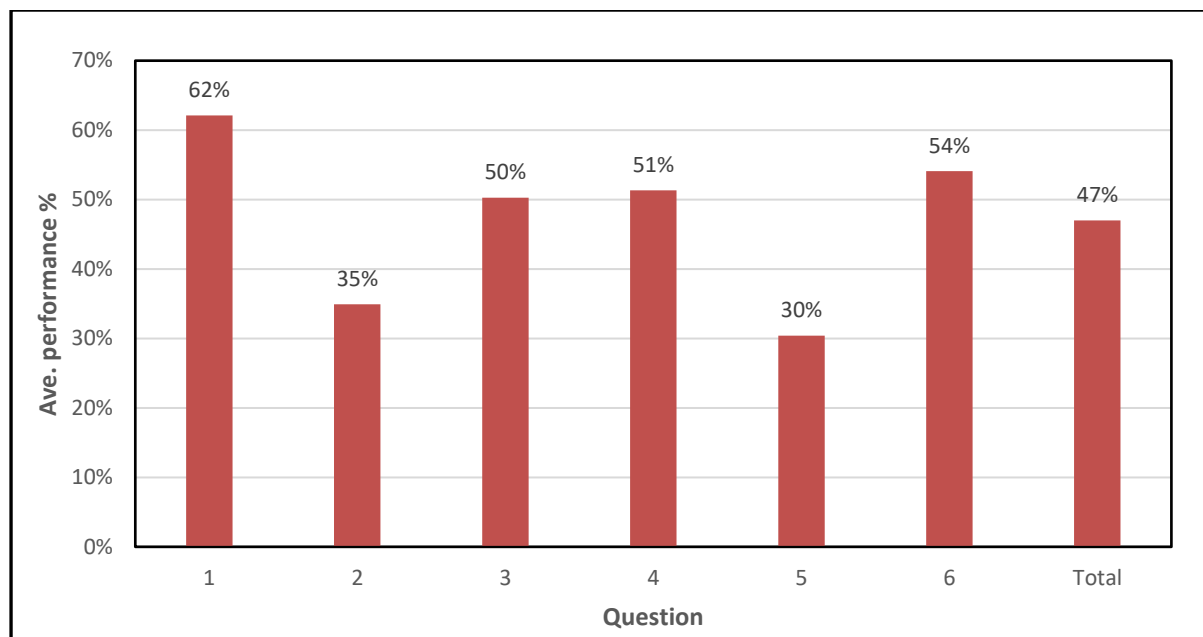
### 4.3 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 1

The graphs presented below are based on data from a random sample of candidates in the different provinces. While these graphs might not accurately reflect national averages, it is useful in assessing the relative degree of challenge of each question, as experienced by candidates.

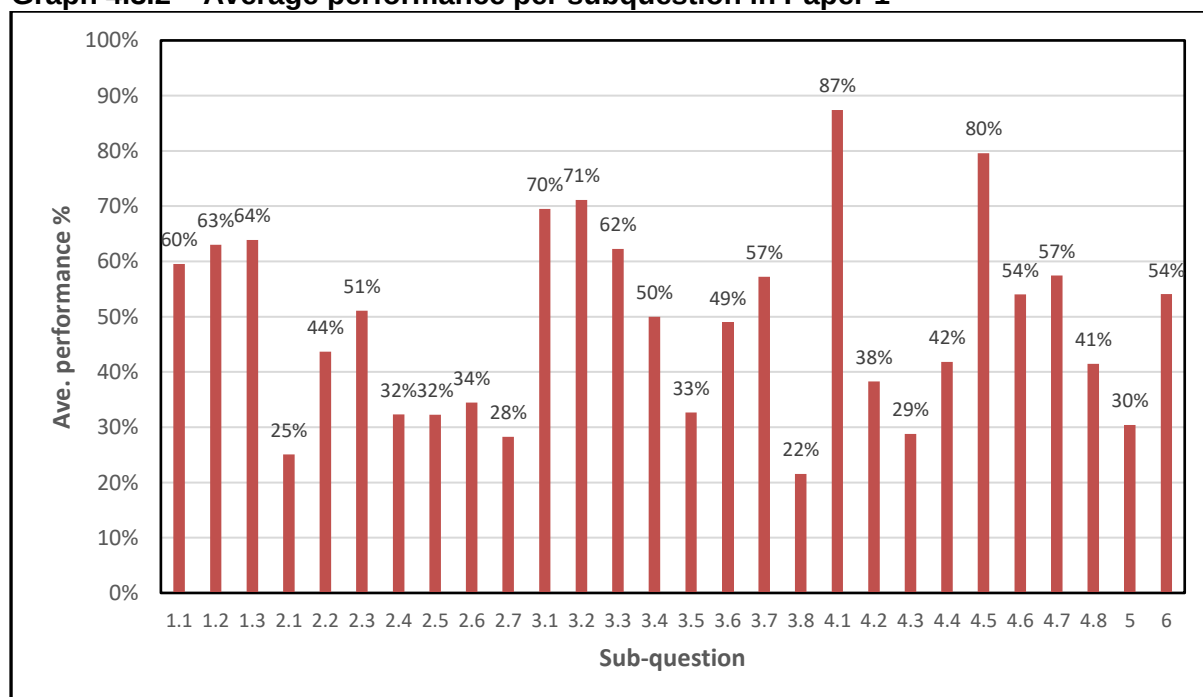
**Paper 1** consists of five subtopics that range from easy to high levels of complexity. Many candidates still did not perform well in questions based on *Business Environments* (Q2) and *Quality of Performance* assessed in Paper 1 even though all questions tested on this topic appeared in past NSC papers. These topics may have contributed to the decline of learner performance on this paper.

A decline in performance was noticeable in Q1, which assessed the topics of *Business Environments* and *Business Operations*. Questions on *Business Environments* were also covered in Q2 and Q5 while some subquestions in Q3 and all subquestions in Q6 assessed *Quality of Performance*. However, the decline in learner performance was greater in the topic *Business Environments*.

Good performance was noted in the *Business Operations* topics covered in Q3 and Q4. This was also evident in subquestions that assessed *Human Resources* and certain aspects of *Quality of Performance*. This may be due to good revision of such questions that appeared in the 2021 NSC November examination.

**Graph 4.3.1 Average performance per question in Paper 1**

| Q | Topic                                        |
|---|----------------------------------------------|
| 1 | Short Questions                              |
| 2 | Business Environments                        |
| 3 | Business Operations                          |
| 4 | Miscellaneous Topics                         |
| 5 | Business Environments (Legislation)          |
| 6 | Business Operations (Quality of Performance) |

**Graph 4.3.2 Average performance per subquestion in Paper 1**

| Sub-Q   | Topic                               | Sub-Q | Topic                                   |
|---------|-------------------------------------|-------|-----------------------------------------|
| 1.1-1.3 | Short questions                     | 3.6   | Bus Opn: Cost of quality by TQM         |
| 2.1     | Bus Env: Defensive strategies       | 3.7   | Bus Opn: Poor implementation of TQM     |
| 2.2     | Bus Env: Diversification strategies | 3.8   | Bus Opn: Impact of TQM elements         |
| 2.3     | Bus Env: PESTLE elements            | 4.1   | Bus Env: Business & extent of control   |
| 2.4     | Bus Env: Strategy evaluation        | 4.2   | Bus Env: Role of SETAs                  |
| 2.5     | Bus Env: National Credit Act        | 4.3   | Bus Env: Intensive strategies           |
| 2.6     | Bus Env: BCEA provision: overtime   | 4.4   | Bus Env: Compliance with the EEA        |
| 2.7     | Bus Env: Porter's Five Forces model | 4.5   | Bus Opn: Employment contract            |
| 3.1     | Bus Opn: Fringe benefits            | 4.6   | Bus Opn: Induction                      |
| 3.2     | Bus Opn: Role of the interviewer    | 4.7   | Bus Opn: PDCA model steps               |
| 3.3     | Bus Opn: Internal recruitment       | 4.8   | Bus Opn: Quality in purchasing function |
| 3.4     | Bus Opn: Termination of contract    | 5     | Bus Env: Labour Relations Act           |
| 3.5     | Bus Opn: Quality financial function | 6     | Bus Opn: Quality of performance         |

#### 4.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN INDIVIDUAL QUESTIONS IN PAPER 1

##### SECTION A: MULTIPLE-CHOICE/SHORT ANSWER QUESTIONS

##### QUESTION 1: COMPULSORY (MULTIPLE-CHOICE, CHOOSING CORRECT WORDS AND MATCHING ITEMS)

Satisfactory performance was noted in this question, irrespective of the slight decline when compared to the 2021 cohort, in all three subquestions. There was an expectation of better performance in this question since only two out of fifteen subquestions were asked for the first time in the NSC Nov Examinations, and the others should have been part of any revision programme.

##### Common errors and misconceptions

- In Q1.1.1 some candidates confused COIDA with the BCEA. This is possibly due to them not reading the entire sentence before responding to this question, as the conclusion of the sentence makes COIDA the necessary response.
- Some candidates did not perform well in Q1.1.5 as many responded on the general *management* function instead of the *administration* function. They were expected to perform well in this question as the business functions form part of Grades 10 and 11 content.
- Most candidates confused the *consumer right to right to fair and honest dealings* in Q1.2.1. It appears that candidates lacked an in-depth understanding of the meaning of each consumer right in terms of the Consumer Protection Act (CPA).
- Some candidates confused the meaning of *forward vertical* strategy with the *backward vertical* strategy in Q1.2.2 despite the clarity offered in past national diagnostic reports on this content.
- Many candidates were not clear about the meaning of *quality management* with *quality performance* in Q1.2.5. This is concerning as candidates were expected to perform exceptionally well in this question as it appeared in 2020 and 2021 NSC November examinations and should have been revised.

### Suggestions for improvement

- (a) It is important that learners first understand the purpose of each Act before addressing the impact and compliance with these Acts. Teachers are advised to design lessons with this in mind as it facilitates a logical flow of the content. This will enable the content to flow logically. An effective approach is to tabulate the differences between these Acts in terms of the purpose impact, compliance, and penalties for non-compliance.
- (b) A baseline assessment of the meaning of each business function is useful to identify content gaps. This can be achieved by giving learners examples of businesses that utilise each business function and then by explaining the quality indicators for each one. Further assessment and remediation must then follow during the academic year. Subject advisors must ensure that this is included in the Grade 10 Term 1 SBA task.
- (c) Teachers must provide learners with real-world examples of consumer rights in terms of the CPA, by making effective use of newspaper articles and video footage of businesses that uphold these consumer rights. In addition, discussions and debates on this content should be encouraged to improve knowledge.
- (d) The skilful use of key words will enable learners to identify each business strategy from given scenarios and statements. These must be linked to a brief explanation of factors that influences a business's choice of strategy. Getting learners to make a collage on each business strategy and critically evaluating their work will enhance their understanding.
- (e) Emphasis must be placed on all concepts related to 'quality' being thoroughly taught as outlined in the *2021 Examination Guidelines*. It must be stressed that quality performance is a prerequisite for quality management because managers are responsible for creating strategies and tactics will be applied by each business function to improve the quality of their products.

### SECTION B: LONGER AND PARAGRAPH QUESTIONS, USING CASE STUDIES AND INFORMATION

#### QUESTION 2: BUSINESS ENVIRONMENTS

Poor performance was noted in this question as reflected by the average of 35% in the sample. Candidates who chose this question did not perform well on six of the seven subquestions on *Business Strategies* and *Legislation* even though some of these questions appeared in the 2020 and 2021 November NSC examinations.

#### Common errors and misconceptions

- (a) Many candidates could not state the types of defensive strategies in Q2.1 as their responses were based on other types of business strategies which were not the required answer. Some wrote *liquidity* instead of *liquidation* as a type of defensive strategy. They forfeited marks for committing this error.
- (b) In Q2.2 many candidates provided and explained the types of diversification strategies instead of outlining the advantages of these strategies. Responses were limited to 'increase in sales' without further explanation. Some confused the advantages of diversification strategies with intensive strategies.

- (c) Good performance was noted in Q2.3 as many candidates were able to identify the PESTLE elements from given statements even though some provided incomplete concepts.
- (d) Some candidates confused the steps in strategy evaluation with the strategic management process and/or problem-solving steps in Q2.4. Candidates were expected to perform well in this question since it was asked in Section C of the 2021 November examinations.
- (e) Many candidates performed well in Q2.5.1 as they were able to quote from the given scenario. Some quoted a distractor that was present in the scenario and forfeited one mark. The impact of the National Credit Act on businesses was confused with the impact of the Consumer Protection Act in Q 2.5.2.
- (f) Poor performance was recorded in Q2.6 as many candidates could not explain *overtime* as one of the provisions of the BCEA. The fact that this question appeared in the NSC November examination for the first time as a stand-alone item may be the cause of the poor result.
- (g) Candidates performed poorly in Q2.7 as they could not provide correct responses to the application of Porter's Five Forces. This challenge persists despite the recommendations made since 2015 to 2019 in national diagnostic reports.

### **Suggestions for improvement**

- (a) Unpack the meaning of 'defensive' in the context of business and impress upon learners that defensive strategies are only used by businesses as a last resort in an effort to continue operating. Adequate assessment and remediation during the academic year must follow. Research on businesses that applied these strategies is also useful to enhance the understanding of this concept.
- (b) Learners must be made aware that the advantages of diversification strategies are embedded in each type of diversification strategy. Teachers must therefore clearly illustrate the relationship between the types of diversification strategies with the advantages of these strategies. Open discussions and debates should be encouraged to enable learners to gain insight in this content.
- (c) Conduct a baseline assessment on the PESTLE elements by requesting learners to explain and give practical examples of challenges posed by each element. Learners can be requested to make recommendations on how businesses can deal with each challenge. Regular short assessment tasks can be used to close the content gaps.
- (d) Practical examples, role plays, and demonstrations must be used effectively to illustrate the steps in strategy evaluation. Learners must be made aware of the importance of evaluating strategies to stimulate interest on how businesses discharge this process. It must be emphasised that the steps in strategy evaluation form part of the strategic management process. Teachers must guide them on how to unpack these steps.
- (e) Lesson plans must be designed to offer a clear distinction that the National Credit Act covers credit sales, and the Consumer Protection Act covers product sales. The content must outline the purpose of these Acts and explain how they differ. Teachers should include these Acts in the SBA task as higher-order questions.

- (f) In addressing the content on *leave*, teachers are advised to provide a detailed description of each type of leave as per the amended BCEA and the DBE notes. Opportunities exist for this topic to be assessed in both direct and indirect questions, including essay type questions, and teachers must exercise some creativity in questioning styles.
- (g) Learners must be given clear instruction on how businesses could apply each force of the Porter's Five Forces to analyse their position in the market. Focus should be placed on what drives the power/force to go up or down. It should be impressed upon learners that businesses apply this model when doing research on the challenges posed by the market environment so that such businesses could be strategically positioned.

### QUESTION 3: BUSINESS OPERATIONS

Generally good performance was evident in this question, as noted by the data of the random sample of candidates. Many candidates performed well in five of the seven subquestions. Exceptionally good performance was noted in subquestions that assessed the *Human Resources function*.

#### Common errors and misconceptions

- (a) Many candidates confused examples of *fringe benefits* with the benefits offered by the *Unemployment Insurance Fund (UIF)*. Some repeated the types of leave allowances and forfeited two marks in such instances. Others confused this subquestion with the aspect that must be included in the employment contract.
- (b) Although good performance was noted in Q3.2 on the role of the interviewer before the interview, some candidates still confused this content with the role of the interviewee during the interview or the role of the interviewer during the interview. This challenge is persistent despite the recommendations made in past diagnostic reports.
- (c) Many candidates performed well in Q3.3.1 and Q3.3.2 as they were able to identify internal recruitment and explain the advantages of this recruitment method. Some, however, confused this content with the advantages of external recruitment. Others approached this question from the perspective of themselves (the learner) as opposed to the business.
- (d) Q3.4 was satisfactorily answered by many candidates, even though some listed the reasons for termination of an employment contract without further explanation. Inappropriate responses such as 'resignation, retirement, retrenchment' were common. This question appeared in the 2020 November examination.
- (e) Most candidates confused the quality indicators of the purchasing function with those of the financial function in Q3.5. Others misinterpreted this question by giving the failure instead of the success factors of the financial function.
- (f) Poor performance was observed in Q3.7 as many of candidates explained ways in which TQM reduces the cost of quality instead of the impact of TQM if it is poorly implemented by businesses, which was a requirement. This challenge continues to challenge candidates despite it having been raised in many past diagnostic reports.
- (g) In Q3.7 many responses on 'TQM if poorly implemented' were vague and incomplete. Some gave positive instead of negative responses for poor implementation of TQM whilst other responses were based on the benefits of a good quality management

system, which was not the required answer. It appears that learners were not attentive when this content was taught and revised.

- (h) The majority of the candidates did not perform well in Q3.8 as they confused the advantages of adequate financing and capacity as a TQM element with the quality indicators of the financial function assessed in Q3.5.

### **Suggestions for improvement**

- (a) Teachers must emphasise that fringe benefits are compensation given to employees above and beyond their remuneration that is subject to taxation. Learners must be made aware that because all leave allowances fall into the same category, all such answers will only receive one mark. Short class tests are useful to fine-tune this content.
- (b) Learners must be clear about the meaning of the *interviewer* and *interviewee* before proceeding to the role of each party. They should also be aware that the interviewer (represents the business) plays two roles; one before and one after the interview while the interviewee (individual who applied for the job) is limited to the interview itself. Effective roleplaying and classroom sketches can be useful to illustrate these differences.
- (c) Learners must first be made aware of the meaning of internal and external recruitment to enable them to understand the differences. They should also know that internal recruitment is the first point of departure when advertising vacant positions which is why internal sources of recruitment are used. The advantages of internal recruitment should focus on what this form of hiring can do for businesses.
- (d) When discussing exam strategies on employment contracts, it must be made clear that the reasons for termination must be explained to obtain full marks. It must also be pointed out that this content is assessed either in Section B and/or Section C because full sentences are expected. Teachers should penalise for vague and incomplete responses so that learners refrain from such practices in examinations.
- (e) Quality indicators for each business function focuses on activities that are conducted by the respective function. A clear distinction must be made between financial function and the purchasing function. Practical examples of different business functions should be brainstormed, and learners must be requested to give other examples to illustrate their understanding of the concepts. This can be done during discussions.
- (f) Learners must realise that the main reason why businesses implement TQM elements is to reduce the cost of quality. They will then be able to explain the relationship between this content with the five TQM elements outlined in the 2021 *Examination Guidelines*. A sound knowledge of these elements will enable learners to understand how TQM elements can be implemented to reduce the cost of quality.
- (g) It is important to teach learners that the impact of poorly implemented TQM requires negative responses as this is the result of failure to implement the TQM elements. Practical examples and videos showing the negative impact of not implementing the TQM in large businesses will assist to enhance understanding.
- (h) Teachers are advised to address the impact of TQM elements on large businesses by using practical examples. Emphasis must be placed on *how* large businesses implement these TQM elements and on the advantages and disadvantages of implementing these elements.

## QUESTION 4: MISCELLANEOUS TOPICS

Candidates' performance was satisfactory. This question assessed both main topics, namely *Business Environments* and *Business Operations* consisting of 20 marks each. Good performance was noted on questions based on *Business Operations* and many candidates did not perform well in questions based on *Business Environments*, as noted in Q2 above.

### Common errors and misconceptions

- (a) In Q4.1 many candidates were able to name the *Business Environments* but struggled to state the extent of control that businesses have over each environment. Some confused the extent of control of the micro-environment with either the market or macro environment. Candidates were expected to easily manage this question as it forms part of the Grade 11 content.
- (b) In Q4.2 some candidates incorrectly based their responses on SETAs being responsible for training employees in the workplaces. Others confused this content with the purpose of the SDA.
- (c) Many candidates could not identify market development as a type of the intensive strategy from the given scenario in Q4.3.1. This affected their responses to Q4.3.2, a follow-on question. Some responses were based on *product development* instead of *market development*.
- (d) In Q4.4 many candidates had difficulty to suggest ways to comply with the Employment Equity Act/EEA. Some confused this content with the purpose of the EEA which was not a required answer. Others obtained only one mark for writing 'promote equal opportunities'. The part on 'fair treatment' was missing in their responses.
- (e) Some candidates quoted the distractor instead of the purpose of induction in Q4.6.1 and forfeited one mark for this error. Others did not read the follow-on question as they wrote other purposes of induction instead of the benefits in Q4.6.2. Teachers should have alerted candidates to this style of setting as it was introduced in 2019.
- (f) In Q4.7 many candidates were able to name the steps of the PDCA model, but they could not explain how businesses could apply these steps to improve the quality of their products. Others used their own words for naming the step of the PDCA model such as 'planning instead of plan, checking instead of check, doing instead of do. They were awarded one mark instead of two marks.
- (g) Poor performance was recorded in Q4.8 as many candidates confused the quality indicators of the purchasing function with those of the financial function. Other responses were incorrectly based on the meaning of the financial function.

### Suggestions for improvement

- (a) Teachers must recap the three business environments and the extent of control that businesses have over each environment. Providing a brief overview on the reasons for extent of control over each environment is suggested to improve learner understanding.
- (b) With regard to SETAs, attention must be focused on a clear understanding of the relationship between the functions of SETAs and how they are funded. It must be emphasised that SETAs are the implementers of the SDA. They do not train employees but oversee the training process.

- (c) Teachers should develop a mind-map/spider diagram illustrating the meaning of the four main types of strategies outlined in the *2021 Examination Guidelines*. Suitable practical examples should be used to enhance the teaching and learning of *Business Strategies*.
- (d) Teaching and learning should focus on ways in which a business should comply with the Employment Equity Act. It must be noted that *promoting equal opportunities and fair treatment* will no longer be accepted in this context, as it is covered under the Labour Relations Act. Teachers are advised to remove this statement from resource material in their possession.
- (e) Teachers are advised to clearly illustrate the link between the purpose of induction and its benefits, for example:

| Purpose of induction                                                                 | Benefit of induction                                                                                                      |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Give new employees a tour/<br>information about the layout of the<br>building/office | New employees will know the layout of the<br>building/factory/offices/where everything<br>is, which saves production time |

It must be emphasised that induction benefits both the businesses and employees as it reduces voluntary turnover and contributes to increased productivity.

- (f) Learners should be informed that the PDCA model forms part of continuous improvement to processes and systems. Teaching and learning should focus on how each step contributes towards improving processes and systems. Practical examples should be given to provide an in-depth understanding of this content.
- (g) The meaning of purchasing function and its importance in ensuring the smooth running of a business must be clearly explained to learners. Teachers are advised to assess all business functions by providing relevant statements and requesting learners to classify each statement according to the relevant business function. It must be noted that the statement *Have a thorough understanding of supply chain management to apply the correct procurement procedures/processes* was revised in the 2022 November examination. This must be adjusted in all relevant resource material presently with teachers.

## SECTION C: ESSAY QUESTIONS

### QUESTION 5: BUSINESS ENVIRONMENT: LEGISLATION

This question, which assessed *Labour Relations Act* under *Business Environments*, was not a popular choice in Section C. Many candidates had difficulty understanding the eight Acts outlined in the *2021 Exam Guidelines* and the CAPS policy statement, hence it has contributed to a general decline in performance.

#### Common errors and misconceptions

- (a) In Q5.1 the introduction by many candidates was based on statements copied from the preamble. Others only wrote the word 'introduction' with no content. They forfeited one mark for this error. This error was reported in the 2021 national diagnostic report with recommendations on areas of improvement.
- (b) In Q5.2 a large percentage of the candidates appeared to confuse the rights of employees in terms of the Labour Relations Act (LRA) with either human rights or

economic rights of employees, which are covered in Paper 2. Some responses were limited to the right to 'join trade unions. Others forfeited one mark for writing the 'right to embark on a strike' as they omitted the word 'legal'.

- (c) Some candidates confused the purpose of the LRA with the impact of the LRA and in some instances repeated the same facts in Q5.3. Some responses were based on the provisions of the BCEA which were not relevant to the question.
- (d) Poor performance was recorded in Q5.4 as many candidates' responses were based on the impact of the EEA instead of the LRA on businesses. This question has appeared in past NSC November examination.
- (e) Some candidates showed little or no understanding of the penalties for non-compliance with the LRA in Q5.6 as many responses were limited to 'heavy fines' or 'imprisonment' without further explanation.

### **Suggestions for improvement**

- (a) Learners must be trained on how to write the conclusion based on any four subquestions that were asked in the question paper. They must also be advised that the introduction and conclusion must not be repeated in the body to avoid forfeiting marks unnecessary.
- (b) Teachers must focus on the purpose of the Labour Relations Act with special reference to how this Act protects the employer and employees, as this will enable learners to gain a better understanding of their respective rights in the workplace. Teachers should also alert learners to the fact that some of the rights of employees are not the same as the economic rights of employees as covered in Paper 2.
- (c) Teaching and learning must focus on the purpose of the Labour Relations Act (LRA) in terms of how this Acts promotes a healthy relationship between the employer and employees.
- (d) Leaners must ensure that they gain a sound understanding of the rights of employers and employees in terms of the LRA as this will enable them to understand the impact of this Act on businesses. This content must be adequately assessed through direct and indirect questions during the academic year.
- (e) Learners must be made aware that penalties for non-compliance with each Act are punitive in nature and that businesses may lose a lot of money on heavy fines and legal costs. They must also understand that penalties for non-compliance with each Act are not the same, as this depends on the nature and context of each Act.

### **QUESTION 6: BUSINESS OPERATIONS: QUALITY OF PERFORMANCE**

Candidates performed well in this question which proved to be popular. This aligns with the data from a random sample of candidates presented above. It appears that many candidates have devoted their time well to study the topic *Quality of Performance*.

#### **Common errors and misconceptions**

- (a) In Q6.2 some candidates confused the meaning of *quality assurance* with *quality control*. Others provided incomplete and vague responses on each concept and lost marks in such cases.

- (b) Good performance was noted in Q6.3 even though some candidates confused the benefits of a good quality management system with the impact of total client satisfaction as a TQM element. Others provide incomplete responses such as increased market share, profitability, and productivity. They were awarded only one mark instead of two marks for each fact.
- (c) In Q6.4 some candidates confused the impact of continuous skills development as a TQM element with the impact of the Skills Development Act on businesses. Others confused the impact of total client satisfaction with the benefits of a good quality management system which were asked in Q6.2. The TQM elements appear to pose challenges to many candidates possibly due to poor revision on these aspects.
- (d) Some candidates confused the role of quality circles with the application of the PDCA model in Q6.5. Other responses were based on ways in which TQM can reduce the cost of quality assessed in Q3.6.

### Suggestions for improvement

- (a) It should be noted that *quality control* must take place first before *quality assurance* is addressed as businesses must set targets and check the quality of raw materials before the actual production is done. Quality assurance focuses on the production processes while quality control focuses on the quality of the final product and corrective measures that must be taken by businesses.
- (b) It should be emphasised that businesses that implement the quality concepts as outlined in the *2021 Exam Guidelines* enjoy the benefits of a good quality management system. Employees also benefit from this system as they attend regular training sessions that contribute to good performance and career development.
- (c) The impact of TQM elements on large businesses should focus on how businesses implement each element. It is necessary to explain the meaning of each TQM element and the overall aim of these TQM elements. The emphasis should be on satisfying customers' needs beyond their expectations through effective implementation of these elements.
- (d) Learner must first know the meaning of quality circles to understand their role in improving the quality of the products. Demonstrations of this content should be done along with opportunities for open discussion among learners. Learners must also understand that quality circles do not only solve problems related to quality, but they also ensure that employers and employees have a healthy working relationship.

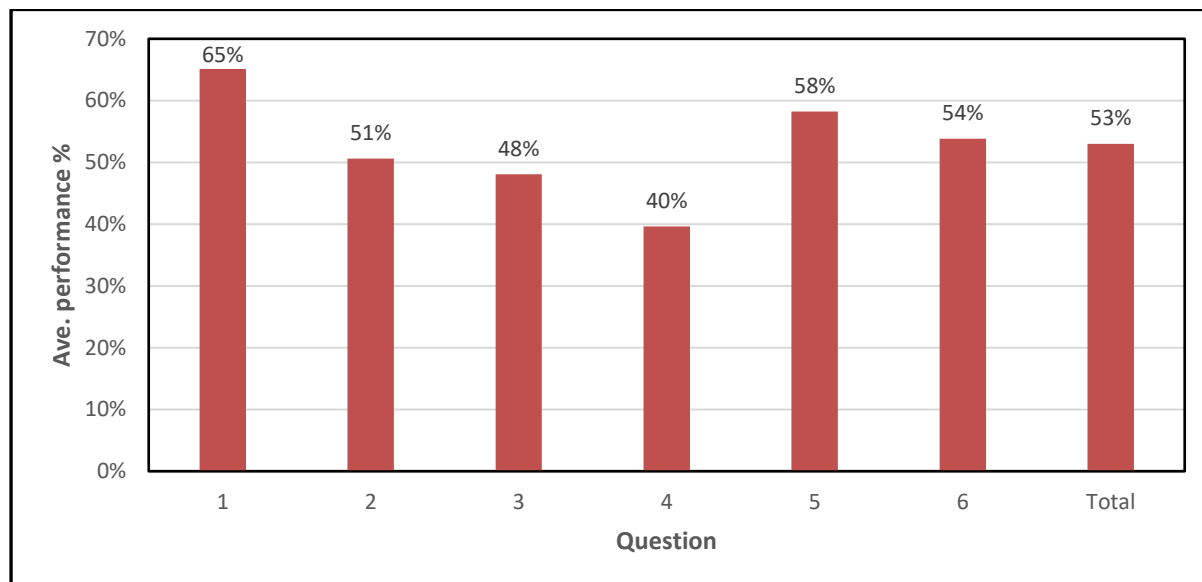
## 4.5 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 2

The graphs presented below are based on data from a random sample of candidates in the different provinces.

The performance of the 2022 cohort has improved in Q1 as the average performance is 65% compared to 63% in 2021. This question consisted of ten subquestions that assessed *Business Ventures* and *Business Roles* and consisted of some relatively simple subquestions that appeared in past NSC examinations. It is noted that the 2022 cohort also performed better in the topic *Business Ventures* which was assessed in Q2 and Q5 when compared to the 2021 cohort. The average percentage for Q2 (Section B) is 51% compared to 39% in 2021. Candidates' performance in Q5 (Section C) is 58% compared to 54% in 2021.

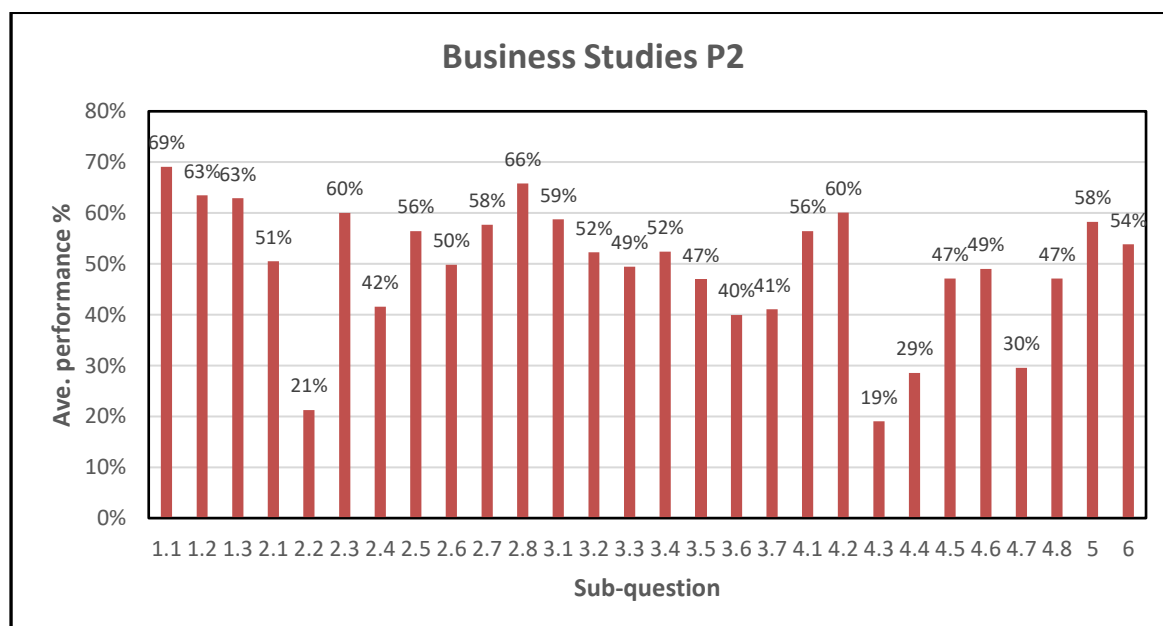
However, candidates' performance declined in Q3 (Section B) which assessed the topic *Business Roles*. However, candidates' performance improved in Q6 (Section C) which assessed the same topic. It appears that candidates devoted their time to studying Creative Thinking and Problem Solving covered under *Business Roles* and probably neglected other subtopics of this main topic.

**Graph 4.5.1 Average performance per question in Paper 2**



| Q | Topic                                                |
|---|------------------------------------------------------|
| 1 | Compulsory                                           |
| 2 | Business Ventures                                    |
| 3 | Business Roles                                       |
| 4 | Miscellaneous Topics                                 |
| 5 | Business Ventures (Management and Leadership)        |
| 6 | Business Roles (Creative thinking & problem solving) |

**Graph 4.5.2 Average performance per subquestion in Paper 2**



| Sub-Q   | Topic                                | Sub-Q | Topic                                            |
|---------|--------------------------------------|-------|--------------------------------------------------|
| 1.1-1.3 | Short questions                      | 3.5   | Bus Roles: Unethical practices                   |
| 2.1     | Bus Vent: Long-term insurance        | 3.6   | Bus Roles: Impact of CSR                         |
| 2.2     | Bus Vent: Meaning of excess          | 3.7   | Bus Roles: Professional and ethical              |
| 2.3     | Bus Vent: Visual aids                | 4.1   | Bus Vent: Investment decision factors            |
| 2.4     | Bus Vent: Limited/Unlimited          | 4.2   | Bus Vent: Compulsory insurance                   |
| 2.5     | Bus Vent: Functions of JSE           | 4.3   | Bus Vent: Partnership                            |
| 2.6.    | Bus Vent: Interest and fixed deposit | 4.4   | Bus Vent: leadership style                       |
| 2.7     | Bus Vent: Advantages insurance       | 4.5   | Bus Roles: CSI focus areas                       |
| 2.8.    | Bus Vent: Handling feedback          | 4.6   | Bus Roles: Stages of team development            |
| 3.1     | Bus Roles: Human rights              | 4.7   | Bus Roles: Nominal group technique               |
| 3.2     | Bus Roles: Health and safety         | 4.8   | Bus Roles: Benefits of diversity                 |
| 3.3     | Bus Roles: Handling conflict         | 5     | Bus Vent: Management and leadership              |
| 3.4     | Bus Roles: Unemployment              | 6     | Bus Roles: Creative thinking and problem solving |

## 4.6 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### QUESTION 1: COMPULSORY (MULTIPLE-CHOICE, CHOOSING CORRECT WORDS AND MATCHING ITEMS)

The performance of candidates in this question was good with an average performance of above 60% in all three subquestions noted. Two of the three subquestions required an application of knowledge. Candidates performed very well in Q.1.1, achieving an average of 69% compared to 53% in 2021.

#### Common errors and misconceptions

- Many candidates confused the application of the *charismatic* with the *transactional* leadership style in Q1.1.1. These concepts were tested in the 2021 NSC examination.
- Q1.1.2 was poorly answered by many candidates as they lacked an understanding of the meaning of *bonus shares*. Most of their responses were based on preference shares and/or founders' shares.
- Many candidates did not perform well in Q1.1.5 as they confused a strategy for dealing with a *complainer* with an *aggressive person*. This content has also been assessed many times in the NSC examinations.
- Some candidates confused the advantage of *unit trusts* with *RSA Retail Savings Bonds* in Q1.2.1. This question was asked for the first time in Section A of the NSC examination.
- In Q1.2.2 the example of *non-insurable risk* was confused with *insurable risk* even though this question has appeared in the NSC examination many times.
- Some candidates did not perform well in Q1.2.3 as they confused the meaning of *professional behaviour* with *ethical behaviour*. It appears that candidates did not read the last part of the statement carefully.

- (g) The majority of the candidates did not know the meaning of *accountability* as they confused it with the meaning of *transparency* in Q1.2.5. Candidates did not take advantage of the hint 'external auditors and financial statements' in the given statement.
- (h) Q1.3.1 was poorly answered as many candidates' responses were based on *transactional leadership* instead of on the *situational leadership theory*.
- (i) Many candidates did not know the difference between *Corporate Social Responsibility (CSR)* and *Corporate Social Investment (CSI)* in Q1.3.4. This is a recurring issue and was raised in past national diagnostic report on learner performance.
- (j) In Q1.3.5 some candidates choose an incorrect description of shared values as one of the criteria for successful team performance. It could have been that this question was asked for the first time in Section A of the November NSC examination.

### **Suggestions for improvement**

- (a) Teachers are advised to use practical examples and demonstration when teaching the application of each leadership style in the workplace. It should be noted that this content informs the impact of each leadership style on businesses. They could also stimulate debates and discussions on this content to improve comprehension.
- (b) Emphasis must be placed on the meaning of each type of share in terms of dividends paid to the shareholders. A clear distinction between the meaning of *bonus shares* in terms of dividends paid to shareholders with a *bonus* as a performance incentive must be made.
- (c) Teaching and learning must focus on a brief overview of each type of difficult personality to enable learners to understand ways in which businesses can deal with each type of a personality.
- (d) Learners must be given the opportunity to conduct research on different forms of investment focusing on the impact of each on investors and present their findings in the classroom. Additionally, all forms of investment outlined in the *2021 Examination Guidelines* must be adequately taught, assessed, and remediated during the academic year.
- (e) Learners need to know the meaning of *insurable risks* and *non-insurable risks* as well as a basic explanation of how businesses classify these risks. This can be achieved by asking them to provide examples of insurable risks and non-insurable risks. It is important that they do not confuse this content with the examples of assets that can be insured.
- (f) Teachers must stress the point that professional behaviour focuses on the employee's conduct and ethical behaviour focuses on the business conduct. Practical examples of each type of unethical and unprofessional behaviour must be provided and discussed to enhance learner understanding.
- (g) Teachers are advised to identify key words that will assist learners to formulate statements on the application of accountability and transparency as King Code principles for good corporate governance. Learners should be requested to present newspaper articles that illustrate adherence to these principles by businesses.

- (h) A detailed description of each leadership theory must be given using demonstrations and practical examples. Learners must be led to understand that leadership theories provide guidelines on the type of leadership style that is applicable to each situation in the workplace.
- (i) Learners should be made aware that CSR focuses on internal initiatives designed to help businesses and their stakeholders while CSI projects focuses on outside initiatives meant to help communities without making a profit. Essentially, CSI projects are informed by CSR programmes.
- (j) Teachers must create opportunities to ensure that learners acquire the meaning of each criterion for successful team performance as well as the activities involved in each. They should also understand that the criteria for successful team performance serves as guidelines on how team members should interact with each other when working together on a project.

## **SECTION B: LONGER AND PARAGRAPH QUESTIONS, USING CASE STUDIES AND INFORMATION (THREE QUESTIONS TO BE ANSWERED)**

### **QUESTION 2: BUSINESS VENTURES**

The performance of candidates who attempted this question was generally average. This was disappointing as such questions have been assessed before, and the level of difficulty of higher-order questions was moderate as only two subquestions consisted of indirect questions.

#### **Common errors and misconceptions**

- (a) Good performance was noted in Q2.1 as many candidates were able to give the examples of long-term insurance even though some responses were based on short term insurance.
- (b) In Q2.2 some candidates confused the meaning of *excess* with *insurance*. This question was assessed for the first time in Section B of the NSC examination.
- (c) Many candidates could not identify the type of visual aids in a given statement in Q2.3. Some confused the meaning of *data projector* with an *overhead projector* and/or *PowerPoint*. Others rewrote the word *hard copies* which was included in the statement in Q2.3.2.
- (d) In Q2.4 many candidates were unable to distinguish between *limited* and *unlimited liability* as they referred to the former as having insufficient capital and the latter as sufficient capital. Some candidates provided vague responses and incomplete responses.
- (e) The majority of the candidates performed well in Q2.5 as the functions of the Johannesburg Exchange have appeared many times in past NSC examinations. However, some were awarded part marks for writing incomplete sentences such as 'raises primary capital'.
- (f) Poor performance was noted in Q2.6.1 as many candidates were not able to calculate simple interest. Some were awarded part marks for showing correct formula and workings. Others showed the correct workings but added the principal amount to the calculations which led to the incorrect answer.

- (g) Many candidates did not perform well in Q2.6.2. Responses were based on the impact of Government Retail Savings Bonds and/or unit trusts instead of fixed. It should be noted that the following facts have been revised in the November 2022 Marking Guideline:  
**Positives:** It is a safe investment as investors are guaranteed of the final payment.  
**Negatives:** The investor may be penalised for any early withdrawals.  
 Teachers are advised to revise the above-mentioned facts in the DBE notes and other credible resource materials.
- (h) Most of the candidates performed well in Q2.7 although some were awarded only one mark for writing incomplete sentences. Some responses were based on the advantages of insurance for individuals instead of businesses.
- (i) In Q2.7 many candidates were not able to suggest ways in which the presenter can handle feedback after a presentation in a non-aggressive and professional way. Others confused this question with the factors that must be considered when doing a presentation.

### Suggestions for improvement

- (a) A clear distinction must be made between the examples of short-term and long-term insurance. Teachers are advised to give a brief explanation of the differences between these concepts.
- (b) A detailed analysis of the meaning of excess should be made and learners must know that excess forms part of the insurance contract. It should also be emphasised that it is the responsibility of the insured to pay excess upfront when a claim for damages is lodged. Subject advisors and teachers should note that the following facts will no longer be credited as they have been revised in the 2022 marking guidelines:
- It is not paid out to the insured when a claim is settled.
  - Excess amounts keep the insurance premium lower and discourage fraud.
- (c) Teaching and learning must only focus on the six types of visual aids stated in the *2021 Exam Guidelines*. Learners must be able to identify and discuss the impact of each visual aid. The use of a collage depicting different types of visual aids should be incorporated in teaching and learning of this content. Furthermore, learners must know key words for each visual aid to enable them to identify different types of visual aids from given scenarios and/or statements.
- (d) Learners must be made aware that only a sole trader and partnership have unlimited liability as they are not legal entities. Subject advisors should ensure that this content is adequately taught and assessed in Grades 10 and 11. Teaching and learning must focus on the meaning of these concepts in a business context.
- (e) Learners must be encouraged to write full sentences on the functions of the JSE to obtain two marks for each fact and be aware of how each fact will be marked using note number 12.2 of the NSC November 2022 marking guidelines.
- (f) Teachers are advised to use the Business Studies formula when teaching simple and compound interest. Learners must know that investors do not accumulate interest on interest when investing on simple interest and that these concepts are also applicable when investing on fixed deposits. Furthermore, learners must also be able to explain the differences between these concepts.

- (g) Teachers must provide a brief overview of the meaning of fixed deposits with specific reference to the impact of this form of investment on investors. Learners must be aware that fixed deposits have more advantages than disadvantages. They must also be given the opportunity to conduct research on this form of investment to validate this argument.
- (h) Learners must be advised that the key word *protect* appears four times and the word *transfers* appears twice but in a different context under the advantages of insurance for businesses. The above-mentioned key words may assist learners to formulate statements on this content. Moreover, teaching and learning should focus on the advantages of insurance for businesses instead of individuals.
- (i) Demonstrations and role play on how presenters should handle feedback in a non-aggressive and professional manner should be done. Learners could be requested to critically evaluate their peers while handling feedback and make recommendations for improvement. The main aim of this exercise is to enable learners to gain insight on this content and to avoid confusing factors that must be considered during the presentation and when handling feedback in a professional manner.

### QUESTION 3: BUSINESS ROLES

Performance in this question was consistent with that of the 2021 cohort. It ranged from average to good. It must be noted that all subtopics assessed appeared in past NSC papers, although the phrasing of some were slightly different. This question only consisted of one follow-on question.

#### Common errors and misconceptions

- (a) Good performance was recorded in Q3.1 as many candidates were able to name human rights in the workplace even though some confused these rights with economic rights.
- (b) Many candidates performed well in Q3.2 although some confused the roles of the health and safety representatives with the responsibilities of employers in protecting the workplace environment. Some responses were only limited to 'identifying potential dangers in the workplace'.
- (c) Many candidates were able to quote the steps in handling conflict from the given scenario in Q3.3.1 even though some quoted the distractor in the scenario. They were also able to describe other steps in handling conflict in the workplace in Q3.3.2. Others repeated the same facts that were quoted in Q3.3.1.
- (d) Good performance was noted in Q3.4 even though some candidates' responses included strategies to deal with poverty instead of unemployment as a socioeconomic issue. Other responses were based on strategies to deal with poverty instead of unemployment.
- (e) In Q3.5 some candidates confused the types of unethical business practices with the types of unprofessional business practices from a given scenario. Some confused tax evasion with tax avoidance. They lost six marks in this instance, as they forfeited marks for motivation since they did not provide the correct concepts.

- (f) Many candidates confused the impact of Corporate Social Investments (CSI) on communities with the Impact of CSR on businesses in Q3.6. This error is persistent despite the recommendations made in past diagnostic reports.
- (g) The majority of the candidates had difficulty to respond appropriately to Q3.7 as they provided general and vague responses on ways in which professional, responsible, ethical and effective business practises should be conducted. Some confused this content with contribution to the wellbeing of employees in the workplace.

### Suggestions for improvement

- (a) Teaching and learning should focus on the six human rights of employees outlined in the *2021 Examination Guidelines*. Learners need to know how businesses should deal with each right in the workplace. Practical examples should be given, and demonstrations should be done during teaching and learning. Learners must be requested to give other examples of these rights in the workplace. It should be noted that learners will no longer be credited for writing the following human right: *Equality* instead of *Equity*.
- (b) Learners must be alerted to the fact that the health and safety representatives form part of the business employees hence they ensure that the employer complies with health and safety requirements in the workplace. They must understand that the health and safety representatives do not provide protective clothing as they are not employers.
- (c) Learners must understand that the steps in handling conflict deals with two employees who have different opinions. Furthermore, these steps can be regarded as an informal process since no minutes of the meeting is kept during the meeting. They should also not confuse the steps in handling conflict with the correct procedure to deal with grievances in the workplace. Teachers are advised to include one distractor in each scenario as part of formal and informal assessment during the academic year.
- (d) Learners must be given the opportunity to suggest various strategies to deal with unemployment as a socio-economic issue. The recommended strategies must be sustainable and long-term in nature. Subject advisors must ensure the strategies to deal with each type of socio-economic issue outlined in the *2021 Examination Guidelines* are adequately assessed and remediated during the academic year.
- (e) Teachers must provide a detailed analysis of the differences between *unethical behaviour* and *unprofessional behaviour*. Learners must know how businesses should deal with each type of unethical behaviour and unprofessional behaviour practice. A sound understanding of the challenges posed by the types of unethical behaviour and unprofessional behaviour will enable them to understand ways in which businesses can deal with each issue. It must be noted that the following type of unethical business practice will no longer be credited: *false advertising* instead of *unfair advertising*.
- (f) The impact of CSR *on communities* should not be confused with CSR *on businesses*. Learners must know that CSI projects can have both a positive and negative impact on communities. Learners must be given a project-based learning task to conduct research on the impact of CSR on communities and present their findings in the classroom. Teachers should stimulate debates on this content to enable learners to gain insight in this content. The following fact will no longer be credited as it forms part of bullet four in the November 2022 NSC Marking Guideline: 'A business often appears

to benefit more from the CSR expenditure than the perceived benefits to the communities.'

- (g) Learners must be able to suggest ways in which professional, ethical, and effective business practice should be conducted in terms of how businesses should deal with employees, take care of the environment, and comply with relevant legislation. Subject advisors must advise the Grade 11 teachers to assess and remediate this content as it progresses to Grade 12.

#### QUESTION 4: MISCELLANEOUS TOPICS

This question assessed the TWO main topics of this paper consisting of direct short questions of 20 marks per main topic. Poor performance has been noted on questions consisting of the topic *Business Ventures*.

#### Common errors and misconceptions

- (a) Many candidates performed well on the factors that should be considered when making investment decisions in Q4.1. Some responses were based on other facts that are no longer in the *2021 Examination Guidelines*. They were, however, not penalised for committing this error.
- (b) In Q4.2.1 some candidates were not able to identify the type of compulsory insurance in the given scenario. Others confused other types of compulsory insurance with the examples of long-term insurance in Q4.2.2 which was a follow-on question. This error is persistent despite the recommendations made in past national diagnostic reports on learner performance.
- (c) Poor performance was observed in Q4.3 as many candidates could not discuss how *capital* and *management* can contribute to the success and/or failure of a partnership. Responses were vague and general to all forms of ownerships and not to partnerships specifically. Other responses were inappropriately based on the characteristics of a partnership.
- (d) A fair percentage of candidates' responses were vague and incomplete in Q4.4. Some confused the application of laissez-faire/free reign with either the situational leadership theory and/or the transactional leadership style. This is a commonly asked question in examinations.
- (e) In Q4.5 many candidates could not adequately provide CSI focus areas. Some wrote *rural areas* instead of *rural development*. Others confused CSI focus areas with the triple bottom line. This is also an indication of poor revision.
- (f) Some candidates did not perform well in Q4.6 as they were not able to identify the stages of team development in the given scenario. They lost marks for the motivations as they provided incorrect concepts.
- (g) Many candidates confused the impact of nominal group technique with brainstorming in Q4.7. It appears that they were not thoroughly prepared for this question.
- (h) Candidates' performance was satisfactory in Q4.8 even though some still confused the benefits of diversity in the workplace with the advantages of creative thinking in the workplace. This error is persistent and was highlighted in the 2021 national diagnostic report on learner performance.

### Suggestions for improvement

- (a) Learners must be able to name and explain all factors that should be considered when making investment decisions as outlined in the *2021 Examination Guidelines*. Furthermore, investment decision factors must be linked to the decisions to invest. Teachers must provide examples of types of investment that are relevant to each investment decision. It must be noted that the following investment decision factors no longer apply and has been excluded from the *2021 Examination Guidelines*:
  - Personal budget
  - Investment planning factors
  - Volatility/Fluctuations on investment markets
- (b) A clear distinction of the differences between compulsory and non-compulsory insurance should be made. Learners must know that the examples of short- and long-term insurance form part of non-compulsory insurance. They should also be conversant with the content for each type of compulsory insurance to enable them to identify them from given scenarios and/or statements.
- (c) The meaning of each criterion that contributes to the success and/or failure of each form of ownership must first be addressed in lessons before proceeding to explain how each criterion contributes to the success and/or failure of each form of ownership. Teachers are advised to recap the advantages and disadvantages of each form of ownership before introducing this content as they inform the success and/or failure of each form of ownership.
- (d) Teachers are advised to first explain the meaning, application, and impact of each leadership style. Learners must know that leaders can only apply the laissez-faire leadership style to well experienced subordinates with skills to perform tasks.
- (e) A clear distinction must be made between the meaning of *CSR* and *CSI*. Learners should know that businesses must first plan and develop *CSR* programmes to be able to identify *CSI* focus areas in given scenarios. They must also know that the *CSI* focus areas are aimed at benefiting communities, employees, and the environment. Support material and lessons must be designed to accomplish this.
- (f) Teachers must recap stages of team development as taught in Grade 11. Practical examples and demonstrations can be used to explain activities that take place in each stage of development. Learners should undertake a project that requires them to work in teams. They must then be asked to reflect on their experiences in working with others, making special reference to each stage of team development. They must also be exposed to source-based and direct middle-order questions on this content.
- (g) Learners need to know the meaning of the nominal-group technique in terms of the role an individual plays in generating new ideas and then sharing them with the group. Teachers should ensure that learners first understand how businesses can apply the nominal group technique to solve complex business problems. This will enable them to understand the impact of problem-solving techniques in dealing with issues.
- (h) The benefits of diversity in the workplace should not be confused with the advantages of creative thinking. The latter focuses on positive strategies and ideas that diverse employees bring in the workplace while the former focuses on the advantages of innovative ideas for businesses. Teachers must provide examples of the benefits of diversity in the workplace and request learners to conduct research on businesses that benefit from a diverse workforce.

## SECTION C: ESSAY QUESTIONS

### QUESTION 5: BUSINESS VENTURES: MANAGEMENT AND LEADERSHIP

Q5 was not a popular choice for many candidates. It was surprising as it consisted of four subquestions that were assessed in past NSC November examinations, and in the 2021 common preparatory examination paper. Those who answered this question performed well in three out of four subquestions asked.

#### Common errors and misconceptions

- (a) Many candidates repeated the statement that was used in the question paper as their introduction to an essay in Q5.1. Some candidates' responses were not relevant to any of the four subquestions asked in this question. This challenge persists and was reported in the 2021 national diagnostic report.
- (b) Poor performance was noted in Q5.2 as many candidates confused the differences between *management* with *leadership*. The reason for this error could be that some resources first start with the differences between *leadership* and then *management*.
- (c) Some candidates could not explain the transformational leadership theory as many responses were based on the transactional leadership style which was not a required answer in Q5.3.
- (d) Poor performance was observed in Q5.4 as many candidates' responses were on the description of the democratic and autocratic leadership styles instead of their impact on businesses.
- (e) Many candidates were able to discuss the role of personal attitude in successful leadership in Q5.5 although some provided inappropriate responses such as ways in which leaders should motivate employees in the workplace. This subquestion has been assessed many times in past NSC papers.
- (f) Candidates forfeited marks for the conclusion as they either repeated facts mentioned in the introduction and/or body in Q5.6. Other responses were vague and incomplete.

#### Suggestions for improvement

- (a) Learners must realise that an introduction should reflect any one or two subquestions of the questions that were asked in the paper. They must be advised to refrain from repeating the same statements that were used in the question paper. Instead, they must be encouraged to write creative responses that will not be repeated either in the body or conclusion.
- (b) A detailed description of leadership and management must be given using demonstrations and practical examples. Learners must be made aware that leaders communicate with employees and that managers communicate through management as the line function. Subject advisors should advise teachers to swap the notes on this content so that they start with management then leadership.
- (c) Teachers should assist learners in defining the word *transform* so that they can have a conceptual understanding of the meaning of a *transformational leadership theory*. It must be emphasised that this theory provides guidelines on how leaders should manage their followers in a dynamic business environment that requires turn-around

strategies. Learners could be requested to conduct research on this theory and present their findings in the classroom through open discussions.

- (d) The differences between the *democratic* and *autocratic* leadership style must be explained in terms of how each leadership style takes decisions, line of communication and employees' confidence. Learners must know that both leadership styles can be applied in the workplace depending on circumstances. The following fact under *management* will no longer be credited as it is the same as *leadership*: 'Focuses on the bottom line/short/medium/long term.' This fact should read as: 'Focuses on short/medium term.'
- (e) Teaching and learning must focus on *how* the leaders' personal attitude contributes to successful leadership. It must be emphasised that a *positive attitude* leads to a successful leadership as leaders are able to overcome any business challenge. It should be noted that the following fact will no longer be credited as the focus is on a positive attitude for successful leadership: 'A leader's bad attitude can influence the failure of the businesses'.

Teachers are advised to revise the following facts in the DBE notes and other credible resource materials:

- Positive attitude releases leadership potential for personal growth.
  - Enthusiasm produces confidence in a leader and inspires them to work even harder.
- (f) Teachers should assist learners on how to write a proper introduction and an essay without repeating statements contained in the preamble.

## QUESTION 6: BUSINESS ROLES: CREATIVE THINKING AND PROBLEM SOLVING

Many candidates performed well in this question. Their responses ranged from average to good. This question, including all four subquestions, were asked in past NSC papers either as contextual or essay questions.

### Common errors and misconceptions

- (a) Q6.2 was well answered by many candidates even though some swapped the differences between *problem-solving* and *decision-making*. Others provided incomplete responses to each concept.
- (b) Q6.3 was fairly answered by many candidates as they were able to explain ways in which businesses can create an environment that promotes creative thinking in the workplace. Some responses were based on the advantages of creative thinking in the workplace which were asked in Q6.4. Other responses were based on the ways in which professional, responsible, ethical, and effective business practices should be conducted, assessed in Q3.7.
- (c) Many candidates provided vague and incomplete responses to the advantages of creative thinking in the workplace in Q6.4. Some responses were limited to 'employees come up with new and innovative ideas'. Others confused this question with the benefits of diversity in the workplace.

- (d) Poor performance was noted in Q6.5 as many candidates either explained the meaning or the impact of the Delphi and force-field analysis techniques instead of ways businesses could apply these problem-solving techniques in the workplace.

### **Suggestions for improvement**

- (a) An in-depth analysis of the differences between *problem-solving* and *decision-making* is essential especially on the word *alternative*. Learners must know that decision-making only considers alternative solutions without a critical evaluation of each solution compared to problem-solving. Learners must be able to write complete sentences on each concept to avoid forfeiting marks for vague and incomplete responses.
- (b) Learners must know that businesses can only enjoy the benefits of creative thinking if they create an environment that promotes creative thinking. Practical examples must be given, and demonstrations done to enhance learners' understanding of this content. Learners could be requested to give examples of businesses that promote creative thinking in the workplace and justify their argument.
- (c) Teachers should focus on *how* new ideas benefit businesses when explaining the advantages of creative thinking in the workplace. Learners must know that creative thinking also benefits employees as they are also given the opportunity to contribute towards problem-solving. Subject advisors must ensure that this content is adequately taught and assessed in Grades 10 and 11 as it progresses to Grade 12.
- (d) Role plays and demonstrations could be done when teaching the application of the Delphi and force-field analysis techniques. Learners must know that businesses distribute questionnaires three times to experts before preparing a final feedback report. They should also know that the force-field analysis deals with change in the workplace and that businesses can only develop an action plan after a decision has been made on the viability of a project.

# CHAPTER 5

## ECONOMICS

The following report should be read in conjunction with the Economics Paper 1 and Paper 2 question papers for the NSC November 2022 examinations.

### 5.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who wrote the Economics examination in 2022 increased by 1 534 compared to that of 2021.

Although the table below indicates a negative trend over the years 2018-2021, a significant improvement in results was evident this year.

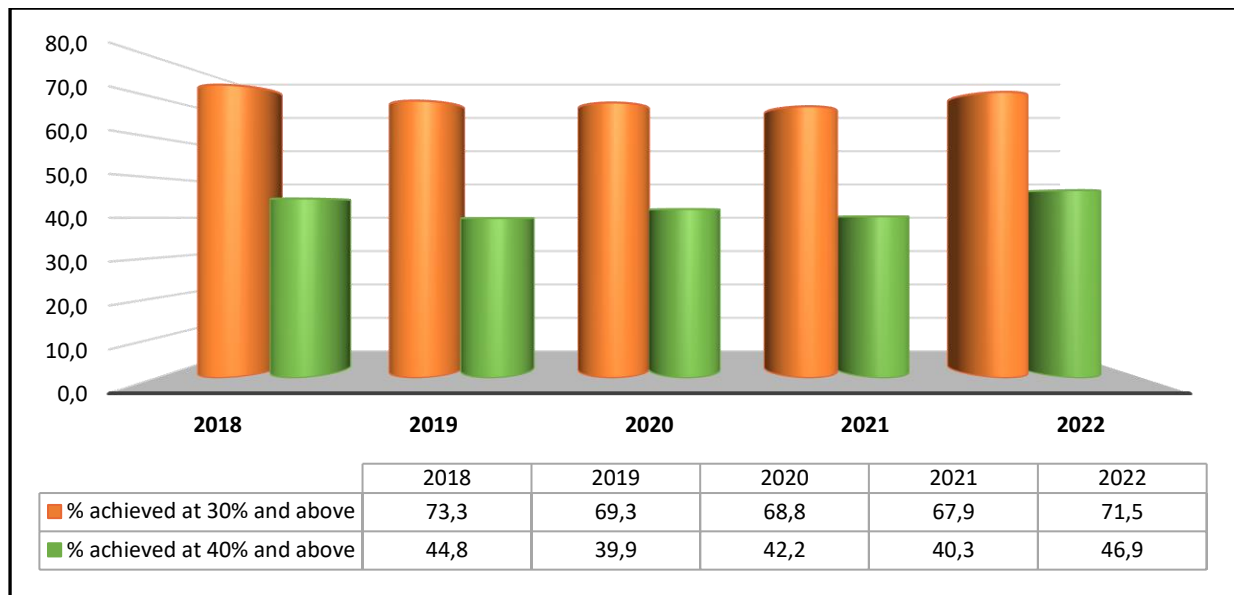
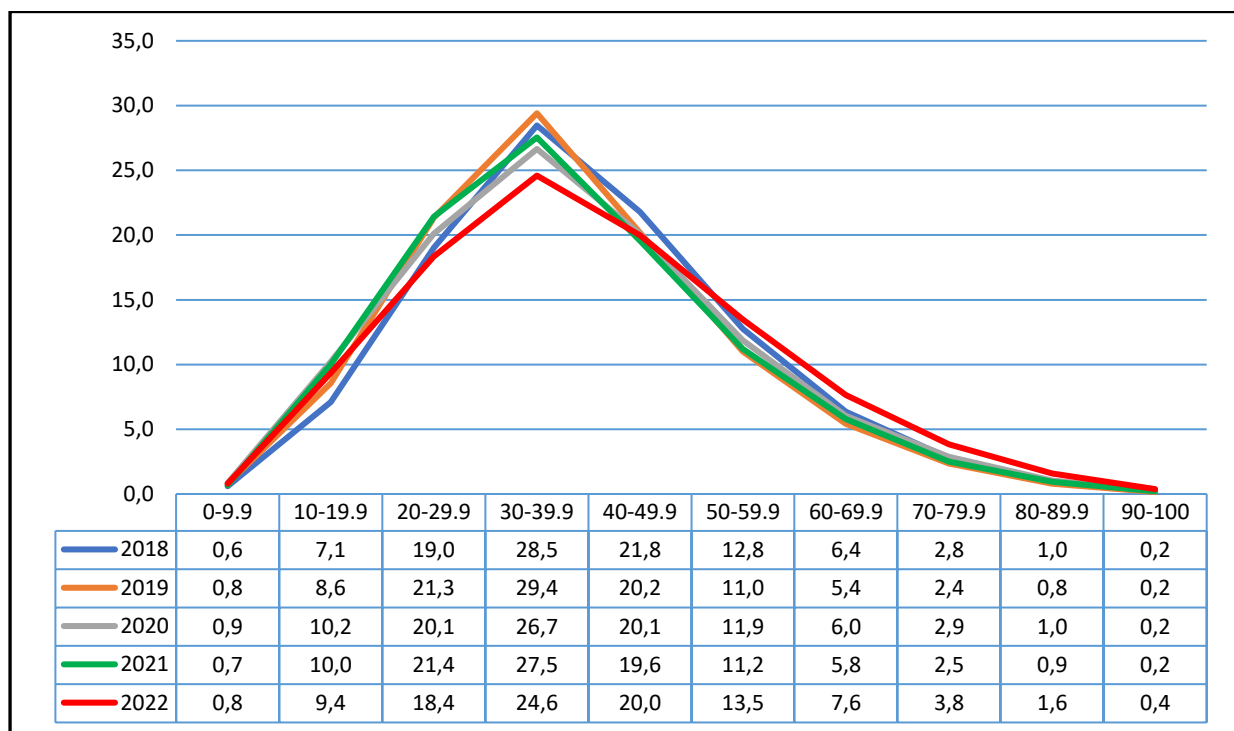
Candidates who passed at 30% (Level 2) improved from 67,9% in 2021 to 71,5% in 2022. There was also a significant improvement in the pass rate at 40% (Level 3) over the past two years from 40,3% to 46,9%.

There was also an increase in the percentage of distinctions over 80% (Level 7) from 1,1% to 2,0%. Given the higher number of candidates, this converts into an increase in the total number of distinctions by 1 222 candidates.

In view of disruption to academic programmes in 2020 and 2021, the results achieved by this cohort are particularly commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the favourable overall performance.

**Table 5.1.1 Overall achievement rates in Economics**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 115 169   | 84 395                        | 73,3                        | 51 609                        | 44,8                        |
| 2019 | 107 940   | 74 796                        | 69,3                        | 43 054                        | 39,9                        |
| 2020 | 118 484   | 81 536                        | 68,8                        | 49 958                        | 42,2                        |
| 2021 | 139 191   | 94 479                        | 67,9                        | 56 145                        | 40,3                        |
| 2022 | 137 657   | 98 414                        | 71,5                        | 64 559                        | 46,9                        |

**Graph 5.1.1 Overall achievement rates in Economics (percentage)****Graph 5.1.2 Performance distribution curves in Economics (percentage)**

## 5.2 OVERVIEW OF CANDIDATE PERFORMANCE: PAPERS 1 AND 2

### General comments

There has been a marked improvement in the writing of essays in recent years. While there has been a marginal improvement in the 8-mark higher-order questions, the candidates were still challenged by the additional part of the essay and the drawing of graphs. The results will only improve in a meaningful and sustainable way if these areas are addressed.

A thorough understanding of concepts in all topics will greatly enhance performance in both papers, as they form the basis of understanding subquestions pitched at various cognitive levels. This has a direct impact on Section A, Section B (where concepts are tested as definitions and also applied in a particular context), and Section C (where concepts form part of the introduction for the essays.) An excellent knowledge of Economics terminology will result in correct interpretation and answering of indirect questions in Sections B and C. Teachers need to reinforce concepts through regular assessment in class.

It is imperative that the content of all topics be covered adequately and timeously to ensure sufficient opportunity for revision. When teachers fall behind in content coverage, topics under *Economic Pursuits* and/or *Contemporary Economic Issues* tend not to be taught thoroughly. Candidates who attempt questions on these topics perform poorly, in comparison to questions about other topics. Teachers should structure assignments, projects and case studies in Grades 10 and 11 in a manner that will develop learners' writing skills, thus preparing them to cope with the Grade 12 content. This is also an area in which teachers must be supported by subject advisors.

In Grades 10 and Grade 11, learners' knowledge of topics/content should be extended wherever possible so that a strong foundation is set to cope with the demands of the Grade 12 curriculum. Grade 10 topics such as the *Circular Flow and Quantitative elements, Budget, Government Intervention, Business Cycles, Quantitative Elements* and Grade 11 topics such as *Calculation of the GDP, Market Structures, Cost and Revenue Curves, Income Inequalities, Indicators, North/South Divide, Globalisation and Environmental Sustainability* have strong links to Grade 12 topics and should be emphasised.

While there has been a general improvement in the drawing of graphs over the years, the technical aspects need to be reinforced. This would include the correct shape, positioning and labelling of cost and revenue curves in the perfect and imperfect markets. Teachers need to address graphs with learners by drawing the cost and revenue curves step by step. As each step is done, it needs to be explained. After the teacher draws each step, learners should do likewise in their workbooks. Emphasis must be placed on the average cost curve (i.e. *smile*) which must always be drawn before the marginal cost curve (i.e. *tick*). This will ensure that the MC always intersects the AC at its minimum point. It is extremely important that teachers recognise the integration of topics from Grade 10 to 12. Graphs relating to cost and revenue curves must be dealt with thoroughly in Grade 11 as this is the foundation for the more complex graphs in Grade 12. Learners still lack the skill of interpreting the graphs they have drawn. Teachers should equip learners with interpretation skills while taking them through the steps of drawing graphs.

Simple calculations and formulae need to be reinforced and assessed regularly as these skills are tested frequently in the NSC examination papers, e.g. *national account aggregates, moving averages, tax burden, the multiplier, BoP, exchange rates, profit and loss, production cost, CBA, percentage changes* and the *inflation rate*.

## Specific findings

- (a) A good understanding of tables, extracts, news articles, figures and graphs enabled many candidates to perform well.
- (b) Although candidates were able to complete each paper within the allocated time, it seemed that they had limited time to review their work as evidenced by some subquestions in Sections A and B being omitted.
- (c) The main reasons for underperformance were the following:
  - **Skills:** Poor language skills made it difficult for candidates to understand the requirements of questions and to express themselves clearly, especially in paragraph-type questions which formed a large part of the question paper. Most candidates were unable to solve problems, give their own opinions or evaluate data connected to their study material. Candidates also lacked basic knowledge of the general economic issues of the day.
  - **Content coverage:** It is evident from the poor performance of many candidates that their teachers did not cover some of the topics. Basic Economics concepts/terminology seemed to be lacking among many candidates and there was also a lack of knowledge on current economic issues, notably in the following subquestions:
    - Paper 1 Q2.5 Analyse the problems faced by the South African government in providing public goods and services
    - Paper 2 Q3.3.5 How can crime as a socio-economic issue negatively influence the tourism sector?
    - Paper 2 Q5 Additional Part: Evaluate the success of international agreements in reducing climate change.
  - **Exposure to different types of questions:** Teachers play a crucial role in preparing their learners to deal with a variety of questions with different instruction verbs, such as *why*, *how* and *what* and the unlocking of knowledge in a variety of ways. This gives them the opportunity to develop a variety of higher-order thinking skills in the context of the subject content being taught. Skilled learners are better able to write essays and paragraphs and offer their opinions with confidence. Such learners are able to focus on the information that is relevant to the answering of each question.
  - **Problem-solving skills:** Learners should be challenged to solve everyday problems experienced in their own communities, e.g.:
    - Paper 1 Q3.3.5 How can the government improve the productivity of the South African labour force?
    - Paper 2 Q3.5 How can South Africa reduce imports in order to correct a balance of payments deficit?
    - Paper 2 Q2.1.2 Why does the government provide subsidies to the producers?
    - Paper 2 Q5 How can the government use the fiscal policy to combat demand pull inflation?
  - **Language ability:** Although proficiency in the language of assessment is still a drawback for many second-language candidates, some centres in deep rural areas produce good results compared to others experiencing similar circumstances. Teaching should take place in such a way that learners understand the content.

## General suggestions for improvement

Teachers are advised to build the following practices into their work plan for the year:

- (a) **Use of past NSC and CAPS aligned exam papers:** In preparation for the 2023 year-end examinations, all learners should have access to and make use of past NSC papers, which must include the final examination papers (2017–2022) and the supplementary examination papers (2018–2023) for clear guidance on style, format and different questioning techniques. Teachers should reflect on their own confidence and their ability to deal with each topic in the classroom situation. Furthermore, teachers should use the *2021 Examination Guidelines* as support material when it comes to the scope and depth of content and on how to assess learners' understanding of the specific content matter. Previous question papers and marking guidelines should be used as revision tools but not as a teaching tool, as it encourages spotting of questions for the exams. Interpretation of questions is critical.
- (b) **Basic concepts:** Teachers should ensure that learners understand basic concepts and terminology before engaging in in-depth applications. More time should be spent on improving the reading skills of especially the second-language learners. Learners' understanding of terminology should be assessed on a continuous basis. A *glossary* of all concepts should be compiled for each topic. A quiz bowl, crosswords or team challenges may be interesting tools to assess knowledge of economic concepts. Regular classwork or homework on definitions will ensure that learners stay familiar with these basic concepts.
- (c) **Requirements of questions:** Teachers should ensure that learners understand the requirements of questions that may appear in NSC examination papers. For example, if a question requires the drawing or analysis of a well-labelled graph, this must be done effectively to earn the relevant marks, e.g. Paper 1, Q2.3.5: 'With reference to the graph above, explain the effect of the increase in the demand for US dollar on the value of the rand.' Paper 2, Q5: 'With the aid of three separate graphs, discuss the short run equilibrium positions in a perfect market.'

Teachers should ensure that their learners understand the phrasing of questions, e.g. the *what*, *why* and *how* type of higher-order questions. However, the misconception that if a question begins with *How* then it constitutes a higher-order question, must be clarified. The following example illustrates this point:

*Q3.1.2 How do households benefit from tourism? (2)*

This is a middle-order cognitive level and an easy question. Learners should be guided by the mark allocation in terms of the depth of the answer required.

Regarding higher-order questions (especially Q2.5, Q3.5, Q4.5 and the additional part of the essay questions), candidates need to read the question carefully and highlight the key issues required. Further, it must be noted that answers to these questions are not necessarily found in textbooks but will require an application of content studied within a particular context. Reading the question again will ensure greater accuracy in the candidates' response. The question should be checked constantly to ensure the response logically suits it. It is not wise to respond immediately after the initial reading of the question.

Higher-order questions place advanced cognitive demands on learners and encourage them to think beyond literal answers to questions. Higher-order questions promote critical thinking skills because these types of questions expect learners to apply, analyse, synthesise, and evaluate information instead of simply recalling facts. Higher-order

questions require learners to make inferences, draw relevant and insightful conclusions and use their knowledge in new situations. It also requires them to relate their thinking to other situations and to their own background knowledge. Issues from the real world can be used to either support or refute a point of view.

Time must be devoted to understanding the question clearly, for example:

- Paper 1: Q6 'Discuss in detail South African growth and development policies and strategic initiatives since 1994.' Some candidates who misinterpreted the question discussed in detail South Africa's initiatives (endeavours) in regional development.
- Paper 2: Q5 Additional part: 'Analyse the advantages of perfect competition as a market structure.' Many candidates referred to the characteristics of a perfect market without explaining how it can be an advantage for the consumer or producer.

Teachers need to realise that there are more interesting and creative ways to teach than by simply promoting rote learning. This includes teaching for understanding, decision-making, problem-solving, connecting a part to a whole, detail-to-concept, and concept-to-concept. There is also inference, prediction, analysis for bias and learning for transfer. Each of these techniques and processes requires some form of critical thinking. Opportunities for learners to develop critical thinking processes will not be found in classrooms dominated by the regurgitation of factual content. They are found in classrooms where active learning is an essential component.

- (d) **Comments and explanations:** Teachers should equip learners with the relevant skills needed to express themselves clearly where comments or explanations are required. Learners need guidance on how to express opinions that are relevant to the context of especially higher-order questions, e.g.:
- Paper 1: Q5 Additional part: 'Analyse the challenges that an economic recession poses for the different participants in the economy.'
  - Paper 2: Q2.5: 'How does mutual dependence influence the behaviour/actions of firms in the oligopoly market?'
- (e) **The importance of formative testing:** Teachers should build the confidence of learners using short, informal formative tests and tasks. These tasks should be used to ascertain whether learners are able to apply their knowledge, placing emphasis on their own opinion and understanding. This will encourage learners to take ownership of the learning process.
- (f) **The structure of the paper:**
- **SECTIONS A AND B:** The demands of these sections should be explained to learners to enable them to organise their answers properly. Leaving lines between subsections, using the correct numbering system, and not omitting question numbers are examples of techniques that make assessment more effective. In Section A, learners must be made aware that no marks will be awarded when they provide more than one answer to a short question. It is important that they cancel an incorrect answer in Q1.1, Q1.2 and 1.3 and write the correct one next to it, instead of writing over the incorrect answer. There is a misconception that answers to 1-mark questions in the *Data Response* items must come directly from the data. A question may require an application of knowledge when the answer appears in the data, or it may be a simple question related to the data.
  - **SECTION C (Essay):** Teachers must stress the importance of the layout of the essay, i.e. introduction, body (main and additional part) and conclusion. There should be a clear distinction between the various aspects with line spacing between them. Using subheadings is crucial as this earns marks and provides structure to the response. Learners should structure the essay according to the outline provided in the question paper. Learners must be made aware that no

marks will be earned if any part of the introduction or body is repeated in the conclusion. Instead, it should include the learner's own opinion/an alternative viewpoint/any fact to support the body or a summary of the discussion.

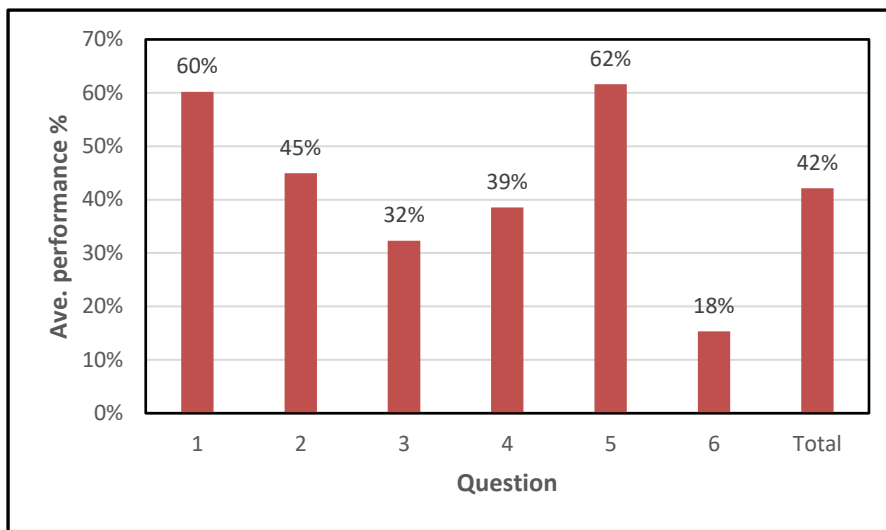
Teachers must allow learners the opportunity to practise the answering of essay questions. When a topic or chapter is finished, an essay question should be given as a test or homework. If given as homework, the essay can be assessed in terms of the following important aspects (detailed assessment is not necessary):

- Relevant introduction
- Subheadings in the main part
- The appropriateness of the additional part
- Relevant conclusion
- Teachers are advised to use a variety of resources (including textbooks) to prepare notes that supplement material available to learners. This is necessary where a prescribed textbook does not adequately cover aspects stipulated in the *2021 Examination Guidelines*. Subject advisors and networking within cluster groups can help to cover content gaps that are evident in the various resources.
- Topics earmarked as possible essays in the *2021 Examination Guidelines* should be thoroughly prepared for the examination. Spotting of questions underprepares candidates and leads to poor performance. In both Paper 1 and Paper 2, various essay topics content was covered in lower-order, middle-order and higher-order questions,
  - Paper 1 reflected a total of 57 marks of these questions (see Q1.1.2, Q1.1.6, Q1.3.4, Q1.3.5, Q1.3.6, Q2.1.1, Q2.5, Q3.1.2, Q3.2.4, Q3.3, 3.4, Q3.5, Q4.3.3 and Q4.4).
  - Paper 2 reflected 65 marks of these questions (see Q1.1.3, Q1.1.8, Q1.2.3, Q1.2.4, Q1.2.5, Q2.1.1, Q2.3, Q2.5, Q3.1.2, Q3.5, Q4.2, Q4.3.5, Q4.4, and Q4.5).

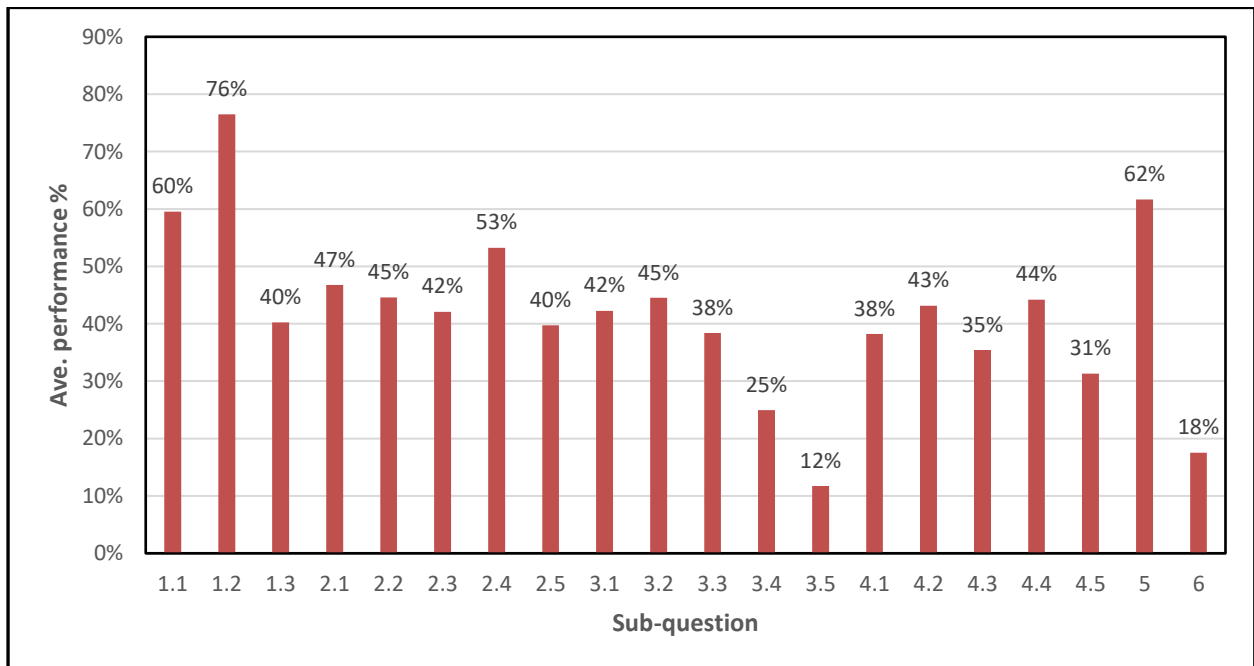
### 5.3 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 1

The performance in Section A increased slightly when compared to that of 2021. There was a decrease in Q1.1 and an increase in Q1.2 and Q1.3. In Section B, candidates performed better in Q2. In Section C, while Q5 showed a good increase, there was a significant drop in Q6.

The following graph is based on data from a random sample of candidates. While this graph may not reflect national averages accurately, it is useful to assess the relative degrees of challenge of each question as experienced by candidates.

**Graph 5.3.1 Average performance per question in Paper 1**

| Q | Topics              |
|---|---------------------|
| 1 | Objective questions |
| 2 | Macroeconomics      |
| 3 | Economic pursuits   |
| 4 | Macro and pursuits  |
| 5 | Macroeconomics      |
| 6 | Economic pursuits   |

**Graph 5.3.2 Average performance per sub-question in Paper 1**

| Sub-Q | Topic                                       | Sub-Q | Topic                                    |
|-------|---------------------------------------------|-------|------------------------------------------|
| 1.1   | Multiple choice                             | 3.3   | Social and Economic indicators           |
| 1.2   | Matching                                    | 3.4   | Methods of export promotion              |
| 1.3   | Give a term                                 | 3.5   | Regional development policies            |
| 2.1   | Public sector & national account aggregates | 4.1   | Effects of international trade and BBBEE |
| 2.2   | Taxes and tax calculations                  | 4.2   | Multiplier                               |
| 2.3   | Foreign Exchange market                     | 4.3   | Industrial development policies          |
| 2.4   | Circular Flow                               | 4.4   | Social Indicators                        |
| 2.5   | Problems in public sector provisioning      | 4.5   | Balance of Payments                      |
| 3.1   | Economic integration and migration          | 5     | Business Cycles                          |
| 3.2   | Protectionism                               | 6     | SA growth and development policies       |

## 5.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 1

### QUESTION 1: MACROECONOMICS AND ECONOMIC PURSUITS

Most candidates achieved moderate results in Q1. The performance of candidates ranged from excellent to poor. Some candidates attained full marks, while others did not even attempt to answer some of the questions. This question was compulsory.

#### Common errors and misconceptions

- (a) In Q1.1 a lack of content knowledge led to many candidates being unable to choose the correct alternative. Poor performance was recorded in Q1.1.3, Q1.1.4, Q1.1.5 and Q1.1.6.
- (b) The majority of candidates performed well in Q1.2 and there were candidates who managed to obtain full marks for this question. The implication is that candidates perform better if they are provided with few concepts from which they may choose. However, poor performance was recorded for Q1.2.2 and Q1.2.5 where many of candidates could not match the correct description for *Keynesian approach* and *diversification* in 1.2.2 and 1.2.5 respectively.
- (c) In Q1.3 most candidates performed poorly compared to Q1.1 and Q1.2. However, there was an overall improvement in this question from 2021.

Many candidates confused the concept *gross national product* with *gross domestic product* in Q1.3.1 and *employment rate* was confused with *unemployment rate* in Q1.3.6.

#### Suggestions for improvement

- (a) It is imperative that learners first attempt to determine the correct answer to a multiple-choice question before considering the given options. Short formative tests on basic concepts are recommended to ensure that learners become familiar with Economics terms and concepts.
- (b) The confidence of learners should be built by spending more time on the understanding of Economics terminology, definitions and concepts to ensure expanded knowledge over the broad spectrum of the subject.
- (c) Commence each lesson by testing concepts that were taught in the previous lesson. Continuous and constant revision of concepts and terminology is strongly advised. *Mind the Gap* and any other supplementary material, which has terms and concepts at the beginning of each topic, can be used. The material can be accessed with the assistance of subject advisors. Learners should attempt to answer all items in Q1.1 and Q1.2 where the options are provided. Learners should also concentrate on more detailed preparation in respect of concepts and terminology to ensure that they attain higher marks for Q1.3.
- (d) Learners must create their own glossary for each chapter. They can keep a separate book/document containing all the terms they have to study. Use informal concept tests to reinforce these concepts. Make use of flashcards, '30 seconds game' for Economics, etc.
- (e) Although multiple-choice questions provide possible answers, they require full content knowledge. Constant revision of terminology is strongly advised.

- (f) Subject advisors should monitor the teaching of Economics concepts by checking activities in learners' books during their school visits.

## QUESTION 2: MACROECONOMICS

### Common errors and misconceptions

- (a) Q2.1.2 was poorly answered as most candidates misinterpreted the question by responding in terms of *cost of production* instead of *market price*. Some responses indicated that candidates could not differentiate between 'subsidies on *products*' and 'subsidies on *production*'.
- (b) Most candidates performed poorly in 2.2.1 and 2.2.5, which were based on tax calculation.
- (c) Poor performance was evident in Q2.3.3. Candidates confused *devaluation* and *depreciation* by excluding government intervention in the description.
- (d) In Q 2.3.4 candidates confused *surplus* on the current account with a *deficit*. Responses focused more on methods to solve the deficit on the current account.
- (e) Candidates' responses in Q2.3.5 showed a misinterpretation of the exchange rate graph as responses focused more on the *appreciation* of the rand against the dollar instead of the *depreciation*.
- (f) A fair percentage of the candidates misinterpreted Q2.4 and explained the role of each participant instead of the interrelationship between households and businesses in the circular flow model.

### Suggestions for improvement

- (a) Teachers should focus on differentiating between *subsidies on products* and *subsidies on production* when discussing the three methods used to calculate the GDP. *Subsidies on production* include employment subsidies (learnerships) and subsidies to prevent pollution. *Subsidies on products* include export subsidies and bread subsidy. In converting *basic prices* to *market prices*, taxes are added, increasing the price and subsidies are subtracted, reducing the price.
- (b) Teachers need to use the examination guidelines when preparing lessons to cover all aspects that are required for a topic. The information on tax tables and calculations is available in the StatsSA publication and must be used in class for practical purposes.
- (c) Concepts related to exchange rates, such as *appreciation* and *depreciation*, must be taught thoroughly by linking them to the graph on exchange rates. Various scenarios relating to the supply and demand of foreign exchange can be given as assessments to enhance learners' understanding of these terms.
- (d) A clear distinction of roles and interactions of participants in the economy must be emphasised when teaching the *circular flow model*.
- (e) Teachers need to engage learners in interpreting questions and to answer according to the cognitive demand of the question.

- (f) Subject advisors need to support teachers by developing content-based documents that address challenging topics. Workshops based on content knowledge should be organised for newly appointed teachers and for those with content knowledge gaps.
- (g) The use of print media is highly recommended in the classroom to acquaint learners with contemporary economic issues.

### QUESTION 3: ECONOMIC PURSUITS

#### Common errors and misconceptions

- (a) In Q3.1.1 candidates performed poorly as they confused *economic integration* with *economic growth policies* and *strategic initiatives*.
- (b) Many candidates misinterpreted the data-response questions (Q3.2 and Q3.3). Most subquestions demanded reading with understanding and interpretation. In Q3.2.4 candidates did not understand the term *infant industry* but rather explained infant in general as a new-born baby.
- (c) Q 3.2.5 indicated that candidates lacked content knowledge and the understanding of *free trade* as a policy, hence they could not explain its benefit. This was also evident in Q3.3.4 and Q3.3.5 where candidates could not explain the impact of *labour productivity* as an economic indicator.
- (d) Candidates' responses were too generic and lacked factual knowledge in Q3.5 and could not evaluate the suitability of South Africa's development policies in terms of the benchmark criteria, due to poor interpretation and application skills.

#### Suggestions for improvement

- (a) Teachers should emphasise that data-based questions cover middle-order responses. The action verbs *how* or *why* in this case would not have the same expectations of the candidates as higher-order type questions.
- (b) Teachers should be able to link and relate content with real-life situations. Practising in the classroom will assist learners to adapt to higher-order questions. Content for discussion should be translated into real and/or current economic circumstances. Arguments in favour of free trade can also be classified as benefits or advantages.
- (c) Additional learning material should be given to learners during the academic year. Data provided in data-response questions should be analysed thoroughly before learners attempt to answer any questions set.
- (d) More case-study questions should be discussed in class and given as homework activities. Debates and presentations of certain topics should be conducted regularly. Teachers must endeavour to include the *why* and *how* types of questions to enable learners to think beyond typical textbook knowledge.
- (e) Teachers must use examination guidelines to determine the depth and scope of the topic. Candidates need to be taught how to interpret questions and answer according to the cognitive demand of the question.

- (f) Subject advisors need to support teachers by developing content-based documents that address challenging topics and collate data-response activities that can be used as formative assessment in the classroom.

#### QUESTION 4: MACROECONOMICS AND ECONOMIC PURSUITS

##### Common errors and misconceptions

- (a) Many candidates performed poorly in Q4.1.1 where they had to name any two effects of international trade (see p.19 of the *2021 Examination Guidelines*).
- (b) Poor performance was also evident in Q4.1.2 as responses showed a lack of content knowledge on Broad-based Black Economic Empowerment (BBBEE). This made it difficult for candidates to relate BBBEE to the promotion of industrial development.
- (c) In Q4.2.1 the identification of *mpc* from the given data was a challenge for candidates. They confused *mpc* with *mps* or provided any amount from the diagram. Typical responses were R80m, R100m and R20m.
- (d) A fair percentage of the candidates performed poorly due to the use of an incorrect formula in calculating the value of the multiplier.
- (e) Most candidates lacked knowledge on the role played by the DTIC in regional development; therefore, they could not name any incentive in Q4.3.2.
- (f) In Q4.3.4 candidates misinterpreted the question and limited their responses to the impact of oil prices on households and not on industrial development as per the requirement of the question.
- (g) Many candidates failed to relate the importance of infrastructure to previously neglected areas, as asked in the question. Responses dealt with the benefits of tarred roads and how infrastructure created employment in a generic sense.
- (h) Candidates performed poorly in Q4.5 due to misinterpretation. Imports were discussed in general and were not related to correcting a balance of payments deficit.

##### Suggestions for improvement

- (a) Teachers should use a variety of resource materials to prepare learners adequately for the examination. Current economic issues should be used as examples to illustrate the subject in context. Economics in the classroom should be linked to Economics in real life by exposing learners to actual data, graphs and statistics.
- (b) Teachers should ensure that learners understand what is expected of them when instruction verbs such as *argue*, *analyse*, *evaluate* or *differentiate* are used in a question.
- (c) Questions based on the application of content knowledge must be taught to ensure that learners can adapt any question to its context.
- (d) Learners should be guided on the process of selecting questions from both Section B and Section C as part of exam-answering techniques. It is noted that often all the questions in Section B are answered; this is a clear indication of poor planning and a waste of valuable time. Instructions need to be explained and reinforced to learners during their revision sessions so that they are clear about the requirements.

- (e) Learners need to be trained to answer higher-order questions. Teachers should ensure that learners have a clear understanding of the depth of knowledge expected of them, as explained in the *2021 Examination Guidelines*.

## QUESTION 5: MACROECONOMICS

### Common errors and misconceptions

- (a) Candidates could not be awarded full marks because they misinterpreted the question and explained the phases of business cycles. Listing of facts in different subtopics also led to poor performance in this section. Candidates confused *leading indicators* with *lagging indicators*. The following subheadings were also not well discussed by candidates:
- *Moving Average*
  - *Amplitude*
  - *Trend line*
  - *Extrapolation*
- (b) The additional part of the essay is higher-order and allocated 10 marks. Some candidates failed to obtain the maximum of 10 marks because they merely listed facts and did not analyse the challenges posed by recession in the economy. e.g. ‘households will not have jobs’ and ‘firms will produce less goods and services.’

### Suggestions for improvement

- (a) It is important that subject advisors supplement content on this topic through teacher development workshops to address gaps in teachers’ content.
- (b) Teachers are urged to use the *2021 Examination Guidelines* which clearly indicate all the possible essay questions for the next three years. These essays should be prepared in advance to ensure excellent marks in the introduction and main parts of the essays. The fact that so many candidates included topics not earmarked as essays, such as different types of business cycles and phases of business cycles, is concerning since the policy guidelines have been provided.
- (c) Teachers should expose learners to past question papers to enable wider awareness of different types of questions. Learners should be exposed to questions on all levels of difficulty during class activities, tests and internal examinations.

## QUESTION 6: ECONOMIC PURSUITS

### Common errors and misconceptions

- (a) Most candidates could list the different arguments in favour of *protectionism*, but tended to be generic in their discussions. Candidates included detail on *free trade* and *export promotion*, which was not a requirement of the question. Most candidates misinterpreted the question and discussed industrial development initiatives instead of economic growth and development policies.
- (b) Demand side policies and macroeconomic objectives were also discussed by some candidates.

- (c) Misconceptions also included responses such as mere listing of examples (e.g. RDP houses), factors of production, industrial development policies, macroeconomic objectives of the state and fiscal measures.
- (d) Most candidates performed poorly in the additional part of the question as they could not explain how *supply-side policies* can be used to promote economic growth and development. Some candidates misinterpreted the phrase *supply-side measures* and instead drew the supply curve from microeconomics.

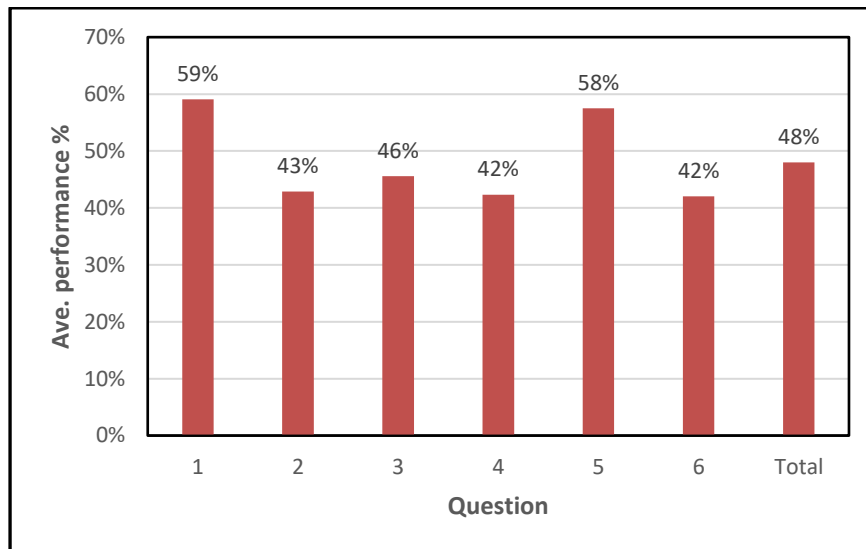
### **Suggestions for improvement**

- (a) Teaching of these topics should be done holistically. Learners should be guided in discussing each fact in detail. This will ensure that they do not omit any crucial aspect of the answer. More time should be used to teach areas that can cause confusion.
- (b) Basic content should not only be covered but should also be linked to the creativity of learners in the practical implementation of each topic. Case studies and class discussions can be used gainfully, in this regard. Learners need to improve their evaluation skills. Knowledge about recent developments can assist in making the module more interesting.
- (c) Different methods of teaching, such as case studies, classroom discussions and debates, can be used to make discussions more meaningful in relation to policies. Teachers should have full knowledge about recent economic information and discuss this regularly in class. Teachers are urged to use the *2021 Examination Guidelines* which clearly indicate all possible essay questions for the next three years. These essays should be prepared in advance to ensure excellent marks in the introduction and main parts of the essays.
- (d) Subject advisors need to monitor that assessment tasks administered contain balanced cognitive levels. Workshops on cognitive levels and levels of difficulty should form part of content workshops conducted during the year.

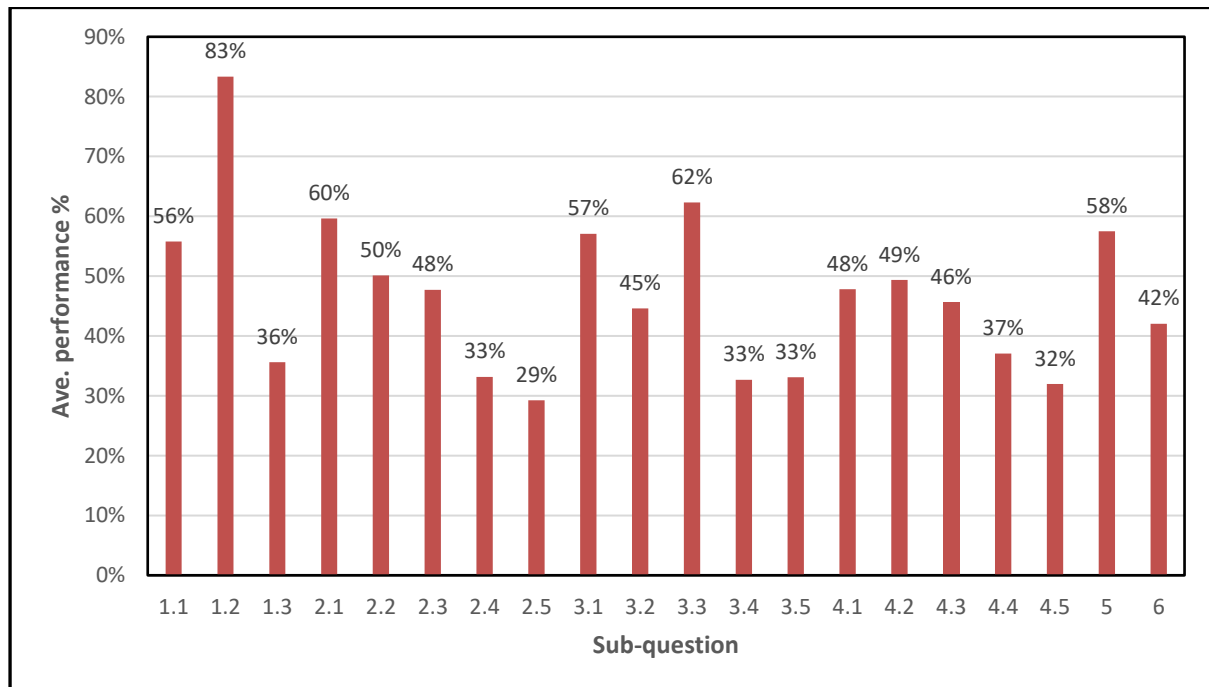
## **5.5 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 2**

The performance in Section A decreased slightly when compared to that of 2021. There was a decrease in Q1.1 and Q1.3. and an increase in Q1.2. In Section B, candidates performed marginally better in Q3 and Q4. In Section C, there was a remarkable improvement in Q5 when compared to 2021. There was a marginal decrease in Q6. The performance in Microeconomics over the years has generally been poor to average.

The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 5.5.1 Average performance per question in Paper 2**

| Q | Topic/s                                         |
|---|-------------------------------------------------|
| 1 | Objective questions                             |
| 2 | Microeconomics                                  |
| 3 | Contemporary economic issues                    |
| 4 | Microeconomics and contemporary economic issues |
| 5 | Microeconomics                                  |
| 6 | Contemporary economic issues                    |

**Graph 5.5.2 Average performance per subquestion in Paper 2**

| Sub-Q | Topic                                   | Sub-Q | Topic                                     |
|-------|-----------------------------------------|-------|-------------------------------------------|
| 1.1   | Multiple choice                         | 3.3   | Reasons for tourism                       |
| 1.2   | Matching                                | 3.4   | Tourism growth                            |
| 1.3   | Concepts                                | 3.5   | Consequence of inflation                  |
| 2.1   | Collusion/producer subsidy              | 4.1   | Merit goods/interest rate                 |
| 2.2   | Monopolistic competition                | 4.2   | Monopoly Graph- Economic loss             |
| 2.3   | Maximum prices/minimum prices           | 4.3   | Inflation                                 |
| 2.4   | Negative externality- graph explanation | 4.4   | Causes of market failure                  |
| 2.5   | Mutual dependence in an oligopoly       | 4.5   | Negative impact of tourism                |
| 3.1   | Consumer inflation/tourism benefits     | 5     | Equilibrium positions in a perfect market |
| 3.2   | Inflation                               | 6     | Sustainability measures by government     |

## 5.6 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### QUESTION 1: MICROECONOMICS AND CONTEMPORARY ECONOMIC ISSUES

Most candidates performed well in Q1, especially in the multiple choice and matching questions. In Q1.3 it was evident that candidates still struggled with concepts and often confused related concepts. Overall, there was slight decline from the previous year. The performance of candidates ranged from excellent to poor. Some candidates attained full marks while a few did not even attempt to answer some of the questions.

#### Common errors and misconceptions

- (a) In Q1.1 many candidates omitted some answers, which resulted in incorrect numbering. In some cases, more than one answer was provided.
- (b) Poor performance can be attributed to the misunderstanding between *allocative inefficiency* and *productive inefficiency* in Q1.3.4 and *preservation* was confused with *conservation* in Q1.1.7.
- (c) In Q1.1.8 a fair percentage of candidates did not know the international agreement that managed the disposal of chemical waste.
- (d) Although most candidates performed very well in Q1.2, many of them chose option 'G' as an answer for implicit cost in 1.2.1.
- (e) In Q1.3.1 most candidates, including the high-performing candidates, performed poorly as answers varied from 'short-run' to 'inflation' instead of *diseconomies of scale*. Some candidates wrote *economies of scale* as the answer.
- (f) Confusion between concepts in Q1.3.1 was evident as many candidates gave *hyperinflation* and *inflation* instead of *demand-pull inflation* as the answer.
- (g) *Transit tourist* was confused with *inbound tourist* in Q1.1.5.

#### Suggestions for improvement

- (a) In Q1.1 learners are expected to write the letter of their choice (A, B, C or D) next to question numbers. However, if for some unknown reason, they decide to write the statements/option, then this would have to be the complete statement, as per the question paper. It is suggested that learners first attempt to determine the correct answer to multiple-choice questions before analysing the given options. Learners must be made aware that no marks will be awarded when they provide more than one answer to a short question. It is important that they cancel an incorrect letter in Q1.1 and Q1.2 and write the correct one next to it, instead of writing over the incorrect letter chosen.
- (b) Teachers should emphasise the difference between *implicit costs* and *explicit costs* and try to relate fixed and variable costs to explicit costs with examples. Teachers could go one step further and explain the difference between *accounting profit* and *economic profit* and how it relates to explicit cost and implicit cost. This will give learners a clear understanding of why *normal profit* in Economics is equated to *break-even point* (considering explicit and implicit costs). This is quite different to *profit* in Accounting, where only *explicit costs* are considered in determining profit.

- (c) Emphasis on distinguishing between the related concepts such as *allocative efficiency and productive efficiency*, *conservation and preservation* is essential in avoiding confusion.
- (d) Definitions and concepts should be emphasised. The use of a glossary should form the basis of the teaching and learning of Economics.
- (e) Revision by means of short, regular formative tests on basic concepts is strongly recommended. Learners should be encouraged to make a list of the key concepts of each topic, especially in Microeconomics.
- (f) Teachers should try to engage learners in terminology games, using media such as Kahoot, Quizziz, Quizlet and Crosswords.

## QUESTION 2: MICROECONOMICS

### Common errors and misconceptions

- (a) In Q2.1.2 many candidates explained *subsidy* as a consumer benefit instead of a producer benefit.
- (b) A fair percentage of candidates mentioned examples of a monopolistic competitive industry such as *Nike* and *KFC* in Q2.2.1, instead of naming the industry.
- (c) Q2.2.3 was answered poorly as most candidates could not explain the term *patent*.
- (d) Most candidates misunderstood the requirements of Q2.2.5 relating to product differentiation and its benefits. Many answers related to product choice.
- (e) In 2.3.3 many candidates confused the explanation of *minimum prices* with that of *maximum prices*. There is also some confusion among teachers who are using the word 'lowest' to describe maximum prices.

*Maximum price* is a limit or cap on a price set by a government or an organisation – it is the highest price that can be set by producers. A price below the maximum is acceptable, and no intervention would follow.

Similar to *minimum prices*, it is the lowest price set by a government or an organisation. Maximum prices are called *price ceilings*, the highest price at which a producer could sell, and minimum prices are called *price floors*, the lowest price at which a producer could sell.

- (f) Lack of application and interpretation of a graph in a certain context was evident in Q2.3.5 as many candidates' responses related to the market/industry instead of the individual business. Candidates found it difficult to explain the impact of one variable on another.
- (g) Most candidates had great difficulty in interpreting and explaining the graph in Q2.4. Externalities were explained in general terms without reference to the graph and its effect on the market.
- (h) Q2.5 was poorly answered. Some candidates did not even attempt to answer this question. Those who did, could not fully relate their responses to mutual dependence in the oligopoly market. Instead of explaining how actions of one firm affect the other, candidates drew a kinked demand curve and explained the two segments.

### Suggestions for improvement

- (a) Misinterpretations of questions seem too common among learners. Attempting past papers and unpacking a question in terms of what is required would help learners to write what is relevant to the question, especially for the 2-mark, 4-mark and 8-mark questions. This should be done without the marking guidelines.
- (b) Teachers must clearly distinguish between examples of *firms* within an industry and examples of an industry. This would help learners to interpret the question correctly.
- (c) A clear distinction must be made between *monopolistic competition* and *monopolistic market* as the latter relates to a monopoly market structure. Teachers must be encouraged to mediate the content, especially related concepts, more effectively to learners.
- (d) While the practice of drawing and labelling graphs is essential to learners' understanding of various concepts and content related to a topic, teachers should also encourage learners to provide explanations of given graphs on a regular basis where credit must be given to explanations related to the graph.
- (e) Teachers must spend time explaining the instructional verbs from the *2021 Examination Guidelines* to learners in detail. Learners should clearly understand the expectations of these verbs, especially in relation to higher-order questions.
- (f) Learners need to be aware of the specific requirements of a higher-order question. In addition, teachers need to assess higher-order questions appropriately so that learners clearly understand the mark allocation. A mere listing of facts without an explanation in context will earn only 2 marks, instead of the full 8 marks.
- (g) Subject advisors, through teacher development, should develop more material on market structures and market failures. It is evident from learners' responses that teachers need support in this section.

### QUESTION 3: CONTEMPORARY ECONOMIC ISSUES

#### Common errors and misconceptions

- (a) In Q3.1.1 candidates were unable to differentiate between *consumer* and *producer inflation*. Instead of naming the types of *consumer inflation*, they mentioned *producer price index* and *the various types of inflation* as an answer.
- (b) Most candidates answered SARS instead of SARB in response to Q3.2.2
- (c) Many candidates' responses to Q3.2.5 indicated they were unable to calculate the inflation rate and were confused by the formula/method. Some candidates calculated the inflation rate of 2018 and 2019. Some failed to multiply by 100 to express the figure as a percentage while others divided by 100.
- (d) In Q3.3.4 some candidates explained the importance of tourism to *consumers*, instead of its benefit for the *government*. Some of the responses included 'Households are able to earn passive income through tourism activities.'
- (e) Candidates discussed the negative impact of crime on the economy as a whole, instead of the impact on the tourism sector in Q3.3.5.

- (f) A large percentage of candidates confused 'Reasons for growth of the tourism industry' with 'Benefits of tourism' in Q3.4. Some candidates wrote on 'how to grow tourism'.
- (g) In Q3.5 many candidates confused oil (crude oil), an input in any production process, with cooking oil.

### Suggestions for improvement

- (a) The teaching of contemporary economic issues is imperative and basic concepts need to be emphasised. Sometimes these topics may not be covered in some centres, possibly due to poor time management. Teachers should cover Environmental Sustainability thoroughly in Grade 11 as a large part of the content overlaps with the Grade 12 topic on the environment. Assessment should also be comprehensive to give learners a head-start in Grade 12. This will allow for more time to cover other topics which are sometimes neglected.
- (b) Learners must be exposed to more data-response questions, i.e. 4-mark questions that require application skills. A thorough understanding of key concepts is necessary to interpret such subquestions. These questions should be discussed in class with the emphasis on using the relevant data to address the requirements of the question. Logical reasoning would enable learners to earn marks, especially if they can show an understanding of the question.
- (c) Many examples with similar calculations within certain topics must be given to learners as practice. Teachers are encouraged to set their own activities by tweaking figures from tables from the various sources. Regular practice will improve the learners' confidence.
- (d) The importance of infusing current economic issues in lessons cannot be over-emphasised, especially where content can be linked to real-life issues. Issues such as the impact of the high price of crude oil and the Russian-Ukrainian war can easily be linked to tourism and inflation.
- (e) In teaching Economics, a crucial element is to motivate learners to think laterally about the topic. Where possible, teachers must relate the different topics to real-world issues. This will help learners prepare for higher-order questions. Learners must gain practice in evaluating, assessing or critiquing issues/topics whenever possible. Teachers are encouraged to set their own higher-order questions, to extend the engagement and knowledge acquisition of the learners in their respective classes. They should realise that textbooks are not adequate in providing all relevant and current responses to questions. Teachers should avoid an over-dependence on textbooks.
- (f) Sufficient informal activities on high-order questions are crucial in preparing learners for subsequent formal assessment tasks. Data-response questions should not merely require learners to re-produce answers from the given data. Learners should be able to apply the data in the appropriate context.

### QUESTION 4: MICROECONOMIC/CONTEMPORARY ECONOMIC ISSUES

#### Common errors and misconceptions

- (a) In Q4.1.2 candidates mentioned that interest rates lead to more borrowing but failed to link borrowing to *inflation*.
- (b) A fair percentage of candidates could not describe the term *natural monopoly* in Q4.2.3. They could not relate it to high start-up costs, instead they related natural monopoly to natural resources in a generic sense.

- (c) Poor performance by candidates in the calculation of economic loss in Q4.2.5 was due to the use of an incorrect formula or incorrect substitution of values.
- (d) Many candidates described the components of fiscal policy in Q4.3.5 without showing how it is used to combat *demand-pull inflation*.
- (e) Candidates gave generic answers on market failures and just explained 'lack of information' and 'factors of production' in Q4.4, without linking their answers to the causes of market failure as required by the question.
- (f) In Q4.5 candidates discussed the negative impacts of tourism in general and not specifically on the environment.

### Suggestions for improvement

- (a) A major contributing factor to poor performance is incorrect interpretation of the question due to a lack of thorough understanding of a particular concept. This negatively influences its application in a context that is relevant to the question and compromises logical reasoning. Teachers should refrain from providing marking guidelines to learners before they attempt challenging questions as this prevents them from thinking critically about the question themselves.
- (b) A clear distinction between *artificial monopoly* (formed due to legal restrictions) and *natural monopoly* (formed due to economic reasons) should be made when teachers are explaining different types of monopoly during lesson presentations. This must be reinforced with examples.
- (c) In calculating profit or loss, the formula ( $TR - TC$ ) should not be altered because it results in an incorrect answer, i.e. a positive instead of a negative in the case of a loss.
- (d) During revision, teachers are encouraged to pay attention to the fiscal policy and its effect within the various contexts (business cycles, public sector, economic growth and inflation). Some of the contexts overlap, such as fiscal policy in the smoothing of business cycles and also in combating inflation.
- (e) When teachers deal with reasons for market failures, they must emphasise how each reason causes the market to misallocate resources either through productive inefficiency and/or allocative inefficiency.
- (f) Learners must be trained to analyse questions. A fundamental problem in misinterpreting questions is a lack of understanding of the terminology contained within the question. This compromises the learners' ability to apply the information within a given context.

### QUESTION 5: MICROECONOMICS

In general, the level of performance in the question was satisfactory. This particular essay was popular. The candidates, however, performed poorly in the additional part.

### Common errors and misconceptions

- (a) The following common errors were identified in the graphs:
- Axes were not labelled or incorrectly labelled.
  - Cost curves were confused with revenue curves.
  - AC curves were wrongly positioned in relation to the demand curve.
  - MC curves did not intersect the AC curve at its minimum position.
  - Monopoly graphs were drawn.
  - Profit maximising point was wrongly identified as either as  $MC = AR$  or  $MC = AC$ .
- (b) Most candidates performed poorly in the additional part of the essay. Characteristics of perfect competition were listed without showing how advantageous they were to either the consumer or the producer.
- (c) Although guided in the question paper as to what a conclusion should entail, the writing of a relevant conclusion was a challenge for most candidates. Some candidates repeated content from the introduction and the body in the conclusion.

### Suggestions for improvement

- (a) Cost curves are considered prerequisite knowledge when dealing with the various equilibrium positions in Grade 12. Cost and revenue curves must be thoroughly addressed in Grade 11. This will serve as a solid foundation when drawing graphs related to perfect and imperfect markets in Grade 12.
- (b) Teachers must assess learners regularly on essay questions by focusing more on the structure of essays. Teachers should focus on the additional part and the conclusion of the essay – these areas have been identified as problematic for many learners.
- (c) Teachers are encouraged to get learners to practise the drawing of graphs and they should assess the accurate labelling of curves, profit maximising point and the shading of economic profit and economic loss. Learners must be reminded that there is no shading for normal profit.
- (d) In the drawing of the various equilibrium positions, irrespective of whether it is a perfect or an imperfect market, the following steps must be emphasised and followed:
- Draw the revenue curves (demand, AR, MR).
  - Thereafter, draw the cost curves. The AC curve must be drawn first followed by the MC curve. Thus, it would be easier to ensure that the MC curve will intersect the AC curve at its minimum point.
  - The positioning of the AC curve in relation to the D/AR curve is important for the various equilibrium positions (minimum point of AC lies below the D/AR curve for Economic profit, above for an Economic loss and tangent in the case of normal profit).
  - In the explanation of the graph the following key points should be the focus:
    - Profit maximising point ( $MR = MC$ ). This is an important reference point as it influences the explanation of the graph.
    - The quantity produced and the price.
    - Compare AR to AC to determine whether a profit or loss has been made.
    - Indicate this profit or loss in the graph by shading the area.
  - The teacher must regularly assess learners to master the interpretation of graphs.
- (e) Subject advisors/Cluster leaders should provide adequate support and material that would help learners to deal with challenging topics, e.g. graphs.

- (f) Teachers must ensure that learners are able to interpret questions correctly to avoid irrelevant information being included in their responses. Teachers are encouraged to expose learners to different questions on the same topic and guide them on the interpretation of questions. In this regard, learners should practise how to structure responses to questions based on key issues.
- (g) Learners must practise answering higher-order questions. Teachers must unpack questions in a way that guides learners to focus on the key issues demanded by the question. Learners should be engaged in classroom discussions, which will promote their ability to interpret content and to think critically because of the feedback they get from others.

### QUESTION 6: CONTEMPORARY ECONOMIC ISSUES

In general, the level of performance in response to this question was satisfactory.

#### Common errors and misconceptions

Additional part:

- (a) Overall, this part of the question was not well answered. Candidates clearly needed more exposure to the topic (environmental sustainability) through regular class discussions.
- (b) Some candidates discussed *the benefits of tourism* instead of *government measures to ensure sustainability*. This suggested a lack of adequate preparation and spotting for the examination.
- (c) Some candidates confused the topic with *environmental problems*.
- (d) A fair percentage of the candidates were confused by the content as they wrote incorrect information to specific headings. e.g. under *environmental tax*, candidates wrote content related to *marketable permits*.

#### Suggestions for improvement

- (a) The teaching of all content should be completed timeously so that more opportunities for revision are created. Poor planning and delivery often lead to some teachers rushing through the last few modules and not spending enough time on contemporary economic topics. Teachers need to ensure that each topic is given adequate attention, as outlined in the ATP.
- (b) In their conclusions, learners should be taught how to structure a response in support of, or against the facts mentioned in the main part. During revision sessions, teachers should constantly remind learners of the guideline in the question paper regarding the conclusion to the essay.
- (c) Learners must be exposed to current affairs/news/events on a continuous basis from Grade 10. SBA tasks should be prepared on current issues to help learners to relate the content to the real world. Teachers must be encouraged to expose learners to the latest developments related to the Contemporary Economic Issues.
- (d) It is essential that teachers closely follow the *2021 Examination Guidelines* and ensure that the content is researched where textbooks do not cover such topics. A case in point is the Paris Agreement, which is an improvement on the Kyoto Protocol, as it limits the

impact of climate change. Reference can be made to the first agreement the *United Nations Framework Convention on Climate Change (UNFCCC)* which failed as it was based on voluntary reduction of CO<sub>2</sub> levels. The structure of the additional part was extracted as it is in the examination guidelines: 'Evaluate the success of international agreements in reducing climate change'.

- (e) Essentially, the goal of the Paris Agreement (2015) is to limit global warming to well below 2 °C, preferably to 1,5 °Celsius, compared to pre-industrial levels. The Paris Agreement is a landmark treaty in the multilateral climate change process because, for the first time, a binding agreement establishes a global common cause to undertake ambitious efforts to combat climate change and adapt to its effects.
- (f) Teachers should make the *2021 Examination Guidelines* available to all learners as this would help to check whether all aspects of the curriculum have been completed and to identify areas where the textbook lacks the relevant information. This will prevent content gaps in the teaching-learning process. It will provide opportunities for the teacher to set small tasks for learners to research and can be given as part of regular homework.
- (g) Most resources are outdated and have not been revised recently. There have been amendments to the *Examination Guidelines*. Teachers must be encouraged to identify content gaps in their sources when interrogating the *2021 Examination Guidelines* and network with other schools, cluster, or the subject advisor in obtaining the relevant content to supplement the resources.

# CHAPTER 6

## GEOGRAPHY

The following report should be read in conjunction with the Geography question papers of the November 2022 NSC examinations.

### 6.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who sat for the Geography examination in 2022 increased by 10 227 compared to that of 2021.

Although the table below indicates a generally negative trend over the years 2018-2021, a significant improvement in results was evident this year.

Candidates who passed at 30% (Level 2) improved significantly from 74,3% in 2021 to 81,3% in 2022. There was a corresponding improvement in the pass rate at 40% (Level 3) over the past two years from 43,2% to 52,2%.

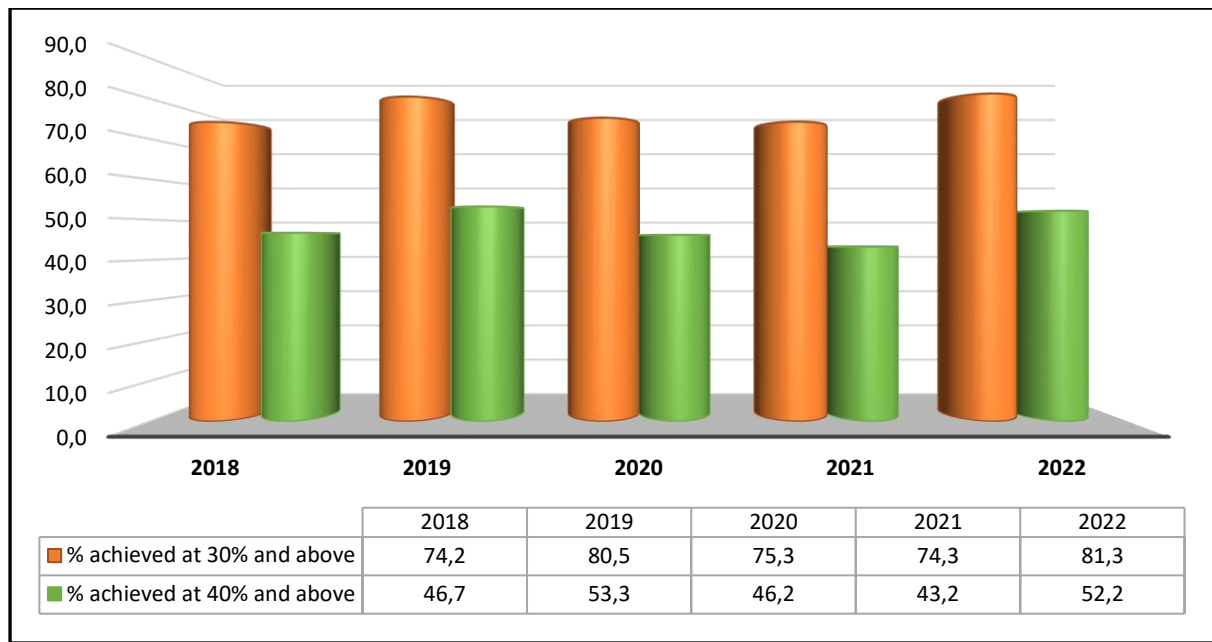
There was also an increase in the percentage of distinctions over 80% (Level 7) from 0,5% to 1,0%, i.e. by 1 896 candidates.

In view of disruption to academic programmes in 2020 and 2021, the results achieved by this cohort are particularly commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the favourable overall performance.

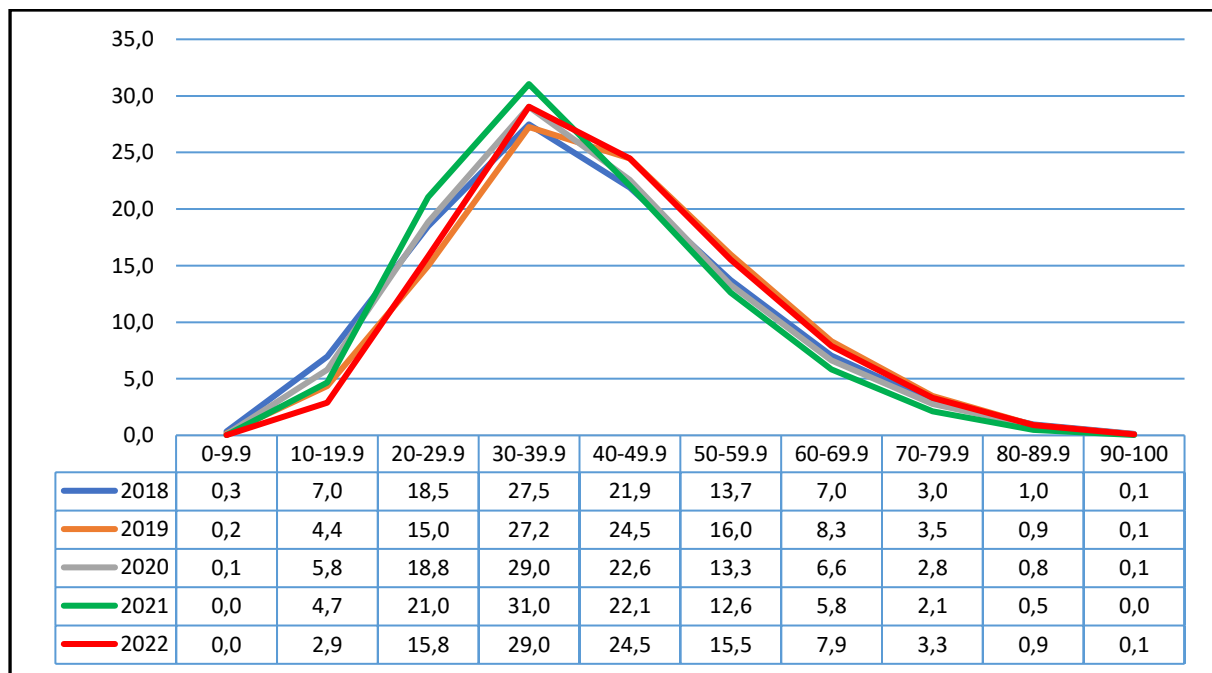
**Table 6.1.1 Overall achievement rates in Geography**

| Year | No. wrote | No. achieved at 30% and above | % Achieved at 30% and above | No. achieved at 40% and above | % Achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 269 621   | 200 116                       | 74,2                        | 126 011                       | 46,7                        |
| 2019 | 271 807   | 218 821                       | 80,5                        | 144 755                       | 53,3                        |
| 2020 | 287 629   | 216 467                       | 75,3                        | 132 955                       | 46,2                        |
| 2021 | 358 655   | 266 402                       | 74,3                        | 155 060                       | 43,2                        |
| 2022 | 368 882   | 299 751                       | 81,3                        | 192 612                       | 52,2                        |

**Graph 6.1.1 Overall achievement rates in Geography (percentage)**



**Graph 6.1.2 Performance distribution curves in Geography (percentage)**



## 6.2 OVERVIEW OF CANDIDATES PERFORMANCE: PAPERS 1 AND 2

The general comments below include points that arose from the November 2022 Geography NSC papers, and relevant advice given in previous *Diagnostic Reports* has been reiterated.

The November 2022 NSC Geography examination marks the second year of having two 3-hour papers of 150 marks each. The two-paper model was again well received as it offered different benefits, which included:

- **More effective management of time:** Candidates had 3 hours to complete each paper; a total of 6 hours to complete a 300-mark examination.
- **Specific content per paper:** Each paper was written on separate days, affording candidates the opportunity to focus their preparations on specific content relevant to each paper.

### General comments

#### (a) Quality of candidates' performance

In general, the overall performance shows a significant improvement which is promising. The averages for both Papers 1 and 2 are very similar, indicating a balance between the learner performance in the two papers. The inclusion of the source material in the question paper and the use of smaller topographic and orthophoto maps made the papers more learner friendly.

It was pleasing to note that candidates in many centres attempted more questions in both Papers 1 and 2, than in the past. As each question is subdivided into several subquestions, candidates would have been able to score some of the marks allocated to the lower-order questions.

There was some improvement in candidates' achievement in the paragraph questions regarding the need to respond in full sentences. The advice and recommendations suggested in past *Diagnostic Reports* seem to have been implemented by teachers to an extent, but there is still room for further improvement.

In the shorter response questions, candidates displayed an improved understanding of geographical processes and were able to provide appropriate responses. Topics that reflected the most improvement were Climate and Weather, Geomorphology and Rural and Urban Settlements. Although a small improvement was noted, candidates continued to experience challenges in both Paper 1 and Paper 2 with the questions on Geographical Skills and Techniques, Map calculations, and particularly GIS, where candidates either answered only a few subquestions or did not attempt the question at all.

The inclusion of part-marking into the marking principles for certain questions has had a positive impact on learner performance, especially in Paper 2 where it was applied to a number of questions. Previously, in questions where a factor and qualifier were required to obtain two marks candidates either received two or zero, depending on whether they gave a full response or not. The inclusion of part-marking allowed markers to award one mark where the candidate had given the correct factor without a qualifier. This marking principle was applied in the following questions: Paper 1: Q1.3.6(c) and Paper 2: Q2.3.5; Q2.4.6 and Q2.5.5.

**(b) Policy documents and LTSM:**

- The revised CAPS document, recovery annual teaching plans (ATPs), the 2021 *Examination Guidelines* and *Diagnostic Reports* are essential planning documents that form the basis of the planning process. Rotational topics must be noted each year.
- Prescribed textbooks, study guides and other departmental publications like *Mind the Gap* continue to be the primary resources available to teachers. However, teachers are expected to adapt the information to ensure it remains current.
- Online resources continue to play a significant role in the teaching and learning process. Educational websites, video presentations, shared resources from different provinces and publishers contribute to the continuity necessary in the teaching and learning of Geography.

**(c) Use of past NSC papers and exemplar papers:**

- Accessing and utilising NSC papers of the past two years are essential for teaching and learning. It is evident that teachers make use of these in lessons and revision. NSC Geography papers serve as a reliable source on questioning patterns and style. New trends in the testing of short objective questions and the use of infographics are reflected.
- The migration to two equal-weighted papers in 2021 did not necessitate any changes to the content. It is, therefore, essential that every learner has access to past papers from 2015 onwards, as these are based on the current CAPS content. Teachers and candidates must familiarise themselves with the specific content for each paper and extract the relevant questions from past examination question papers when preparing for examinations. Special note must be made of the relevant yearly rotational topics in the Economy of South Africa module.
- The exemplar papers with digital maps are also available and will also assist teachers and learners. The A4 maps can be printed on regular high-quality colour copiers and are now more accessible to all.

**General comments**

Areas of concern that were raised in the 2021 *Diagnostic Report* remain pertinent in the context of the 2022 NSC examination paper. They are emphasised here once again.

- (a) There were sufficient lower-order questions to give all candidates a fair chance to pass. Most of these were found in the short objective questions at the start of each question. Based on the 100-script analysis from all provinces, candidates achieved an average score of above 55% in these questions with the exception of Q2.2 in Paper 1 where a 46% average was achieved. At the start of each subquestion there are also lower-order short objective questions. Answers to these questions could be obtained directly from the source or, as in Q2.3.1 in Paper 2, a definition was required. The part-marking principle was applied here.
- (b) Curriculum coverage was a challenge for a large number of candidates. In Paper 2, Q2.5 (15 marks) the Dube Tradeport was one of the rotational topics tested. New sections stipulated in the *Examination Guidelines* with regard to rotational topics was not covered by a significant number of candidates.

- (c) Questions that required a factor/explanation and qualifier (Q2.3.5; 8 marks, Q2.4.6; 4 marks and Q2.5.5; 6 marks) were poorly answered by a significant number of candidates; they either gave only the factor/explanation or the qualifier.
- (d) Many candidates' responses were still not aligned to the demands of the instructional verbs in the question. Many candidates did not know how to approach these questions or understand what was expected of them. Words such as 'suggest', 'identify', 'describe', 'determine', 'explain how', 'explain why' and 'explain' require different types of responses. This resulted in candidates losing many marks in the middle- to higher-order questions.
- (e) **Short objective questions:** Candidates generally performed well in these questions. A variety of short objective-type questions were used, e.g. multiple-choice questions: Paper 1 (Q1.1-synoptic weather maps; Q2.2-river capture), Paper 2 (Q1.1, settlement terminology, Q2.1, economic concepts), match the columns: Paper 1 (Q1.2, rural and urban climates, Q2.1, drainage basins), Paper 2 (Q1.2, terminology, Q2.2, types of industry). However, candidates did not perform well in Q2.2 in Paper 1.
- (f) **Source-based questions** (1 or 2 marks): Questions that required basic comprehension skills to obtain answers from sources like infographics still remain a challenge for some candidates: Paper 1 (Q1.4.1; Q1.5.3; Q2.3.1; Q2.3.3; Q2.4.2; Q2.5.1; Q2.5.2) Paper 2: (Q1.3.2; Q1.3.3 (b); Q1.5.2; Q2.3.2; Q2.4.1; Q2.4.2; Q2.5.2).
- (g) **Short data response-type questions** (2, 4 or 6 marks): These data response-type questions, where a short explanation and more detailed reference were required, were often poorly answered. Candidates were, in many cases, unsure whether a one-word answer or longer phrase was required as the response: Paper 1: Q1.4.5; Q1.5.4; Q2.5.4; Q2.5.5. Paper 2: Q1.3.5; Q1.4.4; Q1.4.5; Q2.3.4; Q2.4.5; Q2.4.6; Q2.5.4; Q2.
- (h) **Paragraph-style questions** (8 marks): These questions were of middle- to higher-order as is the norm. Candidates' performance in Paper 1 showed an improvement in Q1.5.5 on *inversion layers*, but Q2.4.4 on *meanders* was poorly answered. It seemed as though candidates could not describe the processes that took place to change a *meander* into an *ox-bow lake*. Most candidates struggled to integrate all aspects. The candidates' performance in paragraph questions showed a significant improvement due to the application of part-marking in Paper 2 (Q1.5.5 on informal settlements and Q2.3.5 on food security). Despite this, candidates did not fare well in Q2.3.5 on food security. It appeared that many candidates did not know when to give causes, effects, impacts and solutions when responding to these questions. They were also unable to demonstrate understanding of command/action words, e.g. 'suggest' or 'explain how'.
- (i) Some candidates did not have a sound knowledge of the basic geographic concepts and therefore were not able to answer questions of a higher cognitive demand linked to these concepts. Paper 1: Q1.5 (*inversion layer*), Q2.4 (*fluvial landforms*); Paper 2: Q2.3 (*food security*), Q2.5 (*IDZ Dube TradePort*).
- (j) There appears to be continued lack of practice to master the application skills in the Geographical Skills and Techniques section of the question paper. Candidates did not make good use of the topographic map and orthophoto map to assist them in answering questions. In many instances, it appeared that the General Information page was ignored. It would have assisted candidates in Paper 1 (Q3.2.3) and

Paper 2 (Q3.2.8; Q3.2.9). It was also evident that a thorough integration of map skills and theory was not done by a significant number of candidates. These two aspects of Geography need to be taught in an integrated manner. Regular practise of calculations is important for candidates to prepare correctly for NSC examinations. This section produced the poorest results in both papers with an average of 43%.

- (k) Map application seems to be a perennial challenge. Questions that required candidates to give answers relating to the topographic maps and orthophoto maps or a specific block on the maps provided a challenge: Paper 1: Q3.2.4 (a) (b); Paper 2: Q3.2.2; Q3.3.3(a); Q3.3.3(b). A significant number of candidates gave general answers not found on the maps or blocks.
- (l) Candidates continue to struggle with Geographical Skills and Techniques, which are examined in both question papers. Some candidates lost marks for not indicating the correct steps in calculations: Paper 1 (Q3.1.3, 2 marks), Paper 2 (Q3.1.4, 2 marks). Many candidates did not understand the terminology and the application thereof to answer questions on Geographical Information Systems (GIS) Paper 1: Q3.3.2; Q3.3.4 resolution. Paper 2: Q3.3.3 (*buffering*); Q3.3.4 (*data layer*). Candidates struggled with the paper GIS in Paper 1 (Q3.3.5; Q3.3.6).
- (m) Most of the major topics in Section A in both Papers 1 and 2 and all sections regarding Geographical Skills and Techniques mentioned in the CAPS document, were tested. This benefited the candidates who studied their work comprehensively. This trend could continue in future papers.

### General suggestions for improvement

- (a) Teachers need to emphasise the importance of reading the instructions carefully before answering the question paper. These instructions provide important information with regards to the length of responses (point 9) and to indicate the unit in final answer (point 10).
- (b) The understanding and application of instructional verbs that demand a higher cognitive level continue to be a challenge for learners. Questions containing these instructional verbs should always be answered in full sentences, showing a clear knowledge and understanding of the geographical content tested. The instructional verb that was deemed the most difficult in this examination was: 'explain how'. These questions require answers that include a factor/explanation and a qualifier. Included in the *Examination Guidelines* (2021) is a comprehensive list of typical instructional verbs used in Geography and the response required to meet the intention of the verb. Teachers are encouraged to make this list available to their learners and to use the instructional verbs in class daily to illustrate how questions can be asked.
- (c) Selective reading seems to be a perennial problem and many learners do not fully understand the requirements of the question. Highlighting the instructional verbs and important aspects of the question will definitely assist learners in interpreting the question correctly.
- (d) Most questions require some interpretation technique and understanding of geographical processes. Learners, therefore, cannot merely reproduce content knowledge gained in the classroom. Responses should be extracted from the source material given as well as the learners' own theoretical knowledge.

- (e) Geographical issues are often assessed, and thorough preparation is crucial. Learners should focus on the causes and effects, both negative and positive impacts, as well as possible solutions or sustainable strategies/measures to be implemented to overcome these issues. These questions are set regularly: Paper 1: Q1.4.4; Q2.5.2. Paper 2: Q1.5.5; Q2.3.5; Q2.4.6. In-depth knowledge of such issues is essential. This might well involve informal research by teachers and learners. There are many reliable geographical websites that will provide up-to-date and valid information. Reviewing past question papers will also give learners some idea of how questions are phrased.
- (f) Teachers must note that the short 15-mark questions at the beginning of each of the four questions are not necessarily going to test lower-order thinking skills and straight-forward recall only. Some questions might involve a higher level of cognitive ability. Learners must read the instructions carefully before answering the objective questions.
- (g) Learners must be exposed to different styles of multiple-choice questions. In both papers complex multiple-choice questions are asked. Paper 1: Q1.1.6; Q1.1.8; Q2.2.3; Q2.2.7 and Paper 2: Q1.1.1, Q1.1.5, Q1.1.7 and Q1.1.8 (1 mark each). They also need to be made aware that there are distractors in the options given in the multiple-choice questions. They must be taught to recognise subtle differences in the options given. The questions in some instances have important descriptive words which can lead learners to the correct answer, e.g. *dry-point settlement* (P2 Q1.1.5; 1 mark). Learners must consider all four options before they make their choice. Teachers are advised to update their methods of setting compliant multiple-choice questions which can include lower-, middle- and higher-order cognitive skills testing. Learners should be made aware of the principles underlying multiple-choice questions.
- (h) Knowledge of paragraph writing skills is essential. Learners need to write in full sentences and not use bullets or point form. These questions usually require a degree of critical and analytical thinking, which places them on a higher cognitive level. Since paragraph questions are 4 x 2 marks, four points (if required) must be explained; answers in most instances require qualification. The recommended eight lines should be used as a guide to the length of the paragraph. This is to avoid long-winded answers and wasting of valuable time.

When planning a response, learners should underline or highlight the main topic of the question, the instructional verb and the focus areas of the question. Good practice when writing paragraph responses would be to make at least four points and then elaborate on each point. Poor punctuation and sentence structure make it difficult for markers to assess these questions effectively.

Regular practice of paragraph writing in short informal and formal tests, as well as in internal examinations, will allow learners to improve their skills and confidence when attempting these questions.

- (i) Teachers must ensure that learners know all the geographical concepts and definitions required. Learners should compile a glossary of terms/concepts and an explanation of each in their notebooks for easy reference. This will assist them when describing and defining these concepts and terms and it will extend their geographical vocabulary. Definitions/Concepts are often asked as the first question on a particular topic and carry 2 marks each (P2: Q2.3.1). As these are seen as concepts, they do not have to be explained verbatim. Baseline assessments and other informal tasks which mirror the questions of the NSC Examination should be

practised regularly, focusing on the concepts taught. This should be done on completion of every subsection.

- (j) Geography is a dynamic subject and new information on numerous topics is updated regularly. Recent Climatological and Geomorphological events must be used in teaching and learning. The urban and economic environment is constantly changing. Teachers are therefore encouraged to collect resources on an ongoing basis and link current events to content taught in Grade 12. As life-long learners, teachers must set the right example by staying abreast of new developments in their subject.
- (k) Teachers are encouraged to include compliant source-based questions (like those used in the NSC examinations) in class assignments, tests and examinations. They should make use of relevant and recent reliable resources from the internet and avoid using sources that appear only in textbooks and are familiar to learners. Teachers should expose learners to a variety of sources, e.g. diagrams, sketches, photographs, infographics and graphical data (line graphs, bar graphs and pie graphs). Learners should be taught how to draw on and interpret information from these different sources. Teachers and learners must be aware that different sources may also be combined for examination purposes like in Paper 1: Q1.3; Q 1.4; Q2.3 and Paper 2: Q.1.4; Q2.4. As the interpretation of graphs and tables containing statistics remains a challenge for learners, the more practice they get, the more confident they will feel about answering a question based on a cartoon.
- (l) Infographics are informative and were used as stimuli in both question papers. This information was very valuable and should have assisted the learners in obtaining the correct answers (Paper 1: Q1.3 and Paper 2: Q1.4; Q2.4; Q2.5). It is a visual representation of information or data, e.g. contains written information (extract), a sketch or map, graphs, tables and photographs. Extracts contain valuable information to guide learners to appropriate answers. All the information given should be considered when answering questions. The skill of integrating the visual and written information is one that should be practised regularly. Learners could be asked to quote or state directly from the text in the infographic which then needs to be verbatim and not paraphrased. If learners are asked to provide evidence from the infographic, they are not expected to quote directly.
- (m) Teachers should be fully aware of the relevant subject content to be taught by constantly referring to the CAPS document and the *Examination Guidelines*. Details regarding the choice of agricultural product, mineral, core industrial areas, spatial development initiatives (SDIs) and industrial development zones (IDZs) to be studied for 2023, are included.
- (n) Rotational topics like *Spatial Development Initiatives (SDIs) and Industrial Development Zones (IDZs)* must be covered by teachers and learners. In most cases they are either briefly mentioned or are not found in textbooks at all. Teachers and subject advisors need to conduct additional research into these topics to share with one another. The *Examination Guidelines* have divided these topics into subsections, and it is essential that teachers provide sources based on these subsections to support learners.
- (o) As most prescribed textbooks do not cover all the subject content mentioned in the CAPS and the *Examination Guidelines* to the same degree, teachers should do additional research themselves. Teachers should consult more than one textbook if possible. Information provided in the various textbooks might not always be geographically sound and, when in doubt, additional research should be done on the topic.

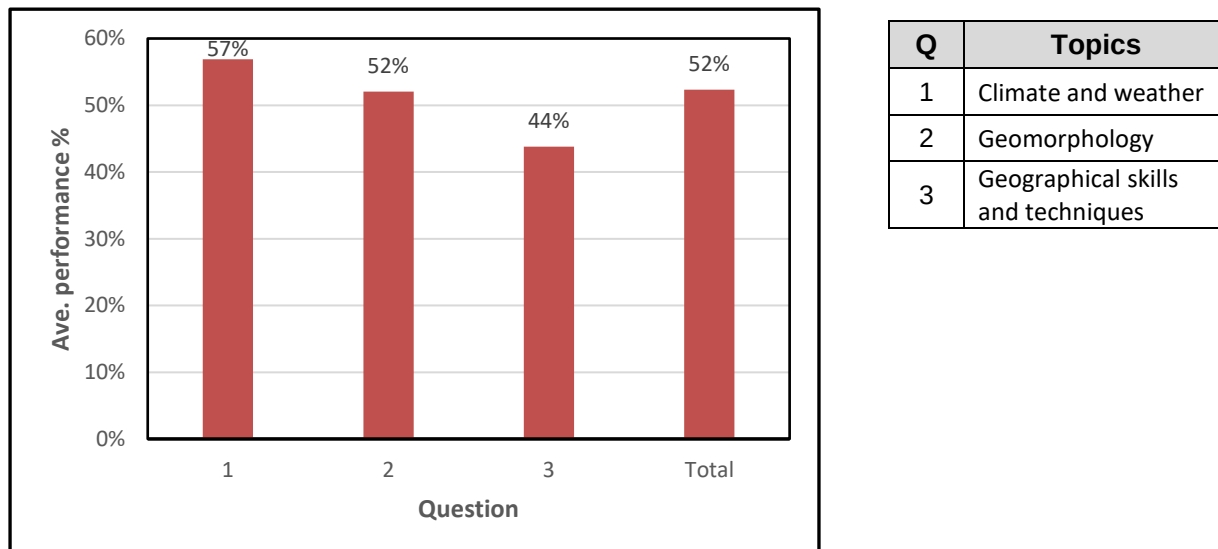
- (p) Teachers should provide each learner with a copy of the new *Examination Guidelines*, highlighting the content that will be taught. This can be used as a checklist to ensure that all content is covered, and to assist in preparing for tests and examinations. The format of each examination paper is also clearly spelt out.
- (q) To improve learner performance, teachers must refer to previous examination papers as a guide to ensure that the standard of questions and the variety of questioning techniques used in assessment at school level is appropriate. This would also assist teachers to show learners how scaffolding of questions occurs, from those testing lower-order cognitive skills to the higher-order questions which address more advanced thinking skills. Previous question papers should not, however, be used as a tool for predicting future papers.
- (r) Teachers must ensure that the distribution of marks in the internal assessment tasks is also compliant with the requirements stipulated in the CAPS document. Blooms' Taxonomy or a similar tool should always be supplied for formal tests, examinations and tasks. The weighting is 25% lower-order, 50% middle-order and 25% higher-order. If too many lower-order questions are asked in the internal assessment conducted at school-level, learners will not be exposed to questions addressing a higher cognitive demand, as asked in the final NSC examination. This will also give learners false notions of the level of performance required in the NSC examination. All tasks should be based on the new 2021/2022 format regarding structure and content distribution.
- (s) Teachers need to emphasise the importance of the units of measurement in the final answers where required. Marks will not be awarded if the correct unit of measurement is not provided in the final answer. Learners should be made aware that this instruction applies to both the theory and mapwork sections of the question paper.
- (t) With regard to improving mapwork results, learners must do a proper analysis of the general information of the map and orientate the topographic map to the orthophoto map before answering Section B on Geographical skills and techniques. This will assist in answering questions in Paper 1 (Q3.2.3) and Paper 2 (Q3.2.8; Q3.2.9; Q3.3.3). When orientating the maps an easily identifiable feature on both maps should be identified. Using features like roads, railway lines, larger features and the shape of built-up areas can also be used to do map orientation. By finding the area indicated by a red/black block on the topographic map, a learner can determine the location of the orthophoto map on the topographic map.
- (u) Learners need to understand the importance of integrating their theory knowledge with Geographical Skills and Techniques. Although most of this integration is in Q3.2 (12 marks) in both papers, it must be noted that it can be found in other questions e.g. P2: Q3.3.4 (b). The frequent use of topographic maps and orthophoto maps as teaching aids in theory lessons will assist learners. Mapwork skills and interpretation exercises should be regularly practised in all types of questions, e.g. multiple choice, map calculations, map application and interpretation and GIS.
- (v) The answers to Section B on Geographical Skills and Techniques are mostly derived from the maps provided. Regular revision using past papers from 2014 to 2022 will assist learners to master this skill. The focus, however, should be on the NSC 2021 and 2022 papers due to the new structure of this section.

- (w) All aspects of mapwork as stipulated in the *2021 Examination Guidelines* must be covered effectively.
- (x) Geographic Information Systems must be taught in detail. Teachers must emphasise the significance and purpose of GIS concepts and how to apply them: P1: Q3.3.3; Q3.3.4 (*resolution*), Q3.3.5; Q3.3.6 (*paper GIS*); P2: Q3.3 (*human-made and natural polygon and line features*); Q3.3.1 (*buffering*); Q3.3.4 (*data layers*).
- (y) Expose learners to the methods/steps for calculations indicated in the *2021 Examination Guidelines* and NSC marking guidelines. It must be noted that while certain methods may be correct in Mathematics, they may not contain all the steps that are required in Geography.
- (z) When practising and setting Mapwork exercises teachers are encouraged to use a variety of maps which reflect the different regions of South Africa, e.g. inland or coastal regions, and from different provinces. This will prepare candidates to answer questions on whichever map they receive in the NSC examination. A variety of maps are available to teachers and learners from past NSC examinations.
- (aa) In addition to making use of previous examination papers and SABC revision programmes to explain and revise important geographical concepts, other useful tools include YouTube live feeds and presentations. Exemplar papers showing the new Geography paper format with the annexure included in the question paper, are also available.
- (bb) Teachers are encouraged to refer to the updated instructions and information page in the 2022 NSC examination paper and to include this in their formal tests and examinations. This will prepare the learners for the final examination. There are general instructions for Section A and specific instructions for Section B (Geographical Skills and Techniques) that should be highlighted. The annexure should also be incorporated into examination papers as was the case in the final 2021 and 2022 NSC question papers. Teachers are also encouraged to implement the new format of testing the short objective questions using a number of related sources instead of just one source as in Paper 1: Q1.1; Q2.1 and Paper 2: Q1.1.

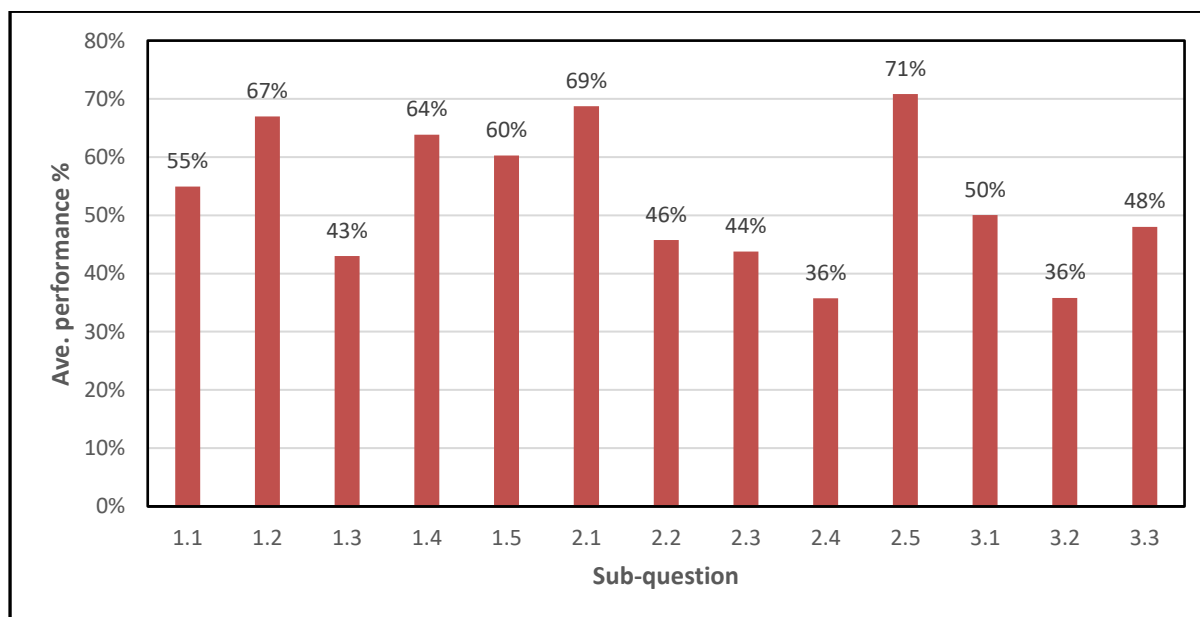
### 6.3 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 1

The following graph is based on data from a random sample of 100 candidates per province. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 6.3.1 Average performance per question in Paper 1**



**Graph 6.3.2 Average performance per subquestion in Paper 1**



| Sub-Q | Topics                    | Sub-Q | Topics            | Sub-Q | Topics                      |
|-------|---------------------------|-------|-------------------|-------|-----------------------------|
| 1.1   | Synoptic weather maps     | 2.1   | Drainage density  | 3.1   | Map skills and calculations |
| 1.2   | Micro-climates            | 2.2   | River capture     | 3.2   | Map interpretation          |
| 1.3   | Mid-latitude cyclone      | 2.3   | Drainage basins   | 3.3   | Geographical info systems   |
| 1.4   | Tropical cyclone Batsirai | 2.4   | Fluvial landforms |       |                             |
| 1.5   | Inversion layer over SA   | 2.5   | River catchment   |       |                             |

## 6.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 1

### SECTION A

#### QUESTION 1: CLIMATE AND WEATHER

##### Common errors and misconceptions

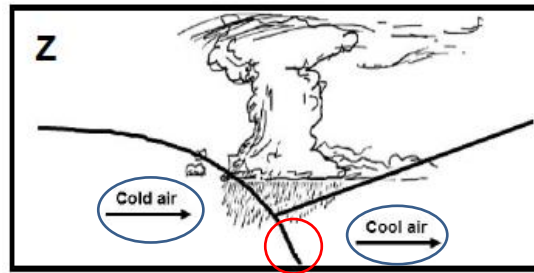
- (a) Many candidates struggled to select the correct answer in the multiple-choice questions (Q1.1; 8 marks). Separate sources were used for a number of subquestions instead of one source with which candidates were not as familiar. Q1.1.6 and 1.1.8 were complex multiple-choice questions where a combination of responses was required to select the correct answer.
- (b) Candidates did not fare well in the short objective questions in Q1.2 (7 marks). Although the content on micro-climates is often tested separately, in this question, candidates needed to compare conditions associated with both rural and urban areas. Application of knowledge remains a concern.
- (c) Question 1.3 on mid-latitude cyclones and cold front occlusions proved to be the most challenging to candidates resulting in the lowest average being attained in this question. Candidates could not differentiate between the *wind belt* and *the name of the wind* in Q1.3.1(1 mark), some even mistakenly referring it as the *thermal belt*. In Q1.3.4 (2 marks) candidates struggled to give a reason why front A moves faster than front B. Candidates struggled to provide evidence for the mid-latitude cyclone being found in the Southern Hemisphere in Q1.3.5 (2 marks). Q1.3.6 a–c (8 marks) on the *cold front occlusion*, proved to be the most challenging with many candidates receiving zero or not attempting the questions.
- (d) In Q1.4 (15 marks) candidates did not make full use of the information in the infographic based on Tropical Cyclone Batsirai. The satellite image, short extract, plotted path of the cyclone and wind speed graph needed to be utilised to answer the questions set. Q1.4.1 (1 mark) which required candidates to give the date on which Tropical Cyclone Batsirai reached its mature stage proved to be more problematic than intended, as the first question. In Q1.4.2 (2 marks) candidates struggled to link the westward movement of Tropical Cyclone Batsirai with the tropical easterly winds. Candidates in Q1.4.4 (4 marks) referred to factors other than physical factors that were impacted on by storm surges. No marks were awarded for mentioning the impact on people, cattle or crops as these are not deemed part of the physical environment.
- (e) Although candidates answered most of Q1.5 (15 marks) well, there were challenges with Q1.5.4 (2 marks) and Q1.5.5 (8 marks). Q1.5.4 required candidates to explain the role played by descending air on the development of the inversion layer over South Africa. Most candidates received zero for this question. It was possibly the least well answered question in the paper. In the paragraph question (Q1.5.5) many candidates were unable to describe the change in the position of the inversion layer in sketches A and B. Candidates did not make specific reference to sketches A and B respectively in their responses and therefore could not be awarded all marks.

##### Suggestions for improvement

- (a) The short questions in Q1.1 tested a variety of concepts associated with synoptic weather maps. In many instances the subquestions had their own individual resource, or one resource was used for a few subquestions. This is a trend which



and/or to look at the temperatures of the air mass behind the cold front and ahead of the warm front. The air mass with the lowest temperature behind the system is the cold front occlusion. Refer to the sketch in the NSC question paper below.



[Source: NSC November 2022]

Q1.3.6(b) asks for a reason for choosing Z. The questions expect the candidates to use both the plan view and cross-section to obtain their answer. On the cross-section one can see that the cold front arm is in contact with the surface while the warm front arm has been uplifted already. The air behind the cold front is colder than the air ahead of the warm front. On the plan view the station model behind the cold front shows that the air temperature is colder (10 °C) than the air in front of the warm front (14 °C). The cold front symbol is at the apex of the mid-latitude cyclone.

Q1.3.6(c) was classified as a higher-order question which tested candidates' ability to explain how a *cold front occlusion* develops. This process requires examining the cross-section Z and explaining the various movements that are taking place. There is the faster forward movement of the cold front due to the air mass behind the cold front being colder and heavier than the air in front of the warm front. The cold dense air lifts up the warm sector air ahead of it as it is light and less dense. It causes the warm sector to narrow. The denser colder air behind the cold front then uplifts the cooler air in front of the warm front leaving only one air mass remaining on the surface which is the 10°C air mass behind the cold front.

Occlusions are part of the CAPS content and are covered in the *2021 Examination Guidelines*, P5. under the heading of associated weather patterns: occluded front conditions.

For the first time the marking principles accommodated part marking in this question. As this question was an 'explain how' question the response required a factor/explanation with a qualifier. Candidates could now be awarded 1 mark for giving the factor if there was no qualifier. This saw candidates now able to achieve part marks.

Teachers are encouraged to use a variety of questions to test learner's conceptualisation and integration of the theory once each topic has been taught. Mid-latitude cyclones are one of the most technical topics in the Grade 12 curriculum. It has been noted that learners seem to struggle with the application of knowledge especially in regard to this topic.

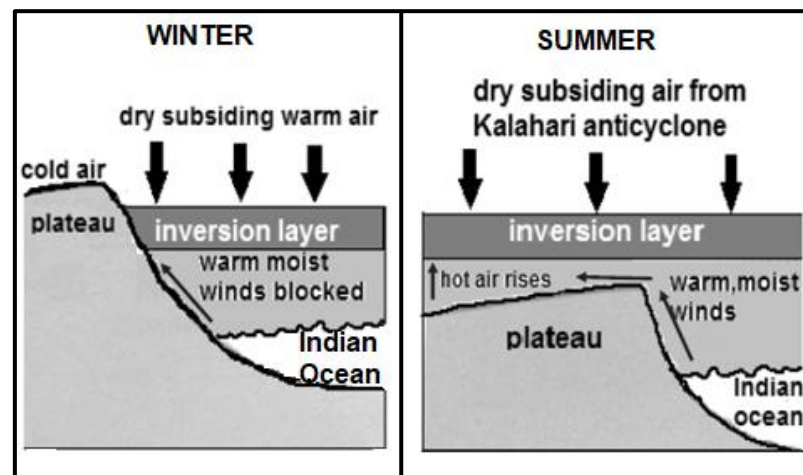
- (d) Although Q1.4.1 seemed straight forward and the date required appeared on three of the four aspects of the infographic candidates were not able to answer the question if they did not know the characteristics of the mature stage of a tropical cyclone. The mature stage of a tropical cyclone is characterised by the following: high wind speeds (205 km/h) which is taken from the graph, the well-developed eye which is evident on the satellite image and the symbol of a mature cyclone shown on the sketch of the path of the tropical cyclone.



In Q1.4.2 candidates could not link the westward movement of the tropical cyclone to the tropical easterly wind belt. Teachers must correlate where tropical cyclones are found (between 5° to 30° N and S of the equator) with the occurrence of the easterly wind belt which is where the *easterlies* or trade winds blow. It is important to note that winds are named from where they originate which is why there might be confusion. In this case the easterly wind is responsible for moving the tropical cyclone in a westerly direction.

The term *physical environment* refers to the *natural environment* in Geography. In Q1.4.4 candidates were expected to say how *storm surges* (the abnormal rise in seawater level during a storm) negatively impacted the east coast of Madagascar by causing soil erosion, destroying habitats of plants and animals and destroying the biodiversity of the natural environment. Teachers must ensure learners are able to make the distinction between the physical (natural) environment and the built environment.

- (e) When air descends it starts to heat up according to the dry adiabatic lapse rate which is 1 °C per 100 m. This is due to the air being compressed. As this warming air descends it comes into contact with rising air from the Earth's surface. The rising cooling air meets with the descending warm air to form an inversion layer. An inversion layer is *a layer in the atmosphere in which air temperature increases with height*. Teachers need to integrate this concept when teaching the influence of anti-cyclones on South Africa's weather. Refer to the sketches below which illustrates the focus of Q1.5.4 and Q1.5.5.



[Source: <https://www.elimuza.com/grade-12/item/1068-geography-grade-12-annexure>]

Q1.5.5 tested the candidate's ability to describe the changing position of the inversion layer between summer and winter and then to say how it affected the amount of rainfall received in the interior in each season. Although this question was accessible to most candidates, many did not use the letters A and B when writing down their response. The lack of ability to describe a climatological process thoroughly and in sequence is a cause for concern. Learners need to spend time practising paragraph questions, looking at layout and delivering a response sequentially using letters or seasons as they are depicted on the sketches.

The role of the inversion layer over South Africa is vital as its changing seasonal position is mostly responsible for the increased rainfall experienced in summer and the lack of rainfall in winter over the interior. Comparisons between summer and winter conditions need to be explained.

Teachers are encouraged to link various topics and not just teach isolated pockets of information. Q1.5.5 integrated the following topics: *inversion layers*, *adiabatic*

*heating*, the influence of the *Kalahari high-pressure cell* and resultant occurrence of *rainfall* over South Africa. Learners need to be able to see how these different aspects work together to impact rainfall over the interior of South Africa.

## QUESTION 2: GEOMORPHOLOGY

### Common errors and misconceptions

- (a) Candidates experienced challenges with the multiple-choice questions testing the concept of river capture in Q2.2 (8 marks). This was the first time that this topic was examined as short objective questions. Q2.2.3 and Q 2.2.7 were complex questions where candidates had to match two responses to select a final answer which increased the degree of challenge. Q2.2.2 and Q2.2.6 posed a challenge as candidates were not familiar with the landforms tested.
- (b) Q2.3 (15 marks) recorded the second lowest average performance in the Geomorphology section at 44%. Many candidates struggled to use the extensive source material to guide them through the questions. The key showing both features on the sketch and the water tables for summer and winter clearly confused candidates. The concept of a drainage basin (and its characteristics) is not foreign to the candidates, yet they struggled to apply their knowledge adequately or integrate the various aspects of river profile, factors influencing runoff, seasonal changes in water table that were tested, in this question. In Q2.3.2 (2 marks) most candidates could not state two characteristics of a river system evident in the sketch. In Q2.3.3 (4 marks) candidates had difficulty giving evidence from the sketch for the greater surface runoff at A as opposed to B.

Candidates struggled to give a reason for the selection of cross-section Z representing point C on the sketch in Q2.3.4(b) (2 marks). Q2.3.5(a) and (b) were not well answered as candidates could not link the impact of a decrease in precipitation to the water table and the resultant type of river.

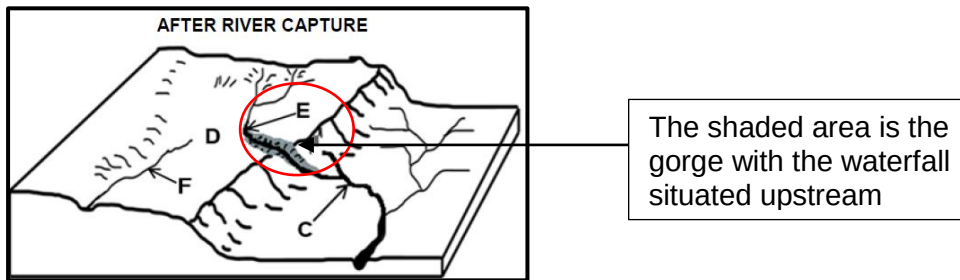
- (c) Q2.4 (15 marks) which tested *fluvial landforms*, recorded an average of 36% which was the poorest performance in a subquestion in this paper. While candidates could identify the *fluvial landform* as a *meander*, many could not draw the rough cross-section from B-C in Q2.4.3(a) (2 marks) and struggled to answer Q2.4.3(b) (1 mark) and (c) (2 marks) thereafter. Most unexpected was the poor result attained in the paragraph question; Q2.4.4 (8 marks). Candidates had difficulty describing the processes that resulted in *landform A (meander)* becoming an *ox-bow lake* (landform D).

### Suggestions for improvement

- (a) In Q2.2.2 (1 mark) candidates had to identify the landform caused by headward erosion (answer to Q2.2.1). A gorge forms when headward erosion occurs along the more energetic river. The additional water flowing down the captor river may accelerate erosion and encourage the development of a gorge. *A gorge is a narrow valley with steep, rocky walls located between hills or mountains.*

The trend of asking complex multiple-choice questions was introduced in the NSC November 2021 examinations and as such, teachers should be incorporating similar style questions in class tests and examinations to prepare their learners. In Q2.2 (7 marks), Q2.2.3 and Q2.2.7 were set as middle to higher-order and required a high level of understanding of the topic tested to reach the correct response. Two options are required before the final selection of the combined response takes place.

The correct response to Q2.2.7, which was classified as a higher-order question, was 'waterfall'. Although it is not one of the frequently asked landforms associated with river capture, it should be taught using sketches like in Q2.2.



[Source: NSC November 2022]

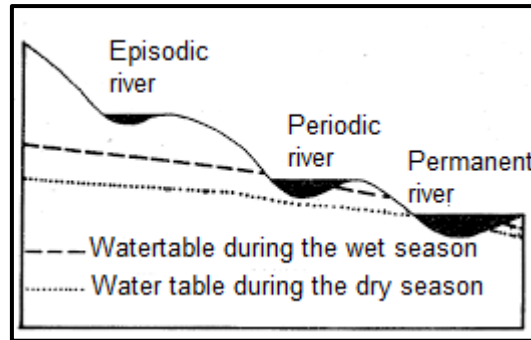
Teachers are encouraged to use a variety of diagrams when teaching river capture. Learners must be able to identify the various features associated with river capture as tested in this question.

- (b) Q2.3 (15 marks) on drainage basins linked a number of topics together: a long profile of a river, factors that influence runoff, types of rivers and the related water table were tested in this question. In Q2.3.2 candidates had to refer to the sketch and state two characteristics of the river system. Candidates did not understand the concept *river system*. A river system is a number of rivers which consists of one main river and all its tributaries which drains into a lake or ocean. Q2.3.3 was deemed to be a middle-order question where evidence from the sketch to prove surface run-off was greatest at A was required. Factors influencing runoff like amount of vegetation present, number of tributaries evident and width of channel needed to be considered.

Teachers must spend time teaching the skill of interpreting sketches with their learners. It is noted that learners struggle with understanding the 3-dimensional sketches often used in Geomorphology. An idea could be to use a photograph and similar sketch side by side so that learners can relate the real feature to the sketch. Many such images are available on the internet.

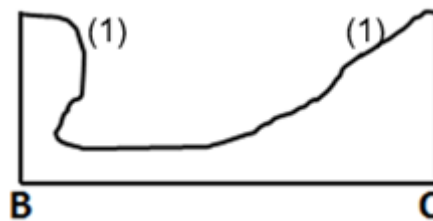
In Q2.3.4(a) and (b) candidates had to show an understanding of the concept *water table*, the boundary between water-saturated ground and unsaturated ground, and how the water table intercepts the riverbed in both the wet and dry season at C (NSC Nov. 2022) forming a permanent river. The river at C is therefore supplemented throughout the year with underground water. Q2.3.5 was deemed a middle to higher-order question as candidates had to show a deeper understanding of the impact a decrease in precipitation would have on the water table and the type of river. This requires the candidate to go beyond the information provided on the sketch to answer correctly.

It is imperative that teachers teach the different types of rivers in conjunction with their cross-profiles showing the wet and dry season water tables.

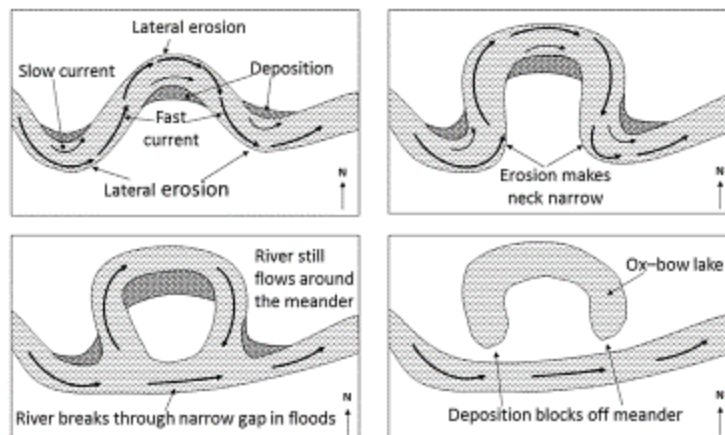


[Adapted from a previous NSC paper]

- (c) It is evident that teachers did not spend sufficient time on teaching the formation of the different fluvial landforms especially meanders and ox-bow lakes. There is still a lack of skill in drawing a rough cross-section as asked for in Q2.4.3(a). A rough cross-section is drawn without measurement and is not drawn to scale. A cross-section is the view of a feature from one riverbank to the other which in this case was shown from B-C on the sketch. Learners need to practise this sketch and must be able to draw the different slopes shape as taken from the marking guideline. Marks were awarded for the correct shape of the slopes as indicated below.



Teachers must practise the skill of teaching the geomorphological processes associated with the formation of fluvial landforms as was required in Q2.4.4. This was the paragraph question and carried 8 marks. Learners must be able to describe the processes that a meander undergoes to change into an ox-bow lake. Learners need to understand that erosion takes place on the outer bank due to the strong current and deposition on the inner bank due to the slow current. The neck narrows due to this and eventually breaks through allowing the river to straighten. Some water still flows around the meander, but deposition eventually blocks the path separating the river from the ox-bow lake. Using a series of sketches is the best way to teach this process.



[Source <https://timeforgeography.co.uk/more/blog/diagram-ox-bow-lake>]

## SECTION B

### QUESTION 3: GEOGRAPHICAL SKILLS AND TECHNIQUES

#### Common errors and misconceptions

- (a) Candidates did not take note or make use of the General Information given on page 16 of the question paper.
- (b) Calculations and Map Skills continued to pose a challenge for many candidates. In Q3.1.2 (1 mark) which was a complex multiple-choice question, candidates struggled to correlate the topographic map scale with that of the orthophoto. In the straight-line distance calculation in Q3.1.3 (2 marks) many candidates did not measure accurately or give the unit of measurement required (metres) in their final answer and so scored zero or one out of two. If the answer was given in kilometres, no marks were awarded as the question asked for metres. Once again candidates struggled with the rough cross-section in Q3.1.4 (2 marks). The answer to Q3.1.5 (1 mark) depended, to an extent, on their cross-section and could have resulted in an incorrect answer if it was not accurate.

The magnetic declination calculation in Q3.1.6 remains a challenge despite both the difference in years and annual change being given. Candidates did not simplify the answer or include the correct units in the final answer. This is instruction 10 on the question paper. In some instances, some of the steps were left out (instruction 16) which resulted in candidates losing marks.

- (c) In Q3.2 (map interpretation) candidates did not make full use the topographic map and the orthophoto map as was required. Many candidates were not able to integrate their Physical Geography knowledge to answer the questions asked in the mapwork section resulting in a 35% average being achieved for this section. The following questions were problematic:
  - Q3.2.2 (b) (1 mark) where candidates had to give a reason for saying why the photograph was taking in the morning. Many did not create challenges for candidates in responding to why there are many perennial water sources and furrows.
  - Candidates continued to struggle with how to determine the use map evidence in their response.
  - The phrase 'climatological reason' used in Q3.2.3 (2 marks) direction of flow of a river as in Q3.2.4(a) (1 mark) where they needed to provide evidence from block B1 on the topographic map for river H flowing in a north-easterly direction. In Q3.2.4 (b) candidates did not provide map evidence as requested.
  - Many candidates did not describe the underlying rock structure in Q3.2.6 (2 marks) but gave an example of the type of rock instead.
- (d) Most candidates struggled to answer Q3.3 (8 marks) on Geographical Information Systems. In Q3.3.1 (1 mark) candidates struggled to match the symbol representing the environmental issue depicted in the photograph. There were still issues with the understanding of the term (high) *resolution* as asked in Q3.3.3 (1 mark) and how it will assist a GIS specialist to find a solution to the environmental issue (Q3.3.4; 2 marks) identified in Q3.3.1.

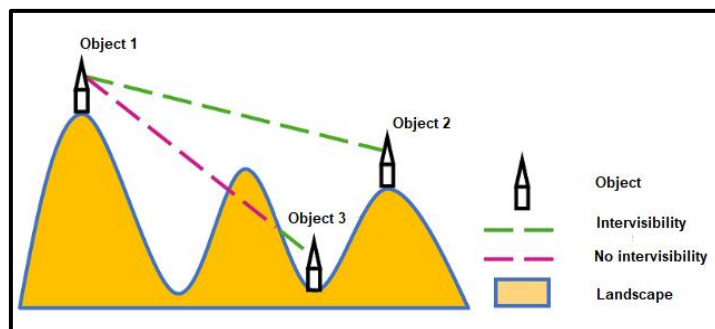
Q3.3.5 (1 mark) and Q3.3.6 (2 marks) were based on identifying the missing infrastructure layer in block C3 on the topographic map. Candidates had difficulty in using a paper GIS to respond to this question and could not indicate on the redrawn block where the power line or building needed to be inserted.

### Suggestions for improvement

- (a) Learners are encouraged to read through the General Information and take note of where the place being tested is located. The small map of South Africa shows exactly where the mapped area (e.g. Merrivale) is located and the province is mentioned in the text. Some of the questions are set from the general information. There are also English terms and their Afrikaans translations provided as some topographic maps might use a combination of Afrikaans and English terms.
- (b) Q3.1.2 is a complex multiple-choice question, testing the candidates' understanding of the different scales of a topographic and orthophoto map. The rule is that a 1: 50 000 map has a smaller scale and therefore shows less detail than a map with a scale of 1 : 10 000. Teachers need to teach this concept with a topographic map and orthophoto map of the same area. Learners can then compare the same feature seen on both maps. Features on a topographic map will be 5 x smaller than the same feature on the orthophoto map.

Teachers are encouraged to follow the mark allocation for steps in all calculations as mentioned on page 20 in the *2021 Examination Guidelines*. Learners cannot be credited for the final answer in a calculation without the relevant unit of measurement being indicated. Teachers must train learners from Grade 10 already to correct these issues. In Q3.1.3 only correct answers in metres were credited as the question was specific about which unit of measurement was to be used. Accurate measurement must be encouraged and only a 1 mm range on either side of the correct measurement should be acceptable.

Teachers must teach learners how to do freehand cross-sections. No accurate measurements are required. The skill of looking at the increasing or decreasing value of the contour lines, spot heights, benchmarks, or trigonometrical station heights, as well as the spacing between the contour lines, will assist with determining the gradient of the slope. Intervisibility (*can one identified object be seen from the other identified object*) as asked in Q3.1.5 between F and G requires candidates to check the cross-section to see if F can be seen from G and vice versa. Learners should look at the topographic map between the same two objects/points to see if there are any higher-lying obstructions between the two objects/points that might obscure intervisibility. If a valley or lower-lying area can be identified between the two objects/points (1 and 2) there is intervisibility as seen below.



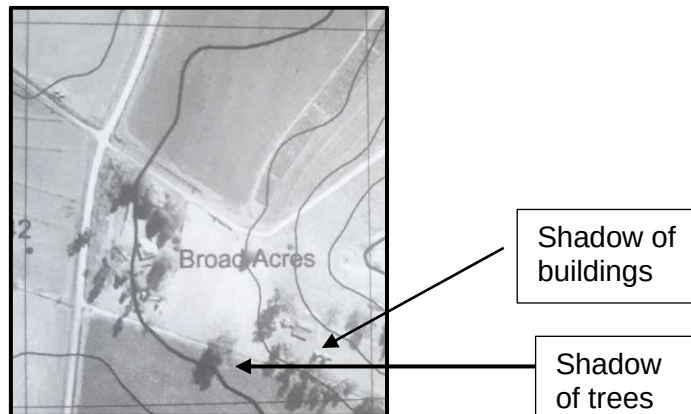
[Adapted source: <https://gistbok.ucgis.org/bok-topics/intervisibility-line-sight-and-viewsheds>]

If a higher-lying area (obstruction) lies between the two objects/points (1 and 3) there will be no intervisibility.

In Q3.1.6 teachers must ensure that learners place the correct unit of measurement at the final answer of the magnetic declination calculation. The correct unit is *west of*

*true north*. As this calculation is not measured, one cannot use west as that is a cardinal point giving a measured direction. Learners must also be trained to simplify their answer to full degrees and minutes.  $24^{\circ}96'$  is not correct as the answer should be  $25^{\circ}36'$  West of True North. Teachers must award marks in the same way as laid out on page 20 in the 2021 *Examination Guidelines*.

- (c) Shadows cast by features on the orthophoto map are the most accurate way of determining whether the photograph was taken in the morning or afternoon. Shadows of buildings and trees lie to the south-west as seen in block D4 on the extract of the orthophoto map of Merrivale below



In Q3.2.3 the climatological reason required in the answer was that rainfall is seasonal. If rainfall only occurs for a part of the year, there will be evidence of perennial water (dams) to store water, and furrows to move the water from the dams to where it is needed. Teachers must alert learners to this type of interpretation when practising mapwork skills.

One of the often tested, but more difficult skills is identifying the general direction in which a river flows on a topographic map. In Q3.2.4 (a) map evidence had to be given to confirm that River H flows in a north-easterly direction.

The following types of evidence on maps should be drilled by teachers in this regard:

- V-shape of the contour lines point towards the high-lying area.
- Use contour heights, spot heights along the path of the river.

The following information can also be used to identify direction of river flow:

- Identify a dam wall as rivers flow out from these walls;
- The acute angle at which tributaries join the mainstream: confluence points in the direction of the mouth of the river.

Refer to <https://study.com/learn/lesson/topography-topographic-maps.html>

Learners need to be able to apply their theoretical knowledge on laminar flow and look for proof on the topographic map as evidence to answer Q3.2.4 (b). The wide spacing of contour lines along river H in block B1 is proof that laminar flow is taking place. This question is considered a middle to higher-order question.

Teachers need to clearly differentiate between the underlying rock structure and rock type when teaching drainage patterns as was required in Q3.2.6. The underlying rock structure which results in a dendritic drainage pattern is that the rock is horizontal or layered and of equal resistance. The type of rock would be sedimentary or igneous rock.

- (d) GIS concepts and application need to be well-taught and practised regularly as these concepts are abstract and not easy for learners to grasp. Choosing from one of four symbols on the reference sheet to match the environmental issue depicted in Q3.3.1 required good integration skills which was a challenge for many. Regular practice is required to hone these skills.

Q3.3.4 and Q3.3.5 were based on the concept *resolution* which refers to the amount of detail in a photograph which is determined by the number and size of pixels that make up the image. The higher the resolution the clearer the image due to the smaller pixel size or a higher pixel count.



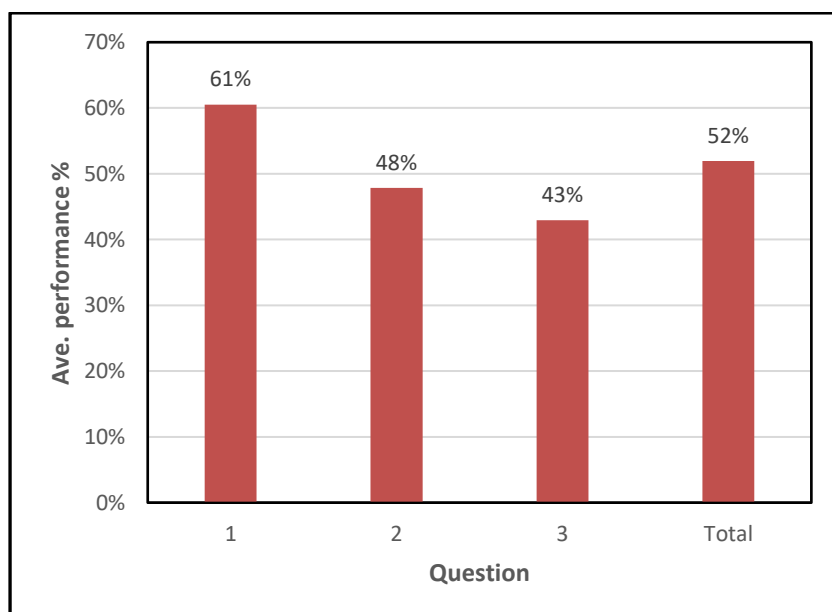
[Source: <https://ccideas.com/chatter/high-resolution-images-vs-low-resolution-images>]

Q3.3.5 and Q3.3.6 tested paper GIS skills. This aspect of GIS needs to be taught thoroughly and practised regularly in class exercises and examinations. Learners should do exercises where they add various layers to build up a sketch so that they develop this skill. This question was deemed to be a higher-order question. It was not well answered, as candidates had to firstly identify the correct layer from the orthophoto and then using the correct symbol, place it in the appropriate place on the sketch.

## 6.5 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 2

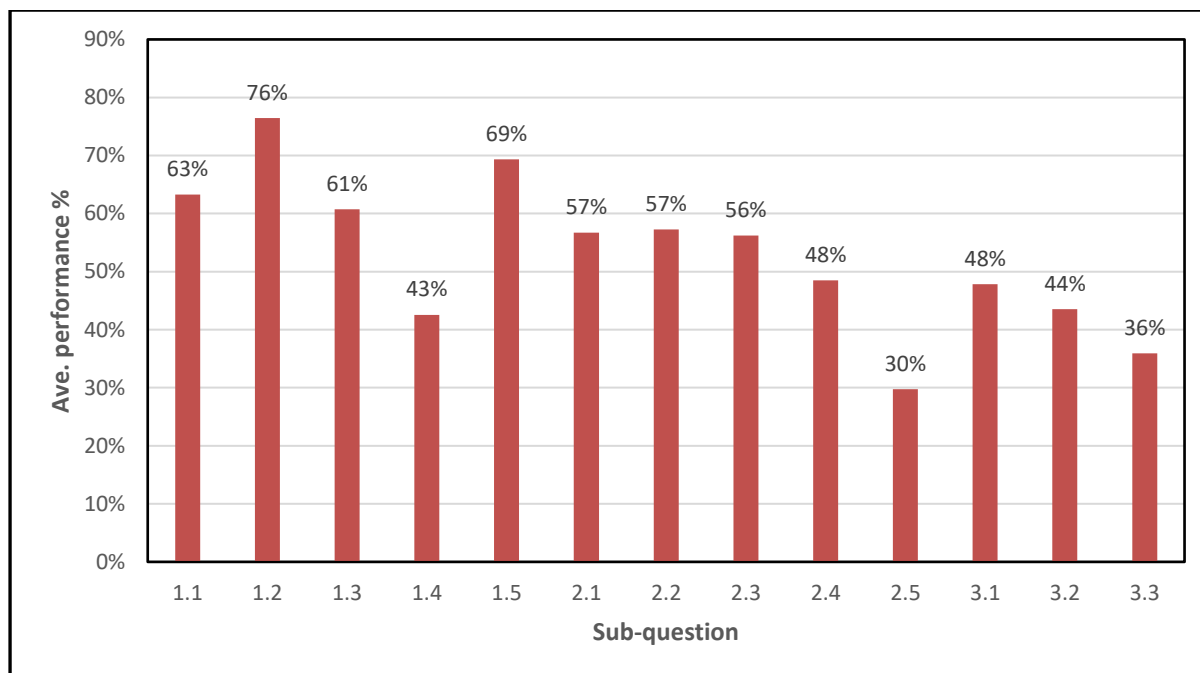
The following graph is based on data from a random sample of 100 candidates per province. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 6.5.1 Average performance per question in Paper 2**



| Q | Topics                             |
|---|------------------------------------|
| 1 | Rural and urban Settlements        |
| 2 | Economic geography of South Africa |
| 3 | Geographical skills and techniques |

**Graph 6.5.2 Average performance per subquestions in Paper 2**



| Sub-Q | Topics                    | Sub-Q | Topics               | Sub-Q | Topics                      |
|-------|---------------------------|-------|----------------------|-------|-----------------------------|
| 1.1   | Settlement terminology    | 2.1   | Economic terminology | 3.1   | Map skills and calculations |
| 1.2   | Settlement terminology    | 2.2   | Types of industries  | 3.2   | Map Interpretation          |
| 1.3   | Land reform               | 2.3   | Food security        | 3.3   | Geographic Info Systems     |
| 1.4   | Central business district | 2.4   | Gold mining          |       |                             |
| 1.5   | Informal settlements      | 2.5   | Dube TradePort IDZ   |       |                             |

## 6.6 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### SECTION A

#### QUESTION 1: RURAL AND URBAN SETTLEMENTS

##### Common errors and misconceptions

- (a) Although Q1.1 (8 marks) was generally well answered there **were** challenges experienced with regard to the following questions:
- A1.1.2 (1 mark) a significant number of candidates had challenges differentiating among the characteristics of a dispersed and nucleated settlement patterns, e.g. they had difficulty in associating an abundance of
  - Q1.1.4 (1 mark) some candidates had difficulty in differentiating between site and situation. They also focused on the first part of the question and therefore did not relate it to the surrounding area. They gave 'site' instead of 'situation' as the answer.
  - Q1.1.5 (1 mark) candidates seemed to focus only on the first part of the option and not the second part. It must be noted that this is a middle-order question and has two parts to the options that need to be considered. Learners gave A because they saw 'far from' (part of the option in A) and did not focus on the second part which indicated 'scarce', which was incorrect. water with a dispersed settlement.

- Q1.1.8 (1 mark) had four options. Candidates had to choose the two options that were correct. They had difficulty applying the skill to answer the question.
- (b) In Q1.2.3 (1 mark) candidates struggled to select the correct option between *centralisation* and *counter-urbanisation* with regard to movement between urban and rural areas. They could not differentiate between the concepts.
- (c) In Q1.3.2 (1 mark) some candidates did not focus on the extract as required by the question and gave general answers.
- (d) Q1.3.4 (4 marks) was a theoretical question where candidates could give answers that they have studied. Unfortunately, a significant number of candidates lacked this content knowledge and were not awarded marks.
- (e) With regard to Q1.3.5 (4 marks) some candidates only gave general benefits instead of focusing on social benefits as was required.
- (f) In Q1.4.1 (2 marks) a number of candidates gave general characteristics of the CBD, but the question required characteristics evident in the photograph.
- (g) A number of candidates experienced challenges in Q1.4.3 (2 marks) with regard to graph interpretation skills to derive the correct answer.
- (h) With regard to Q1.4.5 (6 marks) a significant number of candidates did not seem to understand the concept *out-lying business district* (OBD). They focused instead on the CBD and its negative consequences.
- (i) Q1.5.4 (2 marks): Some candidates gave general answers as to why emergency services could not reach informal settlements (e.g. lack of street names and house numbers). However, the focus of the question was on the lack of proper roads.
- (j) In Q1.5.5 some candidates wrote about measures to assist informal settlements, like providing them with food and other general answers but the question focused on measures to prevent flood related disasters. A significant number of candidates gave only the factor/explanation or qualifier and as a result could not be awarded full marks.

### Suggestions for improvement

- (a) Learners need to understand the difference between concepts like *site* and *situation* as asked in Q1.1.4. They also need to be able to apply concepts to different scenarios using linked characteristics of the concept, e.g. a dispersed settlement pattern is associated with an abundance of water. Using visual stimuli will definitely assist.

Nucleated Pattern buildings are close together



Dispersed/Isolated Pattern buildings are far apart



[ Source: <https://youtu.be/GP71QPyZlzo> ]

The concepts given as options in Q1.2.3 were not well understood. *Centralisation* refers to activities being concentrated in one area and *counter-urbanisation* is the movement of people from urban areas to rural areas.

- (b) Not all short questions are lower-order questions. Some options in multiple choice have two responses in one option, e.g. in Q1.1.5. Learners need to look at both responses before determining which is the most appropriate answer. These are referred to as complex multiple-choice questions.

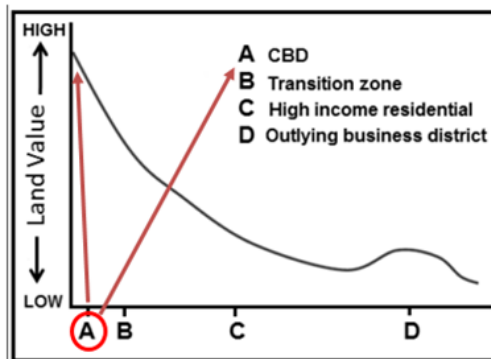
Learners need to understand the different techniques involved in answering multiple-choice questions. In Q1.1.8 learners need to focus on the instructional verbs found in the stem, e.g. physical factors not general factors. They then need to study the four options given and determine which two are the physical factors and finally select the correct option between A, B, C and D.

- (c) In Q1.3 it is important for learners to first read the extract before answering the questions as some of the answers can be taken directly from the extract and other answers will be assisted by the extract. For Q1.3.1 candidates needed to give comprehensive definitions to be awarded full marks. The definition for *Land Reform* is a *policy to bring about equitable distribution/access to land/landownership*. For Q1.3.2 candidates needed to give answers directly from the extract, not general answers.

1.3.2 According to the extract, what is the benefit of well-managed land reform? (1 x 1) (1)

Well-managed land reform can create assets for the poor, stabilise relationships and promote development. On the other hand, if it is poorly managed, it destroys assets, causes poverty to communities and weakens the economy.

- (d) Learners must read the questions properly so that they do not give general answers, e.g. Q1.3.5 focuses specifically on social benefits and not just on general benefits. *Social* refers to people, *economic* to wealth and *environmental* to natural environment.
- (e) In the case of Q1.4 where the source is an infographic, each part must be analysed and understood properly as a significant number of questions are based on the given information. In Q1.4.1 the characteristics of the CBD had to be identified in the photograph e.g. high density of buildings are visible in the photograph.
- (f) By analysing the graph, learners will be able to work out the land-use zone with the highest land value, in Q1.4.2.



- (g) When learners are asked to quote from the extract, as in Q1.5.2, they must write their response exactly as it is written in the extract with inverted commas, e.g. 'Rapid urbanisation' and 'government failure to deliver adequate housing'.

- (h) Learners must always refer to the initial statement at the start of each question as this will indicate which section is being assessed, e.g.
  - Q1.3 Refer to the sources below on land reform in South Africa.
  - Q1.4 Refer to the infographic below on the central business district (CBD) of Johannesburg as a land-use zone.
  - Q1.5 Refer to the extract below on informal settlements.
- (i) Ensure that the learners are exposed to a variety of sources as the paper does not limit itself to one type of resource, e.g. Q1.4 included a graph, an extract and a picture.

## QUESTION 2: ECONOMIC GEOGRAPHY OF SOUTH AFRICA

### Common errors and misconceptions

- (a) In Q2.1.1 (1 mark) some candidates lacked a proper understanding of the difference between the concepts *GDP* and *GNP*. The main factor leading to the answer of *GNP* was 'permanent inhabitants', but candidates focused instead on the total values of goods and services produced and erroneously gave *GDP* as their response.
- (b) Candidates mistakenly took construction in Q2.1.5 (1 mark) to be the same as real estate and classified it as a tertiary (instead of a secondary) economic activity.
- (c) Q2.1.7 (1 mark) referred to the statistics indicated in the table and some candidates had difficulty in interpreting the table.
- (d) In Q2.2.4 (1 mark) candidates confused *footloose* with *ubiquitous* industries as both can be found anywhere but only one is reliant on the market. This is a common error.
- (e) In Q2.3.1 (2 marks) some candidates gave vague definitions, e.g. 'food' instead of 'nutritious food'. Incorrect words were used like 'available' instead of 'accessible'.
- (f) Q2.3.5 (8 marks), Q 2.4.6 (4 marks) and Q2.5.5 (6 marks) used the instructional verb 'explain how' and this required both a factor/explanation and a qualifier in the response. A significant number of candidates only gave factors/explanations with no qualifiers and only received part marks.
- (g) In Q2.4.4 (4 marks) some candidates gave general factors that favoured South Africa's gold production, while the question focused on physical factors which refers to the natural factors.
- (h) Candidates performed poorly in Q2.5 with an average of 30%. From the responses, it was clear that the candidates lacked content knowledge and were not prepared for this section.

### Suggestions for improvement

- (a) Learners must note that short questions do not necessarily need to be purely theoretical but can involve analysis of a source like a table with statistics. In order for candidates to reach the correct answer, statistics need to be properly analysed, e.g. Q2.1.3.

|                        | 2020              | 2021            |
|------------------------|-------------------|-----------------|
| GROSS DOMESTIC PRODUCT | R301,92 billion   | R351,43 billion |
|                        | RANDS IN MILLIONS |                 |
| Utilities              | 100 789           | 104 317         |
| Transport              | 342 443           | 335 001         |
| Services               | 1 052 692         | 10 607 113      |
| Public administration  | 368 503           | 369 805         |
| Mining                 | 213 459           | 220 293         |
| Manufacturing          | 520 826           | 506 343         |
| Construction           | 108 604           | 111 021         |
| Agriculture            | 127 960           | 114 030         |

[Adapted from <https://www.statssa.gov.za/?p=15273>]

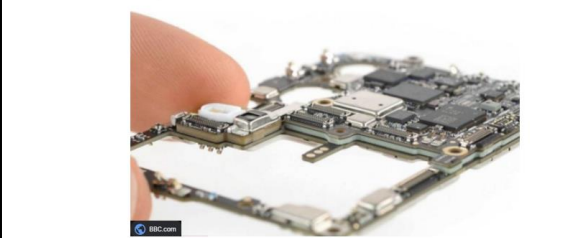
- (b) Learners must be able to apply definitions as asked in Q2.1.1. Just knowing definitions may not be enough if they are asked in a multiple-choice question. In Q2.1.1 the first part of *Total value of goods and services* applies to both GDP and GNP, but *permanent inhabitants* applies only to GNP. Learners need to analyse all four options before choosing one as every multiple-choice question has distracters that look like the correct answer. As all options are economic the learner needs to consider all options carefully.
- (c) In Q2.2 the learners need to have a clear understanding of the differences between the types of industries. When teaching this section, teachers must make learners compare and explain the differences of the different types of industries. Teachers must use examples that learners can relate to and visuals illustrating the different types of industries. This will create clarity as some types of industries have similar descriptions but **are** not the same, e.g. *Footloose industry: location is not determined by any specific factors* and a *Ubiquitous industry: can be located anywhere but is dependent on a market*.

**Ubiquitous industries** are industries that are inseparable from the immediate markets that they serve and are widely distributed/found everywhere. Example telecommunications (Telkom)



**Footloose industries:** are industries that can be located anywhere e.g. manufacturing of micro chips.

- Location is not dependent on factors like raw materials and market.



[Source: <https://youtu.be/SqiYDI3D4fM>]

- (d) When covering sub-sections in class, all parts of the CAPS curriculum must be taught thoroughly. Q2.3.4 tested factors causing food insecurity (based on health pandemics). Learners may have the knowledge and may have studied the information, but they need to know how to apply it to the context given to meet the demands of the question. Q2.3.5 focuses on solutions to achieve 'Zero hunger' but it is specific to assisting **farmers**.
- (e) It must be noted that definitions are now awarded two marks and to get full marks comprehensive definitions need to be given. In Q2.3.1 the definition of *food security* is when nutritious food is accessible to all people in a country. Learners need to know the importance of using the correct verbs e.g. 'accessible' means that all people will be able to obtain nutritious food, whether they are employed or unemployed. The word 'available' means that while there is sufficient food if you do not have the money, you would not be able to buy it.

- (f) Q2.4 clearly illustrates the importance of analysing and having a proper understanding of instructional verbs e.g. Q2.4.4 and Q2.5.5 use the following instructional verbs: 'state' and 'suggest' which require shorter and direct responses. The instructional verb 'Explain how' (Q2.4.6) requires a longer response with a factor/explanation and a qualifier. Learners must be exposed to this type of questioning throughout the year. Learners cannot be awarded marks if they do not give full responses in sentences when required.

2.4.6 Explain how a reduction in South Africa's gold production will impact negatively on its economic growth. (2 x 2) (4)

This is an example of a desired response for Q2.4.6:

*A decrease in the GDP (Factor/Explanation) as there will be a decrease in all economic sectors (Qualifier) (2)*

*Reduced buying power (Factor/Explanation) due to an increase in unemployment (Qualifier) (2)*

Both parts of the answer need to be given to be awarded the 2 marks.

- (g) Teachers need to ensure **that** the rotational topics are researched and taught well according to the guidance given in the CAPS document and 2021 Examination Guidelines. The DUBE TradePort IDZ was the rotational topic tested in this examination and recorded the lowest average. Geography teachers and subject specialists have put information regarding this content on the internet which is accessible to both teachers and learners. There are short video clips, PowerPoints and notes available for all rotational topics.

**Key facts to concentrate on with regard to the prescribed IDZ's and SDI's**

- Definition and difference between an IDZ and SDIs
- Map showing the location of prescribed IDZs and SDIs
- Factors influencing the location of the prescribed IDZs and SDIs
- Main industrial activities
- Factors that favour and hinder the development of the prescribed IDZs and SDIs
- Economic and social impacts

Although the Dube TradePort IDZ is not the IDZ for November 2023, the following source can be used as an example on how to compile your content according to the CAPS document, <https://youtu.be/Oi8casDeAyM>

- (h) Geographical terms/concepts need to be properly explained to learners as this will impact positively on their performance. The following terms/concepts: *accessibility* (Q2.5.3), *infrastructure* (Q2.5.4) and *upskill* (Q2.5.5) should be familiar to learners. It must also be noted that on some occasions specific reference is made in the questions, e.g. Q2.5.2 where not just 'infrastructure' is asked but is specific to 'transport infrastructure'.

## SECTION B

### QUESTION 3: GEOGRAPHICAL SKILLS AND TECHNIQUES

#### Common errors and misconceptions

- (a) In Q3.1.2 (1 mark) it was clearly seen that candidates' focus was on the calculation and not the application of the calculation. Candidates generally struggled to determine the vertical exaggeration that gave the most detailed indication of landforms.
- (b) A significant number of candidates in Q3.1.3 (b) (3 marks) did not know how to work out the minutes and seconds for the grid reference. In some cases, they estimated the seconds and were out of range.
- (c) The calculation of average gradient that was required in Q3.1.4 (4 marks): seems to be a perennial problem. Candidates mixed up the *vertical interval* (VI) with the *horizontal equivalent* (HE). Candidates multiplied by the incorrect map scale, left out steps and expressed the final answer as a fraction and not a ratio as was required.
- (d) In Q3.2 (12 marks) candidates could not apply their theory knowledge to the questions asked in the Map Interpretation section.
- (e) In Q3.2.4 (2 marks) some candidates gave responses not found on the topographic map and/or photograph. A significant number of candidates identified the low-income residential areas but could not give any characteristics.
- (f) With regard to Q3.2.5 (1 mark) many candidates did not understand that the space between the contour lines determined the gradient, e.g. contour lines that are close together indicate a steep slope and those that are far apart, a gentle slope.
- (g) In Q3.2.6 (2 marks) many candidates could not apply the knowledge of the characteristics of different street patterns or why a particular pattern developed. In some instances, candidates did not know what a street pattern was or the types of street patterns.
- (h) The main reason for the poor performance in Q3.2.9 (2 marks) was that it was linked to Q3.2.8. If the candidate got Q3.2.8 incorrect, they automatically got Q3.2.9 incorrect. A considerable number of candidates answered Q3.2.9 incorrectly due to the challenges faced in Q3.2.8. This was the only linked question in the question paper.
- (i) Some candidates did not seem to have the basic knowledge of what a polygon was as was asked in Q3.3.1 (a) (1 mark). Candidates could not differentiate between a *point* and *polygon* feature and gave *building* instead of *cultivated land* as an answer. The focus of the question was on a *human-made polygon* and candidates gave polygon features that were natural, e.g. woodland.
- (j) Although the definition of *buffering* was given at the start of the question to create context, candidates did not understand the term and as such could not apply it to answer Q3.3.3 (a) and (b) (3 marks).
- (k) In Q3.3.4 (2 marks) the understanding of the terms/concept *infrastructure* and topographic *data layer* created the following challenges for candidates:

- In some cases, where candidates misunderstood the concept infrastructure *data layer*, they gave examples not related to international links.
- Candidates experienced difficulties relating the topographic *data layer* to *gradient* which is significant to the ideal location of the Dube TradePort IDZ.

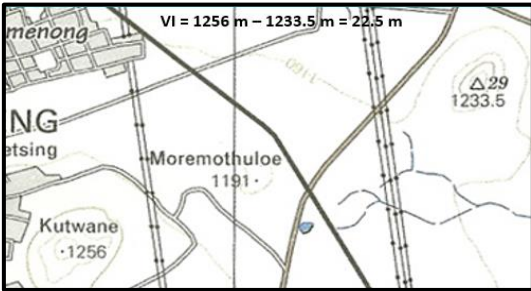
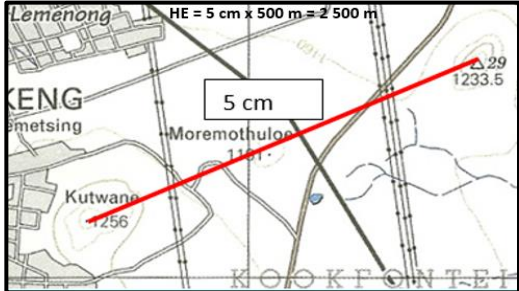
### Suggestions for improvement

- Teachers must remind learners to check which of the two maps is being referred to, the topographic or orthophoto map. Both maps now have an alpha-numeric grid and so it is easy to use the incorrect map. Q3.1.1 made specific reference to the orthophoto map but some candidates gave the answer relating to the topographic map. Teachers should ensure that when calculations are being taught, the application of the calculation must be discussed. It is important that learners understand that the more times the cross section is vertically exaggerated, the more detail will be visible (Q3.1.2).
- All calculations need to be done step by step, e.g. Q3.1.3. Learners must be taught to calculate the seconds and not estimate them. Refer to <https://youtu.be/xBaQUIhtAkq> to assist with grid reference.
- Calculations need to be completed as stipulated in the 2021 Examination Guidelines as each step is allocated a mark. Using other methods from other subjects which do not include certain steps may prevent the candidate being awarded maximum marks e.g. Q3.1.4. Calculation of average gradient seems to be a perennial challenge. Getting back to basics will assist in teaching this calculation e.g. using the formula or making the formula more descriptive. Teachers must expose learners to the 2021 Examination Guidelines so that they can see the steps and mark allocations for all calculations. This will allow them to score maximum marks. Learners should know that the final answer for the gradient calculation must be expressed as a ratio and not a fraction.

Difference in height  $\rightarrow$  Vertical interval

Formula: **Average gradient** =  $\frac{\text{Vertical interval}}{\text{Horizontal equivalent}}$

Actual distance in metres  $\rightarrow$  Horizontal equivalent

Formula: **Average Gradient** =  $\frac{\text{vertical interval (VI)}}{\text{horizontal equivalent (HE)}}$

VI = 1256 m - 1233.5 m  
= 22.5 ✓ m

HE = 5 ✓ cm x 500  
= 2 500 ✓ m

=  $\frac{22.5}{2500}$  ✓ (One mark for correct substitution)

=  $\frac{1}{111.11}$

= 1 : 111.11 / 1 in 111.11 ✓

[Source: <https://youtu.be/T6Z7vJc1P5A>]

- (d) Learners must be made aware that the majority of the mapwork questions that are integrated with Geography theory are found in Q3.2. Therefore, it is of utmost importance that teachers have maps available whenever a theory section can be integrated with mapwork. Teachers will be able to illustrate various concepts and terms that can be identified on the maps to the learners.

Teachers must emphasise that in these theory questions, reference is made to a specific map or to a block using the alpha-numeric grid or points on the map. The answers in most instances must come from the maps and cannot be general answers.

A discussion of the above techniques will create clarity on whether the settlement pattern is dispersed or nucleated on the map as in Q 3.1.1 or whether the residential area is high or low income as asked in Q3.2.3. In certain instances, learners need to use the reference symbols below the topographic map to answer the questions. Teachers must emphasise that the general information must be analysed before looking at the questions. This information would have assisted the candidates in answering Q3.2.8 and Q3.2.9.

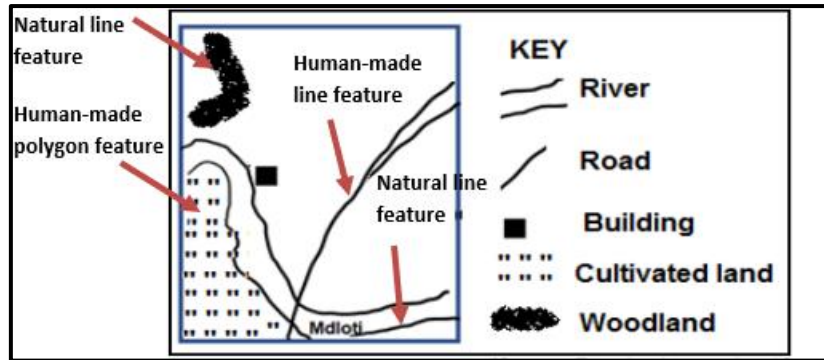
The Hazelmere Dam, just a few kilometres north of Verulam, is the main source of water for the area and is used for a variety of activities, such as watersports and fishing. One of the main rivers that flows through Verulam is the Mdloti River in which

- (e) Teachers must not assume that learners are familiar with mapwork content taught in the previous grades, e.g. Q3.2.5 focused on information relating to contour lines. This content must be revised in Grade 12 as this is part of the curriculum for mapwork as mentioned in the *Examination Guidelines*.

When teaching types of settlement patterns, learners should not only be able to identify the pattern and describe it, but also focus on the characteristics and causes of the street pattern. For instance, steep hilly land results in an irregular pattern being developed.

- (f) Q3.2.7 is a good example to use when emphasising to the learners that they must focus on the specific areas referred to in the question. The question referred to Canelands (6) in block A5 and only answers from A5 were only accepted.
- (g) Q3.2.8 and Q3.2.9 were linked. The tertiary activity answered in Q3.2.8 determined the answer in Q3.2.9. Candidates had to give answers regarding the contribution of the economy based on the correct tertiary activity answered in Q3.2.8. Learners must read the question thoroughly before answering.
- (h) Poor performance in Q3.3 on GIS is due to learners lacking content knowledge and application skills. The bulk of the GIS questions focused on content that learners have been doing from Grade 10 and consolidating each year. *Line* and *polygon* features and *buffering* were the main concepts tested. Other areas of concern are mentioned below.

Learners need to read questions carefully to understand the specific reference being made e.g. human-made *polygon* not just *polygon*, natural *line* features not *line* features.



[NSC November 2022]

- (i) Learners need to differentiate between concepts/terms. In Q3.3.2 *vector* and *raster data* were tested. Q3.3.3 highlights the importance of not only teaching the terminology but showing learners how to apply it. The question provided the definition of *buffering* but learners could not apply it to the questions that followed, thus showing their lack of understanding of the term. Learners need to focus on the stipulated block referred to and not give answers outside the block e.g. residential areas.
- (j) Learners should be taught to apply concepts to different contexts as required in a question such as Q3.3.4. Q3.3.4(a) referred to an infrastructure data layer and Q3.3.4 (b) referred to a topography data layer.



# CHAPTER 7

## HISTORY

The following report should be read in conjunction with the History question papers of the November 2022 NSC examinations.

### 7.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who sat for the History examinations in 2022 increased by 9 879 candidates.

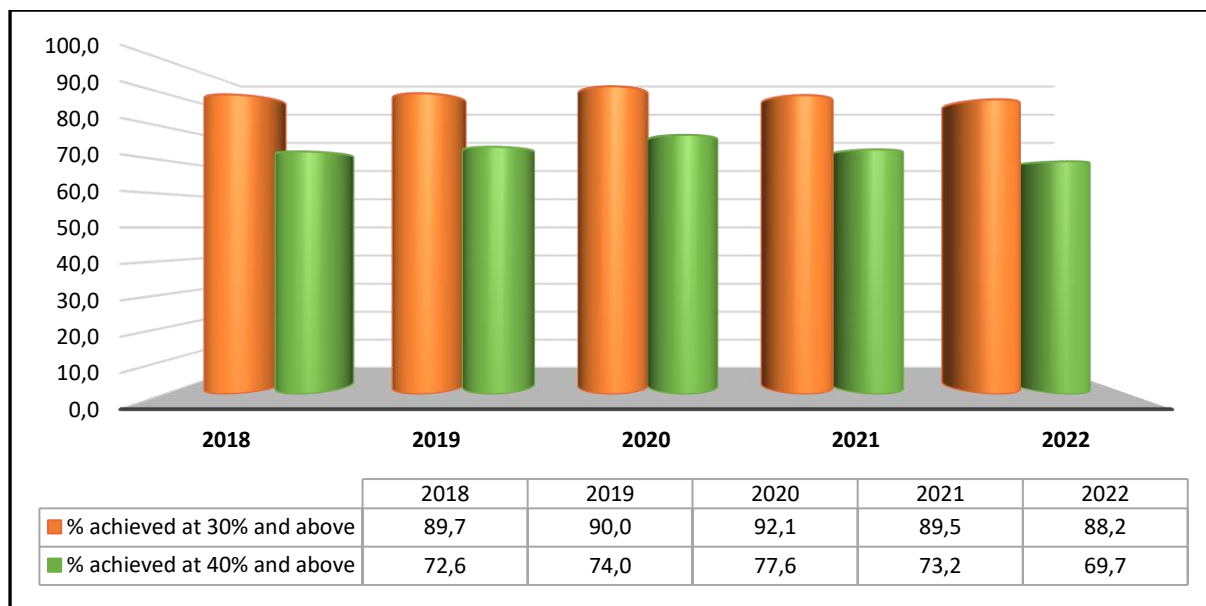
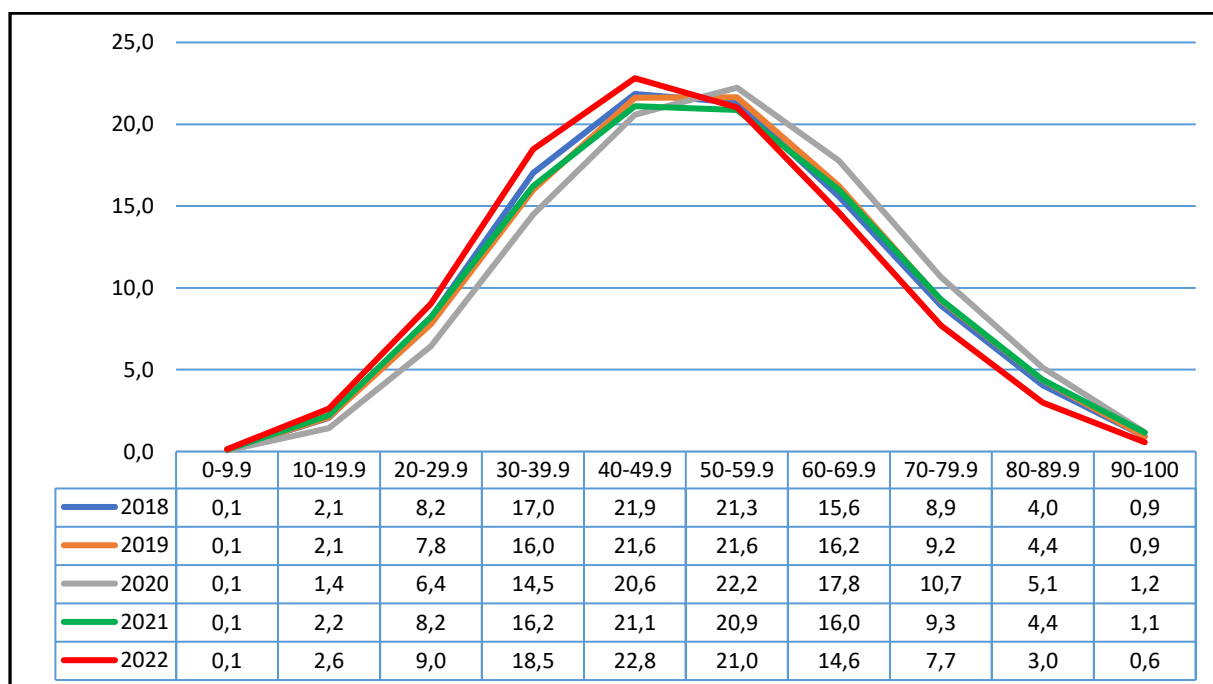
While the table below indicates a positive trend in the pass rate over the period 2018-2021, a decrease in the pass rate was evident this year. Candidates who passed at 30% (Level 2) declined marginally from 89,5% in 2021 to 88,2% in 2022. There was a corresponding decline in the pass rate at 40% (Level 3) over the past two years from 73,2% to 67,7%.

There was also a decrease in the percentage of distinctions over 80% (Level 7) from 5,5% to 3,6%. This converts into a decrease in the total number of distinctions from 12 510 to 8 544.

In view of disruption to academic programmes in 2020 and 2021, and despite the factors mentioned above, the results achieved by this cohort are generally commendable. Strategic intervention programmes at all levels (National, Provincial, Districts, Circuits and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the overall performance.

**Table 7.1.1 Overall achievement rates in History**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 154 536   | 138 570                       | 89,7                        | 112 266                       | 72,6                        |
| 2019 | 164 729   | 148 271                       | 90,0                        | 121 936                       | 74,0                        |
| 2020 | 173 498   | 159 737                       | 92,1                        | 134 610                       | 77,6                        |
| 2021 | 227 448   | 203 473                       | 89,5                        | 166 576                       | 73,2                        |
| 2022 | 237 327   | 209 315                       | 88,2                        | 165 483                       | 69,7                        |

**Graph 7.1.1 Overall achievement rates in History (percentage)****Graph 7.1.2 Performance distribution curves in History (percentage)****General comments on Paper 1 and Paper 2**

The year 2022 marked the second year of a new three-year cycle in the subject History, with the introduction of new topics in both papers.

Paper 1 included source-based questions which examined *Origins of the Cold War in Europe* and the *Civil Rights Movement*, as well as two essays based on the *Extension of the Cold War in Vietnam* and *Independent Africa (the Congo)*.

Paper 2 included two relatively new sections. These are the source-based question, i.e. *Internal Resistance* and an essay, i.e. the *Black Consciousness Movement (BCM)*.

Unlike the previous four years (2017–2020) where candidates were exposed to the same content focus areas, last year's candidates had to contend with the new content focus for 2021–2023; this might account for the decline in the results.

Under these circumstances, it is likely that teachers and learners might have experienced uncertainty regarding the style of examining the new content, which could have affected the performance of candidates this year.

In Section A of both question papers (source-based questions), it was clear that a significant number of candidates were unable to answer middle- and higher-order questions. These questions required candidates to interpret, analyse, evaluate, compare and determine the *usefulness*, *limitations* and *reliability* of evidence in sources. Many other candidates also failed to write logical and coherent paragraphs based on the key question.

In Section B of the question papers (essay questions), most candidates displayed good content knowledge but were unable to take a stance and develop a balanced and independent line of argument. Several essays lacked relevant introductions and convincing conclusions. Nonetheless, performance in essays was far better than in source-based questions.

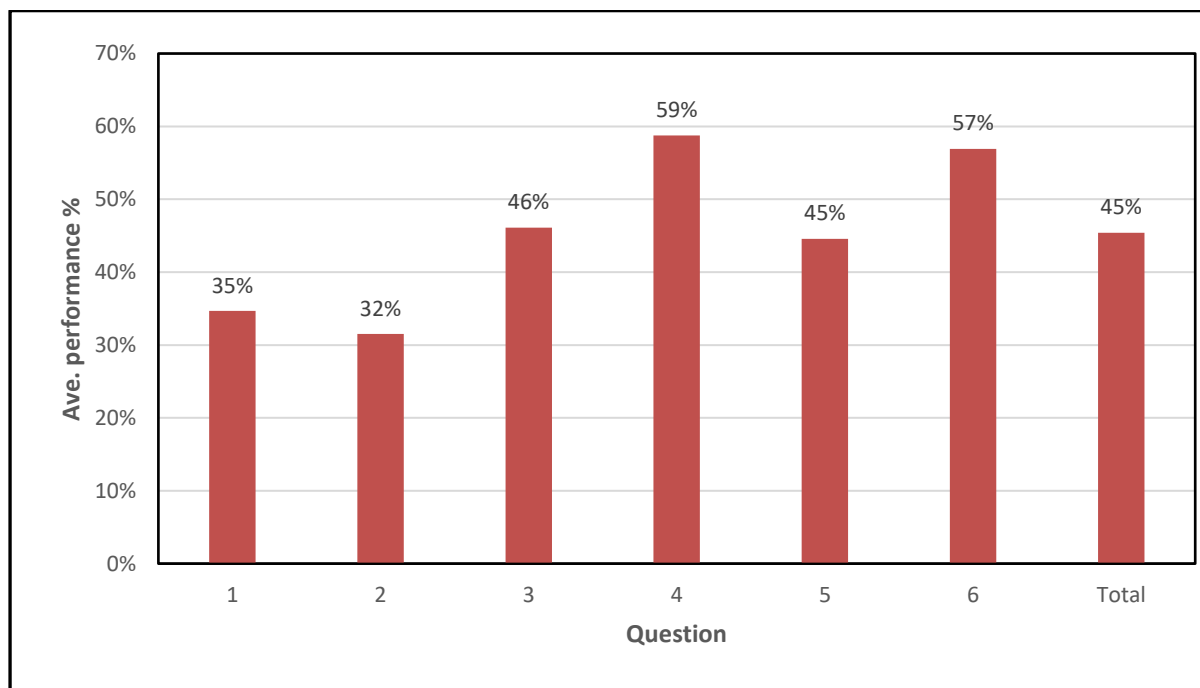
## 7.2 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 1

- (a) Generally, candidates' performance in this question paper ranged from fair to very good. It was evident that some candidates opted for two essay questions and one source-based question. The popular choice questions were Q1, Q3, Q4 and Q6. Few candidates attempted Q2 and Q5.
- (b) In Section A: Source-based questions, many candidates found it challenging to define concepts in their own words or explain concepts in the context of a section. They were unable to interpret statements from the sources effectively. It was also evident that many candidates lacked the ability to extract, select, interpret, analyse, evaluate and synthesise information from the sources that were provided. This resulted in unsatisfactory responses to higher-order questions, where candidates were unable to explain the *limitations*, *reliability* and *usefulness* of sources. Comparison of information from different sources also proved to be challenging.
- (c) A large number of candidates relied mostly on the relevant information in the sources, with little or no reference to their own knowledge. They were unable to write a well-structured paragraph effectively. Candidates simply copied information from the sources.
- (d) Successful candidates were able to interpret, analyse, evaluate and synthesise evidence from the given sources and also use their own knowledge to consolidate their responses. They were also able to comment on the *usefulness*, *limitations* and *reliability* of the sources used.
- (e) In Section B: Essay questions, there was a general improvement in essay-writing this year as most candidates could write and complete a comprehensive essay. Candidates demonstrated the required content knowledge in the essays but could not develop relevant introductions and conclusions or take a stance and defend it with more persuasive lines of argument.

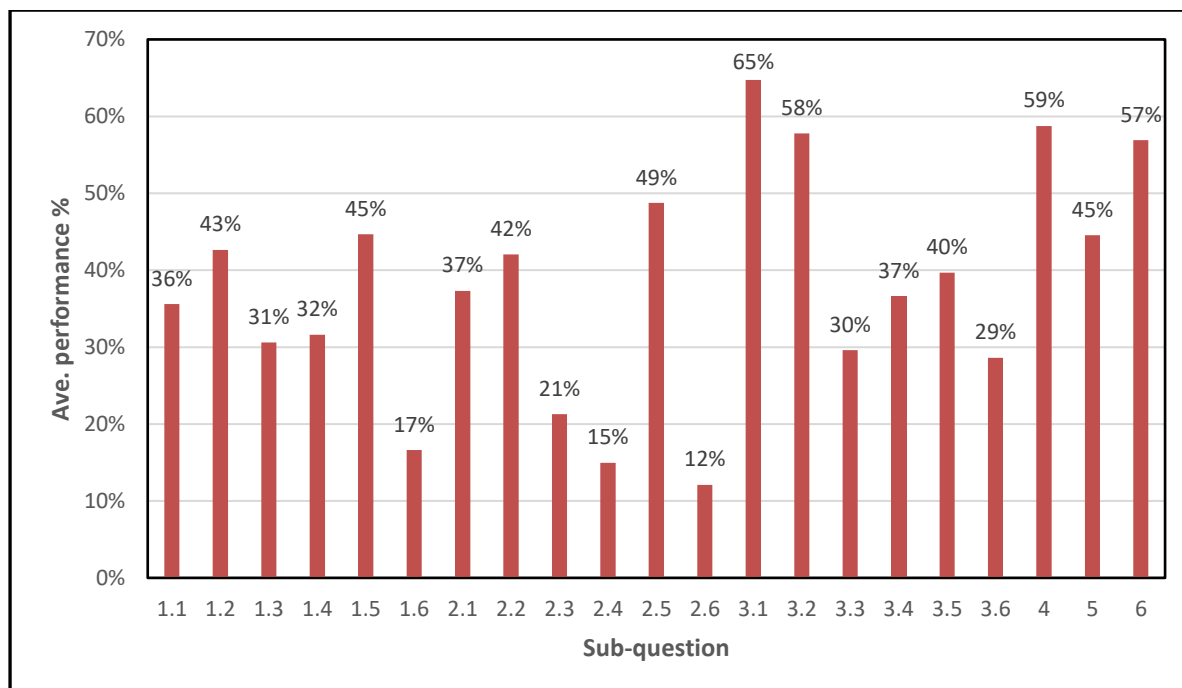
### 7.3 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 1

The following graph is based on data that was gathered from a random sample of candidates' scripts. While this graph might not accurately reflect the national averages, it serves as a useful tool to analyse how candidates performed in specific choice questions.

**Graph 7.3.1 Average performance per question in Paper 1**



| Q | Topic/s                                                                               |
|---|---------------------------------------------------------------------------------------|
| 1 | The Cold War: The Origins of the Cold War                                             |
| 2 | Independent Africa: Africa in the Cold War: Case Study – Angola                       |
| 3 | Civil Society Protests from the 1950s to the 1970s: The US Civil Rights Movement      |
| 4 | The Extension of the Cold War - Case Study: Vietnam                                   |
| 5 | How was independence realised in Africa in the 1960s and 1970s? Case Study: The Congo |
| 6 | Civil Society Protests from the 1950s to the 1970s: The Black Power Movement          |

**Graph 7.3.2 Average performance per subquestion in Paper 1**

| Q   | Skills assessed                                                                      | Q   | Skills assessed                                                                                       | Q   | Skills assessed                                                                                       | Q | Skills assessed |
|-----|--------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------|---|-----------------|
| 1.1 | 1.1.1 Extraction<br>1.1.2 Concept<br>1.1.3 Interpretation<br>1.1.4 Interpretation    | 2.1 | 2.1.1 Extraction<br>2.1.2 Concept<br>2.1.3 Extraction<br>2.1.4 Interpretation<br>2.1.5 Interpretation | 3.1 | 3.1.1 Extraction<br>3.1.2 Concept<br>3.1.3 Interpretation<br>3.1.4 Extraction<br>3.1.5 Interpretation | 4 | Essay           |
| 1.2 | 1.2.1 Extraction<br>1.2.2 Concept<br>1.2.3 Interpretation<br>1.2.4 Extraction        | 2.2 | 2.2.1 Extraction<br>2.2.2 Interpretation<br>2.2.3 Reliability                                         | 3.2 | 3.2.1 Extraction<br>3.2.2 Interpretation<br>3.2.3 Interpretation<br>3.2.4 Extraction                  | 5 | Essay           |
| 1.3 | 1.3.1 Interpretation<br>1.3.2 Limitations                                            | 2.3 | 2.3.1 Extraction<br>2.3.2 Interpretation<br>2.3.3 Concept                                             | 3.3 | Compare Sources                                                                                       | 6 | Essay           |
| 1.4 | Compare Sources                                                                      | 2.4 | Compare Sources                                                                                       | 3.4 | 3.4.1 Extraction<br>3.4.2 Concept<br>3.4.3 Usefulness                                                 |   |                 |
| 1.5 | 1.5.1 Extraction<br>1.5.2 Interpretation<br>1.5.3 Interpretation<br>1.5.4 Extraction | 2.5 | 2.5.1 Extraction<br>2.5.2 Interpretation<br>2.5.3 Extraction                                          | 3.5 | 3.5.1 Interpretation<br>3.5.2 Interpretation                                                          |   |                 |
| 1.6 | Paragraph                                                                            | 2.6 | Paragraph                                                                                             | 3.6 | Paragraph                                                                                             |   |                 |

## 7.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 1

### SECTION A: SOURCE-BASED QUESTIONS

#### QUESTION 1: THE COLD WAR: THE ORIGINS OF THE COLD WAR

##### Common errors and misconceptions

- (a) In Q1.1.2 most candidates could not define the term *containment* in their own words. They provided generalised responses such as 'suppression' and 'repression'.
- (b) In Q1.1.3 a large number of candidates were unable to explain the implication of the statement, *Truman decided it was time for the United States to step into the role of world leader*, in the context of the Truman Doctrine. They simply extracted the information provided in the source, which was not a requirement to answer the question.
- (c) In Q1.1.4 many candidates had difficulty in explaining why communism was regarded as a *threatening 'unstoppable virus' to neighbouring European nations*. They made general statements without providing relevant explanations.
- (d) In Q1.2.2 most candidates could not explain *totalitarian regimes* in the context of the Truman Doctrine. They provided generalised responses such as 'they are governments' and 'are supported by misery and want'. A significant number of candidates did not answer this question.
- (e) In Q1.2.3 the majority of candidates were unable to explain what Truman meant by the statement, *The seeds of totalitarian regimes are nurtured by misery and want*, regarding the spread of communism to Europe. They simply extracted the information provided in the source, which was not a requirement to answer the question.
- (f) Some candidates could not explain the messages that were conveyed in the cartoon. While many candidates responded by rewriting the message from the source without explaining it, others copied the contextualisation from the source.
- (g) In Q1.3.2 most candidates struggled to comment on the *limitations* of Source 1C for an historian researching communism. Responding to a question on the skill of identifying limitation remains a problem for most candidates.
- (h) Q1.4 required candidates to compare two sources. Candidates experienced difficulty in explaining how Source 1B supports Source 1C regarding the assistance that the United States of America had given to Europe in order to contain the spread of communism. Most candidates only provided one comparison.
- (i) In Q1.5.3 some candidates were not familiar with history terminology like *imperialist/anti-imperialist* and *democratic/anti-democratic*. As a result, they were unable to extract information from the source.
- (j) Many candidates responded poorly to the paragraph question (Q1.6), simply copying information directly from the sources. Some were unable to use the information in the sources to write a comprehensive paragraph. The majority of candidates displayed an inability to interpret, evaluate and synthesise information from different sources.

## QUESTION 2: INDEPENDENT AFRICA: AFRICA IN THE COLD WAR: CASE STUDY: ANGOLA

### Common errors and misconceptions

- (a) In Q2.1.2 most candidates could not define the term *provisional government* in their own words. They provided incorrect responses such as a 'provincial government' and 'local government'.
- (b) In Q2.1.4 a large number of candidates could not explain the implication of the statement, ... *the Angolans retreated to Cuito Cuanavale and the South African Defence Force (SADF) laid siege*. They lacked basic interpretation skills.
- (c) In Q2.1.5 most candidates were unable to explain why the Angolan government requested the Cubans to assist them in defending Cuito Cuanavale. This is likely due to their lack of knowledge about the role that Cuba played in the Battle of Cuito Cuanavale.
- (d) Many of the candidates could not explain the *reliability* of Source 2B for an historian researching the losses suffered by the Cuban/FAPLA and SADF forces during the Battle of Cuito Cuanavale. They merely copied the contextualisation of the source and provided this as their response.
- (e) Most candidates could not explain the term *accord* in the context of the end of the Battle of Cuito Cuanavale (Q2.3.3). The words, *in the context of*, were misunderstood and many candidates made general statements without providing relevant explanations.
- (f) The majority of candidates had difficulty comparing the information in Sources 2B and 2C regarding losses suffered during the Battle of Cuito Cuanavale in Q2.4. Many of them could not link the information in both sources.
- (g) In Q2.5.2 some candidates could not explain why objective observers believed that the Battle of Cuito Cuanavale ended as a tactical military stalemate. They lacked the ability to interpret the question. Many of them provided one response, yet the question required two responses.
- (h) A large number of candidates showed very poor competence in paragraph-writing skills and did not answer the question asked. Some candidates tended to look at sources in isolation. Candidates required the ability to utilise the sources to support their response to a question without a strong dependence on using direct quotes from the source itself.

## QUESTION 3: CIVIL SOCIETY PROTESTS FROM THE 1950s TO THE 1970s: THE US CIVIL RIGHTS MOVEMENT

This proved to be a popular question which was attempted by a large number of candidates. The candidates' performance ranged from fair to good.

### Common errors and misconceptions

- (a) Q3.1.2 and Q3.4.2 were poorly answered because many candidates were unable to define the term *voting rights campaign* (Q3.1.2) in their own words, and to explain the concept *civil rights* (Q3.4.2) in the context of the Selma to Montgomery March in 1965.

- (b) In Q3.2.2 a large number of candidates struggled to comment on what was implied by the statement, *There will be no word*. They lacked knowledge and failed to interpret the question correctly. They used information from the source verbatim.
- (c) Some candidates experienced difficulty with the comparison question (Q3.3). They could not explain how Source 3A supported Source 3B regarding how the State Troopers dealt with demonstrators at the Selma to Montgomery March on 7 March 1965. Most candidates only provided one comparison.
- (d) In Q3.4.3 several candidates struggled to comment on the *usefulness* of Source 3C for an historian researching preparations for the second Selma to Montgomery March on 9 March 1965. Responding to a question on the skill of identifying *usefulness* remains a problem for most candidates. They used information from the source verbatim and confused *usefulness* with *reliability* of a source.
- (e) It was evident in Q3.5.1 that some candidates were unable to explain the messages that were conveyed in the photograph. Many candidates responded by rewriting the message from the source without explaining it. Some also used the contextualisation of the source as their response.
- (f) In Q3.5.2 many candidates were unable to explain why the photograph was regarded as an iconic image of the civil rights protests. They lacked knowledge and failed to interpret the question correctly. Moreover, they did not know the meaning of word 'iconic'.
- (g) In Q3.6 some candidates struggled to use the information in the relevant sources and their own knowledge to write a coherent paragraph explaining *the challenges that were encountered by civil rights protestors who participated in the Selma to Montgomery marches in March 1965*.

## SECTION B: ESSAY QUESTIONS

### QUESTION 4: THE EXTENSION OF THE COLD WAR – CASE STUDY: VIETNAM

This was the most popular question and the performance of candidates who attempted this question ranged from satisfactory to good.

#### Common errors and misconceptions

- (a) Some candidates provided an unnecessary and detailed background about Vietnam. Candidates needed to be able to address what was required by the question.
- (b) A significant number of candidates were able to take a stance to indicate agreement or disagreement with the statement, *The USA lost the war in Vietnam (1962 to 1975) due to unconventional strategies of guerrilla warfare and environmental factors on the battlefield*.
- (c) A substantial number of candidates applied the *L* in the *PEEL* method incorrectly. They ended each paragraph by simply repeating the statement provided in the question which did not relate well to the information provided in the preceding paragraph.
- (d) It was also noted that the essays of weaker candidates lacked proper introductions and contained irrelevant background information. In addition, many could not sustain their line of argument or draw convincing conclusions.
- (e) Chronology should be stressed in this question as the line of argument depends on it.

**QUESTION 5: INDEPENDENT AFRICA: HOW WAS INDEPENDENCE REALISED IN AFRICA IN THE 1960s AND 1970s? – CASE STUDY: THE CONGO**

This question was attempted by many candidates. Generally, the performance ranged from poor to satisfactory. After script analysis was conducted, it was clear that this question was not well answered, because candidates gave too much background information and were unable to critically discuss the question.

**Common errors and misconceptions**

- (a) A large percentage of candidates who attempted this question were unable to discuss the statement critically with reference to the effectiveness of Mobutu Sese Seko's socio-economic and political policies in developing the Congo after attaining independence from Belgian colonial rule in the 1960s. The content presented was largely descriptive and there was little attempt to develop a line of argument.
- (b) Many candidates simply discussed the policies without stating whether they were effective. Responses lacked relevant introductions and conclusions and did not develop a line of argument. Some candidates also gave a lot of irrelevant background information from the period before the 1960s.

**QUESTION 6: CIVIL SOCIETY PROTESTS FROM THE 1950s TO THE 1970s –THE CIVIL BLACK POWER MOVEMENT**

Most candidates answered this question and seemed to have prepared for the theme thoroughly. The marks obtained were marginally better than those attained in the other essays. Those who did not perform well appeared to be candidates who had not covered, or were not taught, this theme at school.

**Common errors and misconceptions**

- (a) Many candidates were unable to explain *to what extent various role-players were successful in using the Black Power philosophy to end discrimination against African Americans in the USA in the 1960s*.
- (b) Some candidates just gave a narrative account of Black Power, the philosophy, the leaders and the Black Panthers without mentioning how various role-players were successful in using the Black Power philosophy to end discrimination against African Americans in the USA in the 1960s.
- (c) A few candidates wrote essays that lacked introductions, a logical and sequential body of events as well as persuasive conclusions.

**7.5 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 2**

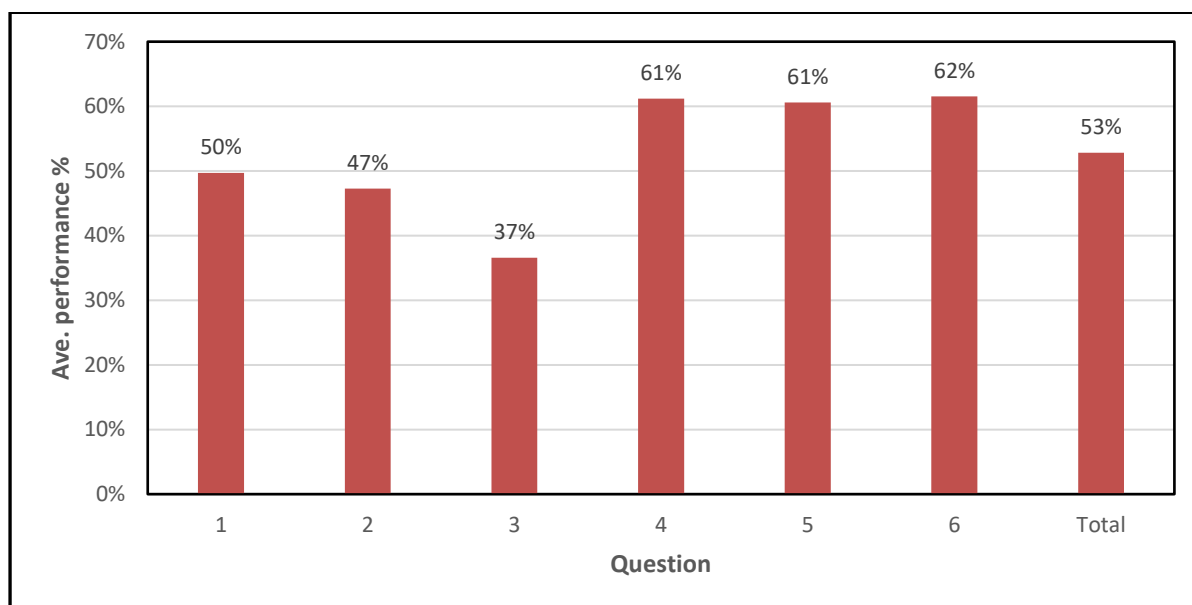
- (a) Generally, candidates' performance in this question paper ranged from poor to good. Most candidates opted to answer two essay questions and one source-based question. The popular choice questions were Q1 & Q2 (source-based), and Q4 and Q5 (essays). There was a general improvement in essay writing this year as some of the candidates could complete a comprehensive essay. Weaker candidates, however, failed to provide properly structured essays with relevant introductions and conclusions.
- (b) Some candidates displayed a slight improvement in paragraph writing, as well as in responding to questions on source-based material and in addressing questions involving comparisons, *usefulness*, *reliability* and *limitations*.

- (c) Those candidates who were successful were able to interpret, analyse, evaluate and synthesise evidence from the given sources and also use their own knowledge to consolidate their responses.
- (d) In the essays, candidates scored good marks because they were able to take a stance and maintain the line of argument in the essay. In the source-based questions a few candidates achieved very good marks.
- (e) Candidates who answered two source-based questions and one essay question generally did not perform well because of poor application of skills in source-based questions.
- (f) The comments made about Section A: Source-based questions in Paper 1 are also applicable to Paper 2. Refer to point (b) on pg. 133.
- (g) The performance of candidates in Question 2 seemed to indicate that they were surprised by the angle that the examiners took on this question. Hitherto the focus had been on different case studies and this proved to be a challenge for candidates. It appeared that candidates had based their preparation on previous exam papers and were unable to adapt to this year's approach; they should, however, have coped with the subtopic if they had prepared the content thoroughly.
- (h) In Section B: Essay questions, there was a general improvement in essay-writing this year, as most candidates could write and complete a comprehensive essay. Candidates demonstrated the required content knowledge in the essays but could not develop relevant introductions and conclusions or take a stance and defend it with more persuasive lines of argument.
- (i) Most essay responses were concluded with regurgitation of content with very little effort to respond to the demands of an argumentative essay.

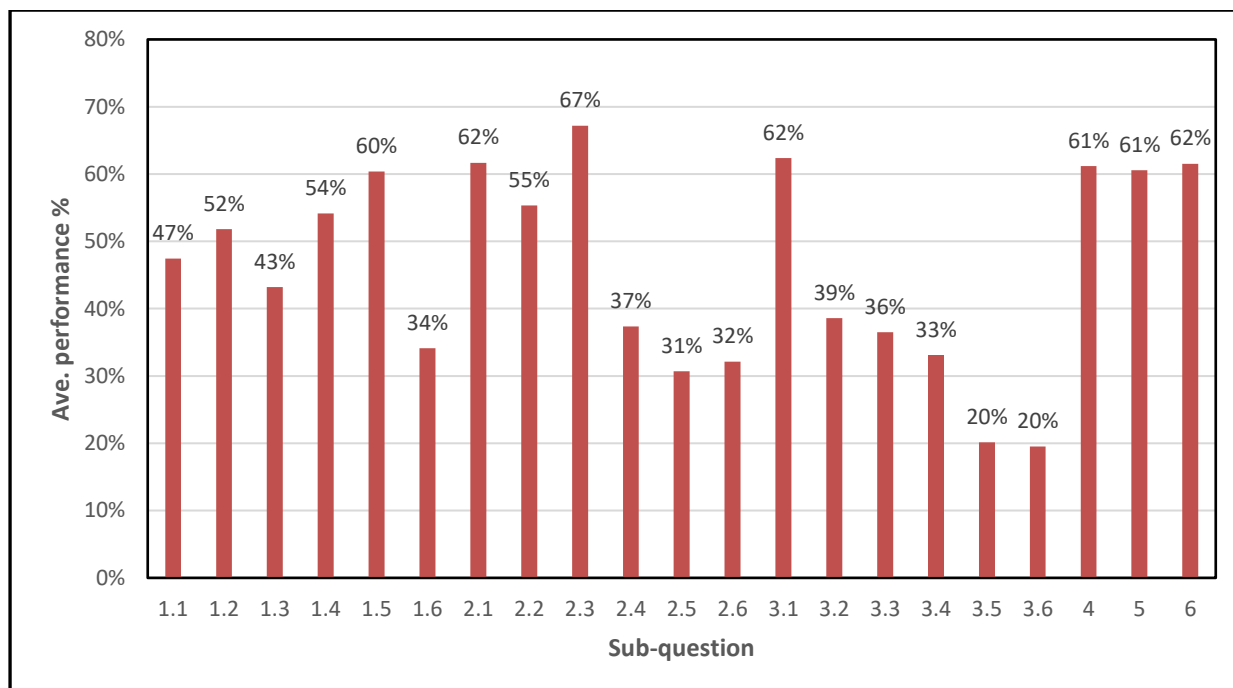
## 7.6 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 2

The following graph is based on data that was gathered from a random sample of candidates' scripts. While this graph might not accurately reflect the national averages, it serves as a useful tool in analysing how candidates performed in specific choice questions.

**Graph 7.6.1 Average performance per question in Paper 2**



| Q | Topics                                                                                                      |
|---|-------------------------------------------------------------------------------------------------------------|
| 1 | Civil Resistance, 1970s to 1980s: SA – The crisis of apartheid in the 1980s: Internal resistance to reforms |
| 2 | The coming of democracy to South Africa and coming to terms with the past – The TRC                         |
| 3 | The end of the Cold War and a New World Order, 1989 to the present – A new world order                      |
| 4 | Civil Resistance, 1970s to 1980s: South Africa: Challenge of Black Consciousness to apartheid state         |
| 5 | The coming of democracy to SA and coming to terms with the past – Negotiated settlement and the GNU         |
| 6 | End of Cold War and a New World Order: Impact of Gorbachev's reforms on Soviet Union and SA                 |

**Graph 7.6.2 Average performance per subquestion in Paper 2**

| Q   | Skills assessed                                                                                       | Q   | Skills assessed                                                                                                           | Q   | Skills assessed                                                                                       | Q  | Skills assessed |
|-----|-------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------|----|-----------------|
| 1.1 | 1.1.1 Extraction<br>1.1.2 Interpretation<br>1.1.3 Concept<br>1.1.4 Interpretation<br>1.1.5 Extraction | 2.1 | 2.1.1 Extraction<br>2.1.2 Concept<br>2.1.3 Interpretation<br>2.1.4 Extraction<br>2.1.5 Interpretation<br>2.1.6 Extraction | 3.1 | 3.1.1 Extraction<br>3.1.2 Extraction<br>3.1.3 Concept<br>3.1.4 Interpretation                         | 4. | Essay           |
| 1.2 | 1.2.1 Extraction<br>1.2.2 Extraction<br>1.2.3 Interpretation<br>1.2.4 Reliability                     | 2.2 | 2.2.1 Interpretation<br>2.2.2 Interpretation<br>2.2.3 Concept                                                             | 3.2 | 3.2.1 Extraction<br>3.2.2 Concept<br>3.2.3 Interpretation<br>3.2.4 Extraction                         | 5. | Essay           |
| 1.3 | 1.3.1 Interpretation<br>1.3.2 Interpretation                                                          | 2.3 | 2.3.1 Extraction<br>2.3.2 Interpretation<br>2.3.3 Usefulness                                                              | 3.3 | 3.3.1 Extraction<br>3.3.2 Interpretation<br>3.3.3 Extraction<br>3.3.4 Extraction<br>3.3.5 Limitations | 6. | Essay           |
| 1.4 | Compare Sources                                                                                       | 2.4 | Compare Sources                                                                                                           | 3.4 | 3.4.1 Interpretation<br>3.4.2 Interpretation                                                          |    |                 |
| 1.5 | 1.5.1 Extraction<br>1.5.2 Concept<br>1.5.3 Interpretation<br>1.5.4 Interpretation                     | 2.5 | 2.5.1 Extraction<br>2.5.2 Extraction<br>2.5.3 Interpretation                                                              | 3.5 | Compare Sources                                                                                       |    |                 |
| 1.6 | Paragraph                                                                                             | 2.6 | Paragraph                                                                                                                 | 3.6 | Paragraph                                                                                             |    |                 |

## 7.7 DIAGNOSTIC ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### SECTION A: SOURCE-BASED QUESTIONS

#### QUESTION 1: CIVIL RESISTANCE, 1970s TO 1980s: SOUTH AFRICA: THE CRISIS OF APARTHEID IN THE 1980s – INTERNAL RESISTANCE TO REFORMS

##### Common errors and misconceptions

- (a) In Q1.1.3 most candidates had difficulty responding to this question. They were unable to explain the implication of the statement.
- (b) The majority of the candidates were unable to explain the term *Civil Disobedience* in Q1.1.4 in the context of activities organised by the MDM in September 1989.
- (c) In Q1.2.1 some candidates could not explain the messages conveyed in the photograph, regarding the MDM's Open Hospital defiance campaign against segregation. Candidates wrote out/copied the messages from the placards, e.g. *Apartheid health kills* and were unable to explain the message conveyed by the source in general.
- (d) In Q1.3 candidates had difficulty explaining how the information in Source 1A supported the evidence in Source 1B regarding the defiance campaigns organised by the MDM against segregation laws.
- (e) In Q1.4.2 most candidates could not interpret what Tutu meant. They were unable to grasp or interpret phrases that were presented in the source. The phrase *in the context of* proved to be problematic. Candidates simply copied information from the source.
- (f) Many of the candidates could not determine the *usefulness* of the source in Q1.4.5 for a researcher studying the events that took place during the peace march to the Cape Town City Hall on 13 September 1989.
- (g) In Q 1.6 candidates could not explain how the MDM challenged the segregation laws of South Africa's apartheid government. The following skills should be reinforced:
  - Ability to identify relevant evidence from sources
  - Writing logical and coherent paragraphs
  - Complying with the length restriction
  - Referencing evidence from sources

#### QUESTION 2: THE COMING OF DEMOCRACY TO SOUTH AFRICA AND COMING TO TERMS WITH THE PAST – THE TRC

##### Common errors and misconceptions

- (a) In Q2.1.1 some candidates failed to extract evidence from the sources. When asked to give the Act that brought the TRC into being, some candidates wrote 'Promotion of National Unity Act' and left out the words 'and Reconciliation'.
- (b) Most candidates could not extract a response for Q2.1.2 that was based on what Tutu had stated as *what should remain absolutely central to the concerns of South Africans in 'the work of our Commission (TRC)'*.

- (c) In Q2.1.3 many of the candidates were unable to explain what Tutu meant by the statement, *no matter on which side we stood. We all stand in need of healing*, in the context of reconciliation and national unity.
- (d) In Q2.2.2 some candidates failed to explain the messages conveyed by the visual source.
- (e) In Q2.2.3 many candidates could not comment on the *limitations* of the source for an historian researching the submission of the final report of the Truth and Reconciliation Commission.
- (f) More than 50% of candidates who attempted Q2.3.1 could not define the term *perpetrator* in their own words, with some not attempting it at all.
- (g) In Q2.4.2 the majority of the candidates could not comment on the statement, *the TRC was never given a full mandate to address the group effects of apartheid*. This signifies a lack of content knowledge.
- (h) Many of the candidates who attempted Q2.4.3 could not explain the term *redress* in the context of what the government wanted to achieve through the TRC.
- (i) In Q2.5 most candidates were still unable to compare sources, in this case, how the information in Source 2C supported the evidence in Source 2D regarding the work of the TRC.
- (j) Paragraph writing (Q2.6) remains a serious challenge for many of the candidates. They could not explain what South Africans can learn from the role played by Archbishop Desmond Tutu, chairperson of the Truth and Reconciliation Commission (TRC), from 1995 to 1998.

### **QUESTION 3: THE END OF THE COLD WAR AND A NEW WORLD ORDER, 1989 TO THE PRESENT – A NEW WORLD ORDER**

#### **Common errors and misconceptions**

- (a) In Q3.1.2 many candidates could not define the concept *climate change* in their own words.
- (b) Most of the candidates who attempted Q3.1.4 could not state TWO activities in the source that were delegated to the Inter-Ministerial Committee on Climate Change in preparation for the hosting of the UNFCCC in May 2008. Candidates failed to extract (quote) full evidence from sources in Level 1 questions, e.g. Q3.1.4, and provided the first portion of extraction, which fell short of answering the question.
- (c) In Q3.2.2 the majority of the candidates could not explain why they thought rich countries failed to honour their pledge to reduce their emission of greenhouse gases.
- (d) Many candidates who attempted Q3.2.4 could not explain the *reliability* of the source for an historian researching the climate change issues between 2000 and 2021.
- (e) In Q3.4.1 some candidates simply copied the context from the source given in the question.

- (f) In Q3.5 candidates were unable to explain how the information in Source 3B supported the evidence in Source 3D regarding the implementation of the climate change agreements.
- (g) Paragraph writing (Q3.6) continued to be problematic. Many of the candidates could not explain the commitment of developing countries of BRICS (Brazil, South Africa and India) to global climate change issues between 2000 and 2021. They just copied from sources.

## SECTION B: ESSAY QUESTIONS

### QUESTION 4: CIVIL RESISTANCE, 1970s TO 1980s: SOUTH AFRICA–THE CHALLENGE OF BLACK CONSCIOUSNESS TO THE APARTHEID STATE

#### Common errors and misconceptions

- (a) Question 4 was the most popular essay question. Many candidates were comfortable answering this question and most of them were able to take a stance and maintain a line of argument throughout the essay. This question was well answered and it boosted the marks of most candidates who answered it.
- (b) The candidates who scored poorly in this essay were those who had a content gap and were therefore unable to answer the question
- (c) The weakest area identified in the essays is the formulation of the introduction and conclusion. Learners should be trained on how to take a stance and support it. When a point is introduced in a paragraph, it must be explained by means of the *PEEL* method.
- (d) Content (in the body of the essay) was sometimes reproduced without any attempt to respond to the argumentative demands of the question. A general and predetermined LOA was imposed on each paragraph even if it is not in line with the discussion in the said paragraph.
- (e) A few candidates gave too much background information.

### QUESTION 5: THE COMING OF DEMOCRACY TO SOUTH AFRICA AND COMING TO TERMS WITH THE PAST – NEGOTIATED SETTLEMENT AND THE GNU

- (a) The question was well answered by many candidates. Some candidates were able to take a stance and sustain the line of argument throughout the essay.
- (b) Candidates who scored poorly in this essay were not familiar with the content.
- (c) Some candidates gave detailed background information about PW Botha's reforms (Tricameral Parliament), which was not required.
- (d) In this question, candidates had challenges with regard to chronology, dates and names.
- (e) Candidates confused CODESA 1 and CODESA 2 when explaining what happened during these meetings.

- (f) The manner in which the conclusion was approached was the same deficient way as the introduction. There was no effort to provide apt summation of the preceding argument.
- (g) It was evident that the majority of candidates had content knowledge but lacked the skill of demonstrating the extent to which compromise by different role players (political parties) remained a key feature that paved the way for a new political dispensation in South Africa in 1994.

**QUESTION 6: END OF THE COLD WAR AND A NEW WORLD ORDER – IMPACT OF GORBACHEV’S REFORMS ON SOVIET UNION AND SOUTH AFRICA**

**Common errors and misconceptions:**

- (a) The way this question was set focused on how the policy changes led to disintegration of the Soviet Union and brought about positive results in South Africa. Some candidates did not show how the changes led to the disintegration; they just discussed the positive results in South Africa. Some candidates had sufficient content knowledge but could not sustain the line of argument.
- (b) This is a very interesting topic and each year more and more candidates respond to the question. With enough support, the question provided a base for good marks.

**7.8 SUGGESTIONS FOR IMPROVEMENT IN PAPER 1 AND PAPER 2**

**Teachers should:**

- (a) Ensure that the prescribed content, as contained in the *CAPS, Abridged Section 4 of CAPS* and the *2021 Examination Guidelines* document, are aligned to the Recovery Annual Teaching Plan (ATP) or any revisions to this plan and is comprehensively covered within the timeframes.
- (b) Incorporate suggestions for improvement in relation to teaching and learning from this Diagnostic Report into teaching and intervention as well as learning strategies.
- (c) Teach learners the relevant themes using interactive, user-friendly teaching methodology and relevant notes so that they have an in-depth understanding of the content focus areas.
- (d) Refer to past NSC question papers for practical examples on how the challenging skills (specific skills relevant for history learners, as outlined below) can be tested and be incorporated into lessons.
- (e) Expose learners to a variety of sources (e.g. visual, written, statistical, graphical and electronic) and provide opportunities to develop the related source-based skills such as extraction and selection of relevant information, interpreting, analysing, evaluating, comparing and contrasting sources, and ascertaining *limitations*, *usefulness* and *reliability* of sources.
- (f) Develop a list of concepts pertaining to each theme that must be covered and apply these concepts throughout the year as each topic is addressed. This will assist learners to refine skills to answer interpretative questions.
- (g) Equip learners with the necessary skills related to the definition or explanation of concepts in an historical context: extraction, interpretation, analysis, ascertaining the

reliability, limitations and usefulness of historical sources to improve their historical consciousness. These foundational historical skills must be underpinned by the teaching and learning of the prescribed content that has been delineated into source-based and essay questions. It is strongly advised that all historical concepts applicable to a specific topic be thoroughly unpacked and explained at the beginning of each topic, while the content focus is gradually unlocked.

- (h) Capacitate learners with the necessary and essential skills on how to work with sources such as interpretation, analysis, usefulness and comparison as well as how to integrate information from both the sources and their own knowledge.
- (i) Use item analysis, especially after formal tasks, for positive feedback.
- (j) Integrate ITC in their lessons to help learners understand the content and make provision for adaptation. Different sources of information like videos and written sources should be provided to enhance the knowledge of the candidates. Develop user-friendly resource materials, especially for the new content areas such as Origins of the Cold War, Vietnam, Congo, Internal Resistance to Reforms and The Challenge of the Black Consciousness to the apartheid state.
- (k) Essay writing skills should be given priority. Different aspects of the essay like the introduction can be practised as an informal task to polish the skill. Different questions on the topic should be used to practise and consolidate the knowledge of the topic.
- (l) Teach all the bullets that are indicated in the *Examination Guidelines*. The impact of the collapse of Communism in the *Examination Guidelines* indicates that South Africa and the USSR must be done in depth.
- (m) Administer at least three daily assessment activities/tasks on source and essay skills per week, taking into consideration the quantity and quality of informal assessment tasks.
- (n) Share model marked scripts with all learners to entrench an understanding of the demands of argumentative essays. This should be done to expose them to a high standard of argumentative essays, not to encourage rote learning.
- (o) Broaden and acquire more knowledge, by referring to more than one textbook while preparing for lesson presentations. Teachers should be aware of new resource materials such as media articles or newspaper supplements. Radio and TV features can also be used productively. Visit the DBE website as it has useful information on *Working with Sources* and the SBA document. Go to [www.dbe.gov.za](http://www.dbe.gov.za) and follow the links for the NSC.
- (p) Provide more exposure to the TRC, especially on how to work with sources and the use of case studies. It is advisable for teachers to visit the SAHO and the SAHA websites, which use case studies to highlight the workings of the TRC.
- (q) Sharpen paragraph-writing skills by ensuring that learners do the following:
  - Read the question and underline the key words.
  - Study all sources to gain a thorough understanding of them.
  - Underline the words in written sources and incorporate them in the writing of paragraphs.

- Start a paragraph with an opening statement that affirms or opposes the question and conclude the paragraph with a closing statement that supports the opening statement.
- Learners should not copy directly from the sources but use their own words, e.g. *According to Source 1A ...*
- Ensure that responses are concise and to the point by structuring short sentences to frame the paragraph.
- Always refer to the question when writing a paragraph.

(r) Encourage learners to use the 5 Ws in all historical inquiries, as listed below:

| Question to ask | Purpose                                     |
|-----------------|---------------------------------------------|
| Who?            | To gain knowledge about historical figures  |
| What?           | To equip learners with historical knowledge |
| When?           | To study historical periods                 |
| Where?          | To learn where historical events took place |
| Why?            | To learn why historical events took place   |

On completion of a topic or specific content focus, learners should be trained to acquire skills by asking them to apply the 5 Ws to the content being taught.

- (s) Set skills-focused tasks to assess specific cognitive levels, such as making comparisons between sources so that answers can be structured in the context of the question posed.
- (t) Develop the requisite essay-writing techniques by:
- Coaching learners on how to unpack the question posed by identifying four key aspects namely, the action verb used (e.g., explain to what extent, do you agree or critically discuss), content focus, context of the content focus and time frame.
  - Underlining the key words in the question. If the question demands that a stance be taken, this must be stated in the introduction.
  - Using the *PEEL* writing template listed below to teach learners how to write an argumentative essay:
    - **Point:** State the main point by indicating a line of argument. Each paragraph should include a point that sustains the major point (line of argument) that was made in the introduction.
    - **Explanation:** Explain the point or line of argument by demonstrating how it relates to the question posed (line of argument).
    - **Example (Evidence):** Select and provide appropriate evidence (examples) to support the line of argument.
    - **Link:** Ensure that the concluding statement in each paragraph is linked to the line of argument taken at the introduction.
- (u) Guide learners to respond to an argumentative essay by focusing on a strong introduction (with a line of argument or an independent line of argument), paragraphs that sustain the line of argument and a conclusion that ties up the line of argument and links well with the introduction.
- (v) Teachers should attend content and assessment workshops, subject meetings or subject briefing sessions to firstly familiarise themselves with the requirements and demands of the CAPS and the *2021 Examination Guidelines* document and, secondly, to use recent and relevant teaching and learning methods in classrooms.

- (w) Instil the following steps when teaching learners how to compare evidence in two sources to answer questions on either similarities or differences:
- Read the question thoroughly and underline the main point.
  - Study the contextualisation of both sources and underline its main context.
  - Check the author and the purpose of each source, after studying the contextualisation of the source. This will give a clue about the perspective and intention of the source, which could then be compared to the other source.
  - Detect opposing viewpoints by identifying the rival organisations or ideologies that the two sources represent. If opposing viewpoints form part of the contextualisation of the two sources that are compared, learners must underline both viewpoints, because the different perspectives displayed by each source will already provide the learner with one option of the answer.
  - Familiarise themselves with the information in the sources mentioned. The 5 Ws stated earlier can be used in this regard.
  - Have a clear understanding of what a visual source entails by finding dates, numbers, historical figures, facial expressions, text or any other object relevant to the question.
  - Provide the required responses for the question in either of the following ways:
    - In answering a question about similarities, learners could use the following: *Both sources refer to ... or Source 1A mentions ... and Source 1B shows ...*
    - When comparing information for differences or contrast, it is crucial that learners state: *Source 1A says ... WHILE Source 1B states ... or In Source 1A we read... WHILE Source 1B shows ... or Source 1A is written from a communist perspective (Russian point of view) WHILE Source 1B is written from a capitalist perspective (US point of view).*
  - Highlight the point that will be credited for each response that refers to both sources, i.e. 2 (two) marks. To get full marks in a question with a mark allocation of (2 x 2) (4), learners should provide TWO responses that refer to both sources, but on two different aspects.

**Subject advisors should:**

- (a) Assist teachers with an understanding of the expectations of the *Examination Guidelines*.
- (b) Thoroughly study and understand the *2022 Diagnostic Report* and then conduct workshops with teachers on the findings of the *2022 Diagnostic Report*, as well as challenging topics.
- (c) Plan, prepare and conduct intensive content and assessment workshops on problematic areas as identified in this report with FET History teachers.
- (d) Conduct assessment training on how to mark higher-order source-based questions (usefulness, compare and paragraph writing) and essay questions. A sample of learner responses should be used to train teachers on how to use the levels of rubrics and matrix to assess paragraphs and essays correctly. Orientate teachers on the principles and criteria of how to mark source-based, paragraph and essay questions which are found on pages 2 to 6 of the NSC November 2022 Marking Guidelines.
- (e) Vigorously monitor and quality assure the assessment tasks administered by teachers.
- (f) Develop appropriate resource materials with which both teachers and learners can interact on an on-going basis.

- (g) Train teachers on how to set quality assessment tasks in order to standardise tasks that constitute the Programme of Assessment.
- (h) **SUGGESTIONS FOR TEACHING RELIABILITY, USEFULNESS AND LIMITATIONS OF A SOURCE:**

For proper evaluation of these aspects, the **contextualisation of the source** should include the following:

- Who the **author (owner) of the source** is: this should not only address the name but also the position or profile of the author;
  - Identifying the author as a chairperson of an institution or as the President of a country enhances the *reliability* or *usefulness* of a source.
  - A source by an author from a rival ideology might suggest *NOT Reliable* or *NOT Useful* depending on the context of the question/If the source is biased/one-sided, that would be a *Limitation* of the source – again, depending on the context of the question.
- **When (date) the source** was written:
  - It is important to check whether the date of the source was at the time of the event or whether it was too far away from the time of the event; The date of a source that is very close to the event can be regarded as *Reliable* or *Useful*.
  - A source far removed from the time of the event, giving a perspective different from the original event might be a deliberate distortion, suggesting *Limitation* of the source.
- **Where (the venue/place) the source** was created:
  - A speech given in an official venue, e.g. Parliament or at an official function held anywhere, would give credit to the *Reliability* or *Usefulness* of a source.
  - Publications in the US national newspapers like *New York Times* or *The Washington Post* might be pro-capitalist (*Reliable* or *Useful*) or be anti-communist (*Not Reliable* or *Not Useful*) depending on the context of the question).
  - Publications in newspapers from the Soviet Union, e.g. *Pravda* or *Izvestia*, might be pro-communist or anti-capitalist, depending on the context of the question).
- The **purpose** of writing (publishing) the source or **why** the source was published:
  - If the purpose is to strengthen internal processes of an institution or to outline how a project would unfold, that could be positive (*Reliable* or *Useful*) but if it paints a negative picture about a rival ideology, it could be considered as full of propaganda and therefore be regarded as biased (having *limitations*).
  - Is the source an opinion piece of writing (biased) or factual information (*Reliable/Useful*)?
- Can the information be independently **corroborated**?
  - Refer to other sources within the Addendum that corroborate information in the question (*Reliability* or *Usefulness*).

**Teacher Development should:**

- (a) Identify teacher needs and gaps as outlined in the *2022 Diagnostic Report*.
- (b) Assist teachers on how to plan, prepare and present interactive History lessons.
- (c) Ensure that new teachers are supported and guided on subject content and teaching methodology.

- (d) Equip teachers with the latest teaching trends and techniques in History.
- (e) Prepare teachers to apply the principle of Language Across the Curriculum (LAC) in their subject.
- (f) Train school principals on the implementation of quality management systems (QMS) that, among others, would include management of curriculum implementation.

# CHAPTER 8

## LIFE SCIENCES

The following report should be read in conjunction with the Life Sciences question papers of the November 2022 NSC examinations.

### 8.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who wrote the Life Sciences examinations in 2022 increased substantially by 14 791 candidates.

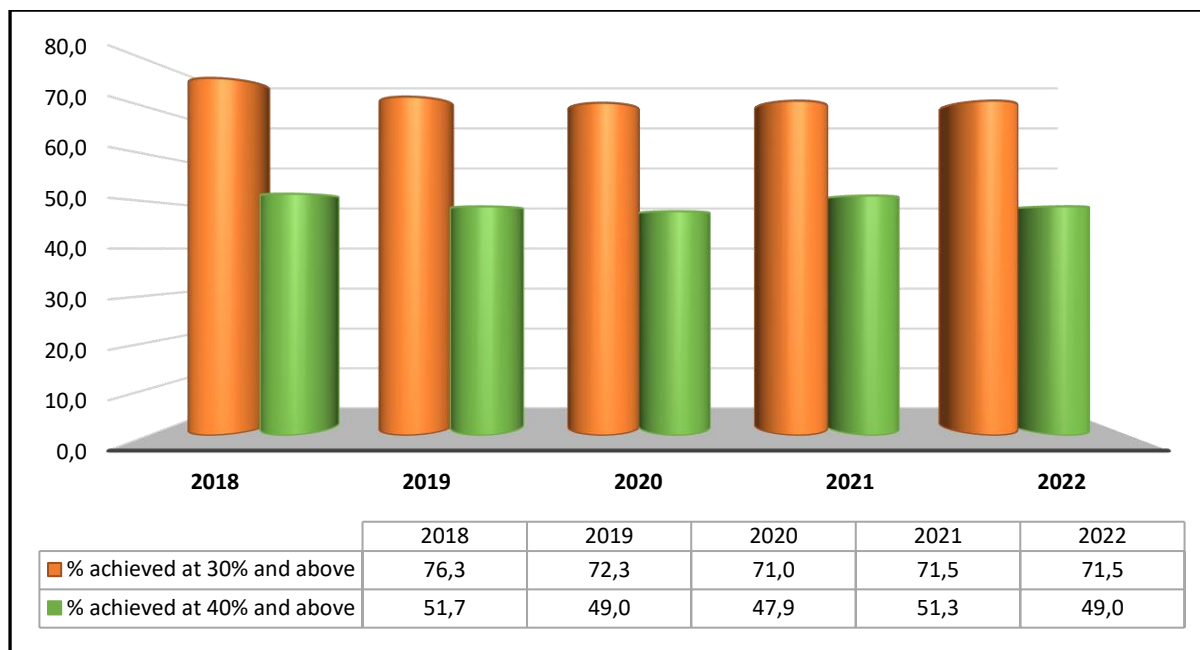
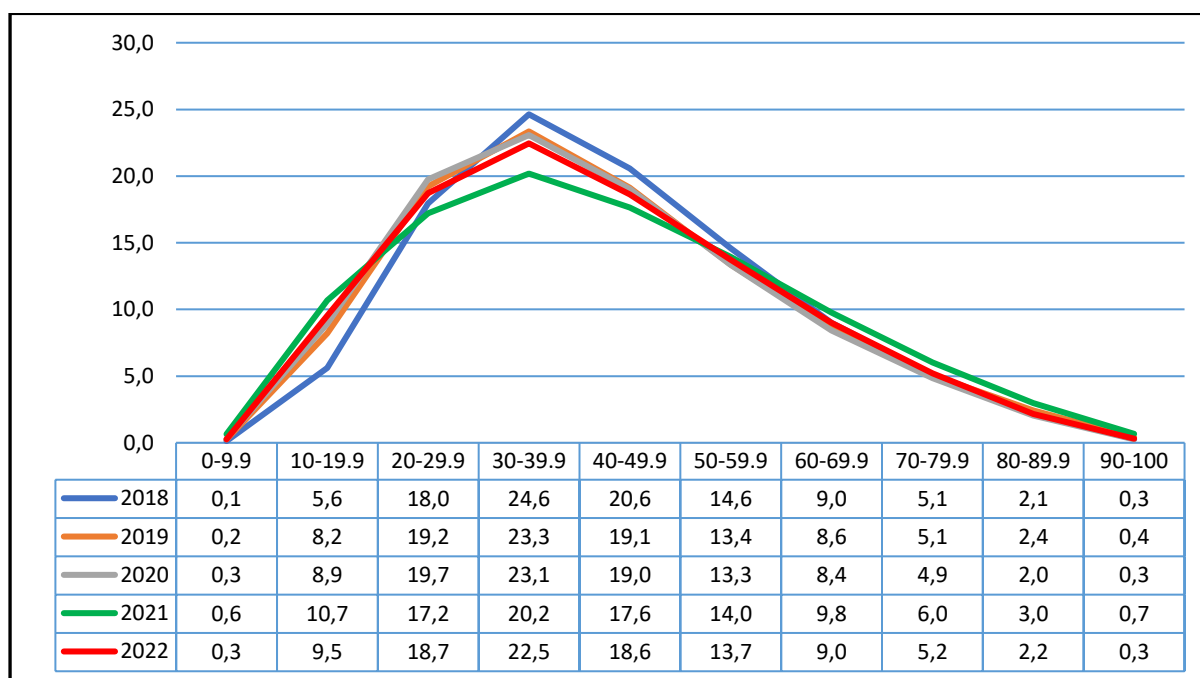
The table below indicates a negative trend in the pass rate over the period 2018-2020 with an improvement in 2021. While candidates who passed at 30% (Level 2) remained static at 71,5% in 2021 and 2022, there was a decline in the pass rate at 40% (Level 3) over the past two years from 51,3% to 49,0%.

The percentage of distinctions over 80% (level 7) decreased from 3,7% in 2021 to 2,5% in 2022. This converts into a decrease of 4 241 in the total number of distinctions.

In view of disruption to academic programmes in 2020 and 2021, and despite some factors mentioned above, the results achieved by this cohort are commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the overall performance.

**Table 8.1.1 Overall achievement rates in Life Sciences**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 310 041   | 236 584                       | 76,3                        | 160 208                       | 51,7                        |
| 2019 | 301 037   | 217 729                       | 72,3                        | 147 436                       | 49,0                        |
| 2020 | 319 228   | 226 700                       | 71,0                        | 153 028                       | 47,9                        |
| 2021 | 384 216   | 274 584                       | 71,5                        | 197 017                       | 51,3                        |
| 2022 | 399 007   | 285 217                       | 71,5                        | 195 620                       | 49,0                        |

**Graph 8.1.1 Overall achievement rates in Life Sciences (percentage)****Graph 8.1.2 Performance distribution curves in Life Sciences (percentage)**

### Life Sciences: background information

The NSC examination in Life Sciences specifically sets out to assess the acquisition of scientific skills as outlined in the CAPS. It is therefore encouraging to note that the majority of candidates have developed the necessary skills in data analysis, data presentation (drawing of graphs) and calculations. The application of knowledge to practical situations is an area that appears to require strengthening, as does the evaluation and design of scientific investigations. This form of assessment is found in Papers 1 and 2 and it would serve the learners well if they could master it. Scientific investigations are introduced in Grade 10 and should be thoroughly reinforced before Grade 12. Training and teacher support on these concepts must be given from Grade 10.

With the implementation of the amended section 4 of the CAPS (NSC November 2021), topics on 'Reproduction and Responding to the Environment' in Paper 1 have a greater weighting than before. In Paper 2, there is a higher mark allocation (21 marks) toward the topic of 'Meiosis'. This section is assessed more rigorously than before. More time and resources should be provided to strengthening these sections.

There is no longer an essay in the Life Sciences examination papers, but the ability of candidates to successfully articulate their responses is still assessed in questions that require extended writing. The very nature of Life Sciences research is based on the ability to read and extract relevant information. Any valid assessment will involve lengthy texts. Learners need to be trained not to be intimidated by these questions and should practise active reading and comprehension skills.

## 8.2 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 1

### General comments

- (a) The Grade 12 NSC qualification in Life Sciences is a three-year qualification and content from the preceding years may be examined within the context of the Grade 12 curriculum. Teachers need to identify the links between content in Grades 10, 11 and 12 on cellular respiration, blood circulation, nutrition, gaseous exchange and excretion. These sections must be properly emphasised in Grades 10 and 11 and revised in Grade 12, with particular reference to *homeostasis* and the *negative feedback mechanism*. The topic, *Human Impact on the Environment*, is no longer assessed in Grade 12, leaving sufficient time for teachers to revise the aforementioned content.
- (b) Candidates struggled to meet the response demand of instruction verbs such as 'name', 'state', 'describe', 'explain', 'tabulate' and 'calculate'. Misinterpretation of the instructional verbs reflects in low learner performance.
- (c) Many candidates undermine the importance of correct spelling. In Life Sciences, the incorrect spelling may change the meaning of the response and candidates may lose marks. For example, correct spelling is necessary when writing terms such as *glucagon* and *glycogen*; *chorionic* and *chronic*; *uterus* and *ureter*.
- (d) Particular attention must be paid to the disorders of human physiology. The questions based on *goitre*, *diabetes mellitus*, *spinal cord injuries* and *the use of grommets* in treating middle-ear infections were problematic for many candidates. It is evident that teachers merely brush over these disorders as they are encountered at the end of each topic. Knowledge of how the structures and processes of a system operate are best explained when application is made to the disorders.
- (e) Learners must use the correct biological terminology in their responses, e.g. reference must be made to *higher blood glucose levels* and not *high blood sugar*.
- (f) Teachers should emphasise the difference between commonly confused terms, e.g. the difference between *sperm* and *semen*; between an *egg* and an *ovum*; between the *uterine wall* and the *uterine lining*; *the corpus callosum* and *the corpus luteum* and between the *amnion* and *amniotic fluid*.
- (g) Some candidates gave more responses than were required by a question. If only two answers were required, only the first two answers were marked according to Marking Principle 2 of Life Sciences.

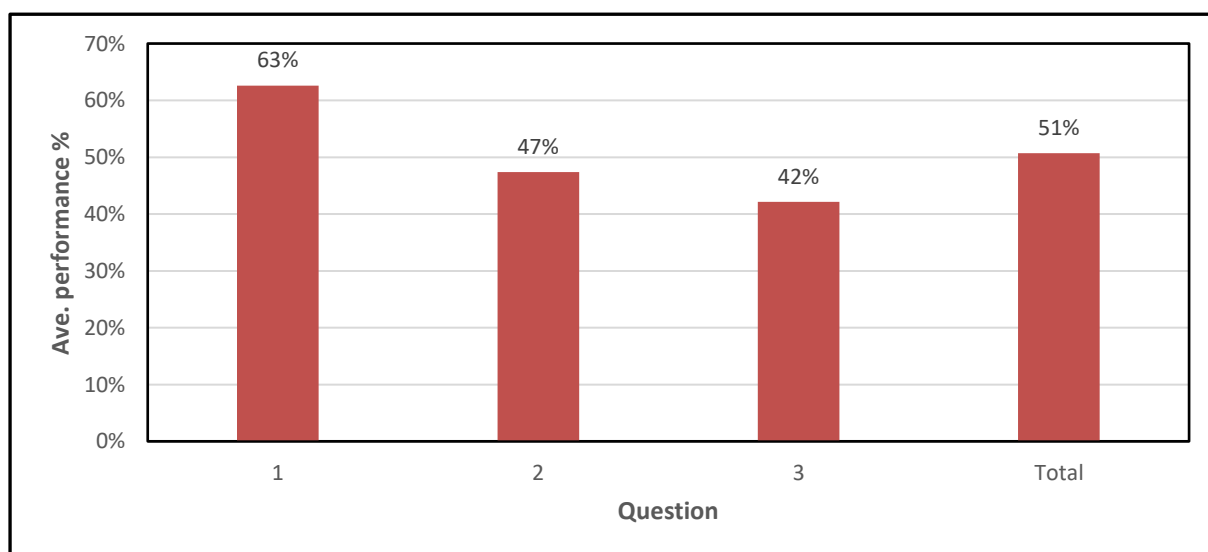
- (h) Poor performance was still evident in questions based on scientific investigations despite the support provided in the diagnostic reports of previous years. Candidates gave generic responses to questions that required specific information on the investigation provided.
- (i) There was unusually poor performance in the sections on *reproductive strategies* and *plant responses to the environment*. These are relatively short topics and it appeared as if some teachers merely glossed over them. It is important to concentrate on these two topics, as they are a compulsory inclusion in the paper for 8 and 13 marks respectively.
- (j) Poor performance was also noted in questions on *Human Reproduction* and *Homeostasis*. When teaching these topics, teachers must emphasise the importance of stating the levels of each substance and where these occur, e.g. *more CO<sub>2</sub> is transported to the lungs; the CO<sub>2</sub> levels in the blood decrease*. Candidates will not be credited for merely stating *more CO<sub>2</sub> is transported and levels drop*.
- (k) When using past papers for revision, teachers must ensure that learners do not regard the marking guidelines as definitive or exhaustive on a particular topic. Scenarios might be different, depending on the demands of a question. Learners need to identify the requirements of a question to respond to it appropriately.

### 8.3 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 1

Based on the Rasch analysis, the weakest performance by candidates was recorded in the subquestions on the *Female Reproductive Hormones* and the *Menstrual Cycle*, *Homeostasis*, *Hearing and Balance in the Human Ear* and *Plant Responses to the Environment*.

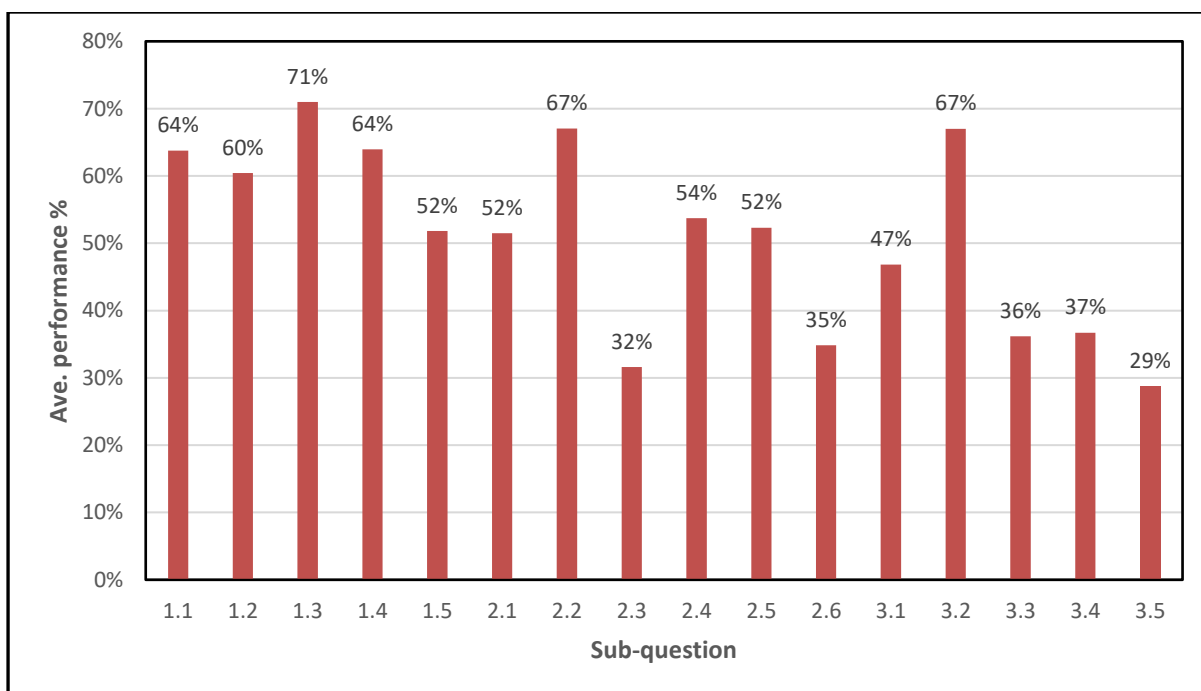
The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 8.3.1 Average performance per question in Paper 1**



| Q | Topics                                                                                                                                         |
|---|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Multiple choice, Terminology, Matching Items, Receptors – the human eye, negative feedback mechanism (thyroxin)                                |
| 2 | Male reproductive system, gametogenesis - sperm, menstrual cycle, reproductive strategies, homeostasis - negative feedback mechanism (glucose) |
| 3 | Brain, receptors - the human ear, scientific investigation and plant responses to the environment.                                             |

Graph 8.3.2 Average performance per subquestion in Paper 1



| Sub-Q | Topic                        | Sub-Q | Topic                                  |
|-------|------------------------------|-------|----------------------------------------|
| 1.1   | Multiple-choice question     | 2.4   | Endocrine – ovarian hormones           |
| 1.2   | Terminology                  | 2.5   | Reproductive strategies                |
| 1.3   | Matching items question      | 2.6   | Homeostasis of blood glucose           |
| 1.4   | Receptors – the human eye,   | 3.1   | The brain                              |
| 1.5   | Thyroxin - negative feedback | 3.2   | Data response (brain injuries)         |
| 2.1   | The male reproductive system | 3.3   | Receptors - the human ear              |
| 2.2   | Structure of sperm           | 3.4   | Scientific investigation (homeostasis) |
| 2.3   | Menstrual cycle – FSH and LH | 3.5   | Plant responses                        |

## 8.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 1

### QUESTION 1: MULTIPLE-CHOICE, TERMINOLOGY, MATCHING ITEMS, THE HUMAN EYE AND THYROXIN (NEGATIVE FEEDBACK)

#### Common errors and misconceptions

- (a) In Q1.1 candidates performed well except for Q1.1.9 and Q1.1.10. Candidates had to apply their knowledge of *auxins* to *geotropism*, rather than to *phototropism*, which is

frequently asked. The question was based on an investigation and was pitched at a higher cognitive level.

- (b) Q1.2 required candidates to provide the correct biological terminology for a phrase or description. This section required accuracy and attention to detail. Candidates:
  - Used the abbreviation *PNS* instead of the full term *Peripheral Nervous System* in Q 1.2.6. It is also evident that the branches of the nervous system are not well taught.
  - Lost marks in this question due to poor spelling, e.g. Q1.2.7 *chronic villi* instead of *chorionic villi*.
  - wrote *umbilical cord* instead of *umbilical artery* in Q 1.2.4 and *amnion fluid* instead of *amniotic fluid* in Q1.2.9.
- (c) In Q1.3.3 many candidates selected 'none' as the option. Although some other hormones (GnRH and FSH) *initiate* the sexual development, both *testosterone* and *oestrogen* stimulate the changes that are a part of puberty.
- (d) The poor performance in both Q1.1.6 and Q1.4 was a clear indication that candidates confused the *pupillary mechanism* and *accommodation*. Too many candidates wrote *ciliary muscles* instead of *circular muscles* in response to Q1.4.3(a) and (b).
- (e) Q1.5 was the most poorly answered question in Section A. The concept of the mechanics of the *negative feedback mechanism* was elusive for many candidates. The disorder of the thyroid gland, *goitre*, is a new inclusion in the revised examination guidelines. This may be the reason that this disorder was unfamiliar to many candidates.

### Suggestions for improvement

- (a) It is important to teach learners to write only one letter in multiple-choice questions. If a learner gives more than one letter for a response, he/she should not be awarded any marks.
- (b) School and district-based assessments must cover the full range of instructional verbs to afford learners the opportunity to master the type of responses required. These should include, inter alia, verbs such as 'name', 'state', 'describe', 'explain', 'tabulate' and 'calculate'.
- (c) Learners must be exposed to past question papers, not only to familiarise themselves with the format, but also to familiarise themselves with actual content. The answer to Q 1.2.6 was in Q1.3 of the November 2021 question paper.
- (d) When there is overlapping or confusing content like the *pupillary mechanism* and *accommodation*, it is best to give learners a side-by-side visual – this could be in the form of labelled drawings on a chalkboard and worksheets or a digital presentation.
- (e) Biological terminology can only be mastered through practice after each section is taught. Learners must be made to write these terms out, so that they can perfect their spelling skills.
- (f) Learners should be encouraged to read questions with proper understanding. There may be subtle differences that alter the response required, e.g. Q1.2.1 *the part that protects the brain* could be the *meninges*, but the question asked, *the part of the skull that protects the brain*, which changes the response entirely.

- (g) Learners must strategize their approach to the investigation and data-response questions, e.g. Q1.1.9, Q1.1.0, Q2.3, Q2.5 and Q3.4. They must analyse the source material thoroughly during the allotted reading time.
- (h) Learners must be sensitised to the fact that the diagrams they encounter in the examination may differ slightly from the ones they encounter in the classroom. This should not derail them; the strategy is to find the familiar, and work backwards from there, e.g. in Q1.5 the first step would have been to identify the *thyroid gland* (gland B) and its secretion *thyroxin* (hormone C); thereafter, attempt to answer the questions.
- (i) Teachers should use the information provided by the *Examination Guidelines* and not teach unnecessary content which might be provided in some textbooks, e.g. *Grave's disease* is not a prescribed disorder to be taught and some candidates lost marks when referring to it in Q1.5.3.

**QUESTION 2: THE MALE REPRODUCTIVE SYSTEM, STRUCTURE OF SPERM, MENSTRUAL CYCLE, OVARIAN HORMONES, REPRODUCTIVE STRATEGIES, HOMEOSTASIS OF BLOOD GLUCOSE**

**Common errors and misconceptions**

- (a) Candidates struggled to identify the male *accessory glands* in a diagram. This might be because they had only encountered the front view of the structures and could not identify them in a side view.
- (b) The terms *semen* and *sperm* were used interchangeably by candidates in response to Q2.1.2, even though there is a distinct difference between the two.
- (c) Q2.1.3 required that candidates understand the definition of an *endocrine gland*, by asking two reasons why the labelled structure was *not* an endocrine gland. Most candidates were only able to give one correct reason.
- (d) In Q2.1.5 candidates were not only expected to describe the secretions of the male *accessory glands* but had to explain how the secretions increased the chances of *fertilisation*. Most candidates lost the two marks as they failed to give the explanation.
- (e) Q2.2.4 required candidates to evaluate the suitability of two differently structured *sperm cells* for *fertilisation*. The question asked why sperm 1 was better suited than sperm 2. Candidates had to understand which structures made a sperm cell well-structured for fertilisation and essentially attribute these structures to sperm 1. Candidates struggled to do this, as was the case with most application-style questions.
- (f) Q2.3 and Q2.4 were both based on the female reproductive hormones and the *menstrual cycle*. The average performance in Q2.3 was 22% lower than the candidates' performance in Q2.4. This may be because Q2.3 required the application of content learned to a new scenario, whereas Q2.4 required simple recollection of content.
- (g) In Q2.3 many candidates identified the hormones on the graph as *oestrogen* and *progesterone* because they did not read the stem of the question which stated that the graph shows the levels of *pituitary* hormones. The pituitary hormones were *FSH* and *LH*. Once candidates identified the incorrect hormone, the subsequent questions could not possibly be answered correctly.

- (h) Q2.3.2 and Q2.3.3 required the application of content knowledge to a new scenario which ~~always~~ challenged the average candidate.
- (i) Although Q2.4 was a lower-order question, many candidates failed to obtain maximum marks as their descriptions just fell short of being complete, e.g. candidates were able to identify the *ovarian hormones*, but could not explain their role in the *menstrual cycle*.
- (j) Q2.5 was based on an extract that described the *reproductive strategies* of two fish species and a graph that showed the survival rate of each species. This data-response question is a classic example of how candidates are challenged by higher-order questions. Q2.5.2 required that candidates had to have knowledge of what constituted *oviparous* development and then they had to evaluate why the species represented were considered to be oviparous. This application-style question was challenging for many candidates. Candidates were confusing the location of *fertilisation* with the location of *embryonic development*.
- (k) Many candidates lost marks in Q2.6.2(a) as they incorrectly stated that *insulin* converted *glucose* to *glycogen*, instead of stating that insulin *caused* the *liver* to convert *glucose* to *glycogen*. Furthermore, the terms *glycogen* and *glucagon* are still confused and used interchangeably.
- (l) Poor performance in Q2.6.3 can be attributed to the fact that candidates ignored the comparison to hormone C (*glucagon*) and simply stated the functions of *adrenalin*.

### Suggestions for improvement

- (a) Certain sections of work, especially those that involve structure and function, such as the reproductive system in Q2.1 and the sperm in Q2.2, are best taught using diagrams. Teachers should give learners multiple opportunities to label and annotate diagrams. Furthermore, learners should be given the opportunity to draw the diagrams themselves - this facilitates a better understanding of specific features of structures. Different views of structures must also be made available for learners, e.g. the side view and the front view.
- (b) Classroom activities need to centre on exercises that allow learners to write down a description of a given term, e.g. *semen*, *oviparous*, *ovarian hormones*, *pituitary hormones*, *endocrine glands* etc. This is an exercise that is the corollary of Q1.2, where a description is given in the question paper and the term must be provided as a response.
- (c) The concept of *structural suitability* needs to be emphasised in the classroom as it is an integral concept in Life Sciences. In order to effectively respond to this type of question, the candidate must first describe the feature of the structure itself and then explain how it is suited to its specific function, e.g. the sperm has a long, whip-like tail to facilitate rapid movement.
- (d) Learners must be exposed to more data-response type questions in school-based assessments. Deciphering the hormones on a graph or identifying which data corresponds with theoretical information is a skill that can only be acquired with practice.
- (e) In the classroom distinguish between the *ovum* and the *egg*. The ovum is the unfertilised female *gamete*, whereas the egg is the structure where *embryonic development* occurs.

- (f) Distinction must also be made between the *location of fertilisation* and the *location of embryonic development*. The tables below may provide further clarity:

| Location of fertilisation | Type of fertilisation  |
|---------------------------|------------------------|
| Inside the female body    | Internal fertilisation |
| Outside the female body   | External fertilisation |

| Location of embryonic development                 | Term        |
|---------------------------------------------------|-------------|
| Inside the female body                            | Vivipary    |
| Inside an egg, but <i>outside</i> the female body | Ovipary     |
| Inside an egg, and <i>inside</i> the female body  | Ovovivipary |

Learners must also be taught the features, advantages and disadvantages of all the *reproductive strategies*.

- (g) Informal and formal assessments must incorporate higher-order questions that require learners to analyse and evaluate diagrams, graphs and data tables that they may not have encountered before.
- (h) Teachers must expose learners to more questions that require comparison and differentiation. Learners must be able to identify the deviation from the normal state and describe this change. It is not correct to state that the blood glucose level in a *diabetic* patient is high, as this may be the case in a *non-diabetic* after a meal – it must be stated that the blood glucose level will *remain* high.
- (i) When asked to compare scenarios, structures or effects, learners must be taught that their responses have to be qualitative and comparative, e.g. 'longer', 'faster', 'higher', 'more', etc.

### QUESTION 3: THE BRAIN, DATA RESPONSE (BRAIN INJURIES), THE HUMAN EAR, SCIENTIFIC INVESTIGATION (HOMEOSTASIS) AND PLANT RESPONSES TO THE ENVIRONMENT

#### Common errors and misconceptions

- (a) In Q3.1.1 some candidates confused the *corpus luteum* and the *corpus callosum*
- (b) In Q3.1.2 and Q3.1.3(b) most candidates only obtained 50% of the marks. The instructional verb in these questions was *Explain* and candidates generally presented poorly expressed explanations. In Life Sciences, to assess if candidates understand the function of a structure, a question is posed to indicate what would happen in the absence of the structure. Candidates, therefore, have to start their response with a description of the function of the structure in question and then move on to explain what would happen if those functions ceased. In Q3.1.3(b) the examiners were assessing if candidates understood which part of the brain *controlled* the *skeletal muscles*.
- (c) The skill of drawing of graphs has definitely been well developed over the years and many candidates scored well in this area. Some responses to Q3.2.3 did not get maximum credit due to the:
- Incorrect type of graph being drawn – too many candidates presented a histogram instead of a bar graph;
  - Caption being incomplete or containing only one variable;

- X-axis not being labelled 'Regions of the World';
  - Label on the Y-axis not including a unit;
  - Incorrect plotting of the co-ordinates; and
  - Inconsistent drawing of the bars.
- (d) Q3.3 highlighted the fact that candidates were capable of responding to lower-order questions based on a source but faltered when the questions demanded a higher cognitive ability. Q3.3.1; Q3.3.2 (a) and Q3.3.2 (b) were well answered, but candidates struggled with Q3.3.3 and Q3.3.4. The following common errors were observed:
- In Q3.3.3 most candidates assumed that the build-up of wax was in the *auditory canal* as was the case in previous question papers. This question, however, referred to a build-up of wax at the *tympanic membrane*. Candidates referred to the inability for sound waves to pass through to the tympanic membrane, rather than the inability of the tympanic membrane to vibrate.
  - Q3.3.4 was very poorly answered by most candidates. The understanding of various disorders and their treatments was an area that was unfamiliar to most candidates.
  - Candidates were unable to distinguish between the stimuli in each division of the ear. They used the terms *sound waves*, *vibrations* and *pressure waves* interchangeably.
  - In Q3.3.5 most candidates were able to name the *receptors* involved as the *cristae* but could not explain their role in the maintenance of balance. Also, when referring to *receptors*, candidates failed to identify their role in converting a *stimulus* into an *impulse*.
  - Many candidates failed to include the role of the *auditory nerve*, *cerebellum* and the *skeletal muscles* in the maintenance of balance. Some candidates also named the *auditory canal* instead of the *auditory nerve*.
- (e) In Q3.4.1 many candidates gave an incomplete description of the variables. In Q3.4.1(a) they simply wrote *facemasks* instead of *the wearing of facemasks* and in Q3.4.1(b) wrote *CO<sub>2</sub> levels* instead of *CO<sub>2</sub> levels in the blood*.
- (f) Candidates who did not score well in Q3.4.3, Q3.4.4 and Q3.4.5 gave generic responses instead of responses that are specific to the investigation. Candidates were expected to analyse the design of the investigation given and critically comment on it. The generic responses of *to increase sample size*, *repeat the investigation* and *to ensure validity* were not acceptable responses in this context.
- (g) Q3.4.6 was a lower-order question and candidates should have scored well, but did not. This question was based on the *negative feedback mechanism*, which clearly still posed a challenge to the candidates.
- (h) From the responses obtained in Q3.5.1, it was clear that candidates knew the location of *auxins* in *stems*, but not their exact location in *roots*.
- (i) The responses to Q3.5 provided further evidence that candidates coped well with lower-order questions based on a source but struggled when the questions based on the same source had a higher cognitive demand.
- (j) In Q3.5.2 candidates could not recognise that *auxins* caused an increase in stem length by stimulating *cell division* and *cell elongation*. This might seem an obvious deduction, but candidates failed to express their responses correctly.

- (k) Q3.5.4 was poorly answered as it required candidates to extrapolate from the extract given. The term *propagation* was explained in the extract and candidates had to make the link to *nature conservation*. A Life Sciences learner at Grade 12 level should be *au fait* with the concept of nature conservation.

### Suggestions for improvement

- (a) The structure of the brain is best taught using annotated diagrams. Learners must be issued with unlabelled diagrams of the brain. They must label all the relevant structures and write in the respective functions alongside them.
- (b) The skill of active reading must be reinforced. In Q3.1.3(b), the operative word was *control*. When engaging with the stem of the question and the question itself, candidates must highlight or underline potentially operative words or phrases.
- (c) Learners must be given the skills to respond to higher-order questions. Examiners often describe the removal or damage of a structure and candidates must then explain the consequences of this, e.g. Q3.1.2 describes damage to part C. To respond to this application-type questions, candidates must:
- First identify the structure and name it;
  - State its function under normal circumstances;
  - Describe the consequences in the absence of this function;
  - Describe how the body may respond to compensate for the change (where applicable).
- (d) Learners' graphing skills need to be refined. Learners must be given exercises that require them to draw *all* graph types. Teachers must then provide learners with the marking rubrics, copied from the *National Marking Guidelines*, and allow for peer-marking and self-assessment. Once learners see the necessary criteria, they will be able to achieve them in the examination.
- (e) The distinction between the different stimuli for each part of the ear must be emphasised in the classroom. The following summary could assist in this regard:

| Region of the Ear | Stimulus       |
|-------------------|----------------|
| Outer ear         | Sound waves    |
| Middle ear        | Vibrations     |
| Inner ear         | Pressure waves |

- (f) Learners must be cautioned against pre-empting the question and actively read what is required, e.g. in past examination papers the effect of *cerumen* build-up in the *auditory canal* was required; in this question paper, the question asked the effect of *cerumen build-up* at the *tympanic membrane*.
- (g) Although the essay question no longer features in the examination, learners must still be trained in extended writing. Marks are not given for merely listing the terms and phrases. Responses have to be described or explained concisely to receive full credit.
- (h) When delivering content on *homeostasis*, teachers need to consolidate and conduct baseline assessments on the following Grade 11 topics:
- Control of *blood glucose* levels;
  - Regulation of *blood carbon dioxide* levels;
  - Water balance (*osmoregulation*);
  - Control of salt levels in the blood.

- (i) The concept of *negative-feedback mechanisms* has application across a number of topics and its consolidation is therefore pivotal to improved learner performance. The mechanics of a *negative feedback loop* needs to be workshopped by subject advisors. This is clearly a topic about which many teachers do not feel confident. Learners should not only know how to describe the *negative-feedback mechanism* but should also be exposed to how these feedback mechanisms work in real situations. In *negative-feedback*, a change of a variable causes the body to effect a correcting change in the opposite direction.

The following summary could be used and adapted to each scenario as is applicable:

|      | <b>Stimulus or variable</b>             | <b>Receptor</b>                                                                              | <b>Control Centre</b>               | <b>Effector(s)</b>                                | <b>Effect</b>                                                     | <b>Outcome</b>                                                                                               |
|------|-----------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| e.g. | Low CO <sub>2</sub> levels in the blood | Chemoreceptors in the carotid artery are stimulated and convert the stimulus into an impulse | The Medulla Oblongata is stimulated | - Heart and<br>- Breathing Muscles are stimulated | - Increased heart rate<br>- Increased rate and depth of breathing | More CO <sub>2</sub> is taken to the lungs and exhaled.<br><br>Decreased CO <sub>2</sub> levels in the blood |

- (j) Teachers should focus on *all* the functions of *auxins* in plants, i.e. *phototropism*, *geotropism* and *apical dominance*. Wherever possible, the investigations involving plant responses to the environment should be done. If not, then learners must be exposed to the practical design of these experiments through worksheets, notes, slides and past examination papers. Learners must be familiar with the use of *coleoptiles* and *clinostats* in the design of these investigations.
- (k) Teachers must place more emphasis on scientific investigations and their design. Learners should be taught the value of each design element and how it contributes to a valid investigation. The use of generic responses in investigations must be discouraged and candidates need to respond to the specific context of the investigation presented. Every formal assessment should assess Specific Aim 2, to familiarise learners with the scientific skills.
- (l) Learners must be able to identify variables using the aim of the investigation and not the results. Furthermore, they must be able to express them in full, e.g. *carbon dioxide levels in the blood*, not just *carbon dioxide levels*.
- (m) Candidates must be able to present, inter alia, the 'aim', 'results', 'conclusion' etc. for an investigation.
- (n) A greater emphasis on practical work and practical tasks of good quality in Grades 10 and 11 will also assist in preparing learners more adequately for questions based on scientific investigations. Learners must be challenged to design their own investigations, given an 'observation', 'hypothesis' or 'aim'. This is especially important since knowledge of scientific investigations is assessed in both Paper 1 and Paper 2.

## 8.5 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 2

### General comments

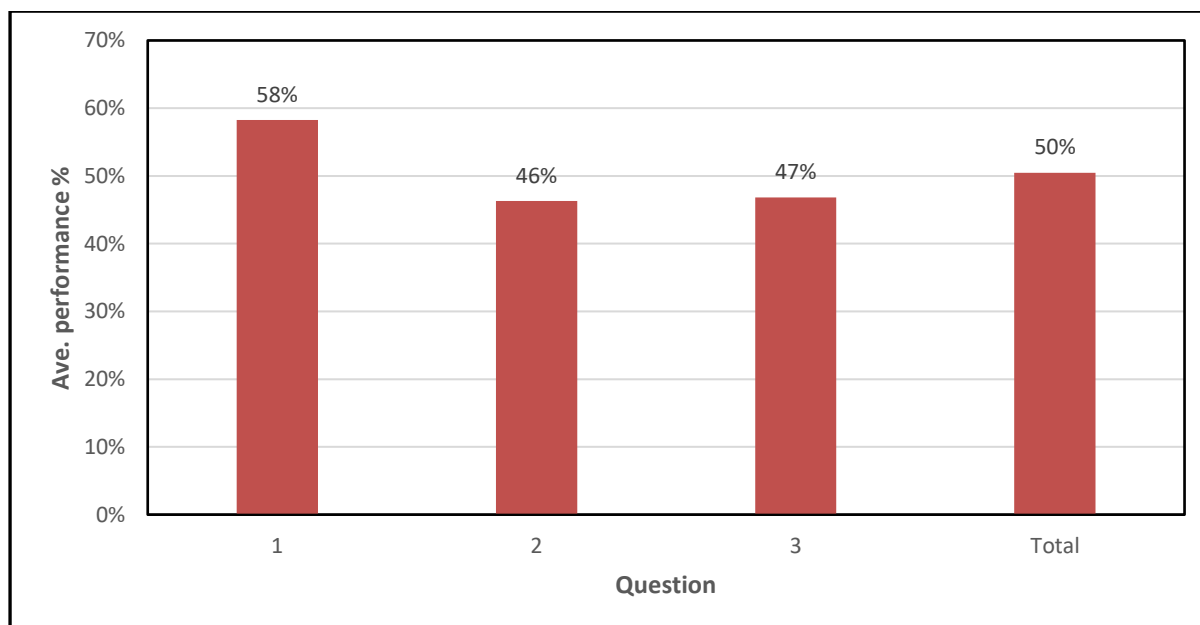
- (a) The poor responses of some candidates emanated from their not reading and following instructions in each question/subquestion.
- (b) Learners should be provided with the official *2021 Examination Guidelines* before a particular topic is taught. This will give them clear guidance on the terminology to be mastered per subtopic, as well as the content that they should know and learn for examinations. Candidates often refer to terminology, e.g. *point mutations*, that is outside the parameters of the *Examination Guidelines* and the *CAPS* document.
- (c) Teachers must always consult Chief Markers' reports and Diagnostic Reports and apply all the necessary recommendations in their teaching.
- (d) Teachers must emphasise the schematic outline of the human life cycle to show the role of *meiosis*, *mitosis* and *fertilisation* as part of the introduction to human reproduction. This will assist learners to understand the concept of prevention of the *doubling effect of fertilisation* and the *maintenance of constant chromosome number* from generation to generation.
- (e) Many candidates underestimate the importance of correct spelling. In Life Sciences, incorrect spelling may change the meaning of the response and candidates may lose marks, e.g. *cross over* cannot be accepted as the correct terminology for *crossing over*.
- (f) Teachers should emphasise the difference between commonly confused terms, e.g. the difference between *centromere*, *centrosome* and *centriole*.
- (g) Candidates give generic responses to questions that require specific information on questions, e.g. *natural selection*.
- (h) Candidates must always use the correct genetic notations given in the paper and not their own, e.g.  $X^B$  and not  $X^D$ .
- (i) Teachers should re-visit the IM report as well as the Diagnostic Report when starting to teach a new topic.
- (j) The 2020 and 2021 Diagnostic Reports are also an invaluable tool in planning and teaching content.
- (k) Subject advisors should organise more content workshops (every term) to share best practices and to update teachers who were not at the marking centres, on the latest developments in the subject.

## 8.6 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 2

Based on the Rasch analysis, the weakest performance by candidates was recorded in the subquestions on *Transcription*, *Gene Mutation*, the *Sex-linked Monohybrid Cross* and *Evolution*.

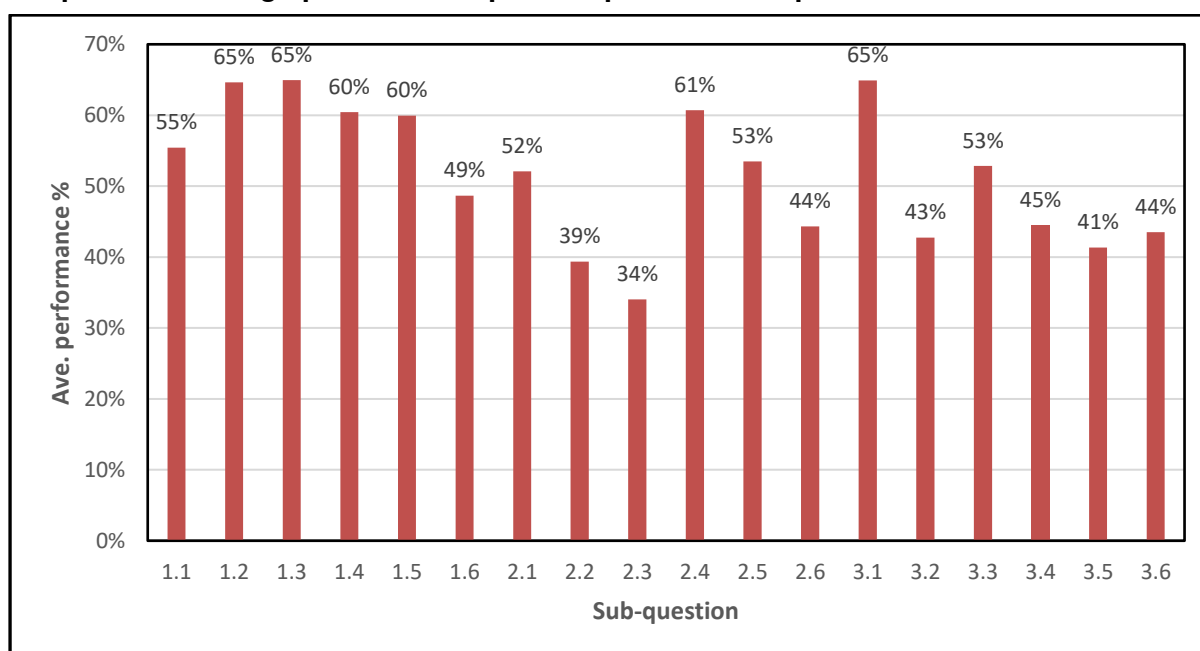
The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 8.6.1 Average performance per question in Paper 2**



| Q | Topics                                                                                                                                                                                                      |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Multiple choice, Terminology, Matching items, Pedigree diagram, Karyotype – abnormal meiosis, dihybrid cross                                                                                                |
| 2 | Transcription, Mutation on mRNA, Meiosis, Blood groups, DNA profile, Monohybrid cross (sex linked)                                                                                                          |
| 3 | Stem cells, species – reproductive isolation, human evolution, phylogenetic tree, evolution – transitional species and Lamarck, scientific investigation - evolution in present times and natural selection |

**Graph 8.6.2 Average performance per subquestion in Paper 2**



| Sub-Q | Topic                        | Sub-Q | Topic                                                                       |
|-------|------------------------------|-------|-----------------------------------------------------------------------------|
| 1.1   | Multiple-choice question     | 2.4   | DNA profile                                                                 |
| 1.2   | Terminology                  | 2.5   | Blood groups                                                                |
| 1.3   | Matching items question      | 2.6   | Monohybrid cross (sex linked)                                               |
| 1.4   | Pedigree diagram             | 3.1   | Stem cells                                                                  |
| 1.5   | Karyotype – abnormal meiosis | 3.2   | Species – reproductive isolation                                            |
| 1.6   | Dihybrid cross               | 3.3   | Human evolution                                                             |
| 2.1   | Transcription                | 3.4   | Phylogenetic tree                                                           |
| 2.2   | Mutation on mRNA             | 3.5   | Evolution (transitional species and Lamarck)                                |
| 2.3   | Meiosis                      | 3.6   | Scientific investigation - evolution in present times and natural selection |

## 8.7 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### QUESTION 1: MULTIPLE-CHOICE, TERMINOLOGY, MATCHING ITEMS, PEDIGREE DIAGRAM, KARYOTYPE – ABNORMAL MEIOSIS AND DIHYBRID CROSS

#### Common errors and misconceptions

- (a) In Q1.1 candidates performed well, except for Q1.1.8 and Q1.1.9. Candidates could not differentiate between the events of *meiosis I* and *mitosis*. Q1.1.9 was a calculation based on the application of *complementary-base pairing*. Many candidates failed to read the information given in the stem of the question.
- (b) Spelling errors in Q1.2 caused candidates to lose marks. Candidates frequently:
  - Confused the terms *centromere* and *centrosomes/centrioles* in Q1.2.3 and Q1.2.6;
  - Confused *homologous structure* with *homologous chromosomes*. They even gave answers such as *modification by descent* and *analogous structures* in Q1.2.7;
  - Stated the site of *translation* in a cell instead of the *organelle* in which translation occurs in Q1.2.9.
- (c) Q1.4.2(b) on the pedigree diagram was not well answered. Some candidates incorrectly wrote the format required for a sex-linked trait as  $X^R X^r$  instead of the correct *autosomal trait annotation* of Rr.
- (d) In Q1.5.1 and Q1.5.2 some candidates could not differentiate between *autosomes* and *gonosomes*. In Q1.5.3 they also confused the *disorder* (Down syndrome) with the *process* causing it (*non-disjunction*).
- (e) Q1.6 on the *dihybrid cross* was the most poorly answered question in Section A. Many candidates could not give the *phenotype* correctly. They had to write the complete characteristic namely *Brown fur rabbit with long ears* but only wrote *brown* and *long* which were not credited. In Q1.6.2(c) candidates confused the *genotype of an individual* with the *genotype of a gamete*. Many candidates wrote the correct *allele* combination but did not leave a space to indicate that the alleles were separated into the gametes.

**Suggestions for improvement**

- (a) Learners should be provided with the *2021 Examination Guidelines* before a particular topic is taught. This will give them clear guidance on the terminology to be mastered per subtopic, as well as the content that they should know and learn for the examinations.
- (b) Teachers should extract topic-focused questions from previous years' question papers and use these in informal assessments. This will enable learners to acquire question interpretation skills and to mitigate against confusing related terminologies.
- (c) Teachers must be trained on how to approach the topic of *meiosis*, highlighting the events that take place during *meiosis I* and *II* as well the differences between the two divisions.
- (d) Teachers must give learners more exercises on *dihybrid* crosses. They must emphasise the difference in the writing of the *genotypes* and *gametes* for a *dihybrid cross* as compared to how they would write it for a *monohybrid cross*. Genotypes of parents, gametes and offspring should not be confused. Teachers should emphasise the difference between *genotype of an organism* and the *genotype of their gametes*.
- (e) Teachers must use diagrams, including that of a *karyotype*, when teaching *abnormal meiosis* so that learners can relate the terminology to the actual process. Emphasis should be on *Down syndrome* being a consequence of *non-disjunction* and that the terms cannot be used interchangeably.
- (f) Teachers should make an effort to pronounce terms correctly. Often learners are spelling the words as they hear them. Learners should practise writing out the terminology.
- (g) There is clear evidence pointing to the fact that candidates studied their content, but lost marks due to poor examination skills. Thorough analysis of the question phrasing and mark allocation must be taught as a skill to the learners.
- (h) Learners must be exposed to past question papers, not only to familiarise themselves with the format, but also to familiarise themselves with actual content.

**QUESTION 2: TRANSCRIPTION DURING PROTEIN SYNTHESIS, MUTATIONS ON A MRNA MOLECULE, MEIOSIS, BLOOD GROUPS, DNA PROFILE, SEX-LINKED MONOHYBRID CROSS**

**Common errors and misconceptions**

- (a) The question was generally answered fairly well, although there seemed to be challenges in questions that needed explanations and descriptions.
- (b) In Q2.1.3 some candidates did not present their answers in the form of a table. They wrote the whole process of *transcription* and could not extract the differences in the

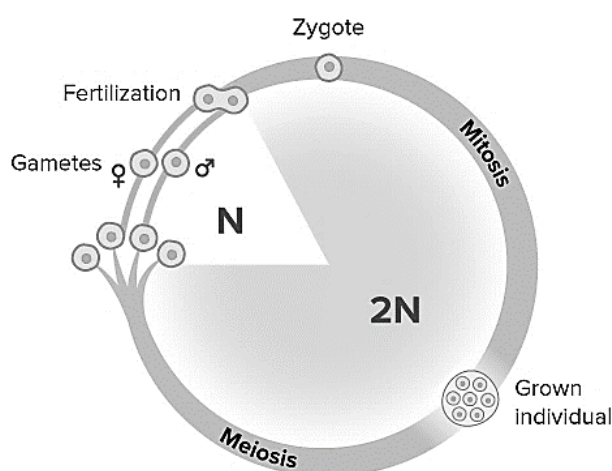
processes. They gave general differences between RNA and DNA. Also, for *DNA replication*, they indicated that an identical strand of DNA is formed instead of *two identical DNA molecules*. Some wrote a *double stranded molecule* is formed which refers to one DNA molecule.

- (c) In Q2.2.1 many candidates answered the type of mutation incorrectly as a *genetic mutation* rather than a *gene mutation*. They also referred to the causes of gene mutations, such as point mutation which, although applicable to this question, was outside the parameters of the *2021 Examination Guidelines* and the *CAPS* document.
- (d) In Q2.2.2 and Q2.2.3 (c) candidates gave general answers, not specific to the question. Instead of describing the exact change and position of the mutation that occurred in the *codon/anticodon*, they generalised without referring to the specific *codons/anticodons*. Some knew that there was a change in amino acids but did not specify the amino acids that changed.
- (e) Many candidates incorrectly wrote 6 as an answer to Q2.2.3(a). They counted the number of amino acids but did not exclude the repeated amino acids, which were coded twice by 2 different codons.
- (f) In Q2.3.2 candidates incorrectly stated that the *gamete or sperm cell* underwent *meiosis* rather than *diploid cells underwent meiosis to form the sperm*.
- (g) Many candidates wrote in Q2.3.4 that *chromosomes* were pulled to the opposite poles without indicating that the *homologous pairs* first separate or are pulled apart, and then move to the opposite poles. Some answered that *spindle fibres constrict* which showed that they confused the terms *contract* and *constrict*.
- (h) Poor performance in Q2.4.3 was evidence that many candidates could not distinguish between *alleles* and *genes* and they were not familiar with the concept of *multiple alleles* in blood grouping. They could identify that the child was blood group AB, but failed to describe how the child inherited this blood group. They incorrectly stated that the child *inherited blood group A from the mother* instead of stating that the child *inherited the allele ( $I^A$ ) for blood group A from the mother*.
- (i) In Q2.5.2 many candidates wrote that the *DNA matched* instead of saying that the *DNA profile bands/bars of the children matched with that of their parents*.
- (j) In Q2.5.3 many candidates failed to correctly state other uses of DNA profiling than those mentioned in the question, i.e. paternity testing and identification of relatives. Some wrote *to develop cures for genetic disorders* instead of *identifying genetic disorders*. Some wrote *to identify criminals* or *to solve crimes* instead of *identifying suspects in crime*. They also wrote *for organ transplants* instead of *identification of matching tissues for organ transplants*.

- (k) In Q2.6.1 candidates could not deduce that *white teeth* was caused by a *recessive allele* because *brown teeth* was caused by a *dominant allele*. Many explained their response in terms of the dominant allele causing brown teeth rather than the recessive allele.
- (l) In terms of Q2.6.2 some candidates had difficulty stating the *phenotype* in the *monohybrid cross*. They had to write that *a male with brown teeth was crossed with a female with white teeth* and not *just brown teeth were crossed with white teeth*.
- (m) Some candidates gave their own genetic notations, e.g.  $X^D$  when showing the genetic cross.
- (n) Candidates gave the phenotypic ratio as *2 females with brown teeth : 2 males with white teeth* instead of stating *the smallest ratio of 1 : 1* and lost the compulsory mark.

### Suggestions for improvement

- (a) Learners should be given sufficient exercises on how to do *base-pairing* in protein synthesis from *DNA* → *mRNA* (codons) → *mRNA* (anticodons) → amino acids as well as in the reverse process.
- (b) Learners must only be taught what is required with reference to *chromosome* and *gene* mutations, as stipulated in the *2021 Examination Guidelines* and the *CAPS* document. Further details on gene mutations such as *point* and *frame-shift mutations* are not required.
- (c) Teachers should expose learners to questions on genetics, based specifically on *sex-linked inheritance* where *dominant/recessive alleles* are involved.
- (d) Learners should clearly understand the *haploid* and *diploid* status of cells. They must be able to differentiate whether the question refers to the *halving of the chromosome number during meiosis* or to the *duplication of chromosomes during fertilisation*. A cyclical diagram like the one shown below could help to consolidate this content.



- (e) Learners must be taught to use the correct terminology when interpreting DNA profiles, e.g. *bars*, *bands* and not *lines* or *barcodes*. They must emphasise that for a paternity

test, each band of the child's DNA profile should match with either that of the biological mother or that of the biological father.

- (f) In a criminal case, however, the DNA profile of the suspect must be an exact match of the DNA profile taken from the crime scene as evidence. Teachers should emphasise that using language precisely is important in the answering of questions, e.g. criminals are not identified or traced using DNA profiles because a suspect must first be arrested and identified before they are convicted and become criminals.
- (g) When asked to explain why males/females have more/less of a specific trait, the *gonosomes* or the *recessive/dominant allele* must be used in the explanation.
- (h) Learners must use the correct notations given for a genetic cross, i.e.  $X^B$  and not  $X^D$ .
- (i) When discussing the inheritance of traits, learners must state that *alleles* are inherited and not *genes*. Refer to page 163 of the 2021 *Diagnostic Report* which outlines the general format of how to answer this type of question.
- (j) Generic answers from previous 2021 *Examination Guidelines* cannot be given when a question provides contextual data. Teachers must ensure that learners practise adapting the generic versions to the specifics in the questions.

**QUESTION 3: STEM CELLS, SPECIES, POPULATIONS AND REPRODUCTIVE ISOLATION, HUMAN EVOLUTION, PHYLOGENETIC TREE, EVOLUTION (TRANSITIONAL SPECIES AND LAMARCK) AND SCIENTIFIC INVESTIGATION (EVOLUTION IN PRESENT TIMES AND NATURAL SELECTION)**

**Common errors and misconceptions**

- (a) In Q3.1.1 some candidates lacked the skill to extract information from the text. Instead of naming the sources of *stem cells*, some rewrote the paragraph as it was. In Q3.1.2 they also referred to using stem cells for replacing organs instead of cells/tissues.
- (b) In Q3.2.1 many candidates failed to give all three parts of the definition of a *population* and mixed up the definition of a *population* with that of a *species* or a *community*.
- (c) In Q3.2.3 candidates lost marks for describing examples of *reproductive isolation* instead of listing the *mechanisms of reproductive isolation*. Many candidates listed examples of reproductive isolation that were not in the 2021 *Examination Guidelines*.
- (d) In Q3.3.1 candidates wrote *to share a common ancestor* instead of *to show a common ancestor*. Candidates stated that opposable thumbs are for *holding things* rather than *providing a power grip* or *a precision grip* in Q3.3.2.
- (e) The following common errors were observed in Q3.4 on the evolution of some *hominids*:
  - Some candidates could not spell *phylogenetic* and were not credited for writing *polygenetic tree* or *phylogenic tree* in Q3.4.1.
  - In Q3.4.2 many candidates did not know the difference between a *genus* and a *species*. They counted the number of species instead of counting the *genera*. Many did not know that the term *genera* is the plural form of *genus*. Some gave the number of generations instead.

- Many candidates only wrote the species name instead of the full scientific name of the genus and species in Q3.4.4.
  - In Q3.4.5 some candidates confused evidence to support the theory of evolution versus evidence to support trends in human evolution. Some of the responses given included *modification by descent*, *biogeography* and *genetic evidence from mitochondrial DNA*. Many gave examples of evidence for evolution, e.g. *mitochondrial DNA* and rock paintings and artefacts, instead of *genetic* or *cultural* evidence.
  - In Q3.4.7 many candidates were not clear on how fossil evidence supported the *Out of Africa hypothesis*. The *2021 Examination Guidelines* state which fossils were only found in Africa and which oldest fossils are found in Africa implying that the youngest have been found outside of Africa. Many wrote *fossils/organisms originated in Africa* instead of *modern humans originated in Africa*.
- (f) In the responses to Q3.5 candidates were able to identify *Ambulocetus* as a *transitional species* between *Pakicetus* and whales but could not explain why this is the case. This indicated that they were not familiar with the characteristics of a transitional species.
- (g) In Q3.5.2 some candidates failed to extract the relevant characteristics from the table to support their answer. In Q3.5.1 they omitted *flipper-like* or *large* in describing the feet of *Ambulocetus*.
- (h) In Q3.5.4 many candidates referred to *Lamarck's law of use and disuse* but could not apply it to the given question. Most wrote that *the legs were not needed* instead of *the legs were used less*.
- (i) In Q3.6.1 some candidates could not extract the variables from the aim of the investigation. This was thoroughly discussed in the *2021 Diagnostic Report*.
- (j) In stating the conclusion in Q3.6.3, many candidates could not show the relationship between the independent and dependent variable, i.e. describe how the change/increase in the number of missed treatments affected the probability of the HI virus developing resistance to antiretroviral medication. Some incorrectly described the relationship between the variables as being directly proportional, implying that the increments were equal, which was not the case when analysing the given data. The conclusion often did not include all aspects (variables, virus name, drug name and the relationship shown).
- (k) Most candidates had difficulty answering Q3.6.4. This question was based on a scientific investigation into evolution in present times; candidates instead focussed on the resistance of the HI virus to antiretroviral drugs. They gave a generic description of *natural selection* without contextualising it. They also mentioned variation without describing the variation as it applied to the resistance of the HI virus to antiretroviral medication.

### Suggestions for improvement

- (a) Instructional verbs such as 'state', 'explain' and 'discuss' should be thoroughly explained to the learners. Learners should be exposed to questions requiring them to discuss/explain to enable them to know the difference.
- (b) Learners must be cautioned against using slang language when writing examinations. No marks are awarded for writing *K-nines* instead of *canines*.

- (c) Open-ended questions should be asked in both formal and informal tasks for learners to understand that where their opinions are asked, these should be relevant.
- (d) Teachers must use questions from past papers to expose learners to exam-type questions pitched at different cognitive levels and levels of difficulty and not only rely on textbook activities.

The principles of taxonomy need to be thoroughly covered in Grade 10 in preparation for *human evolution* in Grade 12. The terms *genus*, *species* and *genera* as the plural of *genus* must be consolidated thoroughly. Learners must be instructed in scientific nomenclature; that when a species name is requested that both the *genus* and species must be named.

- (e) Teachers must teach the concept that a *transitional fossil* displays the intermediate phenotype between a predecessor (which comes before it) and the current organisms of which it is part OR that it has the intermediate phenotype of those organisms that it is currently part of and a descendant (which comes after it). This knowledge should then be applied to information given in the source-based question.
- (f) Teachers must support learners to learn the sequence of steps describing *natural selection* found in the *2021 Examination Guidelines*. Learners should also be able to apply this generic process to the source material provided. Special care needs to be taken to tailor their answers to the context of the question given.
- (g) Application questions on the process of *natural selection*, and also on Lamarck's theories, should be extracted from previous years' question papers and given to learners for daily assessment so that they do not provide generic responses to questions that require application.
- (h) Teachers must emphasise that the conclusion of an investigation would come from altering the aim to show the relationship between the variables.

# CHAPTER 9

## MATHEMATICAL LITERACY

The following report should be read in conjunction with the Mathematical Literacy question papers of the November 2022 Examinations.

### 9.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who sat for the Mathematical Literacy examinations in 2022 increased by 8 938 compared to that of 2021.

The table below indicates an upward trend over the years 2018-2020 followed by a decline in 2021. However, a substantial improvement in results was evident this year.

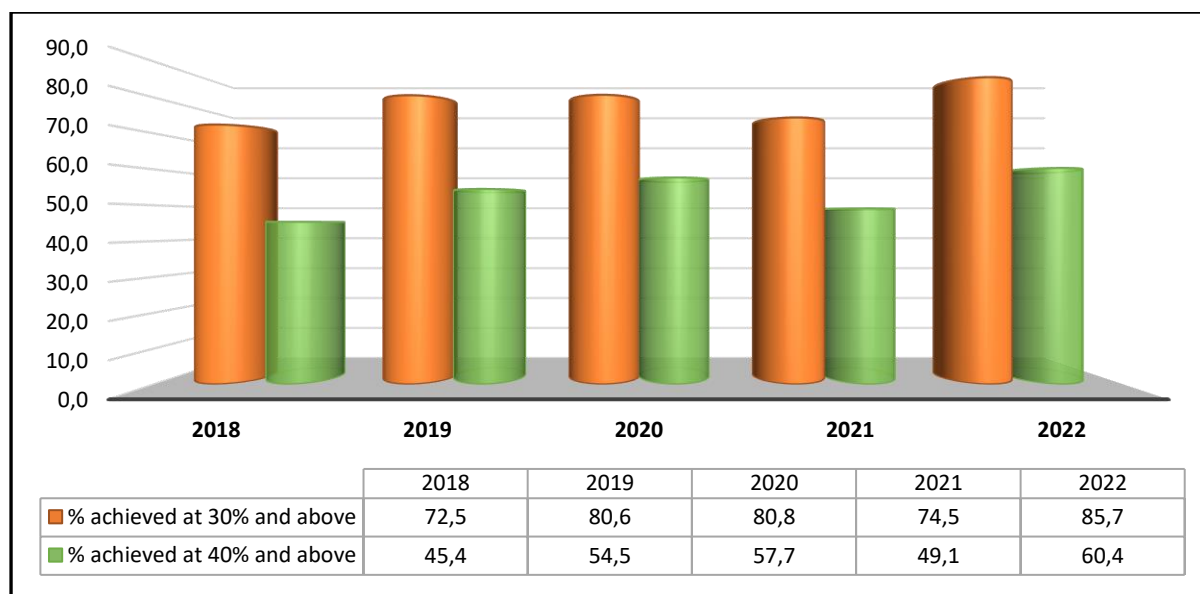
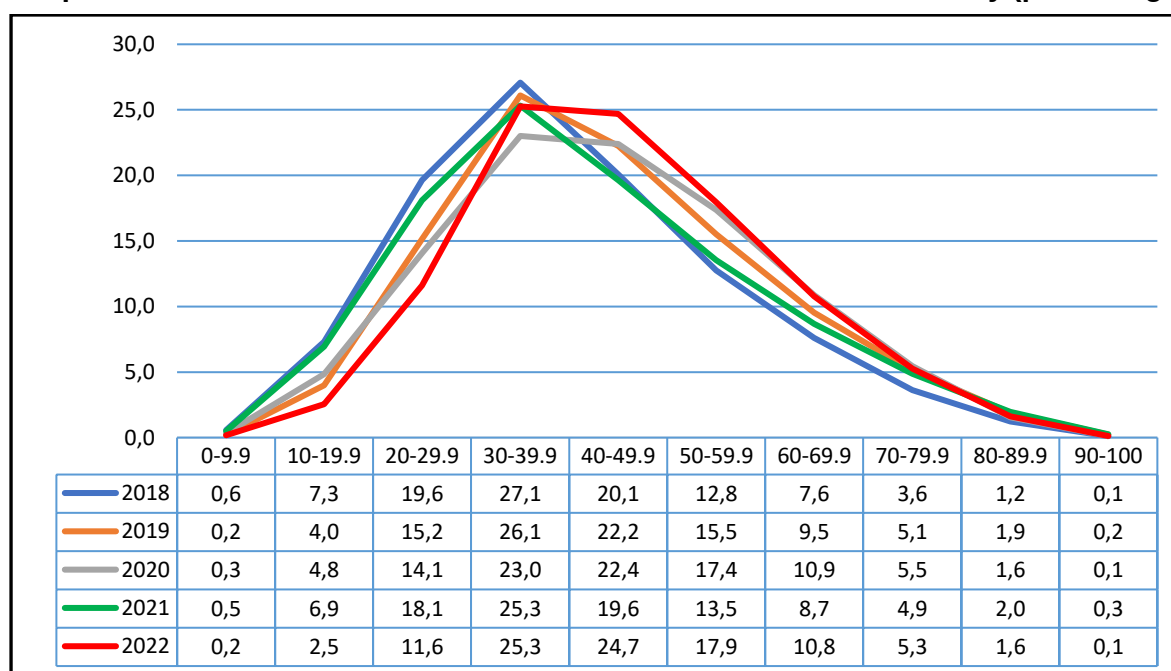
Candidates who passed at 30% (Level 2) improved substantially from 74,5% in 2021 to 85,7% in 2022. There was also a corresponding improvement in the pass rate at 40% (Level 3) over the past two years from 49,1% to 60,4%.

However, the percentage of distinctions over 80% (Level 7) declined from 2,3% to 1,7%. This converts into a decrease of 2 495 in the total number of distinctions.

In view of disruption to academic programmes in 2020 and 2021, and despite the factor distinction factor mentioned above, the results achieved by this cohort are particularly commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the overall performance.

**Table 9.1.1 Overall achievement rates in Mathematical Literacy**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 294 204   | 213 225                       | 72,5                        | 133 568                       | 45,4                        |
| 2019 | 298 607   | 240 816                       | 80,6                        | 162 877                       | 54,5                        |
| 2020 | 341 363   | 275 684                       | 80,8                        | 197 131                       | 57,7                        |
| 2021 | 441 067   | 328 382                       | 74,5                        | 216 692                       | 49,1                        |
| 2022 | 450 005   | 385 515                       | 85,7                        | 271 830                       | 60,4                        |

**Graph 9.1.1 Overall achievement rates in Mathematical Literacy (percentage)****Graph 9.1.2 Performance distribution curves in Mathematical Literacy (percentage)**

## 9.2 GENERAL COMMENTS ON PAPER 1 AND PAPER 2

- (a) **Terminology:** *English Across the Curriculum* should be emphasised. Learners should be taught the definitions of commonly used terms in Mathematical Literacy such as *radius* and *median*. Learners should compile a topic-wise glossary of terms at the back of their notebooks with a brief but clear definition next to each term. A separate notebook may also be kept for this purpose. By the end of the year, all learners should have a comprehensive glossary of all the relevant terms.
- (b) **Enhance learners' skills in interpreting level 4 questions and using information that is relevant:** Teachers are advised to read through and interpret the requirements of each question with learners. Learners should also be guided on how to extract

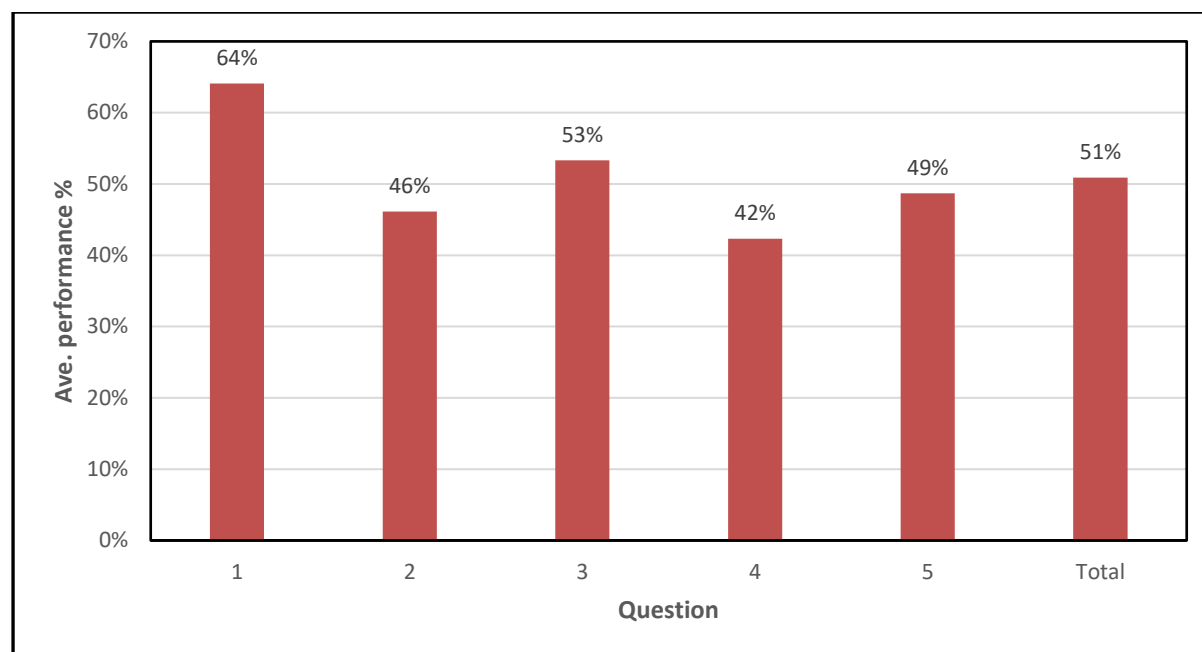
relevant information and how to identify the information that is relevant to each subquestion.

- (c) **Use of past NSC papers:** Firstly, it must be noted that past examination question papers should not be used to teach new content. They must be used for revision purposes only. Past papers cannot replace the CAPS document and *2021 Examination Guidelines*. Teachers can, however, adapt certain questions for use in class. Secondly, teachers should ensure that learners revise questions that define mathematical terms, especially in each context.
- (d) **The importance of formative testing:** Short, informal formative tests must be used to build the confidence of learners in all topics. If learners do their own corrections, it provides them with immediate feedback and an understanding of the mark allocation. The less challenging sections in each of the questions in the NSC Mathematical Literacy papers, particularly Questions 1, 2 and 3 can be used as confidence-boosters.
- (e) **Previous recommendations:** Teachers should consult past Diagnostic Reports to establish if there are topics or concepts that are repeatedly indicated as problematic to most learners. For example, it has been noted over time that learners' basic mathematical knowledge is problematic, and this includes learners' inability to work with large numbers or understand the concept of time.

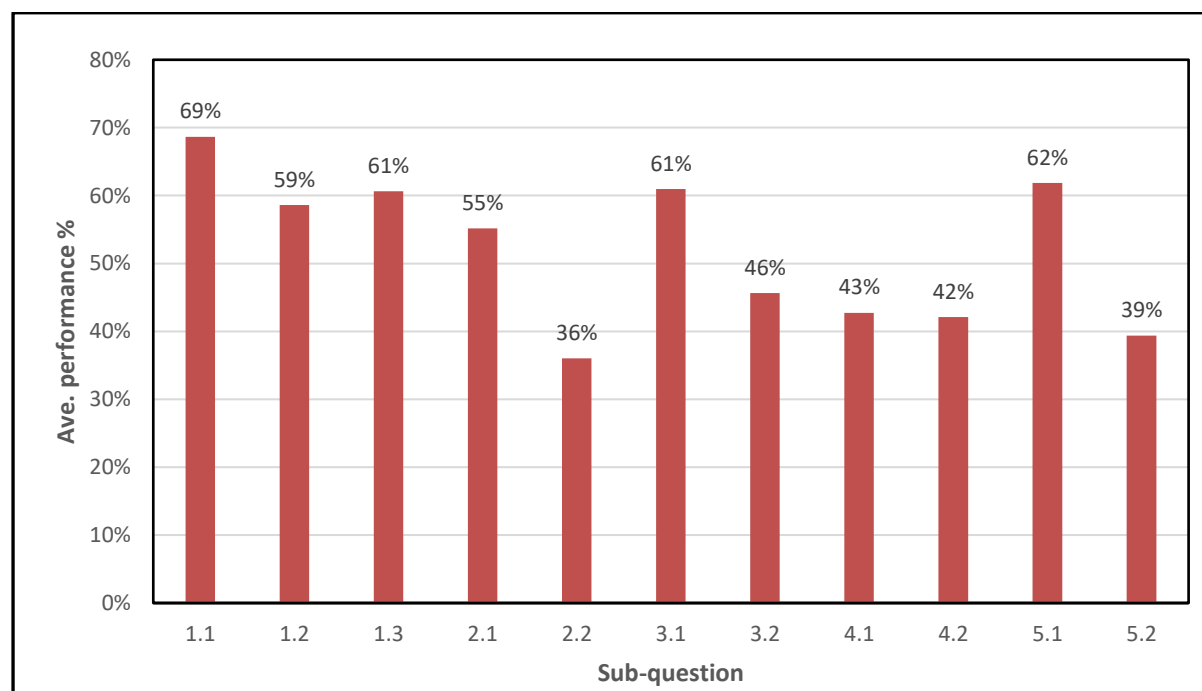
### 9.3 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 1

#### General comments

- (a) The 2022 question paper was set according to the current *2021 Examination Guidelines*. Application Topics tested in Paper 1 are Finance, Data Handling and Probability. Q1 was based entirely on short contexts with all questions pitched at level 1.
- (b) Teachers are advised that the format of the examination paper will not change in 2023. Teachers are encouraged to use the exemplar papers of 2021, November 2021, May/June 2022 and the November 2022 papers as a guide.
- (c) The *2021 Examination Guidelines* need to be studied by the teacher and be made available to Grade 12 learners at the beginning of the academic year.

**Graph 9.3.1 Average performance per question in Paper 1 2022**

| Q | Topic                                  |
|---|----------------------------------------|
| 1 | Finance, Data Handling and Probability |
| 2 | Finance                                |
| 3 | Data Handling                          |
| 4 | Finance, Data Handling and Probability |
| 5 | Finance and Data Handling              |

**Graph 9.3.2 Average performance per subquestion in Paper 1 2022**

| Sub-Q | Topic                                  | Sub-Q | Topic                                  |
|-------|----------------------------------------|-------|----------------------------------------|
| 1.1   | Finance, Data Handling and Probability | 3.2   | Data Handling                          |
| 1.2   | Finance                                | 4.1   | Finance and Probability                |
| 1.3   | Data Handling and Finance              | 4.2   | Data Handling, Finance and Probability |
| 2.1   | Finance                                | 5.1   | Data Handling and Finance              |
| 2.2   | Finance                                | 5.2   | Finance                                |
| 3.1   | Data Handling                          |       |                                        |

## 9.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 1

The three Application Topics and the sequence of questions, where Q1 was based on short, contextual questions, benefitted candidates. Q1 was the best answered question.

### QUESTION 1: SHORT CONTEXTS (INTEGRATED LEVEL 1 QUESTIONS ONLY)

#### Common errors and misconceptions

- (a) Candidates could not identify that the information given in the table was an example of Numeric or Categorical Data in Q1.1.1.
- (b) In Q1.1.6(a) candidates could not define the term *probability in context*; instead, they gave a *general definition of probability*.
- (c) Some candidates could not convert a decimal value to a percentage in Q1.1.6(b).
- (d) In Q1.2.1 candidates could not give the correct definition for the word *investment*.
- (e) Candidates calculated the interest of plan A and the interest of plan B in Q1.2.3 instead of calculating the difference of the values given in Plan B.
- (f) Some candidates did not understand the meaning of *difference* in Q1.2.4 in a mathematical sense, in that they did not subtract the values from each other, although they were able to identify the values.
- (g) In Q3.1.1 candidates could identify the graph as a bar graph, but struggled to name the specific type.
- (h) Candidates could not round the answer to the nearest 50 cents in Q1.3.3; most rounded to the nearest 10 cents.

#### Suggestions for improvement

- (a) The glossary on terminology provided by the DBE and Provincial Education Departments should be utilised during teaching and learning activities at the beginning of a new section.
- (b) Teachers should give learners enough practice in defining Mathematical Literacy definitions in context. As per CAPS, Mathematical Literacy must be taught and tested in an authentic context.

- (c) Mathematics is a component of Mathematical Literacy, therefore teachers are encouraged to integrate the Basic Skills topics within the Application Topics as part of learners' daily classwork and homework activities.
- (d) Rounding in Mathematical Literacy forms part of the Basic Skills topics and this concept is taught at Grade 10 level. The different types of rounding should be revised daily in all FET Grades when a calculator has been used to generate an answer.

## QUESTION 2: FINANCE

### Common errors and misconceptions

- (a) In Q2.1.3 most candidates did not know what to do with the discount. They added the discount instead of subtracting it.
- (b) Most candidates added the R7 000 excess to the quotation from the panel beaters in Q2.1.4, not realising that the quotation was inclusive of the excess amount.
- (c) In Q2.1.5 candidates calculated 15% VAT of the total monthly premium, not realising that the total monthly premium was 15% VAT inclusive.
- (d) Most candidates managed to identify the cost of sanitation in Johannesburg in Q2.2.1 if a property size was 175 m<sup>2</sup>; however, they could not round off to the nearest 10 cents.
- (e) Many candidates multiplied R443,96 with the area, where they just had to add the 15% VAT to the tariff in Q2.2.3.

### Suggestions for improvement

- (a) Teachers should make use of a variety of financial documents in different contexts beside the CAPS document throughout the FET phase. Subject Advisors should conduct workshops with teachers on how to design authentic tasks by making use of financial documents.
- (b) Teachers should focus on the different calculations for VAT inclusive and VAT exclusive calculations. Learners within a school should use one simple method to differentiate between the two basic calculations. When learners use multiple methods for these two basic calculations, they tend to easily confuse the two calculations.
- (c) Teachers and subject advisors are advised to integrate current tariff systems and not always use tariffs systems given in textbooks and study guides. Mathematical Literacy uses real-life contexts and tariff systems change over a period.
- (d) When learners calculate VAT for a tariff system, they do not take all information into consideration before they give the final answer. Teachers are encouraged to inculcate the skill of extracting all information from the given context before learners give their final solution.

## QUESTION 3: DATA HANDLING

### Common errors and misconceptions

- (a) Candidates were unable to give a reason why the table value would be different from the actual value in Q3.1.2. Candidates still struggle to interpret table values given in ten thousands.

- (b) In Q3.1.3 most candidates wrote 365,9 as indicated in the table and did not consider that the data from the table was given in ten thousands, i.e. 3 659 000.
- (c) Candidates miscalculated the mean in Q3.1.5. Many candidates still get confused with the difference between *mean* and *median*.
- (d) In Q3.2.2 candidates could not convert 7,6 million to 7 600 000 and added the 7,6 to the 183 000 without converting the number.
- (e) Candidates could not calculate the difference between the two percentages; they calculated the percentage increase and could not read off the value of Quarter 3 in Q3.2.3. This suggests that candidates find it difficult to read off values from the graph.
- (f) In Q3.2.4 candidates struggled to conclude that the employment and unemployment added up to 100% given the number of unemployed as 7,6 million. Most candidates calculated 34,4% of 7,6 million.

### **Suggestions for improvement**

- (a) Teachers must provide learners with data tables where large numbers are expressed in ten thousands, millions, etc. Questions on why data is expressed in ten thousands, millions, etc. must be integrated in short tests and classwork activities. The difference of table values and actual values when dealing with large numbers should be regularly practised in class.
- (b) Measures of central tendencies (mode, mean and median) should be practised with large and very small numbers. After learners have been tested on these concepts on a regular basis, written feedback should be given to learners by the teacher where misconceptions occur.
- (c) The basic skill topic of percentages should be revised on a regular basis in a variety of contexts integrated throughout all the Application Topics at Grade 12 level. Teachers must integrate percentages into daily informal tasks and note learner misconceptions to design activities to assist learners.

## **QUESTION 4: FINANCE, DATA HANDLING AND PROBABILITY**

### **Common errors and misconceptions**

- (a) Candidates could substitute values correctly into the given formula but struggled with the order of operations (*BODMAS*) in Q4.1.1.
- (b) In Q4.1.2 many candidates could not interpret the monthly tax tables. Candidates calculated the monthly tax payable.
- (c) Candidates, struggled to write large numbers in words in Q4.2. Symbols and letters were mixed in their answers.
- (d) In Q4.2.2 candidates could identify the correct values, but could not write the ratio in the correct order.
- (e) Determining the median as required in Q4.2.3 is still a challenge for some candidates. Candidates did not arrange the data in order before trying to find the median.

- (f) In Q4.2.4 candidates identified and added the correct values but did not subtract it from the total number of vehicles sold.
- (g) Many candidates struggled to use a reverse calculation in Q4.2.5 to calculate Q3 and were unable to change the subject of the formula.

### **Suggestions for improvement**

- (a) Order of operations (*BODMAS*) must be practised in class and learners should not be overly dependent on calculators that are programmed with *BODMAS*. Where basic mathematical calculations are done in any application topic, *BODMAS* must be emphasised.
- (b) When teaching personal income tax, teachers should incorporate all tax tables, i.e. monthly, weekly, and annual tax tables in the classroom activities. Material selection should not be limited to the tables used in past DBE question papers.
- (c) Writing large numbers in words is introduced in Grade 10. In Grade 11 and Grade 12 writing large numbers in words should be integrated into learners' daily activities and tested in short informal texts and SBA tasks where applicable.
- (d) Reverse calculations can be integrated into all topics where a formula is given. Regular exposure to this type of calculation will be beneficial for learners, when preparing for SBA tasks and examinations.

## **QUESTION 5: DATA HANDLING AND FINANCE**

### **Common errors and misconceptions**

- (a) In Q5.1.2 some candidates, struggled to write 8,3 billion as 8 300 000 000. Instead of calculating the total percentage for the agricultural sector which is 11%, they just multiplied by 4%.
- (b) In Q5.2.1 candidates could not identify the currency that was weakest against the rand. Some candidates gave an answer of British pound as the weakest currency against the rand instead of the Japanese yen.
- (c) Candidates did not understand the concept of exchange rate given in the table in Q5.2.2 and could not calculate the inverse of an exchange rate.
- (d) Most candidates did not know when to multiply or when to divide when converting from one currency to another currency (Q5.2.3). For example, some learners multiplied by R11,785 instead of dividing.
- (e) In Q5.2.5 some candidates were able to calculate the amount earned after two years but could not calculate the interest earned for the last 8 months.

### **Suggestions for improvement**

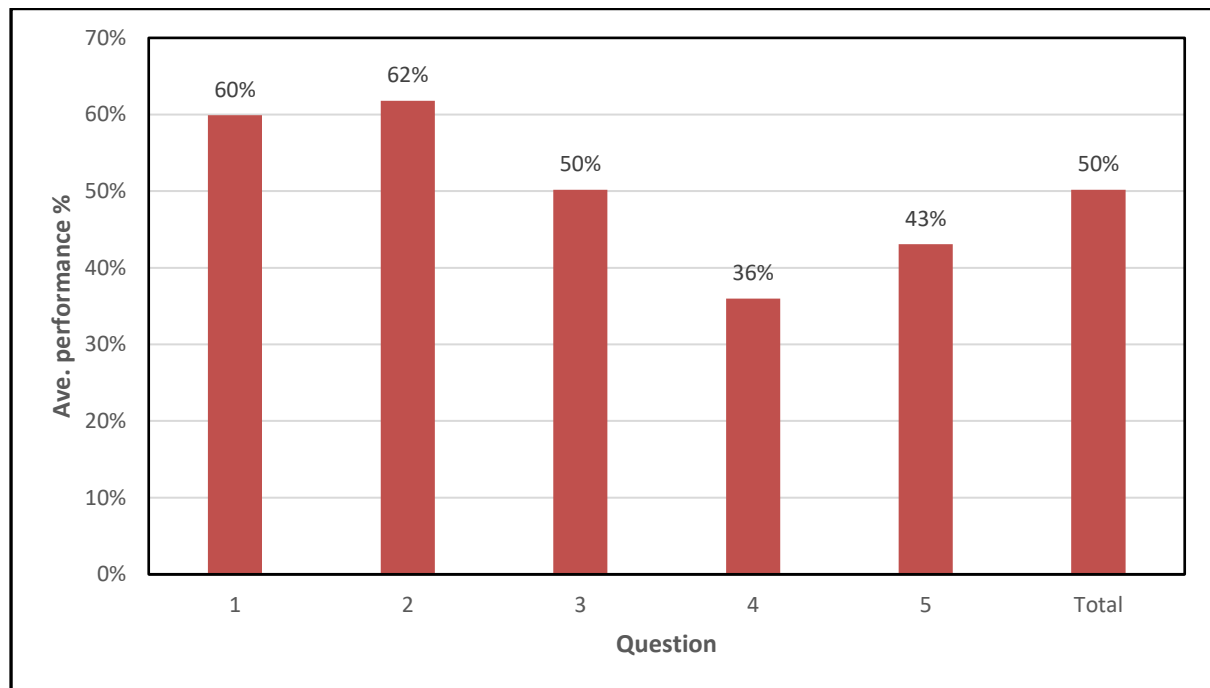
- (a) The focus on exchange rates is on developing an understanding of the value of a currency in relation to other currencies and on the value of a particular currency in relation to the cost of living/business venture in a country, rather than on repetitive calculation using mathematical content and procedures.

- (b) The basic mathematical procedure should be on how to convert from one currency to another. The DBE booklet should be utilised as it explains how ratio can be used to convert from one currency to another. This method pre-empts confusion about when to divide and when to multiply when dealing with currency conversions.
- (c) The compound interest formula is not prescribed in the *Mathematical Literacy CAPS*. Learners must be able to do all mathematical calculations in Mathematical Literacy using a basic calculator. A manual calculation needs to be done when calculating interest (compound).

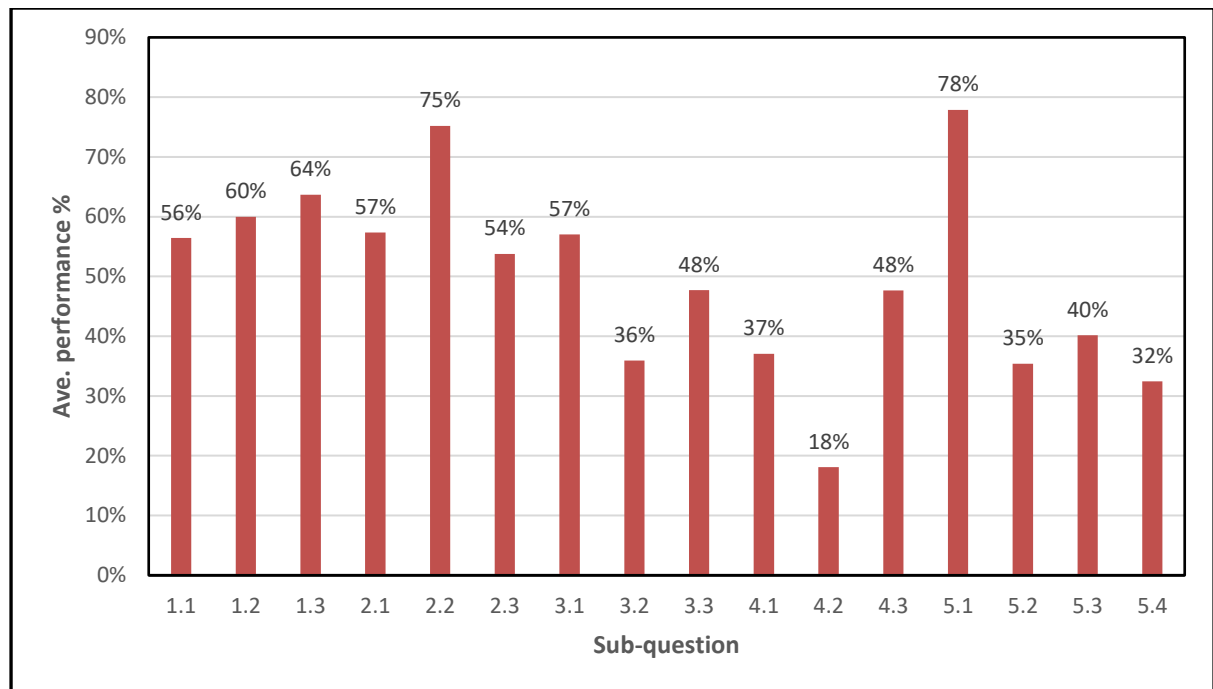
## 9.5 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

The following graph is based on data from a random sample of candidates. While this graph may not accurately reflect national averages, it is useful in assessing the relative degree of challenge of each question as experienced by candidates.

**Figure 9.5.1 Average performance per question in Paper 2 2022**



| Q | Topic                                          |
|---|------------------------------------------------|
| 1 | Measurement, maps, plans and other             |
| 2 | Probability, maps, plans and other             |
| 3 | Measurement and probability                    |
| 4 | Measurement, maps, plans and other and finance |
| 5 | Measurement, maps, plans and other             |

**Figure 9.5.2 Average performance per subquestion in Paper 2 2022**

| Sub-Q | Topic                 | Sub-Q | Topic                              |
|-------|-----------------------|-------|------------------------------------|
| 1.1   | Measurement           | 3.3   | Measurement                        |
| 1.2   | Maps, plans and other | 4.1   | Measurement, maps, plans and other |
| 1.3   | Maps, plans and other | 4.2   | Measurement                        |
| 2.1   | Maps, plans and other | 4.3   | Finance                            |
| 2.2   | Probability           | 5.1   | Maps, plans and other              |
| 2.3   | Maps, plans and other | 5.2   | Measurement                        |
| 3.1   | Measurement           | 5.3   | Measurement                        |
| 3.2   | Measurement           | 5.4   | Measurement                        |

## 9.6 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### QUESTION 1: SHORT CONTEXTS (INTEGRATED LEVEL 1 QUESTIONS ONLY)

#### Common errors and misconceptions

- (a) Q1.1.1 was well answered by most candidates.
- (b) In Q1.1.2 some candidates referred to instruments of time and not time format.
- (c) Many candidates did not include that the time was in the afternoon and lost one mark in Q1.1.3. Alternatively, candidates could not differentiate between *to* and *past* in writing time in words.
- (d) In Q1.1.4 many candidates were unable to identify the total number of clocks that indicate the time in the afternoon.

- (e) In Q1.1.5 some candidates divided by 60 instead of multiplying to convert hours to minutes.
- (f) Many candidates could not interpret the assembly diagram correctly in Q1.2, hence did not know how many nuts were used or left over in each step.
- (g) Most candidates answered Q1.3.1 and Q1.3.3 well, as it was direct reading from the map.
- (h) In Q1.3.2 many candidates were unable to use the compass directions accurately and answered *Limpopo* instead of *Gauteng*.
- (i) Most candidates wrote only one national road, the N14, and left out the N18 since they know that the road running through Vryburg and Mahikeng is the N18.
- (j) In Q1.3.4 many candidates wrote down the names of the towns when the question asked for the number of towns.
- (k) In Q1.3.5 many candidates could not measure the distance between two points on a map. This is a basic skill which is taught in Grade 10. Some wrote their answers in cm and lost a mark.

### **Suggestions for improvement**

- (a) The importance of laying a thorough foundation in Grade 10 cannot be emphasised enough. Teachers should spend the first week of the year simply addressing basic skills which will make it easier for learners to navigate the curriculum since these skills are found in different contexts throughout.
- (b) Basic definitions should be taught in a clear and succinct manner.
- (c) Question by question analysis should be encouraged after assessments so that challenges can be identified and addressed as early as possible.
- (d) Teachers are encouraged to expose learners to various types of assembly diagrams so as to demystify this section.
- (e) Time formats and conversions of time should be practised throughout the year.
- (f) Teachers should cover all the units of time, namely seconds, minutes, hours, days, weeks, months, years, when teaching time conversion.
- (g) Learners must be taught to use the given resource and not rely on their natural surroundings to answer questions.

## **QUESTION 2: MAPS AND PLANS AND PROBABILITY**

### **Common errors and misconceptions**

- (a) In Q 2.1.1, 2.1.4 and 2.3.3(a) and (b) were poorly answered because of the language barriers, which was evident in the manner in which candidates expressed themselves.

- (b) Some candidates misread the number of seats available in Q2.1.2 or rushed their answers and left out the seats at a particular table.
- (c) In Q2.1.3 many candidates focused only on the chairs closer to the south wall and arrived at an answer of six instead of 13.
- (d) In Q2.1.4 a few candidates gave answers that were not in context and lost valuable marks.
- (e) In Q2.1.5 many candidates used left and right or up and down instead of the compass directions and lost three marks.
- (f) Q2.2.1 was poorly answered. Far too many candidates could not give the correct answer, *tree diagram*. Wrong answers such as *spider diagram*, *probability diagram*, *step-and-leave diagram* or a *pedigree diagram* for hybridisation in Life Sciences.
- (g) Q2.2.2. to Q2.2.4 were well answered.
- (h) In Q2.3 many candidates still struggled with basic definitions like *route map*.

### Suggestions for improvement

- (a) Teachers should spend sufficient time on basic concepts like **maps, plans** and **scales**.
- (b) Learners should be tested on definitions regularly.
- (c) Teachers should write the name of the diagram on the chalkboard during the lesson. They should emphasize correct naming of the diagram during both formal and informal assessment.
- (d) Teachers should expose learners to questions that require them to determine the probability as a percentage.
- (e) Teachers should integrate Geography when teaching the topic of Maps and Plans.
- (f) Teachers should highlight that left is west and right is east and should be read as 'WE' in the compass drawings. When north is not facing up, the learners should rotate the map or the plan for north to face up.
- (g) Teachers should refer to the CAPS document and not just rely on past papers.

## QUESTION 3: MEASUREMENT

### Common errors and misconceptions

- (a) In Q3.1.1 many candidates multiplied the sides instead of adding them to calculate perimeter. Some candidates omitted the bracket resulting in the common error: Perimeter =  $2 \times 239 + 89$  mm instead of Perimeter =  $2 \times (239 + 89)$  mm. Some candidates gave the answer as a square unit.
- (b) Many candidates worked with mixed units (mm and cm) in the same calculation in Q3.1.2.
- (c) In Q3.1.3(a) some candidates still confused diameter and radius.

- (d) Q3.1.3 (b) was poorly answered. Candidates could not change the subject of the formula. Most candidates did not round the final answer to the nearest gram.
- (e) In Q3.2.1 many candidates did not know that February had 28 days. Most candidates only gave partial answers, either forgetting to multiply by 5 or 1,6.
- (f) Q3.2.2 was poorly answered. Candidates did not answer the question in context and gave general ways of saving water.
- (g) In Q3.3.1, many candidates could not differentiate between *cooking time* and *preparation time* when reading the recipe. Most candidates could not subtract time.
- (h) Most candidates answered Q3.3.2 well, but some candidates could not round off to the nearest ten.
- (i) In Q3.3.3 some candidates could not work with fractions or convert from millilitres to litres. Many multiplied instead of dividing.

### **Suggestions for improvement**

- (a) Teachers are advised to encourage learners to read questions carefully, i.e. find out whether a question should be rounded to one or two decimal places.
- (b) Teachers should focus on and use past papers to help learners understand key words like: CONVERT, CALCULATE, SHOW, DETERMINE, DESCRIBE, STATE, EXPLAIN, GIVE, VERIFY.
- (c) Learners need to practise writing answers with the correct units and should be taught how to convert from one unit to another.
- (d) Teachers must emphasise that all dimensions must be in the same unit before calculating.
- (e) Teachers should explain to learners the difference between *show* and *verify* in the context of a question.

## **QUESTION 4: MAPS AND PLANS, MEASUREMENT AND FINANCE**

### **Common errors and misconceptions**

- (a) Q4.1.1 was a challenge. The worst performance in the entire paper was in this subquestion. Most of the learners could not spatially align themselves with the various aspects of the birdhouse and were unable to respond. Some candidates added a second 25 instead of a 23, resulting in the answer being 143 cm. Some candidates converted from cm to mm incorrectly.
- (b) In Q4.1.2 the request to verify was misinterpreted in the given context.
- (c) In Q 4.1.3 many candidates used *diameter* instead of *radius*. Some could not square the radius/diameter. Most of the candidates added the last part instead of subtracting. Candidates failed to use the formula.

- (d) Q4.2 was poorly answered. Instead of dividing by 10 and 14 most candidates multiplied. They failed to apply the **ratio** to calculate the number of litres required, given the spread rate. Most of the candidates do not know the meaning of the term *coat* with reference to painting.
- (e) Q 4.3.1: Most candidates thought that R287,40 was the cost of wooden boards for 1 bird house and not for 6, as given in the question. As a result, most did not divide the R287,40 by 6 before adding the costs.
- (f) Q4.3.2: Candidates could not explain *break even* using the given context. They gave partial answers using words like 'meet', 'intersect', 'join', 'merge'.
- (g) Q4.3.3: Some candidates failed to read from the table and took the value from Q4.3.1 to use it in Q4.3.3.

### **Suggestions for improvement**

- (a) Teachers should give more attention to assembly diagrams and their interpretation.
- (b) Simple exercises can be done to increase learners' spatial awareness by allowing them to bring 3D items to class that can be analysed and interpreted.
- (c) Learners should be taught to distinguish between the units of area volume and perimeter.
- (d) Learners should be given multiple opportunities to practise calculator skills and *BODMAS*.
- (e) Teachers should expose learners to various formulae whereby the unknown is not necessarily the subject of the formula, e.g. finding the radius using the formula for calculating the volume of a cylinder.

## **QUESTION 5: MAPS AND PLANS AND MEASUREMENT**

### **Common errors and misconceptions**

- (a) Q5.1 was well answered, but some learners responded with the names of the elements and not the numbers.
- (b) In Q5.2.1 the concept of decade was not understood by most of candidates. Some candidates divided by 100 instead of ten. Other candidates just subtracted the years and wrote the answer as decades.
- (c) Most candidates answered Q5.2.2 correctly. A few candidates did not know how many days there are in November. Other candidates did not understand the concept 'elapsed' and included the starting day.
- (d) Q5.3.1 was well answered. Many candidates could not simplify the ratio or wrote the ratio as a fraction or wrote the ratio in reverse.
- (e) Q5.3.2 was either correct or wrong. Some candidates multiplied instead of dividing to convert from feet to metres.

- (f) Q5.3.3 was challenging as most candidates used the incorrect values to calculate the percentage discount. Some candidates used the incorrect denominator.
- (g) Candidates could not interpret Q5.4 well. They were unable to change the subject of the formula. They also struggled with calculating the actual time used for the journey.

**Suggestions for improvement**

- (a) Learners should be taught to read questions with due diligence. By simply glossing over questions, candidates are overlooking critical information contained in the questions.
- (b) More time should be spent on conversions and time as these aspects seem to have been neglected.
- (c) Teachers should bring along calendars to class when they teach time in months, weeks and days. Learners should perform calculations practically, using the calendars. The emphasis should be on the day from and days in between.
- (d) The first value mentioned in the question statement should appear first in the ratio, followed by the second value.
- (e) Teachers should give learners sufficient practice questions on the calculation of time elapsed and conversion thereof.

# CHAPTER 10

## MATHEMATICS

The following report should be read in conjunction with the Mathematics question papers for the NSC November 2022 examinations.

### 10.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who sat for the Mathematics examinations in 2022 increased substantially by 10 591 candidates.

While the table below indicates a decline in the pass rate over the period 2018-2020, improvement was evident in 2021.

Candidates who passed at 30% (Level 2) declined slightly from 57,6% in 2021 to 55,0% in 2022. There was a corresponding decline in the pass rate at 40% (Level 3) over the past two years from 37,6% to 36,0%.

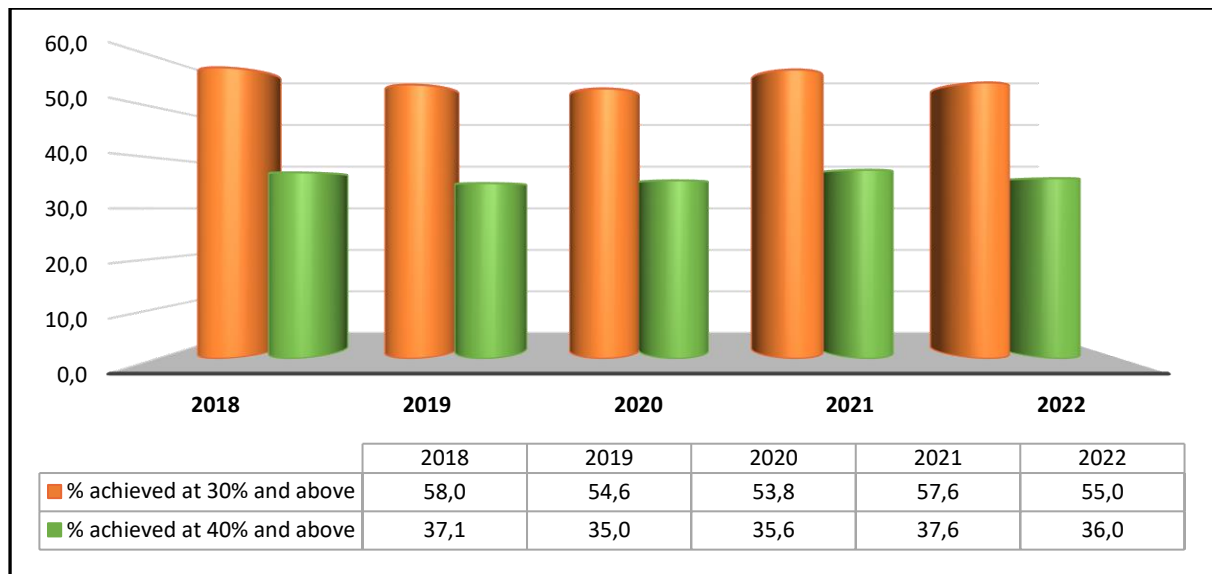
There was also a marginal decline in the percentage of distinctions over 80% (level 7), which decreased from 2,9% in 2021 to 2,7% in 2022. This converts into a decrease of 232 in the total number of distinctions.

In view of disruption to academic programmes in 2020 and 2021, and despite the factors mentioned above, the results achieved by this cohort are commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the overall performance.

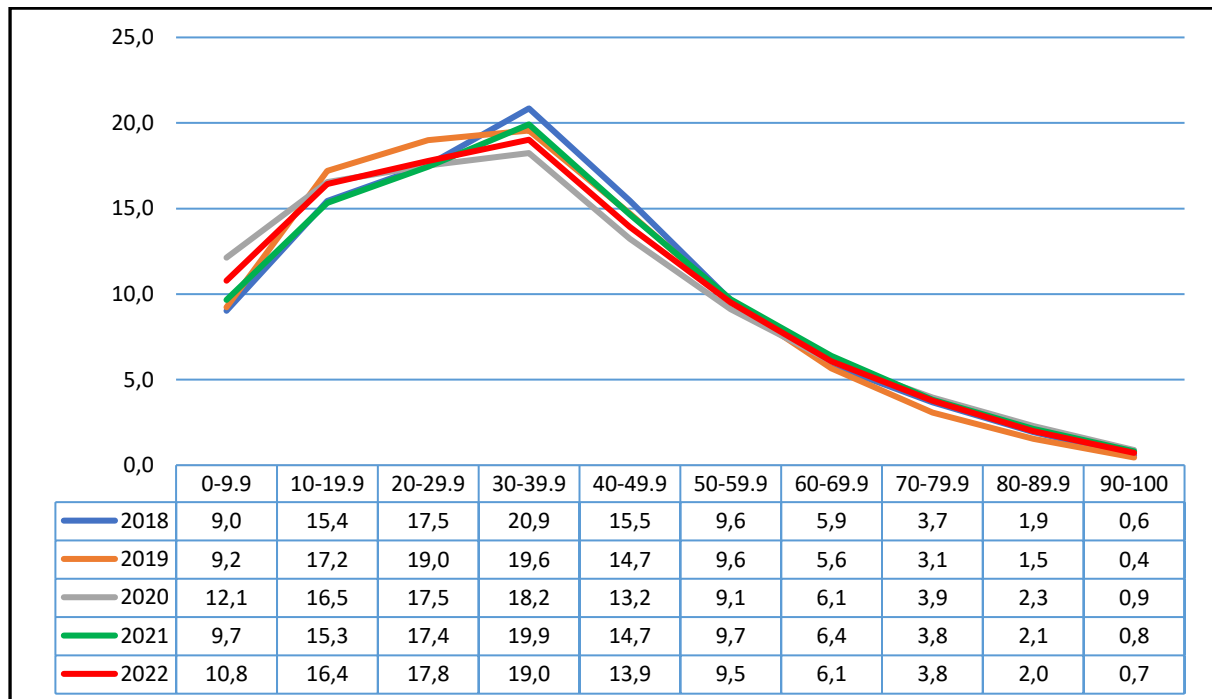
**Table 10.1.1 Overall achievement rates in Mathematics**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 233 858   | 135 638                       | 58,0                        | 86 874                        | 37,1                        |
| 2019 | 222 034   | 121 179                       | 54,6                        | 77 751                        | 35,0                        |
| 2020 | 233 315   | 125 526                       | 53,8                        | 82 964                        | 35,6                        |
| 2021 | 259 143   | 149 177                       | 57,6                        | 97 561                        | 37,6                        |
| 2022 | 269 734   | 148 346                       | 55,0                        | 97 041                        | 36,0                        |

**Graph 10.1.1 Overall achievement rates in Mathematics (percentage)**



**Graph 10.1.2 Performance distribution curves in Mathematics (percentage)**



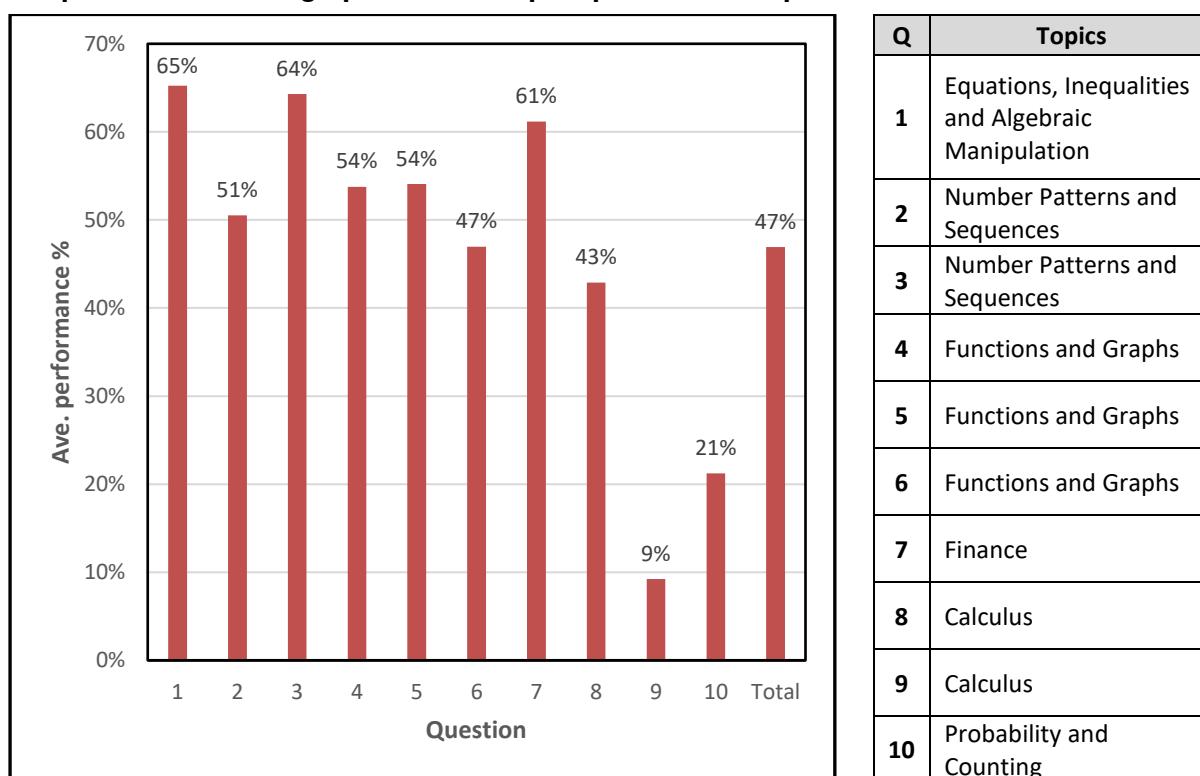
## 10.2 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 1

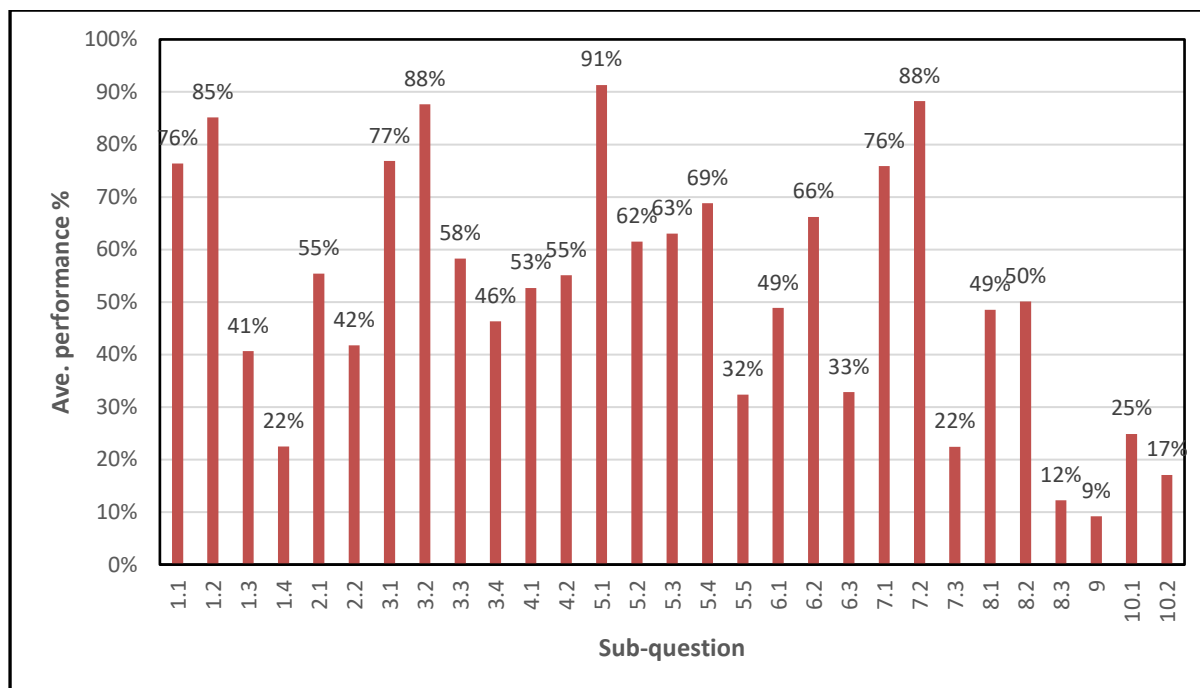
- (a) Many candidates were able to answer the knowledge and routine questions correctly and scored some marks in a majority of the questions. This suggests that the candidates were well prepared to deal with these questions in the paper.
- (b) The algebraic skills of the candidates are poor. Most candidates lacked fundamental and basic mathematical competencies which should have been acquired in the lower grades. This becomes an impediment to candidates when answering complex questions.
- (c) While calculations and performing well-known routine procedures form the basis of answering questions in a Mathematics paper, a deeper understanding of definitions and concepts cannot be overlooked. Candidates did not fare well in answering questions that assessed an understanding of concepts.

## 10.3 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 1

The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 10.3.1 Average performance per question in Paper 1**



**Graph 10.3.2 Average performance per subquestion in Paper 1**

## 10.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 1

### QUESTION 1: ALGEBRA

#### Common errors and misconceptions

- (a) Some candidates multiplied the brackets out but then factorised incorrectly in Q1.1.1.
- (b) Rounding off the answers to two decimal places is still a problem for some candidates. Some candidates still lack basic calculator skills. A few candidates took the value of  $b$  to be 6 instead of  $-6$ .
- (c) In answering Q1.1.3 many candidates treated the inequality as an equation. Their answer read:  $(x+9)(x-10) > 0$  followed by  $x > -9$  or  $x > 10$ . Many candidates struggled to interpret the correct answer from the inequality.
- $$x^2 - x - 90 > 0$$
- $$(x+9)(x-10) > 0$$
- $$\therefore x = -9 \text{ or } x = 10$$
- $$\therefore -9 < x < 10$$
- Some candidates drew a sketch but were unable to use it to write down the required answer. Another common error was the incorrect notation in the answer. Candidates wrote the answer as  $-9 < x > 10$  instead of  $x < -9$  or  $x > 10$ .
- (d) Most candidates had some idea that they had to square both sides of the equation in Q1.1.4. However, few candidates were able to isolate the *surd* before squaring both sides of the equation. The following errors were noted:

$$\begin{aligned} (x - 7\sqrt{x})^2 &= (-12)^2 & \text{or} & & x - 7\sqrt{x} &= -12 \\ \therefore x^2 - 49x &= 144 & & & \therefore \sqrt{x} &= \frac{-12}{x-7} \end{aligned}$$

- (e) While many candidates answered Q1.2 correctly, a few made errors when they wrote the linear equation in terms of one variable. Some candidates disregarded a valid solution. They confused simultaneous equations with surd equations.
- (f) In Q1.3 many candidates were able to write the exponents with separate bases, i.e.  $2.5^n - 5^{n+1} + 5^{n+2} = 2.5^n - 5^n.5 + 5^n.5^2$ . However, some failed to factorise the resulting expression correctly, i.e.  $2.5^n - 5^n.5 + 5^n.5^2 = 5^n(2.1^n - 1^n.5 + 1^n.5^2)$ . The majority of the candidates could not explain why  $5^n(22)$  will be even numbers for all positive integral values of  $n$ .
- (g) Many candidates did not know how to answer Q1.4. They were unable to write both sides of the equation as exponents which had the same bases.

### Suggestions for improvement

- (a) Much of the work in this question is covered in Grade 11. It is therefore important for teachers to set revision tasks on these sections throughout the Grade 12 year.
- (b) Teachers should not take for granted that learners know how to round off a number to the required number of places. Where necessary, this skill should be retaught in Grades 11 and 12. Teachers should penalise learners in SBA tasks when they do not round off to the correct number of places.
- (c) Teachers should take some time, preferably in Grade 10, to focus on teaching learners how to represent inequalities (e.g.  $-3 < x < 5$  ;  $x < -3$  or  $x > 5$  ) on a number line and also how to write an inequality from the illustration on a number line. This will benefit learners as they are required to write inequality solutions for a number of questions in both examination papers. Emphasis on the correct notation is essential when writing down the solutions to inequalities.
- (d) Teachers should explain the difference between *and* and *or* in the context of inequalities. Learners cannot use these words interchangeably as they have different meanings.
- (e) When dealing with *surd* equations, learners should be reminded that they need to isolate the *radical* before they can square both sides of the equation. Teachers must emphasise that implicit restrictions are placed on *surd* equations and that learners should continue to test whether their answers satisfy the original equation.
- (f) Learners need to be exposed to complex questions involving *surds* and *exponents*.

## QUESTION 2: PATTERNS

### Common errors and misconceptions

- (a) Many candidates did not understand what was required in Q2.1.2. They correctly calculated that the sum of the first 13 terms was 114 674. However, they did not subtract 6 from 13 in order to answer the required question.

- (b) While many candidates answered Q2.1.3 correctly, some candidates failed to realise that for a convergent geometric series the value of  $r$  must lie in the interval:  $-1 < r < 1$ .
- (c) In Q2.2 many candidates could not interpret the *sigma notation* correctly. Some of them mistook the general term given in the *sigma notation* to represent a geometric series. A fair number of candidates were unable to calculate the first three terms of the series and consequently were unable to calculate the common difference. A number of candidates failed to make the connection that  $n = k + 1$ . Some candidates incorrectly assumed that  $T_n = 20\frac{1}{6}$  instead of  $S_n = 20\frac{1}{6}$ .

### Suggestions for improvement

- (a) Learners must be taught how to identify the type of sequence they are working with and which formulae are applicable to it.
- (b) Make learners aware of which formulae on the information sheet apply to which type of sequence. It is good practice for learners to use the information sheet in class so that they become familiar with it.
- (c) Teach learners how to identify whether the question requires them to calculate the value of the  $n^{\text{th}}$  term or the sum of the first  $n$  terms.
- (d) Learners must read the questions carefully so that they know what is required of them.
- (e) Remind learners that  $n$  cannot be a negative number, zero or a fraction. When solving for  $n$ , learners should arrive at a natural number solution. If this is not the case, then they should know that they have made a mistake in their working. Learners must be told that if they round off their answers to a natural number, then they will not be awarded any marks.
- (f) It is important to demonstrate, by way of example, the concept of a convergent geometric series, first by taking a value of  $r > 1$  and then taking a value of  $-1 < r < 1$ . This should alert learners to the condition for which a geometric series will converge.
- (g) More attention should be given to expanding a series that is given in *sigma notation* form.

### QUESTION 3: PATTERNS

#### Common errors and misconceptions

- (a) When answering Q3.1, many candidates failed to identify from the question that  $a = 1$  and that  $3a + b = 7$ . Instead, they used  $b = 4$  in their calculations to prove that the first of the first differences is 7.
- (b) Many candidates were able to answer Q3.3 correctly. However, they failed to make the connection between Q3.3 and Q3.4. In answering Q3.4, many candidates incorrectly stated that  $n^2 + 4n + 9 = 157$ .

**Suggestions for improvement**

- (a) When teaching *quadratic number patterns*, it is essential to show learners how the formulae:  $T_1 = a + b + c$ , the first term of the first differences  $= 3a + b$  and the second difference  $= 2a$ , are deduced.
- (b) The sequence of first differences of a *quadratic number pattern* forms an *arithmetic pattern*. This implies that an arithmetic sequence is embedded within a quadratic number pattern. Learners must read the question very carefully in order to establish to which pattern the question is referring. Glossing over words in the question leads to learners making incorrect statements.
- (c) Remind learners that  $n$  cannot be a negative number, zero or a fraction. When solving for  $n$ , learners should arrive at a natural number solution. If this is not the case, then they have made a mistake in their working.
- (d) Teachers should integrate quadratic number patterns with arithmetic patterns and quadratic functions. This will provide learners with the skills to deal with higher-order questions.

**QUESTION 4: FUNCTIONS (HYPERBOLA AND PARABOLA)****Common errors and misconceptions**

- (a) A number of candidates incorrectly stated that  $p = 1$  instead of  $p = -1$  when answering Q4.1.1.
- (b) Many candidates answered Q4.1.2 correctly. However, they did not realise that they had to translate the x-intercept 3 units to the left to obtain the answer for Q4.1.3. The challenge in this question was that candidates were unable to interpret the notation correctly.
- (c) The most common error candidates made when answering Q4.1.5 was to solve the equation algebraically. Candidates failed to relate the given equation to the given hyperbola.
- (d) In Q4.2.2 many candidates were able to calculate the axis of symmetry correctly but were unable to calculate the y-coordinate of the turning point correctly.
- (e) Despite having obtained the value of  $q$  correctly, many candidates were unable to calculate the value of  $a$  correctly in Q4.2.3.
- (f) The common error in Q4.2.4 was notation. Candidates wrote  $y \in (-5; -\infty)$  or  $y \in (-\infty; -5]$  instead of  $y \in (-\infty; -5)$ .
- (g) Very few candidates attempted Q4.2.5 because they failed to interpret the notation used in the question.

**Suggestions for improvement**

- (a) Teachers need to pay more attention to using the correct notation when writing intervals. Pay attention to the meaning of the notation and the difference between the

various notations. Teachers must not condone the use of incorrect notation in SBA tasks.

- (b) A hyperbola can be defined in two ways, viz.  $y = \frac{a}{x+p} + q$  and  $y = \frac{a}{x-p} + q$ .

Teachers are advised to use  $y = \frac{a}{x+p} + q$  as it is referred to in the CAPS. It would seem that learners do not pay attention to the difference in signs between the two equations. However, they are penalised when they write down the value of  $p$  incorrectly.

- (c) *Transformation* is no longer a topic in the curriculum. However, in the section of functions, the basic graph is transformed to obtain all other graphs. The effect of the transformation on the equation of the basic graph must be discussed. In doing so, it is suggested that teachers also pay attention to the effect that the transformation has on the basic equation.
- (d) Learners need to be made aware that an inequality containing fractions cannot be solved in the same way as an equation. When solving an equation, we multiply both sides by the LCD to eliminate the fractions. When solving an inequality, we multiply both sides by the  $(\text{LCD})^2$ .

## QUESTION 5: FUNCTIONS (INVERSE FUNCTIONS)

### Common errors and misconceptions

- (a) In Q5.2 most of the candidates were able to correctly swop  $x$  and  $y$  in the equation. However, some did not make  $y$  the subject of the formula while others made errors when transposing terms. Many candidates did not understand the difference in the notations:  $g'(x)$  and  $g^{-1}(x)$ . Some candidates simply wrote down  $\log$  when they saw  $g^{-1}(x)$ . They believe that the  $\log$  function is the inverse of all functions. This is incorrect.
- (b) When answering Q5.3, many candidates calculated the intercepts with the axes instead of calculating the point of intersection of the two lines. Many candidates did not show any working but simply wrote down incorrect coordinates of  $A$  as if they had read them off the graph.
- (c) Many candidates made incorrect assumptions when answering Q5.5. The most common of these was to assume that  $AB$  was perpendicular to  $BC$ . Some candidates used the incorrect angle when using the area formula:  $A = \frac{1}{2}ab \sin C$ . They used an angle of inclination instead of the angle between the two sides of the triangle.

### Suggestions for improvement

- (a) Teachers should spend some time discussing that all points on the  $x$ -axis have a  $y$ -coordinate of 0 and all points on the  $y$ -axis have a  $x$ -coordinate of 0. The domain is always a set of  $x$ -values and the range is always a set of  $y$ -values.
- (b) Learners must be taught to respond to the question. They should write down the coordinates of a point when the question requires them to do so.

- (c) The section of inverses opens itself to an investigative approach to teaching. Teachers should take the opportunity to allow learners to discover that a function and its inverse are symmetrical about the line  $y = x$  and that the point of intersection, where it exists, of a function and its inverse will also lie on the line  $y = x$ .
- (d) Teachers need to emphasise that the height of a triangle is perpendicular to its base.

## QUESTION 6: FINANCE

### Common errors and misconceptions

- (a) In Q6.1 some candidates used the simple interest formula instead of the compound interest formula. Many candidates used the incorrect value of  $n$  in the compound interest formula. They used 4, 12 or 24 as the value of  $n$  instead of 8.
- (b) Many candidates failed to take into account that Q6.3.2 was based on a deferred payment. Consequently, they did not add interest to the initial loan amount. Common errors were:

$$212500 = \frac{x \left[ 1 - \left( 1 + \frac{0,13}{12} \right)^{-72} \right]}{\frac{0,13}{12}} \quad \text{or} \quad 212500 = \frac{x \left[ 1 - \left( 1 + \frac{0,13}{12} \right)^{-67} \right]}{\frac{0,13}{12}}$$

### Suggestions for improvement

- (a) Learners need deeper insight into the relevance of each of the formulae and under which circumstances each can be used. The variables in each formula must be explained. More practice in Financial Mathematics is necessary so that learners can identify when to use the different formulae.
- (b) Teachers should demonstrate all the steps required when using a calculator. Learners should be penalised for rounding off early in formal assessment tasks at school.
- (c) The difference between *compound interest* and *future value annuities* must be explained.
- (d) The correct Financial Mathematics language should be used in class and learners should read the question with understanding.
- (e) Teachers should expose learners to the various scenarios for loan repayments, viz., deferred payments, missed payments, outstanding balance immediately after a certain number of payments, final payment, etc.

## QUESTION 7: CALCULUS

### Common errors and misconceptions

- (a) In Q7.1 some candidates made the following notational errors:

$$f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{or} \quad \lim_{h \rightarrow 0} = \frac{f(x+h) - f(x)}{h}. \quad \text{They lost a mark for these errors.}$$

Other candidates made errors in substitution, i.e.,

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + (x+h) - x^2 + x}{h} \text{ or } f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + x - (x^2 + x)}{h}.$$

Another common error was incorrect factorisation, i.e.:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{2xh + h^2 + h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) \\ &= 2x \end{aligned}$$

- (b) When answering Q7.3, many candidates calculated the first and second derivatives correctly. However, they substituted incorrectly into the second derivative, i.e.  $6a(-1)^2 + 6 = -7$ . They confused the value of the second derivative at the point of inflection with the minimum gradient of the tangent.

### Suggestions for improvement

- (a) Emphasis should be placed on the use of the correct notation when determining the derivative, either when using first principles or the rules.
- (b) Teachers should explain the need for brackets when determining the derivative from first principles. This prevents the incorrect simplification that follows.
- (c) The concepts of gradient of tangent and point of inflection must be understood. Teachers should explain the difference between the two concepts.
- (d) The focus when teaching *cubic functions* should not only be on calculating the critical points, but also on interpreting the critical points on the graph.

## QUESTION 8: CALCULUS

### Common errors and misconceptions

- (a) Many candidates failed to realise that Q8.1 required them to determine the equation of a parabola given two  $x$ -intercepts and a point on the parabola. Some candidates assumed that the value of  $a$  was  $-1$  because the parabola had a maximum turning point. A few candidates used the function given in Q8.2 to calculate the values of  $m$ ,  $n$  and  $k$ . They were not awarded any marks for this.
- (b) In Q8.2.1 many candidates calculated the  $x$ -coordinates of the turning points by differentiating the given cubic function instead of reading them off the given graph. These candidates could not relate the given parabola to the first derivative of the cubic function.
- (c) Many candidates failed to calculate the  $x$ -intercepts of the cubic function and hence were unable to draw the cubic graph correctly in Q8.2.2.
- (d) Q8.3.1 and Q8.3.2 posed a reading challenge to many candidates. They did not realise that these questions were based on the turning point of the parabola. In

addition, these questions required candidates to make a graphical interpretation. This was beyond many candidates' capabilities.

### Suggestions for improvement

- (a) Learners should be taught to determine the equation of a graph from Grade 10 to 12 in a progressive manner.
- (b) When teaching *factorisation of third-degree polynomials*, teachers should include examples where there is only one real root.
- (c) Teachers should demonstrate the relationship between a *cubic graph* and the *graphs of its first and second derivatives*.
- (d) The application of Calculus lends itself to many applications. Teachers need to expose learners to a wide variety of questions.

## QUESTION 9: CALCULUS

### Common error and misconception

The vast majority of the candidates did not attempt this question because they were unable to determine the distance function from the given information. Some candidates used trial and error to obtain the minimum distance between the points. They were not awarded any marks.

### Suggestions for improvement

- (a) Learners appear to be dependent on the formulae being given when solving *optimisation* problems. It is advisable that learners interrogate the optimum function even when it is given in a question. This should help their conceptual development.
- (b) Teachers should ensure that there is enough time for learners to understand the application of Calculus fully.
- (c) Reading for understanding should be ongoing if learners are to improve their responses to word problems.

## QUESTION 10: PROBABILITY

### Common errors and misconceptions

- (a) Many candidates did not realise that *the probability of at least one of the three events will occur is 0,893* represented the sum of the probabilities of all the regions enclosed by the three circles. Thus they were unable to answer Q10.1.1(a) and Q10.1.1(b) correctly.
- (b) In answering Q10.1.2 many candidates were unable to correctly identify the regions in the diagram that satisfied the condition of *at least two events*.
- (c) In Q10.1.3, a number of candidates read off  $P(A)$  from the sketch instead of  $P(C)$ . They went on to test the independence of events A and B instead of events B and C. Some candidates confused  $P(B)$  with  $P(B \text{ only})$  and incorrectly took the value of  $P(B)$  as 0,183.

- (d) Many candidates were unable to answer Q10.2.1 and Q10.2.2 correctly because they were unable to take the restrictions placed on the digits into account. The problem was compounded by the fact that the digits may not be repeated.

### Suggestions for improvement

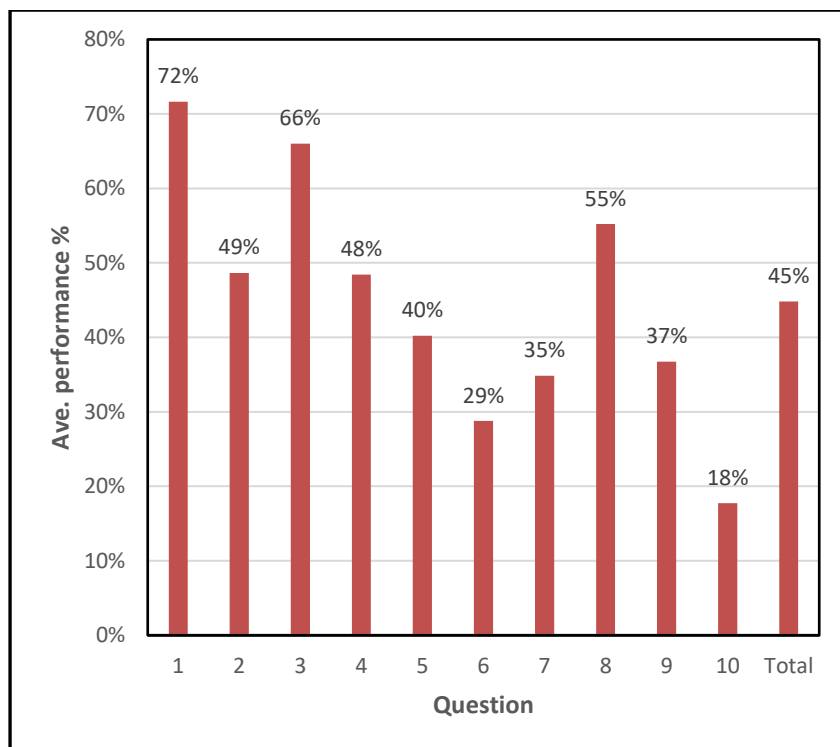
- (a) Teaching basic concepts cannot be overlooked. When learners understand the basic concepts well enough, then the more complex concepts are easier to grasp.
- (b) It must be stressed that the probability of an event  $A$  lies in the interval  $0 \leq P(A) \leq 1$ .
- (c) Reading for understanding must be a regular practice in the classroom. This should equip learners with the skills to deal with word problems in assessment tasks.
- (d) Use Venn diagrams to teach *probability*. It helps with the understanding of the different areas that make up the events, e.g. only  $A$ , only  $B$ ,  $A$  and  $B$ ,  $A$  or  $B$ , not  $A$ , not  $B$ , not  $A$  and not  $B$  and not  $A$  or not  $B$ .
- (e) Teach learners the *Fundamental Counting Principle* in such a way that they will be able to base their answers on their reasoning, rather than on the rule.

## 10.5 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 2

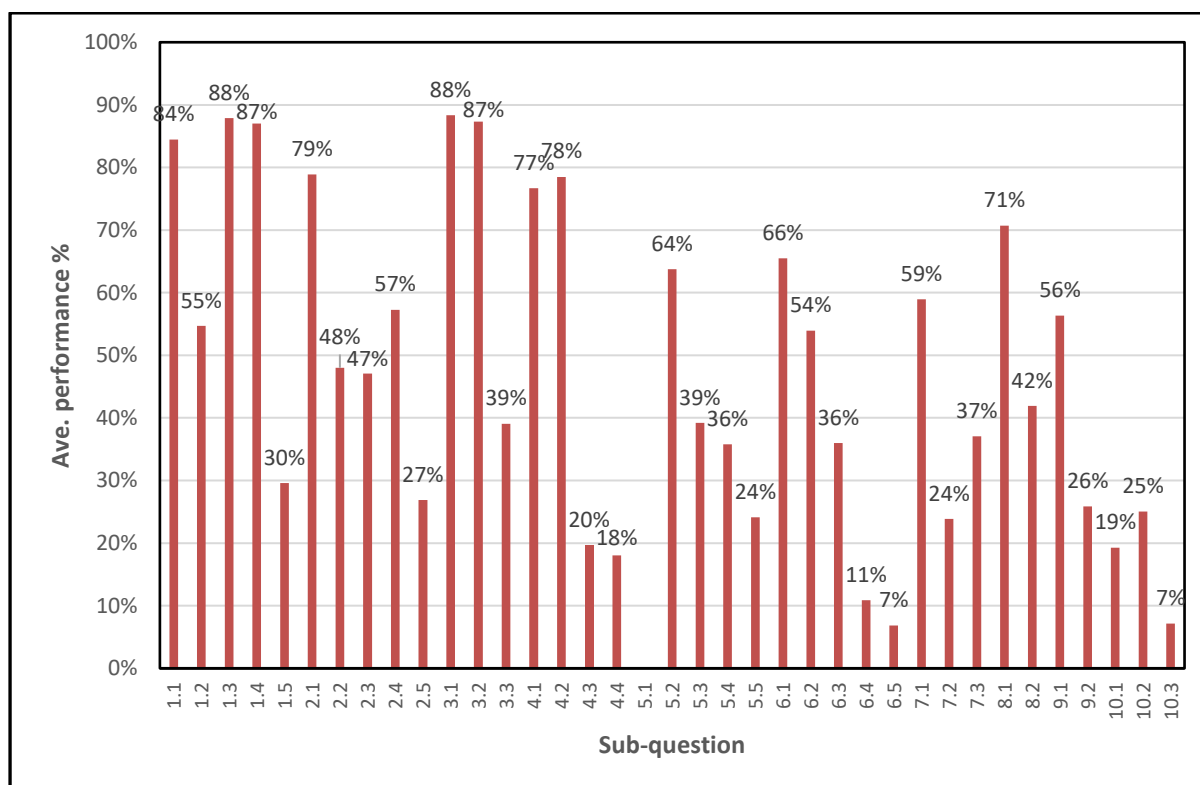
- (a) Integration of topics is still a challenge for many candidates. Mathematics cannot be studied in compartments and it is expected that candidates will be able to apply knowledge from one section to another section of work.
- (b) It is evident that many of the errors made by candidates in answering the Trigonometry questions in this paper have their origins in a poor understanding of the basics and the foundational competencies taught in the earlier grades.
- (c) In general, candidates need to exercise caution with algebraic manipulation skills since overlooking certain basic principles or practices results in the unnecessary loss of marks.
- (d) Although the calculator is an effective and necessary tool in Mathematics, learners appear to believe that the calculator provides the answer to all their problems. Some candidates need to realise that conceptual development and algebraic manipulation are often impeded because of their dependence on a calculator.

## 10.6 DIAGNOSTIC QUESTION ANALYSIS FOR PAPER 2

The following graph was based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 10.6.1 Average performance per question in Paper 2**

| Q  | Topic(s)            |
|----|---------------------|
| 1  | Data Handling       |
| 2  | Data Handling       |
| 3  | Analytical Geometry |
| 4  | Analytical Geometry |
| 5  | Trigonometry        |
| 6  | Trigonometry        |
| 7  | Trigonometry        |
| 8  | Euclidean Geometry  |
| 9  | Euclidean Geometry  |
| 10 | Euclidean Geometry  |

**Graph 10.6.2 Average performance per subquestion in Paper 2**

## 10.7 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### QUESTION 1: DATA HANDLING

#### Common errors and misconceptions

- (a) In Q1.1.1 many candidates did not read the question correctly and calculated the mean of the popularity score instead of the mean number of votes. This may be as a result of them being used to calculating  $\bar{x}$  and not  $\bar{y}$ .
- (b) In Q1.1.2 many candidates calculated the standard deviation for the popularity score instead of the number of votes.
- (c) It is evident that a number of candidates did not understand Q1.2. They calculated the one standard deviation interval but did not conclude whether this was for *invited or uninvited learner*. Some candidates calculated the limit for  $\bar{y} + \sigma_y$  instead of  $\bar{y} - \sigma_y$ .
- (d) When writing the equation in Q1.3, a few candidates interchanged the values of  $a$  and  $b$ . Some calculated the values of  $a$  and  $b$  correctly, but did not write the equation.
- (e) In Q1.4 some candidates used other values besides 72 that was given in the question. A few candidates substituted  $y = 72$  and then solved for  $x$ .
- (f) Candidates were confused when answering Q1.5.1. They provided general responses instead of making use of the *scatterplot* to justify their answer.
- (g) In Q1.5.2 most candidates stated that the prediction would be reliable but could not justify this statement with a comment about the correlation coefficient.

#### Suggestions for improvement

- (a) When teaching *Statistics*, the focus should not be on the calculations only. Teachers should also pay attention to the meaning of the different concepts, e.g. *mean*, *standard deviation*, *correlation coefficient*, etc. The values obtained in the calculations should then become more meaningful for learners.
- (b) Understanding of statistical terminology is developed by using these terms frequently in the class. Using diagrams when explaining the concepts of *standard deviation* and *deviation intervals* from the *mean* should help learners to understand these concepts.
- (c) Practise calculator skills with learners. When calculating the standard deviation, the population standard deviation ( $\sigma_x$ ) should be used and not the sample standard deviation ( $s_x$ ). Learners are advised to become familiar with and use the same brand of calculator in the examinations.
- (d) When determining the equation of the least squares regression line, it is advisable that learners write down the values of  $a$  and  $b$  and then write down the equation of the regression line. In this way, they can get the CA mark for the equation.
- (e) Learners should be able to use the values of their calculations to make predictions and comments about the data. Time should be devoted to interpretation questions.

**QUESTION 2: DATA HANDLING****Common errors and misconceptions**

- (a) A few candidates gave the answer to Q2.1 as 70, the highest y-value on the cumulative frequency axis, instead of the highest value that the graph attained on the cumulative frequency axis.
- (b) In Q2.2 many candidates were unable to state the modal class. They provided a single value instead of an interval.
- (c) In Q2.3 many candidates gave the answer as 34, the cumulative frequency of employees who spent 22,5% or less on fuel. This was incorrect as the question required candidates to subtract this value from 60 in order to get the number of employees who spent more than 22,5% on fuel.
- (d) Some candidates worked out the monthly salary as 7% of R2 400 while others worked out the monthly salary as R2 400 X 7. Both of these were incorrect.
- (e) Many candidates failed to interpret Q2.5 and consequently did not attempt this question. They were unable to relate the increase in fuel price to the *ogive*. Some provided an unrelated response.

**Suggestions for improvement**

- (a) The frequency table should be viewed alongside the *ogive*. In this way, learners should be able to identify the *modal class* as that part on the *ogive* where the gradient is the steepest.
- (b) Greater care needs to be taken when reading off a graph. While the values for the major gridlines are given, learners must take care when establishing the units of the minor gridlines. The minor gridlines are not always 1 unit apart.
- (c) Learners should be exposed to interpretation-type questions as part of their classwork. This should build confidence in them when they are faced with similar questions in tests and examinations.
- (d) Reading for understanding is a fundamental requirement in the *Data Handling* section. This skill needs to be developed in classroom activities.

**QUESTION 3: ANALYTICAL GEOMETRY****Common errors and misconceptions**

- (a) In Q3.1.1 some candidates used the incorrect gradient formula despite it being given on the information sheet. Others substituted incorrectly, i.e.  $m_{AB} = \frac{2 - (-4)}{6 - 4}$  instead of  $m_{AB} = \frac{2 - (-4)}{4 - 6}$ . Some substituted correctly into the formula but arrived at the incorrect answer of  $m_{AB} = -\frac{3}{2}$ .

- (b) In Q3.1.2 many candidates ignored the fact that  $\alpha$  was an *obtuse angle* in the diagram. They correctly wrote down that  $\tan \alpha = -3$  but then calculated  $\alpha = 71,57^\circ$  instead of  $\alpha = 108,43^\circ$ .
- (c) Some candidates assumed that S was the midpoint of AC when answering Q3.1.4. They failed to realise the S was a y-intercept. Others substituted  $x = 0$  into the equation of AC but were unable to work correctly with the fractions to arrive at the correct answer for y.
- (d) In Q3.2 some candidates used the incorrect gradient for the line. Many calculated the gradient as if AB was perpendicular to CD instead of AB being parallel to CD. Some candidates substituted the coordinates of C instead of the coordinates of A.
- (e) In answering Q3.3.1 many candidates correctly calculated the gradient of AC as  $\frac{5}{6}$ . However, they were unable to link the angle of inclination of AC with  $\hat{A}CD$ . Some candidates incorrectly assumed that DC was perpendicular to AB.
- (f) In Q3.3.2 many candidates made the incorrect assumption that POSC was a *trapezium*. Some calculated random lengths of sides and incorrectly used these in the *area formula*.

### Suggestions for improvement

- (a) If learners are not sure, they should consult the information sheet for the correct formula.
- (b) Teachers should request learners to label the coordinates as  $(x_1; y_1)$  and  $(x_2; y_2)$  on the diagram. This should prevent learners from making mistakes when substituting the coordinates into a formula. The order of substitution must be consistent, especially when using the gradient formula.
- (c) Teachers need to emphasise the relationship between the sign of the gradient and the size of the angle of inclination of the line.
- (d) Emphasise to learners that it is not acceptable to make any assumptions, e.g. that a certain point is the midpoint of a line. Even if it looks as if the point is the midpoint, it may not just be assumed and used. These need to be proven first before the results can be used in an answer.
- (e) Teachers should encourage learners to write down the values that they have already calculated (lengths, angles and gradients) on the diagram. This will assist learners when they answer follow-up questions.
- (f) To answer questions in analytical geometry well, learners should master the properties of *quadrilaterals* and *triangles*. Constant revision of *Analytical Geometry* concepts taught in Grades 10 and 11 is essential, as much of the Grade 12 work revolves around these concepts.
- (g) Teachers should develop in learners the skill of dividing a quadrilateral into two triangles.
- (h) Teachers should show learners different orientations of the base and the perpendicular height of a triangle. This should give learners more options when calculating the area of a triangle.

- (i) The different topics in Mathematics should be integrated. Learners must be able to establish the connection between Euclidean Geometry and Analytical Geometry.

#### QUESTION 4: ANALYTICAL GEOMETRY

##### Common errors and misconceptions

- (a) In Q4.1 some candidates substituted incorrectly into the midpoint formula. They calculated the midpoint of the given points. An endpoint and the midpoint were given. The question required candidates to calculate the other endpoint.
- (b) In Q4.2.1 some candidates incorrectly used the coordinates of N as the centre of the circle. Other candidates wrote the equation as either  $(x-3)^2 + (y+5)^2 = 5$  or  $(x+3)^2 + (y-5)^2 = 25$  instead of  $(x-3)^2 + (y+5)^2 = 25$ .
- (c) Many candidates correctly calculated the gradient of the radius in Q4.2.2. However, they used this as the gradient of the tangent, which was incorrect. Other candidates incorrectly used  $-\frac{3}{4}$  as the gradient of the tangent.
- (d) The vast majority of candidates were unable to answer Q4.3 correctly. They were not familiar with the term *secant* used in the question.
- (e) Many candidates did not attempt Q4.4.1 because they could not place points A and B correctly on the diagram.
- (f) While a fair number of candidates had some knowledge of how to calculate the minimum value of AB, the fact that AB was given as a square root posed a challenge to many. Some candidates incorrectly took the square root of each of the terms, i.e.  $AB = \sqrt{2t^2 + 4t + 9}$  was then incorrectly simplified to  $AB = \sqrt{2t^2} + \sqrt{4t} + \sqrt{9}$ . They were then able to calculate the derivative of AB from this simplified expression.

##### Suggestions for improvement

- (a) Teachers should encourage learners to analyse the diagram before attempting any questions. They must first write down any given information on the diagram and then make deductions from this information.
- (b) Teachers need to revise the concept of perpendicular lines and gradients, particularly that the tangent is perpendicular to the radius at the point of contact.
- (c) Learners should be reminded to refer to the information sheet for the relevant formula.
- (d) Teachers should show learners how to visualise and make rough drawings of all extra information given in *Analytical Geometry* questions.
- (e) Teachers should ensure that learners are exposed to assessments that integrate various topics in Mathematics. Learners must also be exposed to higher-order questions in class and in school-based assessment tasks.

**QUESTION 5: TRIGONOMETRY****Common errors and misconceptions**

- (a) It was encouraging that many candidates had some knowledge to answer Q5.2. Some candidates incorrectly reduced  $\cos(90^\circ + \theta)$  to  $-\cos \theta$  or  $\sin \theta$ . Other candidates incorrectly simplified  $\frac{-\sin \theta}{-4 \sin \theta}$  to 4 instead of  $\frac{1}{4}$ . A few candidates incorrectly cancelled terms, i.e.  $\frac{-\sin \theta}{-\sin \theta - 3 \sin \theta}$  was simplified to  $\frac{1}{1 - 3 \sin \theta}$  instead of  $\frac{-\sin \theta}{-4 \sin \theta}$ .
- (b) In Q5.3 a number of candidates multiplied out the factors on the LHS and then attempted to factorise again, but made errors when factorising. Many candidates were unable to solve the equations:  $\cos x + 2 \sin x = 0$  and  $\sin 2x = \frac{1}{3}$ .
- (c) When answering Q5.4.1, many candidates left out the brackets in the expansions. They wrote  $\cos(x + y) \cdot \cos(x - y)$  as  $\cos x \cos y - \sin x \sin y \cdot \cos x \cos y + \sin x \sin y$  instead of  $(\cos x \cos y - \sin x \sin y)(\cos x \cos y + \sin x \sin y)$ . This created challenges when simplifying further. Many candidates did not use the squares' identities when simplifying further.
- (d) Most candidates did not see the link between Q5.4.1 and Q5.4.2. Many ignored the instruction and used a calculator to answer Q5.4.2. They were not awarded any marks for the answer.
- (e) In Q5.5.1 many candidates were unable to factorise the expression correctly into an expression involving a *sine double angle*. Some candidates got as far as  $4 \sin 2x \cos 2x$  but did not recognise that this can be written as  $2 \sin 4x$ .
- (f) The vast majority of the candidates could not answer Q5.5.2 because they were unable to write the expression in Q5.5.1 in terms of one trigonometric ratio. A few who managed to reduce the expression to  $2 \sin 4x$  wrote  $-2$  as the answer. They gave the minimum value of the expression instead of the value of  $x$  for which the minimum will occur.

**Suggestions for improvement**

- (a) Learners find it difficult to recall the Trigonometry taught in Grades 10 and 11. Revision of this work must be ongoing. It is better to revise small sections of work at a time than to give learners a comprehensive revision task.
- (b) Remind learners that the same simplification skills used in Algebra also apply to Trigonometry. Regular practice can remediate the poor algebraic and manipulation skills.
- (c) More emphasis should be placed on solving simple trigonometric equations, particularly that a basic trigonometric equation has two solutions in the interval from  $0^\circ$  to  $360^\circ$ . The key to solving trigonometric equations lies in understanding in which quadrants a trigonometric function is positive or negative.
- (d) Learners should be given exercises to practise solving complex trigonometric equations.

**QUESTION 6: TRIGONOMETRY (GRAPHS)****Common errors and misconceptions**

- (a) In Q6.1 some candidates gave the domain (which is an interval) instead of the period (which is a single value). Some candidates ignored the effect that the coefficient of  $x$  had on the period of the graph. They wrote the period as  $360^\circ$ , which was incorrect.
- (b) In answering Q6.2.1, many candidates did not realise that they could have substituted the  $x$ -coordinate into the equation to calculate the  $y$ -coordinate. Some candidates rounded off the value of  $k$  from 1,73 to 2. This was not accepted as a correct answer as the amplitude of the graph was 2 and  $k$  was not the  $y$ -coordinate of a turning point.
- (c) Although point B was to the left of the  $y$ -axis, some candidates wrote  $120^\circ$  as the answer to Q6.2.2. Others gave the answer as  $-60^\circ$ . Both these answers were incorrect.
- (d) When answering Q6.3, many candidates gave the answer as  $-2 \leq y \leq 2$ , the range of  $g(x)$ , instead of giving the range of  $2g(x)$ . Other candidates incorrectly wrote the range as  $4 \leq y \leq -4$  or  $y \in [4; -4]$ .
- (e) Most candidates were unable to link Q6.4 to the given graphs. Some candidates gave the answer as  $-60^\circ \leq x \leq 0^\circ$  instead of  $-65^\circ \leq x \leq -5^\circ$ . These candidates knew where  $g$  was below  $f$  but did not take into account the translation of  $5^\circ$  to the left.
- (f) Most candidates had little idea how the graphs could be used to answer Q6.5. They could not make the association between  $\sin x \cos x$  and  $\sin 2x$ .

**Suggestions for improvement**

- (a) When teaching the drawing of trigonometric graphs, it is strongly recommended that the approach should be to ensure that learners know the basic graphs – the *mother graphs* – very well. Thereafter, teachers should explain how to draw the required graph by applying knowledge of transformations.
- (b) Although these concepts are discussed in Grade 10, it is necessary for learners to be reminded constantly of the meaning of concepts like *period*, *domain*, *amplitude* and *range*.
- (c) Learners should be told that the period of a trigonometric function is the length of a function's cycle. Since this value is a length, it is a single number and not an interval of values.
- (d) When teaching trigonometric functions, teachers should emphasise the meaning and effect of each of the parameters:  $a$ ,  $k$ ,  $p$  and  $q$  in the equation  $y = a \sin(kx + p) + q$ , for example.
- (e) Teachers should make learners aware of the cyclic nature of trigonometric graphs. This is useful in determining the coordinates of other points on the graph.

**QUESTION 7: TRIGONOMETRY****Common errors and misconceptions**

- (a) In Q7.1 many candidates were able to substitute correctly into the *Theorem of Pythagoras*, i.e.  $AD^2 = (\sqrt{5}p)^2 + (2p)^2$ . However, they made errors when calculating the length of AD. They either gave the length of AD as  $\sqrt{29}p$  or  $7p$ . Some candidates attempted to calculate the length of AD by making use of trigonometric ratios. However, these candidates failed to first calculate the size of  $\hat{BAD}$ .
- (b) In their attempt to answer Q7.2, some candidates used the *sine formula* with side AC and  $\hat{ADC}$ . However, this did not allow them to arrive at the required expression for CD. Some candidates were unable to calculate the size of  $\hat{CAD}$  correctly. They arrived at  $\hat{CAD} = 180^\circ - (45^\circ + x) = 135^\circ + x$  or  $\sin \hat{CAD} = \sin[180^\circ - (45^\circ + x)] = \sin(45^\circ - x)$ . Both of which were incorrect. A fair number of candidates were unsure whether they should use the *sine formula* or the *cosine formula* to answer this question.
- (c) Some candidates incorrectly attempted to calculate the area of  $\triangle ADC$  by using the formula: area of triangle =  $\frac{1}{2}$  base  $\times$  height. They were under the impression that they were working in a right-angled triangle.

**Suggestions for improvement**

- (a) A careful analysis of the information provided will give learners some idea of the concepts required in solving a triangle.
- (b) Teachers need to develop strategies to be used when solving right-angled triangles and triangles that are not right-angled. Teach learners the conditions that determine which rule should be used to solve the question.
- (c) It might be a good idea to give learners an exercise in which they identify which rule is to be used to solve the question. The learners must also give a reason why they think that the rule that they have selected applies to the question.
- (d) Learners should be encouraged to highlight the different triangles using different colours.
- (e) Initially, expose learners to numeric questions when solving 3D problems. This makes it easier for them to develop strategies on how to solve such questions. Once learners have gained confidence with numeric type questions, they should be exposed to non-numeric and higher-order questions.

**QUESTION 8: EUCLIDEAN GEOMETRY****Common errors and misconceptions**

- (a) In Q8.1.1 some candidates incorrectly stated that  $\hat{P} = \hat{M}_2$  with the reason that the opposite angles of a cyclic quadrilateral are equal. Other candidates did not state adequate information in the reason. Opposite angles and opposite angles are supplementary were not accepted as correct.

- (b) When answering Q8.1.2, some candidates incorrectly stated that  $\hat{M}_1 = \hat{O}_1$  with the reason that they were angles in the same segment. Some candidates gave the reason right-angled triangle. This was not accepted as correct.
- (c) In Q8.1.3 some candidates failed to see the relationship between  $\hat{M}_1$  and  $\hat{O}_1$  as being angle at centre is equal to twice the angle at the circumference.
- (d) Many candidates were unable to give the correct reason for the lines being parallel. They confused the theorem with its converse. Answers given were the *proportionality theorem* instead of the *converse proportionality theorem*; or the *converse midpoint theorem* instead of the *midpoint theorem*.
- (e) In Q8.2.2 many candidates assumed that  $BG = DE$  even though this was not the case in the diagram given. Some candidates made algebraic errors, e.g.  $4(x + 1) = 4x + 1$ . Other candidates incorrectly wrote down  $\frac{BF}{FC} = \frac{BG}{DE}$  instead of  $\frac{BF}{FC} = \frac{BG}{GH}$ . Some candidates did not mention the parallel lines in the reason.

### Suggestions for improvement

- (a) The key to answering Euclidean Geometry successfully is to be fully conversant with the terminology in this section. To this end, teachers should explain the meaning of *chord*, *tangent*, *cyclic quadrilateral*, etc. so that learners will be able to use them correctly.
- (b) Teachers must cover the basic work thoroughly. An explanation of the theorem should be accompanied by showing the relationship in a diagram.
- (c) Teachers are encouraged to use the 'Acceptable Reasons' in the *Examination Guidelines* when teaching. This should start from as early as Grade 8.
- (d) Learners should be encouraged to scrutinise the given information and the diagram for clues about which theorems could be used when answering the question.
- (e) Learners should be taught that all statements must be accompanied by reasons. It is essential that the parallel lines be mentioned when stating that corresponding angles are equal, alternate angles are equal, the sum of the co-interior angles is  $180^\circ$  or when stating the proportional intercept theorem.

## QUESTION 9: EUCLIDEAN GEOMETRY

### Common errors and misconceptions

- (a) In Q9.1 many candidates omitted the reason that angles on a straight line were supplementary. Some candidates used *similarity* instead of *congruency* to prove this theorem. Many candidates stated that OD was perpendicular to AB in the proof. This resulted in a breakdown as these candidates were unclear about what information was given and what they had to prove.
- (b) In Q9.2.1 many candidates could not identify that the radius was drawn to the midpoint of the chord. Those who could prove that OTBG was a *cyclic quadrilateral* gave the incorrect reason that angles in the same segment instead of converse angles in the same segment.

- (c) When answering Q9.2.2, some candidates stated that BG and OG were equal because they were tangents from a common point. This was incorrect because OG was not a tangent to the circle. A big challenge in this question was the poor labelling of angles. Candidates would refer to  $\hat{T}$  while there are a number of angles around point T.

### Suggestions for improvement

- (a) Learners should be taught that a construction is required in order to prove a theorem. If the construction is not shown, then the proof is regarded as a breakdown and they get no marks. Teachers should reinforce theory in short tests and assignments.
- (b) Teachers should focus on developing learners' skills to analyse the question and the diagram for clues on which theorems are required to answer the questions correctly.
- (c) Learners should be forced to use acceptable reasons in Euclidean Geometry. Teachers should explain the difference between a *theorem* and its *converse*. They should also explain the conditions for which theorems are applicable and when the converse will apply.
- (d) Learners need to be exposed to questions in Euclidean Geometry that include the theorems and the converses. When proving that a quadrilateral is cyclic, no circle terminology may be used when referring to the quadrilateral.
- (e) Learners should be discouraged from writing correct statements that are not related to the solution. No marks are awarded for statements that do not lead to solving the problem.
- (f) Learners need to be told that success in answering Euclidean Geometry comes from regular practice, starting off with the easy and progressing to the difficult.
- (g) Teachers should take some time to discuss the naming of angles. The acceptable methods are  $\hat{T}$  or  $\hat{T}_1$  or  $O\hat{T}S$ . Teachers should also clarify when it is acceptable to refer to an angle at  $\hat{T}$  and when to refer to it as  $\hat{T}_1$ .

## QUESTION 10: EUCLIDEAN GEOMETRY

### Common errors and misconceptions

- (a) A fair number of candidates made incorrect assumptions when answering Q10.1. Among them were that:  $\hat{S}_1 = 90^\circ$  and  $\hat{C}_2 = 90^\circ$ ,  $\hat{P}_1 = \hat{R}_1 + \hat{R}_2$  with the reason *exterior angle of cyclic quadrilateral*,  $\hat{P}_1 = \hat{C}_1$  with the reason *tan-chord theorem* and  $\hat{S}_2 = \hat{R}_2$  with the reason *angles in the same segment*.
- (b) In Q10.2 some candidates attempted to prove the ratios equal by using the *proportionality theorem* instead of similar triangles. A common error made by candidates attempting to prove that  $\triangle ASD$  was similar to  $\triangle ACR$  was to merely state that  $\hat{S}_1 = \hat{C}_2$  without any proof or reasons. This was seen as a breakdown in the answer.
- (c) Q10.3 required candidates to obtain a proportion from the similar triangles in Q10.2, using the *proportional intercept theorem* in  $\triangle RAC$  to establish a second proportion and

then to combine the two. Many candidates failed to establish one or the other proportion and therefore could not make the conclusion.

### **Suggestions for improvement**

- (a) More time needs to be spent on the teaching of Euclidean Geometry in all grades. More practice in Grade 11 and 12 Euclidean Geometry will help learners to understand theorems and diagram analysis. They should read the given information carefully without making any assumptions. This work covered in class must include different activities and all levels of the taxonomy.
- (b) Teachers should require learners to make use of the diagrams in the Answer Book to indicate angles and sides that are equal and record information that has been calculated.
- (c) Learners need to be made aware that writing correct, but irrelevant statements, will not earn them any marks in an examination.

# CHAPTER 11

## PHYSICAL SCIENCES

The following report should be read in conjunction with the Physical Sciences question papers of the November 2022 examinations.

### 11.1 PERFORMANCE TRENDS (2018–2022)

The number of candidates who sat for the Physical Sciences examination in 2022 increased substantially by 12 036 compared to that of 2021.

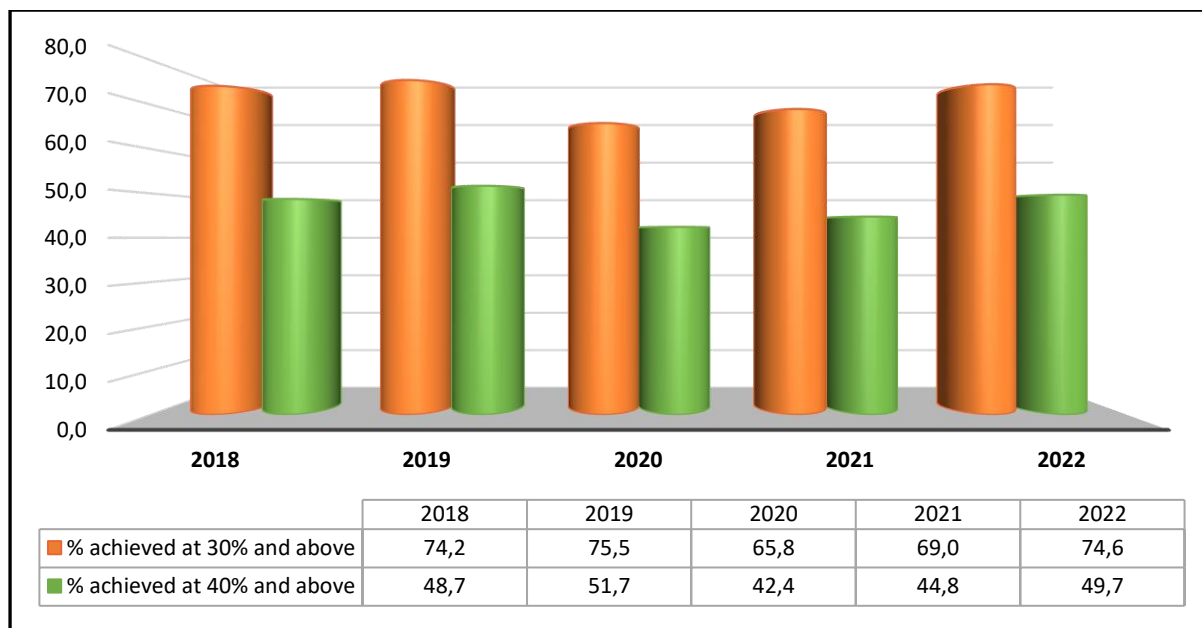
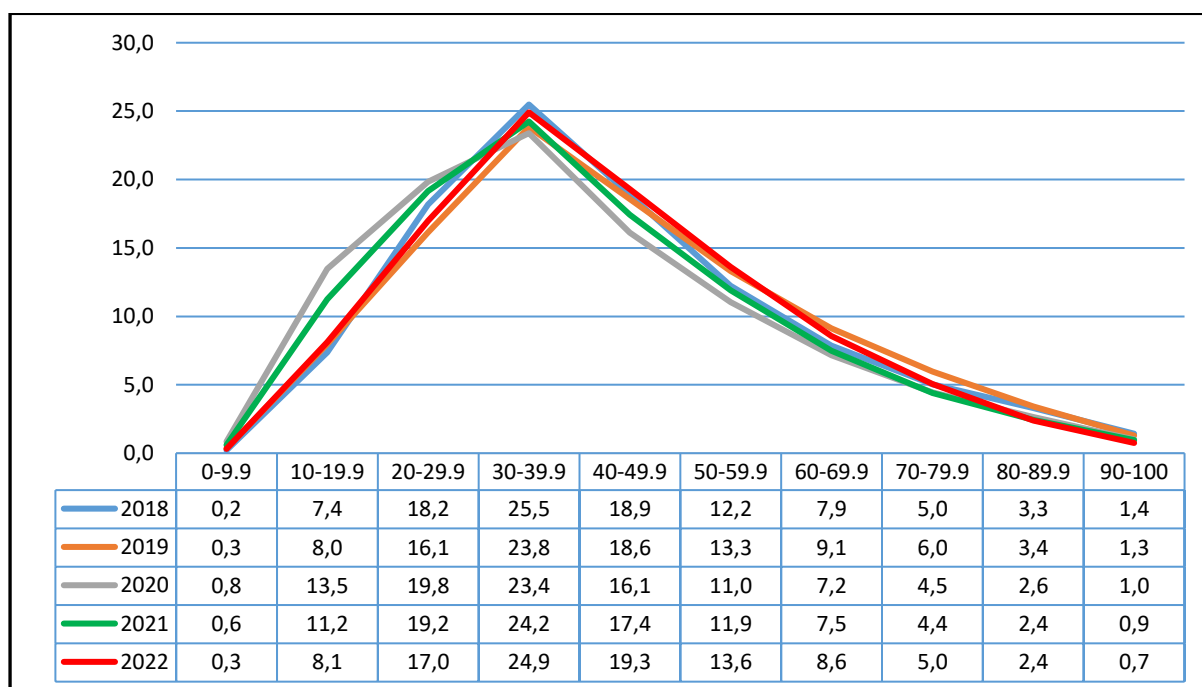
Although the table below indicates variable trends over the years 2018-2022, a significant improvement in results was evident this year. Candidates who passed at 30% (Level 2) improved from 69,0% in 2021 to 74,6% in 2022. There was also a corresponding improvement in the pass rate at 40% (Level 3) over the past two years from 44,8% to 49,7%.

There was a marginal decrease in the percentage of distinctions over 80% (Level 7) from 3,3% to 3,1%. This converts into a decrease of only 21 in the total number of distinctions.

In view of disruption to academic programmes in 2020 and 2021, the results achieved by this cohort are particularly commendable. Strategic intervention programmes at all levels (National, Provincial, Districts and Schools) ensured that learners were adequately prepared. The diligence and perseverance of the above-average candidates also contributed to the favourable overall performance.

**Table 11.1.1 Overall achievement rates in Physical Sciences**

| Year | No. wrote | No. achieved at 30% and above | % achieved at 30% and above | No. achieved at 40% and above | % achieved at 40% and above |
|------|-----------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|
| 2018 | 172 319   | 127 919                       | 74,2                        | 84 002                        | 48,7                        |
| 2019 | 164 478   | 124 237                       | 75,5                        | 85 034                        | 51,7                        |
| 2020 | 174 310   | 114 758                       | 65,8                        | 73 982                        | 42,4                        |
| 2021 | 196 968   | 135 915                       | 69,0                        | 88 164                        | 44,8                        |
| 2022 | 209 004   | 155 877                       | 74,6                        | 103 811                       | 49,7                        |

**Graph 11.1.1 Overall achievement rates in Physical Sciences (percentage)****Graph 11.1.2 Performance distribution curves in Physical Sciences (percentage)****General comments on Paper 1 and Paper 2**

The questions in Paper 1 on Newton's Laws and Projectile Motion were generally well answered. The multiple-choice questions, as well as the questions on the Photo-electric Effect and Static and Current Electricity were poorly answered. Performance pertaining to physical properties of organic compounds and galvanic cells in Paper 2 was good. The question on chemical equilibrium in Paper 2 was very poorly answered.

Questions pertaining to pure recall of content were very poorly answered because key words and phrases were omitted from definitions. Short informal assessment tasks relating to these issues will greatly assist in improving these shortcomings. This can be used to good effect in content relating to definitions and laws listed in the CAPS and the *Examination Guidelines*.

Interpretation of graphs is still a challenge for many learners. Problem-solving exercises that involve graphs should be done in a variety of topics. Identification of the variables in relation to the equation describing the graph should be stressed. Practical work needs more attention in schools to ensure learners are able to apply practical skills, e.g. identification of variables, drawing of conclusions, interpretation of results and drawing and interpretation of graphs.

The application of mathematical principles is still a challenge for many learners. Learners should be given a variety of problem-solving activities that involve mathematical knowledge pertaining to simultaneous equations, quadratic equations, binomials, factorisation, trigonometry and graphs in classwork, homework, tests, and examinations.

## 11.2 OVERVIEW OF CANDIDATES' PERFORMANCE IN PAPER 1

### General comments

Questions 1 and 10 were the worst answered questions. The poor performance in Question 1 may be one of the reasons for the contributing factors to the increase in the percentage of candidates achieving below 30% and the decrease in candidates achieving level 7.

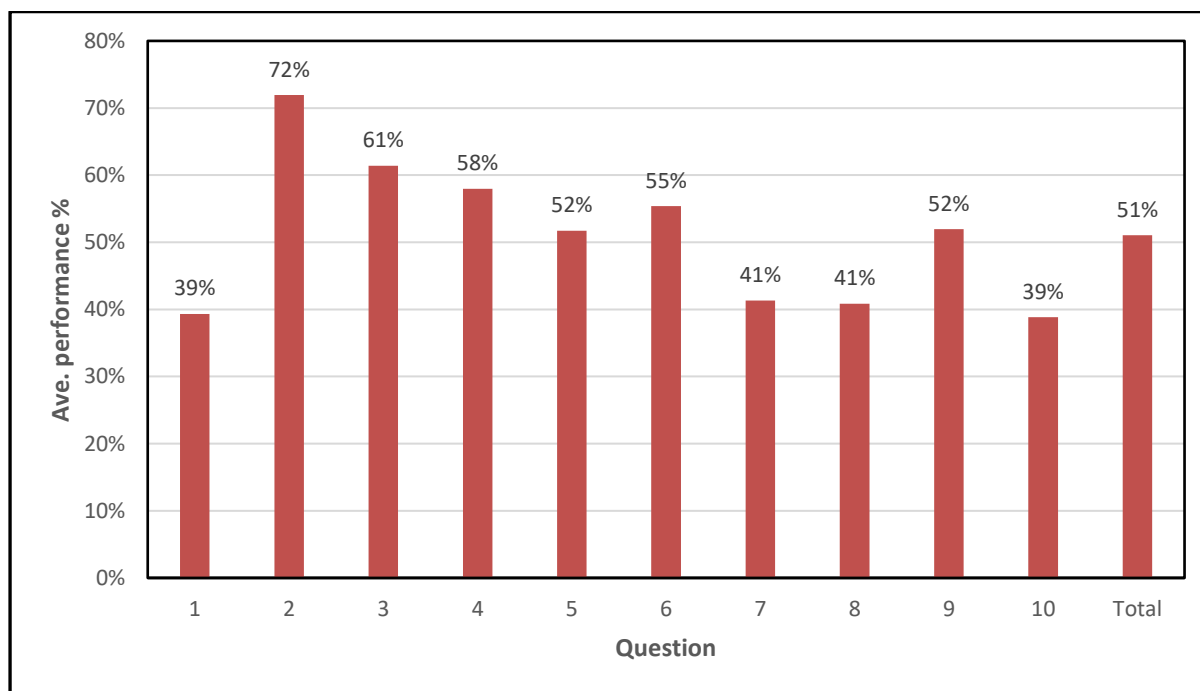
- (a) In 2021, Question 1 (Multiple-choice Questions – All topics) was one of the best answered questions, while in 2022, this was one of the questions answered very poorly. Many candidates were unable to use basic content knowledge of physics to answer these questions. Physical Science is a learning area which promotes critical thinking and higher-level reasoning. Some of the multiple-choice questions tested these two skills. The poor performance cannot be contributed entirely to the fact that there were some level 3 questions. One reason for this may be that many learners are only exposed to the answering of multiple-choice questions in the control tests and the preliminary examinations before they write their final examinations.
- (b) Teachers must differentiate between *vectors* and *scalars* in all topics. Learners must continually be reminded to use the formulae on the data sheet. They must be taught to read the questions slowly and with attention.
- (c) Interpretation and drawing of sketch graphs is a challenge for many learners. Problem-solving exercises that involve graphs should be done in a variety of topics. Identification of the variables in relation to the equation describing the graph should be stressed. The scale of graphs, gradient, ordered-pairs and x and y-intercepts need to be emphasised within problem-solving in science contexts.
- (d) In many questions, candidates made the same errors and displayed the same poor conceptual understanding as in previous years. Provinces should mediate the effective implementation of the recommendations in this diagnostic report.
- (e) The lack of knowledge of topics taught in Grades 10 and 11 can be seen in the poor performance of the candidates in Questions 7 and 8. Electrostatics and electric circuits must be completed in Grades 10 and 11 with only the internal resistance to be done in Grade 12. These two questions had the second lowest averages.
- (f) Many candidates lost marks unnecessarily: for not writing formulae; not showing their substitutions and for not writing the exact formula as given on the data sheet, e.g. leaving out subscripts such as  $F_{\text{net}}$ ,  $E_{k(\text{max})}$ ,  $W_{\text{nc}}$ . Some candidates did not give the correct units for the answers or they did not convert their numerical answers to a minimum of two decimal places and therefore lost the mark for the answer.

### 11.3 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 1

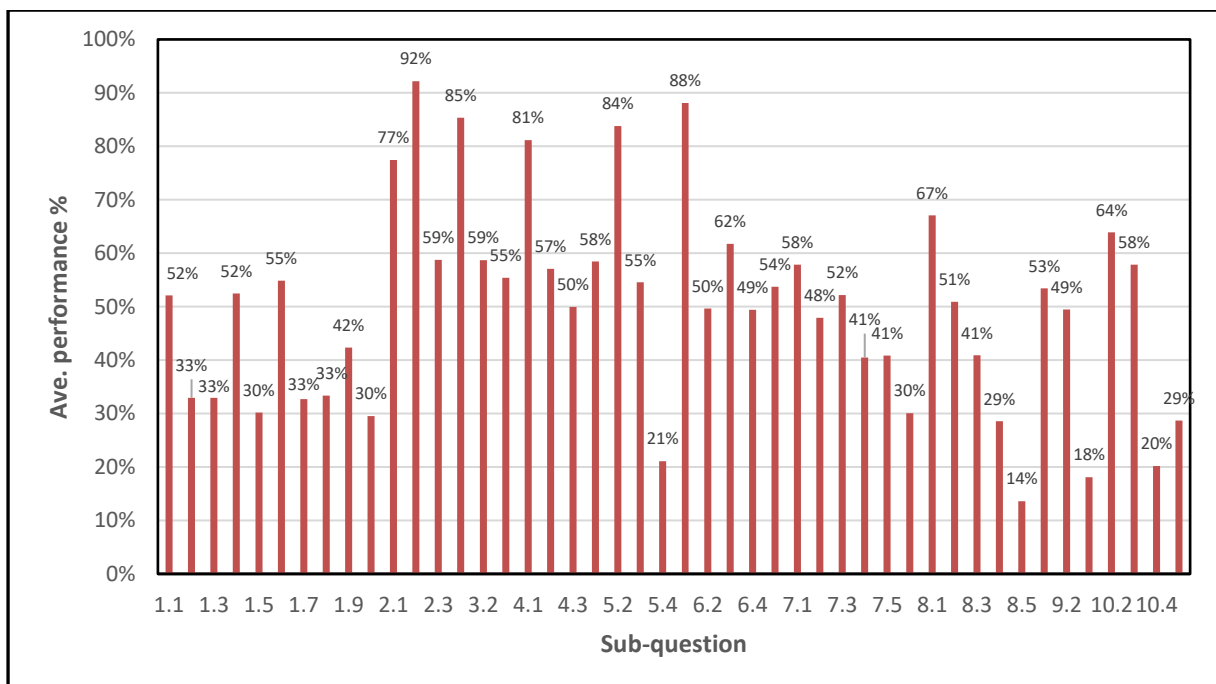
The following graph is based on data from a random sample of candidates. While this graph might not accurately reflect national averages, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

There was poor performance in Question 1 (Multiple-choice questions). It must be noted that the *Examination Guidelines* do not restrict multiple-choice questions to level 1 and 2 questions. The averages also reflect the poor performance in Question 10 and Questions 7 and 8.

**Graph 11.3.1 Average performance per question in Paper 1**



| Q | Topics                     | Q  | Topics                                           |
|---|----------------------------|----|--------------------------------------------------|
| 1 | Multiple-choice questions  | 6  | Doppler effect                                   |
| 2 | Newton's laws of motion    | 7  | Electrostatics (Coulomb's law & electric fields) |
| 3 | Vertical projectile motion | 8  | Electric circuits                                |
| 4 | Momentum and impulse       | 9  | Electrodynamics: Motors, generators, AC and DC   |
| 5 | Work, energy and power     | 10 | Photoelectric effect                             |

**Graph 11.3.2 Average performance per subquestion in Paper 1**

| Sub-Q | Topic                               | Sub-Q  | Topic                                 |
|-------|-------------------------------------|--------|---------------------------------------|
| 1.1   | Vectors and Scalars                 | 6.3    | Graph: proportionality                |
| 1.2   | Newton's Gravitational Law          | 6.4    | Calculating $v_L$ using gradient      |
| 1.3   | Vertical Projectile Motion          | 6.5    | Graph with increased $v_L$            |
| 1.4   | Work and Energy                     | 7.1    | Statement Coulomb's Law               |
| 1.5   | Work and Energy                     | 7.2    | Polarity of charge M                  |
| 1.6   | Doppler Effect                      | 7.3    | Free-body diagram                     |
| 1.7   | Electrostatics                      | 7.4    | Calculating $Q_M$                     |
| 1.8   | Circuit Electricity                 | 7.5.1  | Magnitude $F_{MN}$                    |
| 1.9   | Electrodynamics                     | 7.5.2  | Direction $F_{MN}$                    |
| 1.10  | Optical Phenomenon                  | 7.6    | Calculating $E_{net}$                 |
| 2.1   | Statement Newton's 2nd Law          | 8.1    | Statement Ohm's Law                   |
| 2.2   | Free-body diagram                   | 8.2.1  | Calculating $R_{ext}$                 |
| 2.3.1 | Application Newton 2                | 8.2.2  | Calculating $V_{ext}$                 |
| 2.3.2 | Application Newton 2                | 8.2.3  | Comparison voltmeter readings         |
| 3.1   | Defining free fall                  | 8.3.1  | Defining emf                          |
| 3.2.1 | Calculating time for maximum height | 8.3.2  | Comparison emf and voltmeter reading  |
| 3.2.3 | Calculating final velocity          | 8.3.3  | Explanation emf and voltmeter reading |
| 3.2.3 | Calculating time to pass door       | 8.4.1. | Addition of voltmeter effect on power |
| 3.3   | Velocity-times ketch graph          | 8.4.2  | Addition of voltmeter effect on $V_1$ |
| 4.1   | Statement Conservation of p         | 8.3    | Explanation reading on $V_1$          |
| 4.2.1 | Calculation conservation of p       | 9.1.1  | Component DC generator                |
| 4.2.2 | Calculation impulse                 | 9.1.2  | Direction of current in coil          |
| 4.3   | Elastic/Inelastic collisions        | 9.1.3  | Determining period                    |
| 4.1   | Conservation of p                   | 9.1.4  | Sketching V vs $\Delta t$ graph       |
| 5.1   | Defining non- conservative force    | 9.2    | Calculating average power dissipated  |
| 5.2   | Free-body diagram                   | 10.1   | Evidence particle nature of light     |
| 5.3   | Calculate $W_{nc}$                  | 10.2   | Defining work function                |
| 5.4   | Calculate $\Delta x$                | 10.3   | Calculation max $E_K$                 |
| 6.1   | Naming phenomenon illustrated       | 10.4   | Calculation number of photons         |
| 6.2   | Application Doppler                 | 10.5   | Influence of light intensity          |

## 11.4 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 1

### QUESTION 1: MULTIPLE-CHOICE QUESTIONS

#### Common errors and misconceptions

- (a) In Q1.2 many candidates struggled with the application of *ratio* and *proportion*.
- (b) Two principles – conservation of mechanical energy and the fact that  $g$  stays constant irrespective of the velocity – had to be identified in Q1.3. Candidates could not identify the principles at stake.
- (c) For Q1.4 candidates had to realise that the engine must exert a constant force to overcome the frictional force to stay at constant velocity: The formula  $P = Fv$  is not used often. However, candidates could also have derived the correct answer by using the formula  $\frac{F_{\text{net}}\Delta x \cos\theta}{\Delta t}$  and realising that  $\frac{\Delta x}{\Delta t} = v$ . Both  $F_{\text{net}}$  and  $v$  are constant.
- (d) A few concepts were involved in Q1.5, namely negative acceleration, work done by frictional force and mechanical energy, making this a challenging question.
- (e) In Q1.7 candidates either failed to realise that the electrostatic force ( $F$ ) that one charge exerts on the other is inversely proportional to the square of the distance between them ( $r^2$ ) or that a hyperbola represents an inverse proportionality.
- (f) Candidates generally seem to have a problem with the interpretation of circuit diagrams and the relationship between the voltmeter readings across the series and parallel connections as well as the  $V_{\text{ext}}$ . The poor performance in Q1.8 was repeated in Q8.
- (g) Candidates should have known that the induced emf in a generator is dependent on the speed of rotation of the coil, the number of loops and the strength of the magnet. Although this has been tested repeatedly in past papers, candidates failed to realise in Q1.9 that doubling the speed of the coil will double the induced emf and halve the period.
- (h) The poor performance in Q1.10, a level-2 question, suggests that the teaching of the *photoelectric effect* and the *types of line spectra and their formation* is neglected in many schools as this is the last section to be taught. The poor performance in this question was repeated in Q10, which supports this assumption.

#### Suggestions for improvement

- (a) Multiple-choice questions test learners' understanding of concepts, principles, laws, and the relationship between the dependent, independent, and constant variables. This must be demonstrated through logical reasoning and not by engaging in lengthy calculations. During teaching and learning, carefully selected concrete examples must be worked out to demonstrate these concepts, laws and principles, followed by generalisations and the making of predictions.
- (b) Learners must also be encouraged to refer to the formula sheet because it gives the summaries of laws and principles as well as the relationship between the variables.
- (c) Teachers should include the use of ICT in the teaching of the subject. Multimedia/Software such as *PhET* and *Edukite* as well as YouTube videos should be

used to demonstrate the answers for multiple-choice questions through simulations (virtual experiments).

- (d) Assessment of multiple-choice questions should be included in all class activities and not only in control tests and preliminary examinations. It is recommended that a booklet containing multiple-choice questions from different topics, from different sources, such as previous NSC and provincial papers and textbooks is prepared. The booklet should also have a step-by-step explanation on how to answer multiple-choice questions.

## QUESTION 2: NEWTON'S LAWS OF MOTION

### Common errors and misconceptions

- (a) Candidates omitted key words in the definitions. Some only wrote the first part of *Newton's Second Law*, which carried no marks, and failed to relate *acceleration* to the *resultant force* and *mass of an object*.
- (b) The free-body diagram was very well answered, perhaps because it was on a horizontal plane. Common errors in the free-body diagram were missing labels, missing arrows, arrows not touching the dot, additional forces, the normal force not perpendicular to the plane and the gravitational force not perpendicular to the horizontal. Some candidates drew the free-body diagram for Q instead of P.
- (c) Units were often left out of the final answer of the calculation. Some candidates still used the system approach with the two objects. In Q2.3.2 most candidates failed to calculate the angle because of the substitution of incorrect values due to their inability to resolve forces correctly.

### Suggestions for improvement

- (a) Learners must be exposed to the definitions in the *Examination Guidelines* and the CAPS. Key words must be emphasised in these definitions.
- (b) The importance of drawing free-body diagrams for each object correctly and their usefulness in problem-solving must be emphasised. Encourage learners to use different problem-solving strategies to solve the same problem to ensure that they gain a greater understanding of the problem and solutions.
- (c) Teachers should emphasise that the net force acting on an object is the sum of all the forces acting on the object parallel to the direction of motion of the object.
- (d) Learners must be systematically exposed to different questions in which trigonometric relations must be applied.
- (e) Teachers should expose their learners to all scenarios for this section as described in the *Examination Guidelines*.

## QUESTION 3: VERTICAL PROJECTILE MOTION

### Common errors and misconceptions

- (a) Some candidates omitted the word *only* in Q3.1. Candidates must distinguish between the terms *free fall* and a *projectile*.

- (b) Not applying the correct sign convention to each vector quantity was a problem throughout Q3.2. There are many different displacements, velocity, and time values. A number of candidates failed to substitute the correct values for the different stages of the motion. In Q3.2.2 most candidates omitted the direction of the vector at the final answer.
- (c) Graphs are always a challenge. Many candidates drew a curved graph in Q3.3 instead of a straight line. Some candidates used scales to subdivide the axes and did not show the correct values of the initial velocity and time.

### Suggestions for improvement

- (a) Learners must first make sure that they understand the motion of the projectile and which values are relevant at each position. A rough drawing or diagram indicating the physical motion of the projectiles together with all information given (velocities, displacements etc.) is necessary in all scenarios.
- (b) Advise learners to start every calculation in mechanics, including vertical projectile motion, by indicating the sign convention at the beginning of the problem. Emphasise that the direction of gravitational acceleration does not change in a question but remains constant. They should keep to one sign convention when solving a problem and not change their chosen sign convention within a problem as this could lead to confusion.
- (c) Time cannot be a negative. Learners simply change it to a positive without thinking about it. This creates a substitution error.
- (d) Teach learners the skills of interpreting and sketching graphs especially for projectiles, e.g. to have reference points and to collect all relevant data before calculating any quantity.
- (e) Expose learners to several questions involving different scenarios of projectile motion, including two objects.

## QUESTION 4: MOMENTUM AND IMPULSE

### Common errors and misconceptions

- (a) In Q4.1 candidates lost marks due to the omission of the words *total* or *isolated system*. Note that the correct term is an *isolated system* and not a *closed system*.
- (b) The safest way to start a calculation applying the conservation of total momentum is  $\Sigma p_i = \Sigma p_f$  as was done by many candidates in Q4.2.1. However, some candidates wrote  $E p_i = E p_f$ . Others combined the masses after the collision – this may be due to incorrect reading or assumptions based on previous questions on momentum.
- (c) In applying  $F_{\text{net}} \Delta t = m(v_f - v_i)$  in Q4.2.2 many candidates swapped the initial and final velocities of the trolley or used an incorrect sign convention for the velocities. Many candidates used the incorrect time value which they read off from the graph.
- (d) Several candidates used momentum in Q4.3 to determine whether the collision was elastic or inelastic and then concluded that it was elastic. Candidates who started off with  $\Sigma K_i = \Sigma K_f$  lost one mark. They needed to calculate  $\Sigma K_i$  and  $\Sigma K_f$  separately and then compare the values.

**Suggestions for improvement**

- (a) The vector nature of momentum and impulse should be emphasised during teaching.
- (b) Expose learners to many different scenarios or problems that involve the principle of conservation of linear momentum and impulse.
- (c) Teachers should emphasise to learners not to give the conclusion before starting to calculate the proof when solving questions on elastic and inelastic collisions. Learners should calculate the total kinetic energies before and after the collision separately, compare them and then conclude whether the collision was elastic or inelastic.
- (d) Questions involving graphs should be included in assessments of all topics.

**QUESTION 5: WORK, ENERGY AND POWER****Common errors and misconceptions**

- (a) The average for Q5.1 requiring the definition of a non-conservative force, was disappointingly low. Many candidates omitted key words in the definition. If they did not mention the work done, no marks were awarded. If they omitted the word *force*, only one mark was awarded. Some candidates wrote *independent* instead of *dependent*.
- (b) Candidates lost unnecessary marks in Q5.2 due to writing incorrect labels like  $F_{app}$  or  $F_r$ , showing both weight and components of weight or not drawing  $F_N$  perpendicular to the surface but vertically upwards.
- (c) Some candidates were able to define a non-conservative force in Q5.1 but failed to identify that both the frictional force and the applied force are non-conservative forces in the calculation in Q5.3. Quite a few candidates copied the formula incorrectly from the data sheet as  $W_{nc} = \Delta U - \Delta K$ . They also swapped the initial velocity and the final velocity around.
- (d) Q5.4 was poorly answered. It is a level-4 question and not many candidates got it correct or even attempted it. This was a very innovative question requiring thorough reading and a good understanding of the physics involved.

**Suggestions for improvement**

- (a) Many learners use the work-energy theorem to calculate the work done by non-conservative forces. It would appear that the equation for the work done by non-conservative forces is not taught in all schools.
- (b) When using  $W_{nc} = \Delta E_p + \Delta E_k$  or  $W_{net} = \Delta E_k$ , the identification of forces acting on the object is very important and therefore the use of free-body diagrams is advised. It must be emphasised that frictional force is not the only non-conservative force that can act on an object. An applied force is also a non-conservative force.
- (c) Work done by a force is always equal to the change in energy, e.g.,  $W_{net} = \Delta E_k$ ,  $W_{nc} = \Delta E_p + \Delta E_k$ ,  $W_w = -\Delta E_p$ .
- (d) Clearly explain to learners the difference between  $W_{net}$  (total work done by all forces acting on an object, i.e. both conservative and non-conservative) and  $W_{nc}$  (total work done by only non-conservative forces acting on the object).

- (e) Use *PhET* simulations for the identification of forces acting on objects.
- (f) Create a data base for questions on work, energy and power that can be shared with the learners as homework or use during interventions.

## QUESTION 6: DOPPLER EFFECT

### Common errors and misconceptions

- (a) There was a disappointingly low average for Q6.2 which was a level-1 question. Too many one-word answers and some strange applications were given. Language was an issue here. There is a difference between listening to something and measuring something.
- (b) In Q6.3 many candidates erroneously thought that *directly proportional* and *x increases as y increases* had the same meaning.
- (c) Many candidates struggled with questions involving graphs. For Q6.4 they did not understand that the gradient was the ratio of  $f_L$  to  $f_S$ . Quite a few candidates started with an incorrectly manipulated Doppler formula.
- (d) In Q6.5 some candidates drew either only the graph for B or did not draw the two graphs on the same set of axes.

### Suggestions for improvement

- (a) Teachers need to source and expose learners to a variety of questions relating to the Doppler effect equation as the scope is very broad, given the number of variables in the equation.
- (b) Learners must be taught to write the Doppler formula as it appears on the data sheet. They should not convert it and then start the substitutions. They must always keep the plus and minus there in the first step of writing the formula. Only once substitution starts, should they change the signs.
- (c) Problem-solving exercises that involve the interpretation and drawing of sketch graphs should be done in a variety of topics. Identification of the variables in relation to the equation describing the graph should be stressed. Teachers should note that learners can be asked to draw or interpret a graph for any of the physics topics.
- (d) Expose learners to *PhET* simulations, YouTube videos and the Ten-Fold App to demonstrate the Doppler Effect.

## QUESTION 7: ELECTROSTATICS (COULOMB'S LAW and ELECTRIC FIELDS)

### Common errors and misconceptions

- (a) Candidates struggled to understand the scenario and to answer this question according to *Coulomb's Law*. Candidates did not do well in the question, especially the calculations. The candidates did not differentiate between *electrostatic force* and *electric field* at a point and therefore the calculations in Q7.4 and Q7.6 were affected.
- (b) Q7.1, a level-1 question (stating *Coulomb's Law*), was poorly answered. Quite a few candidates lost marks due to the following mistakes:

- Stating *Newton's Law of Gravitation*
  - Writing *directly proportional to charges* instead of *directly proportional to the product of the charges*
  - Not writing *square* for the inverse proportionality
  - Using *indirectly proportional* instead of *inversely proportional*
- (c) In Q7.2 many candidates did not realise that if sphere N carries a positive charge, then sphere M must have a negative charge so that the two can attract each other.
- (d) Language was an issue in the answering of Q7.3. Many candidates thought that sphere N was hanging from a string and labelled the upward force T. Several candidates labelled the upward force as N whilst others failed to identify the electrostatic forces exerted by M and N on each other.
- (e) In Q7.4 some candidates swapped/mixed the formulae for E and F as follows:  $F = k \frac{Q}{r^2}$  and  $E = k \frac{Q_1 Q_2}{r^2}$ . Candidates did not use the absolute value of the charges when substituting in the formula of Coulomb's Law.
- (f) Some candidates failed to identify  $F_{MN}$  and  $F_{NM}$  as action-reaction forces in Q7.5.
- (g) In Q7.6 many candidates did not understand the superposition of electrostatic forces when there is more than one charge. Candidates did not draw a vector diagram for the electric fields due to each charge and thus did not realise that the magnitudes of the electric fields must be subtracted to determine the net electric field at point X.

### Suggestions for improvement

- (a) Teachers need to explain to learners that calculations of net electrostatic force and electric field are similar in terms of their vector considerations.
- (b) The absolute value of the charge must be substituted in *Coulomb's Law* and the equations for calculating the electric field. (If negative charge values are substituted learners use the negative answer mistakenly as an indication of direction.)
- (c) Teachers should emphasise the difference between *electrostatic force* and *electric field* at a point as well as the equations thereof.
- (d) The application of the *principle of superposition of electric fields* and *electrostatic forces* should be revised.

## QUESTION 8: ELECTRIC CIRCUITS

### Common errors and misconceptions

- (a) Candidates performed very poorly in this question. Many candidates lacked understanding in this question and did not know how to express themselves. Consequently, their explanations were inadequate. The scaffolding of the question is very evident in the performance of the candidates.
- (b) The relatively poor performance of candidates in Q8.1, i.e. the statement of *Ohm's Law* which they should have been taught in Grades 10 and 11, indicates that many schools did not revise current electricity or teach internal resistance in Grade 12.

- (c) Most candidates had difficulty interpreting the circuit. Candidates could not correctly identify the resistors that are in series and those in parallel in Q8.2.1 and Q 8.2.2. Many candidates thought the  $2\ \Omega$  resistor was in series in the circuit. This has been a serious misconception for many years. Candidates seemed to think that parallel resistors must be in lines that are parallel to each other. Candidates struggled to grasp the flow of current through resistors in series and parallel and were thus unable to understand that there is a potential difference across the  $4\ \Omega$  resistor.
- (d) In Q8.3.1 some candidates omitted the keyword *maximum* energy and related the maximum energy to *per time* instead of per coulomb in the definition of emf. The Grade 10 explanation of emf is not the definition.
- (e) In Q8.3.2 and Q8.3.3 many candidates did not understand that  $V_1$  measures  $V_{\text{ext}}$  and  $V_2$  only measures  $V_{\text{II}}$ .
- (f) Q8.4 and Q8.5 both required candidates to understand what the effect is of replacing the switch in the parallel branch with a high-resistance voltmeter. The averages for these subquestions reflect the poor understanding that candidates had of this situation. Most candidates did not refer to the impact of the voltmeter in their answers. A typical incorrect or incomplete explanation given was that  $R_{\text{ext}}$  increases and therefore  $V_{\text{ext}}$  will increase because  $V$  is directly proportional to  $R$  – with no reference to total current or internal resistance and  $V_{\text{LOST}}$  not even referring to internal resistance. This is a serious misconception.

### Suggestions for improvement

- (a) Although the principles of series and parallel circuits are taught from Grade 9, the basic principles should be revisited and practised constantly. The critical features of series and parallel circuits with and without internal resistance must be emphasised.
- (b) Learners must be exposed to the various examples of circuit diagrams where two resistors are included in a parallel branch. Use different scenarios from past question papers to expose learners to these types of questions.
- (c) Ensure learners understand that the  $R_1$  and  $R_2$  in the parallel resistor formula are not necessarily individual resistances but the total resistance of the branch. For example, there could be more than one resistor in a branch in a circuit but the resistance in that branch is equivalent to  $R_1$ .
- (d) The effect on voltmeter and ammeter readings of any short circuit or bridging of resistors, opening and closing of switches, and connecting voltmeters in different positions in a circuit needs to be taught.
- (e) Use *PhET* simulations to demonstrate the relationship between  $V_{\text{ext}}$  and  $V_{\text{int}}$  and the effect of adding resistors or removing resistors in series and parallel can also be demonstrated.

## QUESTION 9: ELECTRODYNAMICS

### Common errors and misconceptions

- (a) The average for this question was relatively low considering that even the level-3 questions were easy. Q9.1.2, Q9.1.3 and Q9.2 were poorly answered. Many candidates either did not read with understanding or were not taught electrodynamics in enough detail.

- (b) In Q9.1.1 candidates had to carefully distinguish between the words *split* and *slit*. Many candidates confused the two words by answering *slipt*, *spilt*, *slipt*, *spleeted*, etc.
- (c) Q9.1.3 was not well answered. Candidates did not understand the relationship between *period* and *frequency*.
- (d) In Q9.1.4 candidates did not relate the position of the coil in the diagram to an induced starting emf of 0 V. Many did not indicate the labels for the 90 V or draw one full cycle or indicate the correct times for  $T$  or  $\frac{1}{2}T$  at the correct position. Quite a few candidates drew the graph for an AC generator.
- (e) In Q9.2 candidates failed to realise that the current electricity formulae may also be used to calculate unknown quantities even though alternating current is being calculated. Conversion of time into seconds was problematic as well. Many candidates were unable to either use the  $W$  formulae directly or to first calculate  $P_{\text{ave}}$  and then  $W$ .

### Suggestions for improvement

- (a) Grade 10 and 11 revision of waves needs to be done before the learners start on this chapter, especially calculations with period and frequency. Skill in graph sketching can be practised more. The Grade 11 work on electromagnetic induction must be revised in Grade 12 when motors and generators are taught.
- (b) Emphasise the use of subscripts in the formulae when rms calculations are done.
- (c) The differences and similarities between  $V_{\text{rms}}$  and  $V_{\text{max}}$ ,  $I_{\text{rms}}$  and  $I_{\text{max}}$ ,  $P_{\text{ave}}$  and  $P_{\text{max}}$  must be explained clearly, and learners must be exposed to sufficient application-type questions.
- (d) Learners must be taught to differentiate AC from DC in words, using graphs, in terms of components used, law or principle used to find the direction of motion or the induced current and the factors responsible for the difference.
- (e) *Khan Academy*, *PhET* and *Edukite* are also very useful resources for teachers especially for topics that have many practical aspects.

## QUESTION 10: PHOTOELECTRIC EFFECT

### Common errors and misconceptions

- (a) Theory and calculations on the photoelectric effect were attempted and done by some candidates. This question was, however, still poorly answered.
- (b) The very low averages in Q10.1, a level-1 question, and Q10.4 are proof that the photoelectric effect is not being taught in many schools. From the answers it was clear that most candidates did not understand the very crux of the photoelectric effect, i.e., that one photon with sufficient energy can only release one electron and that the photoelectric effect provides evidence that light has a particle nature.
- (c) The low average for the definition of work function (Q10.2) again supports the assumption that this last section of the syllabus is either not taught at all or taught after the trial examinations in many schools.

- (d) Many candidates omitted the subscript 'max' in the equation  $E = W_0 + E_{k(\text{max})}$  in Q10.3. Some candidates confused frequency and energy.
- (e) Q10.4 was poorly answered. To relate the equation  $Q = I\Delta t$ , for calculating charge, to the number of photons that strikes the cathode was very difficult for the candidates. The concept that a photon is needed for each electron (causing the current to flow) was a challenge.
- (f) The lack of understanding the concept of intensity of light as the number of photons per unit time and that the increase in current implies an increase in electrons emitted per unit time was the reason for the poor performance in Q10.5, which had an average of 27%. Very few candidates used the terms *per unit time* or *per second*. Most of the candidates got 1 out of 3 marks.

### Suggestions for improvement

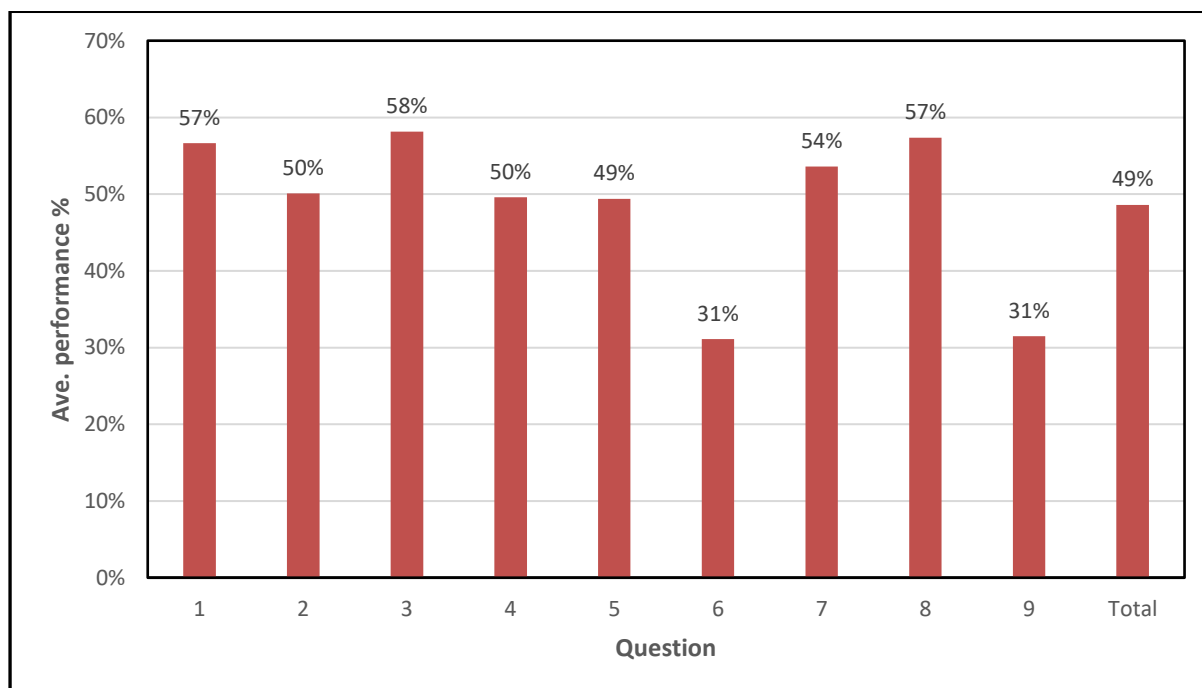
- (a) The importance of the phenomenon of the photoelectric effect in understanding the nature of light and that the particle nature of light was the only way of explaining this phenomenon must be emphasised.
- (b) Learners must understand that the formulae must be copied from data sheet as is and definitions studied from the *Examination Guidelines*.
- (c) Learners must use the formula  $E = W_0 + E_{k(\text{max})}$  as per data sheet (including the subscript max). Make sure that learners understand the meaning of the different expressions in this formula and that they can relate this to a straight-line graph.
- (d) Ensure that learners can explain the influence of the changes in the light (intensity and frequency) incident on a cathode/surface.
- (e) Teachers should use computer simulations (e.g. *PhET*) when teaching the photoelectric effect. This will assist in improving learners' understanding of the concept.
- (f) Teachers should emphasise and ensure that learners understand what causes a change in the number of photoelectrons per unit time and the maximum kinetic energy of the emitted photoelectrons.

### 11.5 DIAGNOSTIC QUESTION ANALYSIS OF PAPER 2

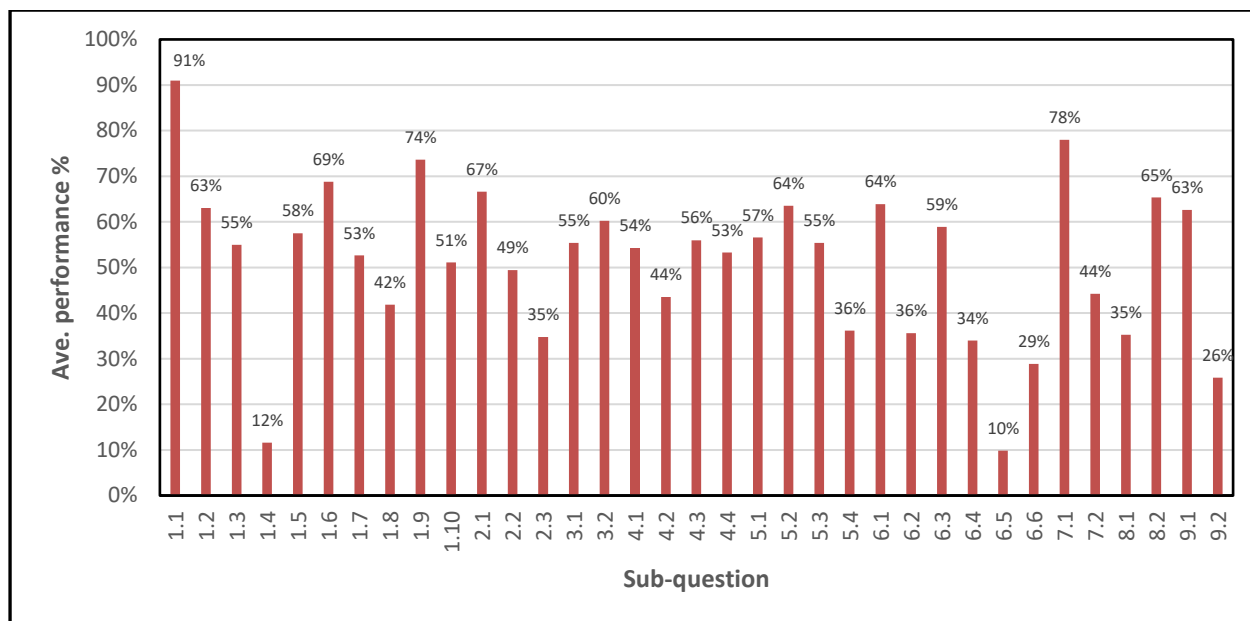
Performance improved in six questions, namely multiple-choice questions (Q1), physical properties of organic compounds (Q3), organic reactions (Q4), rate of reaction (Q5), acids and bases (Q7), galvanic cells (Q8) and electrolytic cells (Q9) as compared to 2021. While there was a significant improvement in the performance in organic reactions (Q4), rate of reaction (Q5), acids and bases (Q7) and galvanic cells (Q8), the performance in Q2 (nomenclature of organic compounds) showed a decline of more than 10%. Performance in electrolytic cells (Q9) improved by 1% from 2021.

The following graph is based on data from a random sample of candidates. While this graph might not reflect national averages accurately, it is useful in assessing the relative degrees of challenge of each question as experienced by candidates.

**Graph 11.5.1 Average performance per question in Paper 2**



| Q | Topic                                    | Q | Topic                |
|---|------------------------------------------|---|----------------------|
| 1 | Multiple-choice Questions                | 6 | Chemical Equilibrium |
| 2 | Organic Nomenclature and Structures      | 7 | Acids and Bases      |
| 3 | Physical Properties of Organic Molecules | 8 | Galvanic Cells       |
| 4 | Organic Reactions                        | 9 | Electrolytic Cells   |
| 5 | Rates of Reaction                        |   |                      |

**Graph 11.5.2 Average performance per subquestion in Paper 2**

| SubQ. | Topic                                                                                                                                                                                                                                 | SubQ | Topic                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1   | Definition of saturated hydrocarbons                                                                                                                                                                                                  | 5.1  | Factors affecting reaction rate; Collision theory                                                                                                                                         |
| 1.2   | Identify intermolecular forces in ethanoic acid and methyl propanoate                                                                                                                                                                 | 5.2  | Effect of temperature on Maxwell-Boltzmann distribution curve                                                                                                                             |
| 1.3   | Reaction of bromine water with alkenes                                                                                                                                                                                                | 5.3  | Volume reading on syringe and stoichiometric calculation                                                                                                                                  |
| 1.4   | Definition of reaction rate                                                                                                                                                                                                           | 5.4  | Interpret mass versus time graph and calculate time using stoichiometry and reaction rate.                                                                                                |
| 1.5   | Factors affecting reaction rate                                                                                                                                                                                                       | 6.1  | State Le Chatelier's principle                                                                                                                                                            |
| 1.6   | Chemical equilibrium                                                                                                                                                                                                                  | 6.2  | $K_c$ calculation                                                                                                                                                                         |
| 1.7   | Hydrolysis of salts                                                                                                                                                                                                                   | 6.3  | Apply Le Chatelier's principle                                                                                                                                                            |
| 1.8   | Acid-base titration and indicators                                                                                                                                                                                                    | 6.4  | Explanation in terms of Le Chatelier's principle                                                                                                                                          |
| 1.9   | Definition of an oxidising agent                                                                                                                                                                                                      | 6.5  | $K_c$ calculation                                                                                                                                                                         |
| 1.10  | Spontaneous redox reaction                                                                                                                                                                                                            | 6.6  | Interpretation of changes made to reaction rate versus time graph                                                                                                                         |
| 2.1   | Isomers, types of isomers, general formula of alkynes, functional group of alcohols                                                                                                                                                   | 7.1  | Lowry-Brønsted theory, weak acids, conjugate bases                                                                                                                                        |
| 2.2   | IUPAC name of a haloalkane, alkyne and aldehyde                                                                                                                                                                                       | 7.2  | Calculate number of moles from concentration, concentration of base from pH; Stoichiometric calculation                                                                                   |
| 2.3   | Esterification, determination of molecular formula of an ester from its empirical formula.                                                                                                                                            | 8.1  | Identify the reducing agent and explanation in terms of relative strengths of the oxidising agents                                                                                        |
| 3.1   | Homologous series – ketones, controlled variable, explain the difference in the melting points of two ketones                                                                                                                         | 8.2  | Galvanic cell: Function of salt bridge, direction of electron flow, calculation of initial emf, balanced equation for net cell reaction, effect of ion concentration on voltmeter reading |
| 3.2   | Definition of vapour pressure, conclusion from results in a table, identification of alcohol with highest boiling point from given vapour pressures and writing of its IUPAC name, change in vapour pressure as temperature increases | 8.3  | Definition of a reducing agent                                                                                                                                                            |
| 4.1   | Tertiary haloalkanes                                                                                                                                                                                                                  | 8.4  | Reducing agent and cell reaction                                                                                                                                                          |
| 4.2   | Elimination of a haloalkane – reaction conditions, type of reaction that takes place, writing of balanced equation using structural formulae                                                                                          | 9.1  | Definition of electrolysis                                                                                                                                                                |
| 4.3   | Hydration of an alkene, conditions needed and structural formula of alcohol                                                                                                                                                           | 9.2  | Interpretation of graph to identify cathode, calculation of total charge transferred, and mass of metal deposited onto cathode                                                            |
| 4.4   | Hydrolysis of a haloalkane and conditions needed                                                                                                                                                                                      |      |                                                                                                                                                                                           |

## 11.6 ANALYSIS OF CANDIDATES' PERFORMANCE IN EACH QUESTION IN PAPER 2

### QUESTION 1: MULTIPLE CHOICE QUESTIONS

#### Common errors and misconceptions

- (a) Q1.1 was the subquestion answered the best. Most candidates were familiar with the term *saturated hydrocarbon*.
- (b) Q1.2 was well-answered. The most common incorrect answer was A, which implied that most candidates thought that the ester, methyl propanoate, also has hydrogen bonds like carboxylic acids.
- (c) Many candidates failed to select the correct answer in Q1. The saturation test to determine whether a hydrocarbon is saturated (alkane) or unsaturated (alkene), was not done in most schools.
- (d) Q1.4 was the poorest answered multiple-choice question. The most common incorrect answer was D. Candidates were tricked by the phrase *per unit time* not knowing that *rate of change* already implies *per unit time*.
- (e) Those candidates who got Q1.5 wrong mainly chose D (adding more Mg powder) as the answer. Such candidates failed to relate addition of more Mg to an increase in surface area. Many candidates did not know that volume has no effect on reaction rate and therefore did not choose A, the correct answer.
- (f) Q1.6 was well-answered. The most common incorrect answer was A. These candidates failed to recognise that at equilibrium the reaction has not stopped, but that both forward and reverse reactions proceed at the same rate. This implies that not all the 2 moles of HI (jar S) would have reacted and therefore it will not form 1 mole of I<sub>2</sub>.
- (g) Q1.7 tested hydrolysis. The salt of a strong acid and a strong base does not hydrolyse. The resulting salt will be neutral and will not affect the pH of water. NaCl is the salt formed when hydrochloric acid reacts with sodium hydroxide.
- (h) In Q1.8 candidates did not understand the phrase *at the equivalence point the pH is 7* and did not link it to a strong acid reacting with a strong base. They were therefore not able to identify the correct acid and suitable indicator.
- (i) In Q1.10 most candidates did not know how to use the Table of Standard Reduction Potentials to determine whether a reaction is spontaneous or not.

#### Suggestions for improvement

- (a) The answering of multiple-choice questions is a skill that needs to be developed. Learners must be guided to eliminate the wrong answers through regular practice and assessment. Practise multiple-choice questions regularly in short tests where learners can self-assess or peer mark. Learners must be told not to leave any multiple-choice question unanswered – they should rather take a guess if they do not know the answer.
- (b) Subject advisors should compile a booklet of multiple-choice questions arranged according to topics for schools. These questions can be used for weekly assessments. Through discussions, learners can then be shown how to approach such questions.

- (c) The types of intermolecular forces in all organic compounds must be thoroughly revised with learners before they are expected to explain why physical properties differ:
- London forces – (all organic compounds) – weakest
  - Dipole-dipole forces – (all polar organic compounds, i.e. haloalkanes, aldehydes, ketones, esters, alcohols, and carboxylic acids) – stronger than London forces but weaker than hydrogen bonds
  - Hydrogen bonds – (only alcohols and carboxylic acids) – strongest
- (d) When defining reaction rate, learners must be made aware that there are two ways of writing the definition, namely *change in concentration of reactants or products per unit time* or *rate of change in concentration of products or reactants*. When the second definition is used, the *rate of change* already implies *per unit time* and stating it again, will result in an incorrect definition.
- (e) When teaching reaction rate, learners should be made aware that increasing the volume of a solution does not affect the concentration and hence does not affect reaction rate.
- (f) The difference between a *reaction that reaches completion* and a *reaction that reaches equilibrium* must be well explained to learners. They must know that a reaction reaches completion because one or both reactants are used up. However, when a reversible reaction reaches equilibrium, the reaction has not stopped, but both forward and reverse reactions take place at the same rate. None of the reactants or products are used up.

In Q1.6 the equilibrium constant is the same for a specific reaction at a constant temperature. Using the process of elimination, option A can be eliminated because it is only true for a reaction reaching completion. To obtain 1 mole of  $I_2$ , 2 moles of HI in jar S must be converted to a maximum of 1 mole of  $I_2$ . Similarly, option B can be eliminated - if all the reactants in jar R are converted to products, then a maximum of 2 moles of HI will form which is not possible in an equilibrium reaction. For the equilibrium constant to be the same at this temperature, the ratio of the concentration of the products to concentration of reactants needs to be the same, as seen from the balanced equation given. Therefore, R and S will each contain the same concentrations of HI,  $I_2$  and  $H_2$ . Thus, jar R and S will contain the same amount of HI(g) at equilibrium.

- (g) When determining if a salt is acidic, basic, or neutral, learners must be taught to identify the positive and negative ion of the salt. Ions that originally come from strong acids and bases will not hydrolyse. The salt solution will be neutral and will not affect the pH of water. Ions from weak acids or bases will react with water (hydrolyse) to form  $OH^-$  (basic) or  $H_3O^+$  (acidic) solutions.
- (h) Learners must be taught which indicator is suitable based on the strength of an acid and a base. The table below gives the pH ranges for the three most common indicators.

| Indicator        | Strength of acid | Strength of base | pH range          |
|------------------|------------------|------------------|-------------------|
| Methyl orange    | Strong           | Weak             | 3,1–4,4 (acidic)  |
| Bromothymol blue | Strong           | Strong           | 6,0–7,6 (neutral) |
| Phenolphthalein  | Weak             | Strong           | 5,3–10 (basic)    |

- (i) The use of Tables 4A and 4B must be explained to learners. Expose learners to enough exercises in which they can practise how to identify the strongest reducing agent and the strongest oxidising agent. Also noteworthy is that oxidising agents appear to the left of the double arrows in both tables and reducing agents to the right. The spontaneous redox reaction will always be between the strongest reducing agent and the strongest oxidising agent.

When marking the relevant half-reactions in Q1.10 on Table 4B, it is easy to see that Mg is the strongest reducing agent followed by Mn etc.

Mg (strongest reducing agent) will reduce  $\text{Mn}^{2+}$  to Mn and Mg will reduce  $\text{Cd}^{2+}$  to Cd. Zn is a weaker reducing agent than Mn and will NOT reduce  $\text{Mn}^{2+}$  to Mn, but Zn is a stronger reducing agent than Cd and will reduce  $\text{Cd}^{2+}$  to Cd. Both Ni and Ag are weaker reducing agents than Mn and Cd and will not reduce  $\text{Mn}^{2+}$  to Mn and  $\text{Cd}^{2+}$  to Cd.

|                                                                                              |       |
|----------------------------------------------------------------------------------------------|-------|
| $\text{Mg}^{2+} + 2\text{e}^- = \text{Mg}$                                                   | -2,36 |
| $\text{Al}^{3+} + 3\text{e}^- = \text{Al}$                                                   | -1,66 |
| $\text{Mn}^{2+} + 2\text{e}^- = \text{Mn}$                                                   | -1,18 |
| $\text{Cr}^{2+} + 2\text{e}^- = \text{Cr}$                                                   | -0,91 |
| $2\text{H}_2\text{O} + 2\text{e}^- = \text{H}_2(\text{g}) + 2\text{OH}^-$                    | -0,83 |
| $\text{Zn}^{2+} + 2\text{e}^- = \text{Zn}$                                                   | -0,76 |
| $\text{Cr}^{3+} + 3\text{e}^- = \text{Cr}$                                                   | -0,74 |
| $\text{Fe}^{2+} + 2\text{e}^- = \text{Fe}$                                                   | -0,44 |
| $\text{Cr}^{3+} + \text{e}^- = \text{Cr}^{2+}$                                               | -0,41 |
| $\text{Cd}^{2+} + 2\text{e}^- = \text{Cd}$                                                   | -0,40 |
| $\text{Co}^{2+} + 2\text{e}^- = \text{Co}$                                                   | -0,28 |
| $\text{Ni}^{2+} + 2\text{e}^- = \text{Ni}$                                                   | -0,27 |
| $\text{Sn}^{2+} + 2\text{e}^- = \text{Sn}$                                                   | -0,14 |
| $\text{Pb}^{2+} + 2\text{e}^- = \text{Pb}$                                                   | -0,13 |
| $\text{Fe}^{3+} + 3\text{e}^- = \text{Fe}$                                                   | -0,06 |
| $2\text{H}^+ + 2\text{e}^- = \text{H}_2(\text{g})$                                           | 0,00  |
| $\text{S} + 2\text{H}^+ + 2\text{e}^- = \text{H}_2\text{S}(\text{g})$                        | +0,14 |
| $\text{Sn}^{4+} + 2\text{e}^- = \text{Sn}^{2+}$                                              | +0,15 |
| $\text{Cu}^{2+} + \text{e}^- = \text{Cu}^+$                                                  | +0,16 |
| $\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- = \text{SO}_2(\text{g}) + 2\text{H}_2\text{O}$ | +0,17 |
| $\text{Cu}^{2+} + 2\text{e}^- = \text{Cu}$                                                   | +0,34 |
| $2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- = 4\text{OH}^-$                              | +0,40 |
| $\text{SO}_2 + 4\text{H}^+ + 4\text{e}^- = \text{S} + 2\text{H}_2\text{O}$                   | +0,45 |
| $\text{Cu}^+ + \text{e}^- = \text{Cu}$                                                       | +0,52 |
| $\text{I}_2 + 2\text{e}^- = 2\text{I}^-$                                                     | +0,54 |
| $\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^- = \text{H}_2\text{O}_2$                    | +0,68 |
| $\text{Fe}^{3+} + \text{e}^- = \text{Fe}^{2+}$                                               | +0,77 |
| $\text{NO}_3^- + 2\text{H}^+ + \text{e}^- = \text{NO}_2(\text{g}) + \text{H}_2\text{O}$      | +0,80 |
| $\text{Ag}^+ + \text{e}^- = \text{Ag}$                                                       | +0,80 |

## QUESTION 2: ORGANIC NOMENCLATURE

### Common errors and misconceptions

- (a) In Q2.1.4 many candidates wrote the name of the homologous series instead of the name of the functional group. It was also evident that many candidates did not know the difference between a *hydroxyl group* ( $-\text{OH}$ ) and the *hydroxide ion* ( $\text{OH}^-$ ).
- (b) When writing IUPAC names in Q2.2, many candidates failed to apply the rules correctly. Hyphens, commas, and alphabetical order were not adhered to, and marks were forfeited. Some candidates forgot to put a hyphen between a number and a letter or a comma between two numbers.

Most learners used incorrect numbering in Q2.2.1. The halogen was incorrectly given preference over the alkyl groups. When determining the parent chain, many candidates did not consider the longest chain and named the compound 3-bromo-2-ethyl-2-methylpentane. These candidates forfeited 3 marks.

In Q2.2.2 many candidates forfeited a mark because they did not give preference to the functional group resulting in incorrect numbering. Some used *pentyl* instead of *pentyne*.

Some candidates numbered the functional group of the aldehyde in Q2.2.3 and forfeited a mark.

- (c) Although most candidates identified the reaction in Q2.3.1 correctly as an *esterification*, many did not even attempt Q2.3.2. Candidates seemed to be unfamiliar with the concept *empirical formula*. Many of those who attempted the question, obtained the molecular formula of compound S, the ester, and stopped there. They did not read/understand the entire question correctly because they had to use the molecular formula of compound S (the ester) to determine the molecular formula of the carboxylic acid. Such candidates forfeited two marks.

### Suggestions for improvement

- (a) When writing general formulae of the different homologous series, learners must be made aware that the correct subscripts are crucial. Instead of writing  $C_nH_{2n-2}$ , many candidates wrote one of the following incorrect answers:  
 $C_nC2N-2$ ,  $CnC2N-2$ ,  $CxH2x - 2$ ,  $Cn + H2n+2$ ,  $Cn 2n-2$ ,  $Cn H2-2$ ,  $CHn-2$ ,  $CNH2N-2$
- (b) When writing IUPAC names, the following should be emphasised:
- Correct use of hyphens (between a number and a letter) and commas (between two numbers).
  - The prefixes, *di*, *tri*, etc. are used to indicate more than one of the same type of substituents.
  - In aldehydes and carboxylic acids, the functional group is always on C-1 and numbering must not be used to indicate the position of the functional group, e.g. it should be *butanal* and not *butan-1-al*.
  - The functional group of haloalkanes do not get preference over the alkyl groups. Therefore, numbering should not necessarily start from the side closest to the halogen. Instead, numbering should start from the side giving substituents, halogen and alkyl, the lowest total number.
  - Numbering should always be from the side giving substituents the lowest number, for example when deciding between 4-bromo-3,3-dimethyl or 3-bromo-4,4-dimethyl, the numbers of each should be added to determine the lowest total. 4-bromo-3,3-dimethyl:  $4 + 3 + 3 = 10$ ; bromo-4,4-dimethyl:  $3 + 4 + 4 = 11$ . Therefore, the lowest total is 10 and numbering should be such that the name is 4-bromo-3,3-dimethyl.
  - Substituents must be written in alphabetical order in IUPAC names regardless of their position in the longest chain. Numbers of substituents cater for the position of substituents in the longest chain.
- (c) When drawing structural formulae, learners should be encouraged to count the number of bonds drawn around atoms to eliminate unnecessary errors. They need to be reminded that a carbon atom cannot have less than or more than 4 bonds around it.

- (d) Frequent informal tests should be used to ensure that learners write definitions correctly. Often, learners write different interpretations of a definition, and they usually end up with incorrect or partially correct statements.
- (e) Emphasise the difference between empirical formulae and molecular formulae.

### QUESTION 3: PHYSICAL PROPERTIES OF ORGANIC COMPOUNDS

#### Common errors and misconceptions

- (a) Q3.1.2 involved identification of variables and was not well answered. *Temperature* was the most common incorrect controlled variable selected. Other incorrect responses were *molecular weight* or *chain length* or *type of compound*.
- (b) Lack of skills to interpret negative values for melting points (Q3.1.3) cost many candidates marks. They did not know that  $-92\text{ }^{\circ}\text{C}$  is lower than  $-72\text{ }^{\circ}\text{C}$ . Many stated general facts instead of explaining the answer by comparing compounds C and D. In most instances they contradicted themselves and forfeited marks.  
Other common errors were:
  - Referring to *boiling point* instead of *melting point*;
  - Stating the correct intermolecular forces without comparing their strengths;
  - Referring to *breaking of bonds* instead of *overcoming of intermolecular forces*;
  - Omitting the last step that refers to the energy needed.
- (c) Some candidates wrote the definition of *boiling point* instead of *vapour pressure* in Q3.2.1. Those who wrote the definition for *vapour pressure*, omitted key words such as *equilibrium* or *closed system*.
- (d) The term *conclusion* (Q3.2.2) seemed to be unfamiliar to many candidates. Some wrote an aim or a hypothesis or even an investigative question. Some of those who wrote a conclusion, stated an inverse proportionality between chain length and vapour pressure. Others identified the dependent and independent variables incorrectly, e.g. giving *molecular mass* as the independent variable instead of *chain length*.
- (e) Common errors when writing the IUPC name of the alcohol in Q3.2.3 were: hexan-1-ol, hex-1-ol, hexane-1-ol, hexanol.

#### Suggestions for improvement

- (a) Most definitions are stated in the *Examination Guidelines* and teachers must ensure that learners study the definitions through regular informal tests.
- (b) Practical skills need to be taught from Grade 10 to Grade 12. Learners lack skills such as identification of variables and writing of conclusions. The difference between an investigative question, a hypothesis and a conclusion should be thoroughly explained.

Advise learners to explain the relationship between variables, such as *boiling point* and *vapour pressure*, in terms of *increase or decrease* and not in terms of *proportionality*. When mentioning a direct proportionality, it means if values are plotted, a straight line passing through the origin must be obtained which is not the case in these relationships.

- (c) Teachers must expose learners to tables with negative values for temperatures and explain to them that a greater negative value implies a lower temperature.
- (d) When discussing intermolecular forces, the concepts *atoms*, *molecules*, and *ions* should be revised. Emphasise the differences between: bonds between atoms in molecules (intramolecular) and forces between molecules (intermolecular) and that the strength of intermolecular forces is responsible for the different phases. Interatomic or intramolecular forces, which are much stronger than intermolecular forces, are formed or broken during chemical reactions when new compounds are formed. Intermolecular forces are overcome (not broken) during phase changes.
- (e) When writing explanations related to physical properties of compounds, learners should be taught to follow the following steps:
  - Comparing two compounds from the same homologous series:
    - Compare the surface areas/chain length of the molecules.
    - Compare the strength of intermolecular forces
    - Compare the energy needed to overcome intermolecular forces
  - Comparing two compounds from different homologous series:
    - State the type of intermolecular force in each compound
    - Compare the strength of these intermolecular forces
    - Compare the energy needed to overcome intermolecular forces

#### QUESTION 4: REACTIONS OF ORGANIC COMPOUNDS

##### Common errors and misconceptions

- (a) In Q4.1 candidates referred to the compound as a *tertiary alcohol* instead of a *tertiary haloalkane*. Instead of stating that a tertiary haloalkane is where the halogen/bromine/functional group is bonded to a C atom that is bonded to three other C atoms, some candidates stated that the halogen is bonded to three C atoms. Many referred to *halide* instead of *halogen* or *bromide* instead of *bromine*.
- (b) Reaction conditions of organic reactions (Q4.2.1, Q4.2.3 and Q4.4.2) were a challenge to most candidates. Those who knew that the inorganic reactant in Q4.2.1 and Q4.4.2 is NaOH, omitted the words *concentrated* and *dilute* respectively and forfeited the two marks.
- (c) Apart from choosing the incorrect inorganic reactant in Q4.2.3, other common errors when writing the balanced equation were:
  - Drawing the structural formula of a product with 5 C atoms in the chain (with no substituent) whereas the reactant had 4 C with an alkyl substituent;
  - Omission of the arrow;
  - Writing structural formulae for only reactants or only products;
  - Writing the structural formula of the minor product;
  - Using molecular or condensed structural formulae instead of structural formulae in the equation;
  - Omitting the inorganic reactant in the equation and thus writing the reaction as haloalkane  $\rightarrow$  alkene + HBr;
  - Placing the double bond in the product after C-1 instead of after C-2.

### Suggestions for improvement

- (a) Explain to learners that (-Br) is not a haloalkane, but a halogen. The halogen together with the C it is bonded to (with correct number of bonds), form the functional group. The presence of a halogen makes the whole molecule a haloalkane. This must also be explained for tertiary alcohols. The terms *primary*, *secondary* and *tertiary* refer to the number of C atoms (one, two and three respectively) bonded to the C atom of the functional group.
- (b) Reaction conditions needed for organic reactions to take place need more attention. The following table summarises reaction conditions for some organic reactions.

| Type of organic reaction | Name of reaction    | Reactants                         | Reaction conditions                                                                                             | Products                                |
|--------------------------|---------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Substitution             | Halogenation        | alkane + halogen                  | hf/ $\Delta$                                                                                                    | haloalkane + hydrogen halide            |
|                          | Hydrolysis          | haloalkane + base (NaOH/KOH/LiOH) | <ul style="list-style-type: none"> <li>dilute strong base</li> <li>mild heat</li> </ul>                         | alcohol + halide salt                   |
|                          |                     | haloalkane + water                | <ul style="list-style-type: none"> <li>add water</li> <li>mild heat</li> </ul>                                  | alcohol + hydrogen halide               |
|                          | Substitution        | alcohol + hydrogen halide         | no water                                                                                                        | haloalkane + water                      |
| Addition                 | Hydration           | alkene + water                    | concentrated $\text{H}_2\text{SO}_4$                                                                            | alcohol                                 |
|                          | Hydrogenation       | alkene + hydrogen                 | Pt, Pd, Ni; dissolved in non-polar solvent                                                                      | alkane                                  |
|                          | Halogenation        | alkene + halogen                  |                                                                                                                 | haloalkane                              |
|                          | Hydrohalogenation   | alkene + hydrogen halide          | no water                                                                                                        | haloalkane                              |
| Elimination              | Cracking            | alkanes                           | high temperature and high pressure OR catalyst                                                                  | alkene + hydrogen OR alkane + alkene(s) |
|                          | Dehydrohalogenation | haloalkane + base (NaOH/KOH)      | <ul style="list-style-type: none"> <li>concentrated strong base</li> <li>heat</li> </ul>                        | alkene + salt + water                   |
|                          | Dehydration         | alcohol                           | <ul style="list-style-type: none"> <li>concentrated <math>\text{H}_2\text{SO}_4</math></li> <li>heat</li> </ul> | alkene + water                          |

- (c) It is important to explain the difference between a *formula* and an *equation*. Learners sometimes write structural formulae of reactants or products when they are supposed to give the entire equation (using structural formulae). An equation must also have an arrow between reactants and products.
- (d) Difference between *minor* and *major* products must be explained clearly and structural and condensed formulae drawn clearly for learners to see the difference and how they are drawn.

**QUESTION 5: REACTION RATE****Common errors and misconceptions**

- (a) In Q5.1 most candidates correctly identified *B* as the reaction at the higher rate due to the presence of the catalyst but failed to state the function of the catalyst and give a correct explanation in terms of the collision theory. Many candidates omitted key words in the explanation, e.g. instead of *more effective collisions per unit time* candidates wrote the following partially correct phrases:
- 'Effective collisions per unit time';
  - 'More collisions per unit time';
  - 'More effective collisions'.
- (b) Most candidates failed to interpret that when no more oxygen is produced (Q5.4.1), the rate will be  $0 \text{ g}\cdot\text{s}^{-1}$ . Common errors were:
- Using the mass of 0,9 g given on the graph as the rate;
  - Stating that the reaction reached equilibrium (These candidates did not know that only a reversible reaction can reach equilibrium.);
  - The reaction stopped or was completed without stating what happened to the rate of the reaction.
- (c) In Q 5.3.2 many candidates used an incorrect formula to calculate the number of moles. Other common errors were:
- Substituting the volume of  $\text{H}_2\text{O}$ , a liquid, in  $n = \frac{V}{V_M}$  which is only for gases;
  - Using  $V_M = 22,4 \text{ dm}^3\cdot\text{mol}^{-1}$  instead of the given  $24\,000 \text{ cm}^3\cdot\text{mol}^{-1}$ ;
  - Omitting the mole ratio step;
  - Using the volume ratio of water (a liquid) and the  $\text{O}_2$ .
- (d) The calculation in Q5.4.3 was poorly answered. Common errors were:
- Substituting the  $M(\text{O}_2)$  as  $16 \text{ g}\cdot\text{mol}^{-1}$ ;
  - Omission of mole ratio step;
  - Using  $22,4 \text{ dm}^3\cdot\text{mol}^{-1}$  as the molar gas volume and not the value given in the question paper.

**Suggestions for improvement**

- (a) Explanations in terms of the *collision theory* need improvement. Learners must be taught to write a full explanation including the necessary key words.

When a catalyst is added:

- The activation energy is lowered.
- More particles have kinetic energy equal to or higher than the activation energy/More particles have sufficient kinetic energy.
- More effective collisions per unit time.

It is important to note that the catalyst does not increase the kinetic energy of the molecules. The catalyst lowers the energy needed (activation energy) and therefore, more particles have sufficient kinetic energy.

- (b) Emphasis should be placed on the difference between *reversible* and *non-reversible* reactions. Reversible reactions can reach equilibrium, while non-reversible reactions take place in one direction and cannot reach equilibrium. Furthermore, a reversible reaction can only reach equilibrium in a closed system. If gases are released it implies that the container should be closed to prevent the gases from escaping. When written,

equilibrium reactions are indicated with double arrows. Single arrows indicate non-reversible reactions.

- (c) *Stoichiometry* will always be an important part of a Chemistry paper. The understanding of *stoichiometry* should be a continuous theme in the teaching of Chemistry. The more thoroughly it is done in Grades 10 and 11, the more successful candidates will be in Grade 12.

Subject advisors need to support teachers with *stoichiometry* and worksheets should be designed involving calculations on percentage purity, percentage yield and limiting reactants. The revision booklet designed by the DBE can be valuable in this regard. Learners have a poor understanding of stoichiometry and are very uncertain when selecting formulae for a specific calculation.

- (d) It should be emphasised that the molar gas volume of  $22,4 \text{ dm}^3 \cdot \text{mol}^{-1}$  is only applicable for gases to STP.
- (e) Remind learners of how to calculate diatomic substances' molar mass, e.g. oxygen's molar mass is  $32 \text{ g} \cdot \text{mol}^{-1}$  and not  $16 \text{ g} \cdot \text{mol}^{-1}$ .

## QUESTION 6: CHEMICAL EQUILIBRIUM

### Common errors and misconceptions

- (a) When stating *Le Chatelier's principle* (Q6.1) some candidates omitted the phrase *closed system*. Others common errors were:
- The *reaction is disturbed* instead of *equilibrium is disturbed*;
  - Favouring the reaction that will *cancel the motion* instead of *cancelling the disturbance*;
  - Stating that the reverse reaction will be favoured.
- (b) Q6.2 was poorly answered. Common errors were:
- Substituting number of moles of  $\text{CS}_2$  in the  $K_c$  expression;
  - Rounding off the final answer to one decimal place (0,2 instead of 0,23  $\text{mol} \cdot \text{dm}^{-3}$ );
  - No  $K_c$  expression e.g.,  $K_c = \frac{[\text{products}]}{[\text{reactants}]}$ ;
  - $K_c$  expressions containing writing errors, such as  $K_c = \frac{(\text{CS}_2)}{(\text{S})^2}$ ,  $K_c = \frac{[\text{CS}]}{[\text{S}]^2}$  or  $K_c = \frac{[\text{CS}_2]}{[\text{S}]}$ ;
  - Using incorrect  $K_c$  expressions, i.e.  $K_c = \frac{[\text{CS}_2]}{[\text{C}][\text{S}]^2}$  (solid included) or  $K_c = \frac{[\text{S}]^2}{[\text{CS}_2]}$  (swapping reactant and product)
- (c) Most candidates got the answer correct in Q6.3 but could not explain this answer in Q6.4. Candidates failed to link the increase in volume to a decrease in pressure. Therefore, some candidates explained the answer in terms of a change in temperature or a change in concentration. Other common errors were:
- Using the *collision theory* instead of *Le Chatelier's principle* in the explanation.
  - Stating *Le Chatelier's principle* instead of applying it when explaining.
- (d) Q6.5 was the poorest answered subquestion in the paper. Many of those candidates who attempted the question, did not realise that the volume decreased by half and thus the initial concentrations for the new situation must be divided by two. Others did not realise that there were reagents initially present in the container.

- (e) Most candidates failed to interpret the graph in Q6.6. They confused the rate versus time graph with a concentration versus time graph.

A common error when answering Q6.6.2 was not mentioning *increase in concentration or amount of S*. Most only stated *increase in concentration* and forfeited the mark.

In Q6.6.3 many candidates could not link the change at  $t_c$  to a drop in temperature and hence the explanation in Q6.6.4 was also poor.

### Suggestions for improvement

- (a) Explain to learners why solids and pure liquids should not be included in  $K_c$  expressions and expose them to enough exercises to practise writing of  $K_c$  expressions for reactions where reactants and products are in different phases. Avoid using  $K_c = \frac{[\text{products}]}{[\text{reactants}]}$  in class creating the impression that it is a correct  $K_c$  expression.
- (b) Learners should be taught to label formulae when doing multistep calculations, e.g. when calculating the concentration of  $\text{CS}_2$ , it should be indicated as  $c(\text{CS}_2)$ .
- (c) When using a table to solve  $K_c$  calculations, learners should be taught to use correct labels in the first column, i.e.  $n(\text{initial})$ ,  $n(\text{change})$ ,  $n(\text{equilibrium})$ ,  $c(\text{equilibrium})$  and to write the correct values of these for each reactant or product in the correct cell. Use previous marking guidelines to show learners the labelling in such tables.
- (d) Learners should be exposed to many exercises requiring explanations in terms of *Le Chatelier's principle*. This will assist them in expressing themselves in such explanations. When explaining in terms of *Le Chatelier's principle*, learners should be taught to use the following steps:
- Identify the disturbance.
  - State that the system will act to oppose this disturbance.
  - State which reaction (forward or reverse) will be favoured when opposing the disturbance.
  - State the effect on, for example the number of moles of products.
- (e) Ensure learners understand and can interpret the various graphs (rate vs time and concentration vs time) that relate to changes in equilibrium conditions.
- (f) Use the revision booklet designed by the DBE to support learners.

## QUESTION 7: ACIDS AND BASES

### Common errors and misconceptions

- (a) Although the definition of an acid in Q7.1.1 was well answered, some candidates referred to *photons* instead of *protons* or to *hydrogen* instead of *hydrogen ions* or *hydronium ions*. Others wrote the definition for the Arrhenius theory.
- (b) A common error in Q7.1.2 was the use of the phrase *acids that dissolve completely* instead of *acids that ionise or dissociate completely*. Many candidates used one of the following incorrect reasons for ethanoic acid being a weak acid:
- It has a low hydronium ion concentration.
  - It is used in the house or in food.
  - It can be swallowed.

- (c) Instead of writing the formulae of the two bases (Q7.1.3), some candidates wrote the formulae of one of the conjugate acid-base pairs or copied the whole equation without indicating the two bases.
- (d) In Q7.2.1 some candidates calculated *mass* instead of *number of moles*. The unit of the amount of substance, namely *moles*, also seemed to be unfamiliar to some. Many candidates substituted the values correctly in the correct formula, but failed to find the concentration due to lack of calculator skills.
- (e) Some candidates failed to copy the pH formula correctly from the data sheet in Q7.2.2. Other common errors were:
- Using an incorrect pH formula, e.g.  $\text{pH} = -\log[\text{H}_3\text{O}^+]$  or  $\text{pH} = -\log(\text{H}_3\text{O}^+)$  or  $\text{pH} = -\log[\text{H}_3\text{O}^{++}]$  or  $\text{pH} = -\log[\text{OH}^-]$ ;
  - Giving the concentration of  $\text{H}_3\text{O}^+$  as the final answer without calculating  $[\text{OH}^-]$ ;
  - Omitting the unit of concentration at the final answer or using an incorrect unit, e.g.  $\text{mol}\cdot\text{dm}^3$
- (f) In Q7.2.3 most candidates did not calculate the  $n(\text{NaOH})_{\text{reacted}}$ . They could not identify the NaOH as being in excess and assumed all the NaOH present initially reacted and used the mole ratio to obtain  $n(\text{CH}_3\text{COOH})$  from  $n(\text{NaOH})_{\text{initial}}$ . These candidates forfeited five of the six marks. Some tried to use the equilibrium table. Most learners could not calculate the change in number of moles of  $\text{CH}_3\text{COOH}$ .

### Suggestions for improvement

- (a) Learners must be taught to convert units correctly, e.g. from  $\text{cm}^3$  to  $\text{dm}^3$ . Two marks are forfeited in a 3-mark question due to incorrect conversions.
- (b) Learners should be taught to copy formulae correctly from the data sheet. Manipulation of formulae can be done in a second step. Starting with an incorrect formula can result in no marks being awarded.
- (c) Ensure that learners copy the pH as it is from the data sheet and teach the calculator manipulation skills to obtain the correct concentration of hydronium or hydroxide ions when a pH is given. Include exercises involving the use of the  $K_w$  formula in daily assessments.
- (d) Ensure that stoichiometric calculations are properly taught in Grade 11. Expose learners to stoichiometric calculations involving limiting reagents from the beginning of their Grade 12 year to give them enough practice.
- (e) Learners should be taught to label formulae when doing multistep calculations, e.g. when calculating the number of moles of NaOH, the formula should be as follows:  $n(\text{NaOH}) = cV$
- (f) The formula  $\frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}$  must only be used when the question refers to an equivalent point/end point/neutralisation or to a colour change. This equation is usually used during an acid-base titration and not when an acid and a base are just added together and one of them is in excess. Rather use mole ratios and calculations involving  $n = cV$  instead of the titration formula.
- (g) Rounding off should only be done at the final answer of a calculation. Learners should be taught not to round off in each step as it leads to an incorrect answer.

## QUESTION 8: REDOX REACTIONS AND GALVANIC CELLS

### Common errors and misconceptions

- (a) In Q8.1.1 many candidates wrote the formula of an oxidising agent  $\text{Zn}^{2+}$  instead of the reducing agent Zn. Some wrote the oxidation or reduction half-reaction.
- (b) Many candidates failed to use the *Table of Standard Reduction Potentials* to explain why a redox reaction took place in Q8.1.2. The relative positions of oxidising agents and reducing agents on the table was not well understood and Zn was used as oxidising agent in the explanation. Instead of stating, e.g. that  $\text{MnO}_4^-$  is a stronger oxidising agent than  $\text{Zn}^{2+}$ , some candidates were penalised for using words such as *MnO<sub>4</sub><sup>-</sup> is lower than/bigger than ...* Some candidates did not limit their explanation to relative strengths of oxidising agents as required.
- (c) The function of the salt bridge in Q8.2.1 was misunderstood by some. Common incorrect answers were:
- A pathway for movements of electrons to ensure electrical neutrality;
  - Neutralises the cell/electrons/reaction;
  - Connects the two half-cells;
  - Completes the cell.
- (d) When calculating cell potential (Q8.2.3) common errors were:
- Using abbreviations in the formula, e.g.  $E_{\text{cell}} = E_{\text{red}} - E_{\text{ox}}$  or  $E_{\text{cell}} = E_{\text{oxidising}} - E_{\text{reducing}}$
  - Swapping the reduction potential of the anode with that of the cathode when substituting
  - No unit at the final answer
- (e) In Q8.2.4 some candidates wrote the *cell notation* instead of the *cell reaction*. Some candidates wrote oxidising agents on one side of the arrow and reducing agents on the other side. A few candidates used incorrect symbols for the elements, i.e.  $\text{MN} + \text{NI}^{2+} \rightarrow \text{MN}^{2+} + \text{NI}$  or omitted the charges on the ions in an otherwise correct equation. Some of those who wrote the correct equation, left it unbalanced.

### Suggestions for improvement

- (a) Concepts such as reducing agent, oxidation, oxidising agent, and reduction should be taught with understanding. Learners must be able to identify, e.g. the reducing agent in a reaction. Regular assessment on this identification is needed to ensure that learners fully understand these concepts and how to identify them on the *Table of Standard Reduction Potentials*.
- (b) Learners should be taught how to use the *Table of Standard Reduction Potentials* to identify the *anode, cathode, reducing agent, oxidising agent, reduction half-reaction and oxidation half-reaction in a galvanic cell*. Regular assessment on this identification is needed to ensure that learners understand the use of the *Table of Standard Reduction Potentials*.
- Learners should also be taught how to use the *Table of Standard Reduction Potentials* in explanations in terms of the relative strength of oxidising or reducing agents.
- (c) Teachers should ensure that learners study and understand the section on the effect of a change in concentration on the cell potential in the *Examination Guidelines*. The answer to a question like Q8.2.5 is provided in the phrase  *$V_{\text{cell}}$  decreases as the*

concentration of product ions increases and the concentration of reactant ions decreases until equilibrium is reached at which the  $V_{\text{cell}} = 0$ .

## QUESTION 9: ELECTROLYTIC CELLS

### Common errors and misconceptions

- (a) Although the definition of *electrolyses* was well answered in Q9.1, some candidates wrote the definition of an *electrolyte* instead.
- (b) In Q9.2.1 some candidates wrote the word *reduction* instead of the reduction half-reaction that took place at the electrode T. Many candidates wrote the half-reaction for  $\text{Cr}^{2+}$  instead of the half-reaction for  $\text{Cr}^{3+}$ . A double arrow was also used by some of those who wrote an otherwise correct half-reaction.
- (c) Candidates made two vital errors when answering Q9.2.2. Many did not convert the 10 hours to seconds by multiplying it by 3 600 and some omitted the unit of charge in the final answer.
- (d) Q9.2.3 was very similar to the last question in the 2021 paper and those who worked through that paper answered this question well. Common errors were:
  - Using an incorrect or no mole ratio when calculating the number of electrons transferred.
  - Multiplying the number of moles of chromium by the charge on one electron instead of first using the ratio to calculate the number of moles of electrons followed by calculating the number of electrons using Avogadro's number.
  - Substituting number of moles of electrons as  $n$  in  $n = \frac{Q}{q}$  (In this formula,  $n$  is the number of electrons.).

### Suggestions for improvement

- (a) The difference between the definitions of *electrolysis*, an *electrolytic cell* and an *electrolyte* should be emphasised. An *electrolyte* is a solution that conducts electricity through the movement of ions whilst *electrolysis* is a process during which electrical energy is converted to chemical energy.
- (b) When copying either the oxidation or the reduction half-reaction from the *Table of Standard Reduction Potentials*, single arrows should be used to represent either the oxidation or the reduction.
- (c) Learners must be taught how to select half-reactions, especially for those substances with more than one half-reaction, e.g.  $\text{Cr}^{3+}|\text{Cr}$ ,  $\text{Cr}^{2+}|\text{Cr}$  and  $\text{Cr}^{3+}|\text{Cr}^{2+}$ . Ionisation of ionic compounds must be revised. If a learner knows that ionisation of  $\text{CrCl}_3$  forms  $\text{Cr}^{3+}$  and  $3\text{Cl}^-$  then it will be easier to choose the correct half reaction. More assessment must be given using the substances that appear more than once in the *Table of Standard Reduction Potentials*.
- (d) Grade 10 and 11 *stoichiometry* must be revised in Grade 12 as *stoichiometry* is still a huge challenge.
- (e) Use of the formulae  $n = \frac{Q}{q_e}$  ( $n$  = number of electrons),  $n = \frac{N}{N_A}$  ( $n$  = number of moles) must also be revised thoroughly.

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